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Ancestry in the Margins: Justice,
Genomics, and the Ethics of Human
Classification

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This thesis is submitted for the degree of Doctor of Philosophy.

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Abstract

This thesis examines how concepts of race, ethnicity, and ancestry influence categorisation in genomic research, drawing on an ethnographic study conducted at the Wellcome Sanger Institute. Positioned within critical race theory and critical realism, it explores how institutional structures shape both ethical challenges and opportunities for achieving genomic equity. Grounded in the researcher's positionality as a Black genetic counsellor, the study critically interrogates how ideas of human difference are constructed, managed, and operationalised within scientific spaces, and how social justice might be more deeply embedded within genomic research practices.

The research investigates the persistence of racialised ideas despite formal scientific rejections of race as a biological category.

Findings reveal that "data diversity" initiatives, while framed as mechanisms for improving patient outcomes, also confer material, reputational, and economic benefits to institutions. Scientists often navigate competing demands, balancing the pursuit of technical excellence, ethical imperatives, and the institutional structures that shape research priorities. Boundary work to produce genetic ancestry datasets enables both the reproduction of historical inequities and the creation of new forms of exclusion through standardised group-making practices.

Through analysis of interviews, publications, and informal aspects of campus life, the thesis shows that racial ideas are often reintroduced at key stages in the research pipeline, from participant recruitment to dataset construction and clinical application, usually without explicit reflection on their ethical or political implications.

This study argues that achieving equitable genomic outcomes and benefits for all populations cannot be accomplished solely through data diversification. Institutions must recognise the power and autonomy they possess to shape research practices and engage honestly with their own interests and limitations.

By reconfiguring data diversity efforts to focus not only on hopes for patient outcomes but also on structural accountability, transparency, and justice, this work seeks to make the social systems that shape genomic knowledge visible. It invites genomic institutions to go beyond inclusion as a symbolic act and move toward a transformative model of genomic equity.

The thesis makes two contributions: it demonstrates how inequities in genomic science are produced through institutional and methodological practice, and it proposes a framework for embedding justice into genomic research from the outset.

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I am deeply thankful to my village, whose support sustained me through a path I forged by refusing to accept that things must always be as they have been. Those who held my hand and my heart. Their belief, care, and steadiness made it possible to keep going.

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'In my little fig tree, is where I wanna be, so safe and free'

Bunny Wailer, *Fig Tree*

1 Introduction

I grew up in Newham, a working-class borough of East London and one of the most ethnically diverse places in the United Kingdom (Regional Ethnic Diversity, 2022). Cultural, racial and ethnic diversity formed the landscape of my early life. It shaped how I understood belonging, community, and the social conditions that determine whose identities are affirmed and whose are marginalised. As I moved into adulthood, I encountered environments where racial and cultural differences were interpreted in ways unfamiliar to me. What had been normal in Newham became, in other settings, a source of exclusion or misrecognition. These shifts sharpened my commitment to understanding how racialisation operates, and to challenging the forms of inequity that follow.

My professional training as a genetic counsellor deepened these concerns. Within clinical genetics, my presence as a Black woman was unusual, both among colleagues and among the patients we served. The genomic data that informed clinical practice reflected similar absences. Over time, four questions came to animate my work. Where is the science needed to support *Black* and other minoritised communities? Why does the genetic counselling workforce lack ethnic diversity? What does science need to do to serve all communities fairly? And what questions could we ask of culturally diverse communities to support their healthcare? These questions form the basis of the inquiry presented in this thesis.

This study was conducted at the Wellcome Sanger Institute (Sanger/ The Institute) (3.8.1), a globally influential centre for large-scale genomic research. The Institute was one of the central sequencing centres in the Human Genome Project (HGP) and played a decisive role in making the project possible at scale. Sanger's role in generating population-scale genomic datasets and its long-standing involvement in shaping scientific approaches to human variation make it a particularly significant site for examining how population descriptors, scientific responsibility, and notions of equity are negotiated in practice. The Institute provided both the organisational setting for this research and an intellectual environment in which questions about "diversity", "inclusion", and scientific good practice were already under active discussion.

Within a British context, discussions of race in science are often overshadowed by narratives shaped in the United States (US). Although entangled with global forces, race in Britain has its own distinctive formation rooted in empire, migration, statecraft, and resistance. Much of the scholarship on population categories, genetics, and racial inequalities draws from the American experience. A British perspective, therefore, offers an opportunity to examine how racial categories are taken up, resisted, and naturalised in scientific practice outside the dominant US frame.

1.1 Defining Race, Ethnicity, and Ancestry for this thesis

The terms race, ethnicity, and ancestry (REA) require clarity given their conceptual instability. I have drawn together and will be using definitions of these concepts as summarised in Table 1. For context, race refers to social categories organised by perceived physical characteristics and geographic identity (Delgado & Stefancic, 1998). Ethnicity describes cultural and perceived geographic origins connected through shared practices, language, or history (Birney et al., 2021). Genetic ancestry refers to lineage inferred through genomic markers. It is distinct from race or ethnicity, although often correlated in ways that can mislead interpretation (Dauda et al., 2023; A. Lewis et al., 2022). Genetic similarity refers to the measurable degree of overlap in genetic variants between individuals or groups. It reflects statistical resemblance in the genome rather than social or cultural identity. Genetic similarity does not imply discrete biological boundaries (Fujimura & Rajagopalan, 2011; National Academies of Sciences, 2023).

All these terms create socially meaningful categories, but only genetic similarity is biologically measurable. These terms are therefore used critically throughout this thesis, recognising both their social importance and their potential to entrench health inequities if used without reflexivity. Labels used to denote race, ethnicity, and ancestry groups are capitalised and italicised when not used in a personal context, either by the author or within quoted identity descriptors. This formatting reflects the treatment of these terms as socially and politically constructed categories rather than neutral descriptors. The conventions adopted in this thesis recognise that such labels carry historical, colonial, and juridical significance, and that their use is shaped by power relations, institutional practices, and political struggle. An exception is made for ‘whiteness’, which is rendered

in lowercase and without italics to reflect its operation as an unmarked and normative position of power rather than as a discrete identity category. Where terms appear in quoted material, original capitalisation and formatting are retained to preserve the positionality and intent of the source.

Although diversity and inclusion can encompass a wide range of individual and group attributes, given the focus of this thesis, unless otherwise qualified, diversity and inclusion here are generally framed through the lens of race, ethnicity, and ancestry.

Table 1. Definitions of common population terms used in this review, Adapted from Birney et al., 2021; Dauda et al., 2023; Delgado & Stefancic, 1998; Fujimura & Rajagopalan, 2011; A. Lewis et al., 2022; National Academies of Sciences, 2023.

Term	Definition (as used in this report)	Examples / Notes
Race	A socially constructed grouping based on perceived physical or social characteristics, sometimes tied to political identity. Not a reliable biological proxy.	<i>Black, White, Asian</i> (individuals may have disparate geographic origins).
Ethnicity	A grouping defined by shared culture, language, or recent geographic origin; may intersect with but is distinct from <i>race</i> .	<i>Afro-Caribbean, South-East Asian, Indian.</i>
Genetic ancestry	A grouping based on measurable genetic lineage or relatedness; demonstrable through genetic testing. Correlates with but is not equivalent to <i>race</i> or <i>ethnicity</i> .	May indicate the proportion of genomic variants associated with regional or ancestral populations.
Genetic similarity	A grouping based on statistical measures of shared genetic variation between individuals or populations. Used to infer relatedness or structure	Individuals or samples that cluster together in statistical analyses; often used to generate ancestry groupings in genomic datasets.

	without implying social or cultural identity.	
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1.2 Positionality and the Epistemic Commitments of Research

Early in my doctoral research, I met a senior figure in British medical genetics, introduced to me as someone “sympathetic” to questions of inclusion. When I described my project, he warned that positioning “social justice” in genomics was provocative. I explained why that provocation was essential to this type of enquiry. His final words stayed with me: “You are the right person to do this work.”

I did not hear encouragement. I heard permission. A *White* senior scientist, shaped by his own history and institutional authority, had taken it upon himself to affirm my legitimacy. The encounter made visible how questions of justice and, in particular, social justice in genomics are not only epistemic but relational, shaped by the hierarchies that govern who may speak, and with what authority. This experience also revealed a broader truth: within many scientific spaces, the moral boundaries of genomics are guarded by those who presume neutrality, even as they reproduce the conditions that make neutrality impossible.

One other conversation brought this into sharp focus. During a Zoom call with one of the United Kingdom’s first sickle cell pioneers, I mentioned my forthcoming fellowship. She immediately referenced the 2019 accusations that Sanger had commercialised African genomic data without consent (Stokstad, 2019). Although Sanger denied wrongdoing and formal investigations cleared them of any breach, the incident reignited longstanding concerns about extractive research and consent. Such controversies expose the fragility of institutional trust and highlight how promises of open science can obscure uneven distributions of power.

These moments resonate with my own professional encounters with pioneering Black women whose achievements sit alongside deep awareness of the exclusions still present within British science. My mother often remarks that we continue to live in the

era of firsts. Being a “first Black woman” in a scientific domain brings both visibility and a burden of representation. It is a reminder that our inclusion is still conditional.

I come to this work as a dark-skinned, British-born woman of Jamaican descent, shaped by diasporic histories of slavery, economics, and colonialism. These forces influence not only my own identity but the scientific structures that shape genomic research today.

The research is situated within the broader institutional climate surrounding data diversity and the need to expand genomic research involving historically underrepresented populations. During the period in which this thesis was being written, Sanger also published a report outlining principles for equitable international research collaboration (Nicholson et al., 2024) . While that document articulates an institutional commitment to equity through guidance on data sharing, capacity building, and partnership, it operates at the level of policy and best practice.

The conditions under which the research was funded and conducted are set out in Appendix G. The fellowship was established through a joint initiative between Sanger and Wellcome Connecting Science, with academic oversight provided by the University of Cambridge.

1.3 Race, Class and Power in Britain: A Necessary Context

Understanding race in Britain requires attending to the historical and political conditions that shaped its emergence. The emergent dynamics matter for genomics because racial classification in science reflects and reinforces broader political hierarchies. From the early 1950s, Caribbean, African, South Asian, and Irish migrants arrived in large numbers into a society structured by imperial hierarchies. Britain’s post-war need for labour coexisted with entrenched racial attitudes, producing xenophobic responses not only to colonial subjects but also to Irish migrants, who were portrayed as poor or undesirable (Gilroy, 2013; Walter, 2002). These dynamics reveal how whiteness in the UK has never been uniform. White Irish and Traveller communities were frequently racialised and subjected to discrimination, surveillance, and hostility (Bhopal, 2018; Garner, 2003; Judd, 2009). Antisemitism, with deep historical roots, resurfaces during periods of socio-economic instability (Tolan, 2023). Race in Britain, therefore, operates as a hierarchy that is both fluid and internally stratified.

The political struggles of the 1960s and 70s, led by African, Caribbean, South Asian, Jewish, and Irish communities, responded to institutional racism, economic injustice, and state violence (Bunce & Field, 2014; Hall, 2021; Stephenson, 2014; Waters, 2019). During this period, political blackness emerged as a strategic alliance across these groups. It was rooted in collective resistance and solidarity, rather than phenotype alone (Gilroy, 2013; Virdee, 2014).

Yet, race is also shaped through class. Racial minoritisation is often conflated with poverty or lower socio-economic position, reinforcing Stuart Hall's argument that race functions as a key organising principle of inequality. From the 1990s onwards, various coalitions of political blackness declined as policy and public discourse shifted towards recognising ethnic difference. Categories such as *BAME*, *Brown*, or *People of Colour*, many influenced by American racial terminology, became prominent (Starr, 2023). Although presented as inclusive, they lacked the political force of earlier movements and often reduced racial difference to administrative labels (Abbas, 2019; Gill, 2024; Gilroy, 2000). Their limitations are evident in the context of resurgent nationalism. The rise of the British National Party, the English Defence League, and digitally mobilised groups like Britain First, along with the mobilisation of racial anxieties during the Brexit campaign, demonstrate that racial hierarchy persists even as its expression shifts (Solomos, 2020; Valluvan, 2019; Virdee & McGeever, 2019).

Nationalist narratives shape assumptions about belonging and human value, which travel into scientific infrastructures, data practices, and decisions about which populations warrant study or investment.

Racial identities are therefore experienced and understood through layered histories and can shape generational differences in how identity is articulated. In Britain, for example, generations who lived through the most intense periods of British racism may resist the term "Black British" while younger generations often embrace it as a hybrid identity signalling both belonging and critique. These tensions matter because they reflect how categories of identity evolve, shape political consciousness, and determine how people experience the institutions that study and govern them.

Understanding the British racial formation, therefore, sets the stage for examining how race, ethnicity, and ancestry circulate through genomic science today.

1.3.1 Institutional Context: Wellcome Sanger Institute and its Histories of Engagement with Race and Ethics

This research takes place at the Wellcome Genome Campus, home to international leaders in genomic science, including Sanger and Wellcome Connecting Science (WCS), which also funded my PhD fellowship. With nearly three thousand staff, the site is one of the largest genomic research hubs in the world. It is known for its role in the HGP and for pioneering large-scale data infrastructures. Yet the institution has also faced public scrutiny regarding race, data, and ethics.

There is a dearth of UK data on these issues. Though not part of the formal literature review, guided by a science historian, we both reviewed 62 genomics papers (Appendix F) published by Sanger towards the end of the HGP (years 2001 – 2003) to gain an overview of how concepts of race, ethnicity, and ancestry and the collection of human population data formed part of the historical background of Sanger.

Engagement with ethics varied. Many studies relied on legacy samples, often collected under earlier ethical regimes or from tissue banks whose origins were not discussed (Braybrook et al., 2001; Bryan et al., 2002; Howell, 2022; Y. Liu et al., 2002). References to consent were sparse and procedural (Gibbons et al., 2003; Lehmann et al., 2002). The datasets overwhelmingly reflected European populations, drawn from cancer genetics clinics and other longstanding sources such as the Breast Cancer Linkage Consortium (Thompson et al., 2002), The Utah Centre d'Etudes du Polymorphisme Humain (CEPH/Utah) pedigrees (Deloukas & Bentley, 2004), the North Cumbria Community Genetics Project (Meijers-Heijboer et al., 2002) and Icelandic cohorts, valued for their homogeneity.

In the end, papers hardly reflected on these descriptors. Instead, racialised terms appeared as routine labels for population stratification: “Caucasian”, “Ashkenazi Jewish”, “Welsh”, “Estonian” (Antoniou et al., 2002; Bevan et al., 2001; Dawson et al., 2002). Their use was inconsistent and lacked ethical justification, confirming Reardon's (2009, 2017) observation that race in genomics is unaccounted for yet indispensable.

Populations marginal to Britain's demographic reality were central to its scientific knowledge. Others appeared only at the margins. This form of silence became an ethical story, indicating which forms of diversity carried scientific legitimacy and which did not.

1.3.2 Aims and Approach

This thesis unfolds as a cumulative inquiry into how race, ethnicity, and ancestry are produced, negotiated, and stabilised within contemporary genomic science, and how questions of justice emerge from these processes.

This thesis seeks to:

1. Examine how race, ethnicity, and ancestry are conceptualised, operationalised, and contested within genomics.
2. Analyse the institutional, methodological, and structural conditions under which these categories are maintained or challenged.
3. Explore the broader ethical and political implications of classification in genomic science.
4. Develop a framework through which social justice can be embedded into genomic research practice

It moves from conceptual and historical grounding through close empirical analysis of everyday scientific practice to a constructive intervention that seeks to reshape how genomic research is conducted. The thesis begins by situating the study within interdisciplinary debates on race, human difference, and responsibility in science. Chapter 2 traces how population categories have been historically produced and institutionalised across genetics, medicine, and state practice, and how these legacies continue to shape contemporary genomics. Drawing on scholarship from science and technology studies, bioethics, sociology, and critical race theory, it establishes the conceptual terrain for the thesis and identifies a key gap, namely the absence of UK-based empirical work examining how genomic researchers navigate race and ancestry in practice.

The methodological approach is then set out in Chapter 3, which explains how the research was designed, conducted, and analysed. This chapter provides the interpretive context for the empirical chapters that follow, detailing the analytic strategy, ethical

considerations, and limitations of the study. I discuss how my approach is grounded in Critical Race Theory, Critical Realism, and ethnography. These frameworks enable analysis of both empirical practice and the underlying structures that condition it. They allow this thesis to ask not only how race, ethnicity, and ancestry are constructed in everyday scientific practice, but also how these constructions can be reimagined in service of justice.

Attention then turns to how population categories appear in the formal outputs of genomic science. Chapter 4 analyses publications involving Sanger scientists, showing how race, ethnicity, and ancestry are constructed through inherited labels, statistical inference, and harmonisation pressures. It demonstrates how classificatory decisions are often stabilised upstream, travel across datasets and consortia, and are rarely revisited in published accounts, while ethical accountability is framed largely in procedural terms.

Building on this textual analysis, Chapter 5 moves into everyday scientific practice, examining how power operates in communication, methods, and responsibility within the Institute. Through interviews and ethnographic engagement, it shows how race is frequently avoided or displaced. I describe how methodological conventions flatten human diversity and while responsibility for equity is acknowledged it is yet often deferred. Crucially, the chapter reveals how authority, hierarchy, and positionality shape what can be said and done in relation to population categorisation.

Chapters 6 and 7 widen the lens further, situating these practices within broader institutional and systemic contexts. Chapter 6 examines funding structures, collaboration, leadership, and institutional commitment, showing how aspirations for equity are constrained by organisational hierarchies, resource asymmetries, and historical legacies. Chapter 7 then places Sanger within the wider genomic ecosystem, mapping how power and autonomy circulate across funders, collaborators, regulators, industry, and publics, and how ethical ambition is shaped by connection, dependency, and constraint.

These strands come together in Chapter 8, with the analysis arguing that justice in genomics is fundamentally a question of power, rather than a technical problem of missing data alone. Drawing together empirical findings and critical theory, the chapter

shows how racialised logics persist through classificatory practices, methodological standards, and institutional arrangements, while also identifying points of tension, reflexivity, and possibility.

Developed directly from the study's findings, Chapter 9 presents the JUST DATA Framework as a justice-oriented model for genomic research. Rather than treating equity as a retrospective ethical concern, the framework offers a way to embed justice into research design, data practices, communication, and governance from the outset. In doing so, it recognises social justice as the process through which equitable genomic outcomes can be achieved, positioning it as a condition of scientific integrity and a practical contribution to debates about responsibility, power, and inclusion in genomics

Justice in genomics, this thesis argues, must be designed into science from the start.

2 Literature Review

2.1 Background

This literature review explores the evolution of race, ethnicity, and ancestry as categories in genomic research. It begins by tracing the historical development of racial thinking in science, emphasising how these concepts originated within systems of oppression, including slavery, colonialism, and eugenics and gave rise to what is now recognised as scientific racism. These systems imposed artificial, discrete categories onto human diversity, which were reinforced through early data practices in genetics and remain embedded in aspects of contemporary genomic research.

Against this backdrop, it is important to recognise that human genetic variation is naturally continuous, complex, and multidimensional (Fuentes et al., 2019; Lewontin, 1972; Rosenberg et al., 2002), more akin to a constellation than a set of bounded types (see Figure 1).

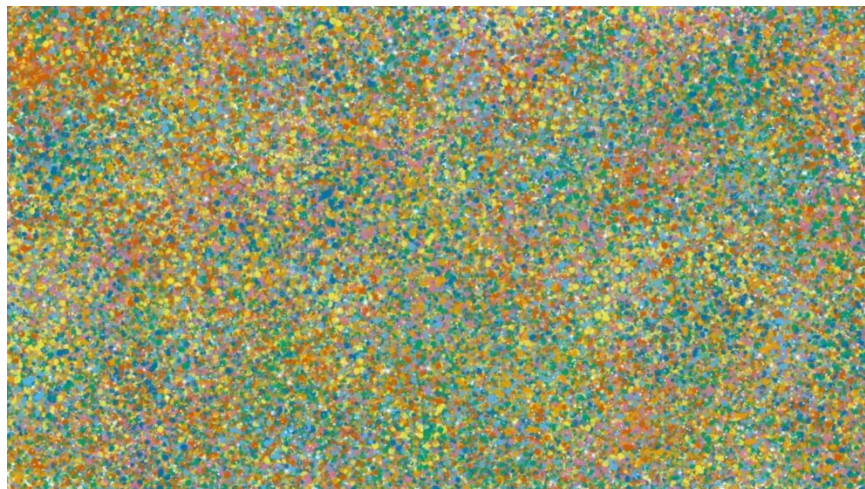


Figure 1: The constellation of human diversity. This multidimensional, complex constellation with difficult-to-interpret boundaries of genetic similarities represents

human population data that is not constrained by time, location or geography. Produced by the author.

While human genetic variation displays patterns of clustering, the categorical boundaries imposed upon it are interpretive constructs mapped onto underlying gradients, as the following discussion will demonstrate. Over time, both conceptual and practical reductions have simplified this human diversity into rigid classifications. These reductions, shaped by social, political, and scientific forces, have contributed to the chronic underrepresentation of African, Indigenous, and some Asian populations in genomics.

Methods such as mtDNA and Y-DNA tracing (tracking genetic lineages through maternal and paternal lines) and later autosomal studies, which examine the full genome, offered increasingly refined means of reconstructing the picture of human ancestry. However, these approaches were constrained by sampling biases and by restricted access to data controlled by powerful institutions. Modern genomic research inherits many of these inequities. Epistemic injustice (Fricker, 2007), understood here as the exclusion of marginalised groups from shaping what counts as legitimate knowledge, continues to determine who is studied, whose data is prioritised, and how it is classified. Policies governing data access, sharing, and harmonisation (using technical solutions to make heterogeneous data interoperable) frequently privilege powerful institutions and populations deemed more convenient or accessible. The metaphor is intentional in this respect: just as constellations are projections we impose upon the stars to create meaning, so too are categories in genetics interpretive rather than natural. To describe variation as “complex and multidimensional” is not empty rhetoric but a deliberate refusal to collapse the many axes of difference into a single scale. While categories can serve pragmatic functions in medicine or genomics, they should never be mistaken for natural essences. Rather, they must be understood as heuristic tools that carry both scientific utility and social consequence, demanding that we hold them with care and transparency.

Drawing on interdisciplinary scholarship from genetics, sociology, and bioethics, this chapter interrogates the assumptions underpinning population descriptors. It examines the methods used to address both the statistical and ethical challenges of describing human population data and ancestral diversity in genomic research. Race, ethnicity, and

ancestry can be associated with health and disease in complex ways (Graves Jr, 2015), and tailored methodologies are needed to monitor and critique research on race and genetics and their impact on health (Ford & Harawa, 2010; Frank, 2007). Hood (2022) and Shim et al., (2014) describe how gene-environment interaction research can use racial and other dimensions of difference as essential clues to the causes of complex diseases.

Through exploring the discourse around these issues, I investigate how the practices of scientists have been presented as solutions to the ethical problems arising from racial and ethnic inequity in research, and ask when, if, and how social justice has formed part of these approaches. The chapter concludes by laying the conceptual groundwork for an epistemology of social justice in genomic research, an approach then examined in greater depth in the methodology chapter.

2.2 The History of Race as a Scientific Category

Humans have long described themselves through physical appearance, language, location, cultural practices, and behaviour, often emphasising how these characteristics distinguish one group from another. The ways in which these distinctions have been constructed and reinforced over time reflect a complex relationship between science, politics, and society. (English Heritage, 2021; Lujan & DiCarlo, 2018).

The concept of race as a way to distinguish people is both enduring and pervasive in its social effects, yet unstable in its definition. As Smedley and Smedley (2005) state, “from its inception, race was a folk idea, a culturally invented conception about human differences” (p. 22). This framing positions race as a historically contingent system of classification designed to organise people into hierarchies of value. Omi and Winant’s (2014) theory of racial formation further suggests that race is continually reshaped through political struggle, media, and institutional practice, rather than emerging from any fixed biological essence. Hall (2021) situates race as a “floating signifier” whose meaning is produced through discourse and social relations, not through biology.

These perspectives underscore how the concept of race evolved to justify inequality, naturalise difference, and stabilise power. Understanding this historical and theoretical

foundation has been essential for interrogating through this literature how contemporary scientific disciplines, including genomics, continue to deploy and reinforce racial categories in ways that reflect and reproduce these legacies.

2.2.1 Turning Difference into Hierarchy

The persecution of Jewish people in the 14th Century by Spanish Christians is one of the precursors of modern ideas of distinct racial differences (Fredrickson, 2016). In early Christianity, conversion was regarded as a pathway to salvation, offering redemption from sin regardless of origin. However, over time, anti-Judaic sentiment evolved into a more rigid and exclusionary ideology, which erupted in massacres of forced conversions across the Kingdoms of Castile and Aragon. By the mid-15th century with the “limpieza de sangre” laws (“cleanliness of blood”) emerged the belief that assimilation across cultural or ethnic boundaries was biologically impossible, marking a key distinction between modern racism and the earlier ethnocentrism that it supplanted (Fredrickson, 2016; Hochman, 2019; Steinberg, 2001).

The notion that the unchristian were soulless influenced how European thinkers interpreted human variation during the era of exploration and empire. (Nimako & Willemsen, 2011). Portuguese writings from the late 15th century provide some of the earliest examples of Africans being described as a distinct race, reducing enormous diversity to a single enslaveable category, “others again were as Black as Ethiops, and so ugly, both in features and in body, as almost to appear (to those who saw them) the images of a lower hemisphere.” (Luzong, 2022, *The Racist Writings of Gomes Eanes De Zurara*, Para 2) . Such racialised language helped frame chattel slavery as a uniquely racial system and distinguish it from earlier, non-racialised forms of captivity. Only as a slave could an African be civilised and Christianised, since before this they were “living in the perdition of their souls.”(Fuentes, 2018 pg. 18). Christianity was thus used to further ideas of predetermined superiority and inferiority of groups based on race or descent (J. K. Carter, 2008; Cox, 2002; Ekechi, 1971; Goetz, 2016).

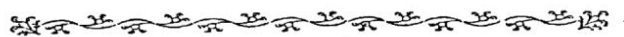
This Christian and European ideology was achieved through increased political, religious and social dominance (Harris, 2004; L. Young, 2009). European superiority has hence been identified as evolving into a racial hierarchy that produced today’s definitions of

White supremacy (Bonds & Inwood, 2016; Jackson, 1998; C. W. Mills & Shelby, 2022); a structural system, rather than simply an ideological position, through which racial hierarchies are produced and maintained, privileging those constructed as White while subordinating other groups across social, political, and economic domains.

These emerging ideas about difference and hierarchy laid the groundwork for the taxonomic systems that would soon formalise race within Western science.

2.2.2 Systemising Human Difference: Western Racial Taxonomies

By the 18th century, the height of the transatlantic slave trade, scientist Carolus Linnaeus divided the human species into four major subdivisions of *Homo sapiens*. Linnaeus described *H. americanus* and *H. africanus*. *H. europaeus* and *H. asiaticus* (Linnaeus, 1758). As shown in a later translation of his work, these divisions were based on appearance and included descriptions of behavioural traits (Figure 2):



MAMMALIA.

ORDER I. PRIMATES.

Fore-teeth cutting; upper 4, parallel; teats 2 pectoral.

1. HOMO.

Sapiens. Diurnal; varying by education and situation.

2. Four-footed, mute, hairy. *Wild Man.*

3. Copper-coloured, choleric, erect. *American.*

Hair black, straight, thick; *nostrils* wide, *face* harsh; *beard* scanty; *obstinate*, content free. *Paints* himself with fine red lines. *Regulated* by customs.

4. Fair, sanguine, brawny. *European.*

Hair yellow, brown, flowing, *eyes* blue; *gentle*, acute, inventive. *Covered* with close vestments. *Governed* by laws.

5. Sooty, melancholy, rigid. *Asiatic.*

Hair black; *eyes* dark; *severe*, haughty, covetous. *Covered* with loose garments. *Governed* by opinions.

6. Black, phlegmatic, relaxed. *African.*

Hair black, frizzled; *skin* filky; *nose* flat; *lips* tumid; *crafty*, indolent, negligent. *Anoints* himself with grease. *Governed* by caprice.

Figure 2: Carolus Linnaeus' classifications of *Homo sapiens*. Translated from his work *Systema natura* by Turton & Linne, 1802

In Linnaeus's classification, physical appearance was directly tied to behavioural traits, constructing a hierarchy that naturalised European superiority and entrenched anti-Blackness. *Homo europaeus* (Europeans) were described as "fair, brawny" and characterised as "gentle, inventive, and governed by laws," implicitly positioning them as the standard of civilisation. In stark contrast, *Homo africanus* (Africans) were defined by their "black" appearance and attributed with traits such as being "indolent, negligent, and crafty," and "governed by caprice," suggesting moral weakness and irrationality. *Homo asiaticus* (Asians) were labelled "haughty" and "covetous," while *Homo americanus* (Native Americans) were described as "choleric" and "regulated by customs," further emphasising cultural inferiority.

Classification in this fashion, given Linnaeus was a botanist, validated the idea that humans, like other species, could be improved by breeding (Chial, 2008; Nimako &

Willemsen, 2011). Johan Friedrich Blumenbach later categorised humans into five categories: American, Malay, Ethiopian, Mongolian, and Caucasian, choosing Caucasians to represent Europeans after his most beautiful discovery of a skull in Russia's Caucasus Mountains. (Blumenbach et al., 1865). It is possible that Blumenbach had not intended his categories to uphold ideas of racial hierarchy when he stated, "All national differences in the form and colour of the human body [. . .] run so insensibly, by so many shades and transitions one into the other, that it is impossible to separate them by any but very arbitrary limits." (Blumenbach 1825) p. 35). Yet what endured was a conceptual image as illustrated by Figure 3 of humanity as divisible into a small number of discrete populations.

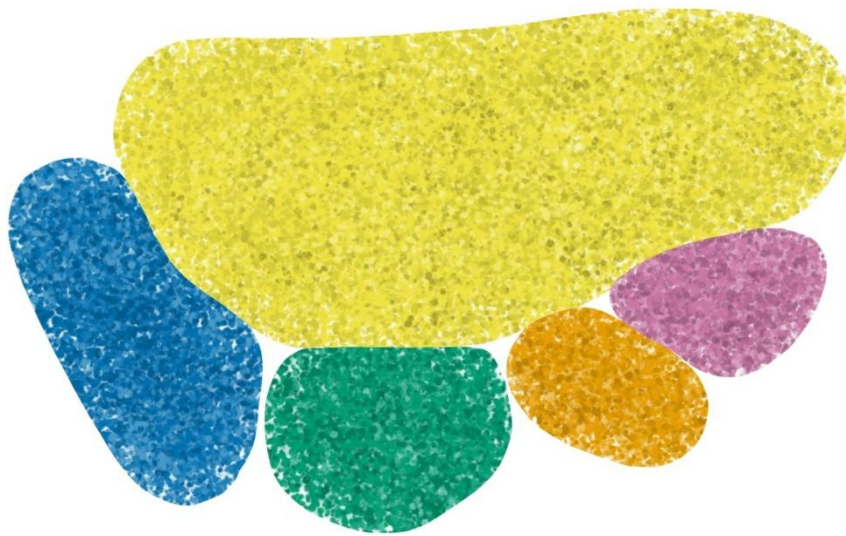


Figure 3. Blumenbach's five racial categories as discrete populations: American is represented by (blue), Malay (orange), Ethiopian (green), Mongolian (purple), and Caucasian (yellow). Caucasian appears larger to represent its position as a site of human origin. The visual separation reflects the historical classification of human difference that took hold conceptually rather than any scientific reality. Produced by the author.

While it is important to acknowledge historical justifications made for Blumenbach for what is now recognised as a racist ideology, the act of categorisation and the power to categorise based on ideological beliefs, such as the idea of European origin, must not be overlooked. As Foucault (1980) suggests, systems of knowledge are inseparable from the power structures that authorise them, meaning that classification itself becomes a

mechanism through which inequality is produced and sustained. This is central to understanding the motivations and directions that scientific thinking would take in the years that followed. The formalisation of these classifications not only naturalised hierarchy but also embedded it within scientific authority, shaping the ideologies that would define whose knowledge counted.

2.3 Ideologies and Epistemic Injustice in Science

While much of the scientific and intellectual history of the 17th and 18th centuries has centred on European figures, there were others whose contributions were known and, to varying degrees, respected by European scientists. For example, Ibn Sina (Avicenna) and Alhazen (Ibn al-Haytham), though active in earlier centuries, were central figures in medieval Islamic science and medicine and their works, such as 'The Canon of Medicine and Book of Optics' continued to influence European medical and scientific thought into the 17th century (Ahmad, 2023; Gutas, 2014). In the 18th century, Anton Wilhelm Amo, a Ghanaian-born philosopher, earned doctorates in philosophy and law in Germany and lectured at the universities of Halle and Jena. His treatises engaged directly with Enlightenment philosophy and were received with respect by some European scholars. Onesimus, an enslaved African in early 18th-century Boston, introduced the practice of variolation (a precursor to vaccination) to Cotton Mather, a Puritan minister and member of the Royal Society. Mather advocated for the technique during the 1721 smallpox epidemic, citing Onesimus as the source of this life-saving knowledge, albeit reaffirming beliefs about the simple-minded ways of Africans in his parody of their voice in 'Angel of Bethesda', the medical treatise written after the epidemic (Koo, 2007; Nakayama, 2022).

Further, knowledge systems from India, China, and the Islamic world, such as herbal medicine, astronomical tables, and algebra, were frequently translated or incorporated into European science, though often without proper attribution (Hobson, 2004; Saliba, 2007). This pattern reflects what decolonial scholars later describe as the systematic marginalisation of non-Western epistemologies within dominant scientific narratives (Maldonado-Torres, 2007; Mignolo, 2011)

These few examples set the scene for a broader, often suppressed history of global intellectual exchange and the resilience of marginalised knowledge systems in shaping modern science (Grosfoguel, 2007). Racialised individuals' contributions to science continued to be marked by exploitation, undervaluation, and the increasing use of their bodies as sites of experimentation and evidence production (Das & Lowe, 2018). As Figure 4 shows, knowledge about human populations was generated and classified unevenly, and against this backdrop, there is a muted constellation of more understanding than is accessed or valued; like outdated thinking about introns in DNA, much of it is treated as non-coding 'junk'.

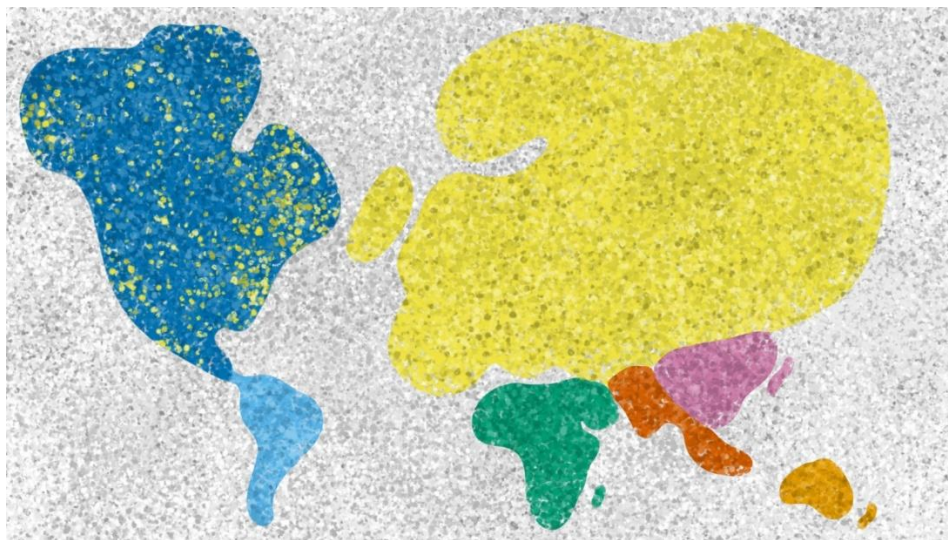


Figure 4: Distorted world map illustrating the geopolitical dimensions of epistemic influence. Regions on the map are sized and coloured to reflect their perceived contribution to knowledge production. Europe appears enlarged, and its influence extends into other regions through traces of yellow, signifying the historical dominance of European epistemic frameworks. Produced by the author.

Figures such as Sarah Baartman, a Khoikhoi woman taken from South Africa to Britain in 1810 at the suggestion of British military surgeon Alexander (often called William) Dunlop, exemplify how *Black* bodies were objectified, sexualised, and used to reinforce *White* superiority. Dunlop claimed she would be exhibited for the study of her physical features. Though not legally enslaved, the racial, colonial, and exploitative systems underpinning slavery were fully at work in the way she was treated: as spectacle, as scientific “evidence” of racial difference, and as property-like. When her popularity in

London declined, she was taken to Paris around 1814, exhibited by an animal trainer, and subjected to the post-mortem examinations of Georges Cuvier, whose anatomical investigations and displays perpetuated her objectification even after death ('The Significance of Sarah Baartman', 2016).

Baartman's treatment was part of broader 19th-century European exhibition culture in which *Black* and Indigenous people were routinely displayed in human zoos, theatres, and scientific demonstrations. These spectacles intersected with the rise of scientific racism, where pseudo-anthropological studies and anatomical dissections ranked non-European bodies within a racial hierarchy (Blanchard, 2008; Sadiya Qureshi, 2011). In medicine, similar exploitation occurred in the work of J. Marion Sims, who in the 1840s conducted repeated gynaecological surgeries on enslaved *Black* women without anaesthesia, justified by racist beliefs about pain tolerance (Spettel & White, 2011). As with the exhibition of Baartman, Sims's experiments demonstrate how 19th-century science leveraged racial hierarchies to legitimise exploitation. Both cases reveal how scientific authority naturalised inequality under the guise of empirical progress, embedding racialised exploitation within the foundations of modern disciplines (Saini, 2019).

More than examples of individual cruelty, these practices formed part of a broader epistemic environment in which the bodies of racialised subjects were treated simultaneously as *objects of study* and as *sources of evidence* for theories of human difference. They exemplify what Ahmed (2002) describes as the way difference is produced through the repetition of embodied encounters, how racialised bodies become both the material on which power acts and the conceptual frame through which knowledge is organised. In this sense, physical and conceptual racialisation are mutually constitutive: the scientific gaze both constructs and confirms the racial categories (Yancy, 2008).

It was within this intellectual and ideological climate that Count Joseph Arthur de Gobineau, often considered one of the founding figures of modern "scientific" racism, drew on such 'knowledge' to compose his *Essay on the Inequality of the Human Races* (Biddiss, 1970). Gobineau argued that human history was driven by racial differences, claiming that the "White race" was superior in intelligence and warned of civilisation decline by race mixing.

Charles Darwin's theory of evolution, *On the Origin of the Species* (Darwin, 1859) was published within this intellectual environment and, while he did not understand the genetic basis of inheritance, his focus on variation, selection, and heritability laid essential groundwork for modern genetics. Importantly, the production of scientific knowledge continued to rely upon contributions from individuals whom Victorian society deemed intellectually inferior to European minds. Darwin had learned taxidermy in 1826 from John Edmonstone (Figure 5), a formerly enslaved man from British Guiana, while studying in Edinburgh (Desmond & Moore, 2009; Freeman, 1978; McNish, 2022).

'Image removed due to copyright restrictions. The original artwork 'In the Workshop of Taxidermist' (1948) can be viewed at the State Darwin Museum '

Figure 5 Artist's impression of John Edmonstone teaching Darwin to preserve birds. By © State Darwin Museum. <https://www.nhm.ac.uk/discover/john-edmonstone-the-man-who-taught-darwin-taxidermy.html>

In a letter to his sister, Darwin wrote: "I am going to learn to stuff birds, from a blackamoor... he only charges one guinea... for an hour every day for two months" (Darwin, 1826, as cited in Cain, 2021). Darwin himself opposed slavery and did not subscribe to polygenism - the idea that races had separate origins. He rejected the notion that *Black* people were a different species. Yet despite these progressive positions for his time, Darwin's own writings were still embedded in Eurocentric hierarchies. He frequently used language such as "civilised" and "savage," and framed certain non-European populations as living representatives of earlier stages of human development. Edmonstone's legacy, like those of many non-European contributors, remains marginalised, lending scientific authority to racial hierarchies. Thus, even where Darwin sought to challenge overtly racist doctrines and slavery, his work contributed, willingly or not, to an intellectual climate that would later enable and legitimise eugenics (Desmond & Moore's (2009).

In this context, knowledge production was inseparable from systems of domination (Maldonado-Torres, 2007; Mignolo, 2011), making epistemic injustice not only about

who is denied recognition, but also about who is objectified and instrumentalised in the very processes that confer scientific authority (Ashby & Machin, 2021).

Together, these histories show that race was never a neutral category but a construct carrying both epistemic and ethical weight. This raises the question of whether such dynamics created the conditions for race science to emerge as a cornerstone of state policy across much of the Western world, an idea the following section explores, particularly in relation to its role as an ideological foundation for eugenic practices that endured well into the 20th century.

2.3.1 Extending Racial Hierarchy into Policy

Francis Galton, a Victorian polymath known for his work in statistics, psychology and the early development of heredity studies, emerged as a central figure in formalising eugenic thought.

Coined by Galton in 1883, eugenics was the idea that breeding should be encouraged among those with desirable traits and discouraged among those with undesirable traits (Galton, 1883). While these categories were not exclusively racial, they frequently intersected with racial hierarchies, with *Black*, *Indigenous*, and other marginalised populations disproportionately targeted and stigmatised as carriers of these undesirable traits (Crossley, 2022; Kevles, 1995).

Working during a period of strong imperial self-assurance, Galton argued that human traits, social standing and even moral character were primarily determined by inheritance. Galton's promotion of eugenics linked ideas of biological superiority to class and race, presenting these claims as scientific principles. His influence demonstrated how the developing field of genetics could be used to justify existing hierarchies and to reinforce the economic and political power structures of empire.

As Britain expanded its imperial reach across Africa, Asia, and the Caribbean, it exported and embedded racial hierarchies foundational to the functioning of the empire. Antisemitism, Islamophobia, and the construction of Blackness as inherently inferior

were pivotal in shaping British racial thought, feeding into a larger imperial logic that presented *White*, European Christian identity as superior and normative.

The eugenics movement entered European and North American society at a time when thoughts around superior and inferior races were already embedded (Stepan, 1982; Turda, 2010).

Although *Black* scholars such as Kelly Miller, W. Montague Cobb, and W. E. B. Du Bois actively contested biologicistic racism, they advocated a form of “eugenics without racism” aimed at sociopolitical uplift by the early 20th century. Influential American sociologists like W. I. Thomas of the Chicago School similarly questioned racial determinism. Failed attempts to contest racial hierarchy reveal the significant institutional power Eugenic ideologies retained within science and public policy (Lothrop Stoddard, 1929; Nuriddin, 2021).

These ideas are accepted as reaching their most devastating expression in Nazi Germany, where eugenics became state policy, used to justify genocide and racial cleansing, culminating in the Holocaust and World War II (Rutherford, 2021). It was only in the aftermath of these atrocities that a widespread rejection of race as a valid biological category began to take hold within mainstream science (Omi & Winant, 2014). Eugenics was eventually overtly discredited not solely for its scientific inaccuracy, but because of its devastating ethical implications and association with fascism (Farrall, 1979; Fogarty, 2015).

Eugenics in Britain at its peak was not a narrow pseudoscience but a broad moral project from which genetics is right to distance itself. Its legacies extend into disability, mental health, intellectual capacity, and reproductive autonomy, the very terrains in which clinical genetics (the application of genetics to medical care (Clinical Genetics Society, 2022)) came to operate into the 21st century. As clinical genetics evolved, eugenic practices persisted in subtler but still coercive forms such as using sterilisation as a way of preventing ‘inferior’ populations from reproducing (Kevles, 1995). In 1996, a Canadian court found that Leilani Muir had been wrongfully sterilised by the Alberta government (Wahlsten, 1997). In 2013, it was revealed that African American and Latina women had been forcibly sterilised in California state prisons (Roth & Ainsworth, 2015).

As recently as 2017, a U.S. judge was reprimanded for offering reduced jail sentences in exchange for sterilisation (Hawkins, 2021).

Darwin's theory of common ancestry and natural variation was gradually co-opted into racial classification, repurposed to justify hierarchy, exclusion, and harm (Kevles, 1995). Scientific ideas, even when not intended to do so, have historically served political agendas. Questions of transparency, legitimacy, and power must therefore be continually re-examined as genomic technologies expand (Clarke & Wallgren-Pettersson, 2019).

Thus, even as figures like geneticist Richard Lewontin challenged the biological basis of race in the 1970s, arguing that "human racial classification is of no social value, positively destructive of social and human relations", (Lewontin, 1972 pp. 381–398), decades later the structures built on earlier misinterpretations of evolutionary theory remained socially and institutionally entrenched. In Britain, these global and scientific currents intersect with class and capitalism, embedding racial ideology within the very structure of economic and social life.

2.3.2 Race in Medicine and Science

These long-standing constructions of race, identity, and hierarchy set the stage for examining how race continues to operate within medicine and science. Contemporary racial disparities in Britain underscore the persistence of structural racism. In this thesis, structural racism is understood as the normalised operation of social, legal, political, and institutional structures that systematically advantage some racial groups while disadvantaging others, thereby reproducing racial inequality across time and context (Crenshaw, 1988; Jones, 2000; Delgado & Stefancic, 2017). In this context the legacies of eugenics continue to shadow debates over health, disability, and genetic research. These disparities are especially apparent in the data obtained through genomic research (qualitative and quantitative), which go on to inform the delivery of clinical genetics services. Clinical genetics research has primarily been carried out on populations described by researchers as 'White European' (Bustamante et al., 2011; Morales et al., 2018; Need & Goldstein, 2009). In addition, treating race as a simple explanatory

category rather than a complex one yields research findings insufficient to establish equitable clinical services (Bonham et al., 2009).

The legacy of colonial medicine continues to shape how racialised bodies are understood and treated in healthcare systems. Racial bias in UK healthcare has deep historical roots, evident in areas such as pain management and the delayed recognition of diseases like sickle cell anaemia in Black patients (All-Party Parliamentary Group on Sickle Cell and Thalassaemia & Sickle Cell Society, 2021). The COVID-19 pandemic revealed stark inequities in the practice and materiality of medicine: pulse oximeters, used to measure oxygen levels, were found to be less accurate on darker skin tones, an example of how racial bias in technology can have life-threatening consequences (Sjoding et al., 2020). Similarly, outdated and poorly evidenced clinical guidelines, such as those adjusting for race in diagnosing and treating hypertension, reflect a flawed biological understanding of race and the persistence of racial thinking in medical practice (Gopal, 2020).

In contrast, the increased carrier frequencies identified for Tay Sachs in the Ashkenazi Jewish population, Cystic Fibrosis in Western Europeans, and Sickle Cell Anaemia in the African and African diaspora populations (African-Caribbean and African American) led to targeted genetic screening for individuals from these populations. Although targeted screening is employed to best meet the health needs of specific populations, we must exercise caution that such uses of race and ethnicity risk deepening health disparities by further marginalising minoritised groups (Dobson & Arjunan, 2021; Kahn, 2008). Indeed, there are examples where decisions made using data classification for targeted health care are lifted, without challenge, into a clinical setting (Gopal, 2020; Lujan & DiCarlo, 2018).

Precision medicine is presented as the future of genetics and as a means of meeting the health needs of all patients. As former UK Chief Medical Officer Dame Sally Davies stated: "[t]he age of precision medicine is now, and the NHS must act fast to keep its place at the forefront of global science" (Von Radowitz, 2017 Para 7). Recent national strategies reinforce this view by positioning genomics as central to the future of UK healthcare. The 2022 *Accelerating Genomic Medicine* report frames genomics as a tool for earlier diagnosis, targeted treatment and more equitable access to care (Hill & Prichard, 2022). The NHS 2025 Ten Year Plan echoes this vision, presenting genomics

as integral to prevention, personalisation and the UK's ambition to lead globally (UK Government, Department of Health and Social Care, 2025). Together, these documents articulate a clear belief that genomics should become a routine and equitable part of NHS practice.

However, the diversity in databases, access to testing and underlying socioeconomic disparities affecting disease outcomes have all been highlighted as limitations to the promise of precision medicine and its ability to reduce health disparities (Geneviève et al., 2020; Madhusoodanan, 2020; Roberts, 2011).

Genetic research, particularly Genome-Wide Association studies (GWAS), have included data from mostly self-identified but also researcher-ascribed categories of *White* European or European (M. C. Mills & Rahal, 2020; Need & Goldstein, 2009). For GWAS, substantial amounts of data are needed from the populations in which they are conducted, and the relative size of a population could skew the interpretation.

Populations underrepresented in genomic datasets are more likely to get uninformative genetic test results (Fraiman & Wojcik, 2021; Kurian et al., 2018; Landry & Rehm, 2018; Richards et al., 2015). The predictive performance of non-European ancestry-derived polygenic scores is low or may not even be transferable to European populations (Babb de Villiers et al., 2020; Curtis, 2018; Martin et al., 2019). Variants may even cause different phenotypic effects in other populations; disease modifiers can go unexplained, or stronger associations inferred (Sirugo et al., 2019). In a clinical setting, these differences result in delayed diagnosis, treatment, and management often in groups with pre-existing poor healthcare outcomes (C. M. Lewis & Vassos, 2020).

However, clarifying the degree of genetic diversity needed poses a statistical challenge (Jacob, 2022). For example, due to numerous complex social and environmental factors, African populations and those of African ancestry exhibit significant yet largely understudied genetic variation (Mathias et al., 2016). Solutions to manage large datasets in genomics will need to remain a priority (Balazka & Rodighiero, 2020; Floridi, 2012) not only to power the interpretative capacity of genomic research but to provide fair outcomes to all (Babb de Villiers, 2019; Hardcastle et al., 2022).

2.3.3 The British Genomic Initiatives and the Racial Politics of Inclusion

The UK has long occupied a leading role in global genetic research, a position reinforced by its imperial legacy, sustained institutional investment, and powerful scientific infrastructure. (Claussnitzer et al., 2020). The HGP, was significantly driven by British institutions such as the Wellcome Trust (Wellcome) and the Wellcome Sanger Centre (later Institute), which quickly became a central node in the global sequencing network early on in the development of the project (Reardon, 2017). The Wellcome Sanger Centre contributed approximately one-third of the human genome's first draft (Wellcome Sanger Institute, 2020). The language around that human genome was often 'the book of humankind,' a metaphor that suggested neutrality and universality (Bodmer & McKie, 1995; Wellcome Sanger Institute, 2000).

It was often noted (and still is) that humans are 99.9% genetically identical, which at the time helped popularise the idea that a single reference genome could represent everyone. U.K. Prime Minister at the time, Sir Tony Blair, echoed this narrative when announcing the success of the HGP, emphasising that human beings differ by only 0.01% at the DNA level (Schatz & Langmead, 2013). As such, scientists pursued the idea of a "universal" genome, one meant to stand in for all humanity. As Robert Waterston, a key figure in the HGP, explained:

"We wanted the human genome... not Joe Blow's genome. It's your genome. It's my genome. It's representative of everybody's genome" (Smart, 2024 para. 20).

In practice, however, the initial reference genome was overwhelmingly based on the DNA of a single anonymous donor, comprising approximately 70–75% of the sequence, with limited contributions from a handful of others. After considerable ethical debate over consent and data use, the human reference genome was stated to be drawn largely from a single donor, whose DNA was described to have 37% African and 57% European ancestry and sourced predominantly from one bacterial artificial chromosome library (Green et al., 2010). The debates and obscurity of identity suggest that, rather than intentional or critically stated choice, the ideal of a universal genome was, instead, built on assumptions of scientific neutrality that overlooked structural privilege and the dominance of *White*, Western population approaches in genomic sampling. Early HGP planners, it seems, gave little attention to racial and ethnic diversity or critical questions

about representation. By the early 2000s, Sanger's publications treated race and ancestry as technical variables rather than social categories (see section 1.3.1).

Outside the UK, efforts such as the Human Genome Diversity Project (HGDP), sought to capture a broader range of human variation (Cavalli-Sforza, 2005) but faced sustained ethical criticism for the ways in which it grouped and engaged with Indigenous and minority communities (Claw et al., 2018), as well as for the potential misuse of findings in discriminatory ways (M'Charek, 2005; Reardon, 2005; Roberts, 2011). This moment marked a shift in awareness: scientists began to acknowledge which populations had been left out, and the ethical necessity of naming, recruiting, and representing them in genomic research (Bentley et al., 2017; Hindorff et al., 2018; Reardon, 2009).

More recently, genomics in Britain has often advanced with limited scrutiny of the racial logics embedded in its practices. Projects such as the UK Biobank, which began in 2006 and now holds data from over 500,000 individuals, remain heavily skewed toward individuals of European ancestry (M. C. Mills & Rahal, 2020). The lack of diversity in early datasets and unclear strategies for inclusion raised questions about whose genomes were being prioritised and why (Bentley et al., 2017; Popejoy & Fullerton, 2016).

The 100,000 Genomes Project, launched by Genomics England, was initially heralded as a transformative initiative linking genomic and health data across the UK population and performed better on diversity than earlier efforts. Our Future Health similarly aligns with the global push to broaden participation in genomic research (Cook et al., 2025). Yet the reliance on broad census categories and limited engagement with social inequalities and the distribution of genetic diversity make this pursuit far more complex than simply increasing representation (Corpas, Guio, et al., 2025)

Genomics England's *Diverse Data*, (2024) initiative aims to address these longstanding imbalances by actively recruiting underrepresented groups into genomic research. Yet even these initiatives have not explicitly engaged with the underlying structures that have historically excluded certain groups from knowledge production.

As Britain continues to invest in genomic science, it must therefore reckon with the socio-political foundations on which this science rests. Scientific objectivity cannot be presumed when the categories used to define and sort human populations are

themselves products of a racialised history, pointing to the need to interrogate how the very groups used to describe and capture diversity are communicated and evolve, and how their use continues to reproduce inequality.

2.4 Language about Human Populations

The growing recognition of bias in human genomic datasets raised complex and consequential questions about how populations are named, conceptualised, and labelled, decisions that are central to the production and interpretation of scientific knowledge (Panofsky & Bliss, 2017). Categorisation, therefore, remains deeply embedded in research design, data analysis, and interpretation. These categories reflect, and in turn reinforce, cultural assumptions about human difference (Bliss, 2011; Lorusso & Boniolo, 2008).

While it is widely accepted that humans share 99.9 per cent of their DNA, this concept has been shaped by geographical and scientific necessities that emphasise how patterned differences along broad ancestral lines still matter for understanding human diversity. Such diversity is often difficult to discern when viewed at a broad scale but becomes clearer with closer and more detailed examination. Figure 6 provides a visual illustration of this point.



Figure 6. We are 99.9 % the same, but ancestry divides us. This conceptual illustration shows a gradient of human diversity in which colours blend across regions rather than forming discrete boundaries. From a distance the pattern appears uniform, but closer inspection reveals subtle variation, representing how ancestral differences emerge only with more detailed examination. Produced by the author.

In Byeon et al., 2021 which explored the use of population categories in the American Journal of Medical Genetics between the years 1949 and 2018 (see figure 7), authors describe the trends they observed, enabling 'ongoing reflection as the terminology used to describe genetic variation continues to evolve'.

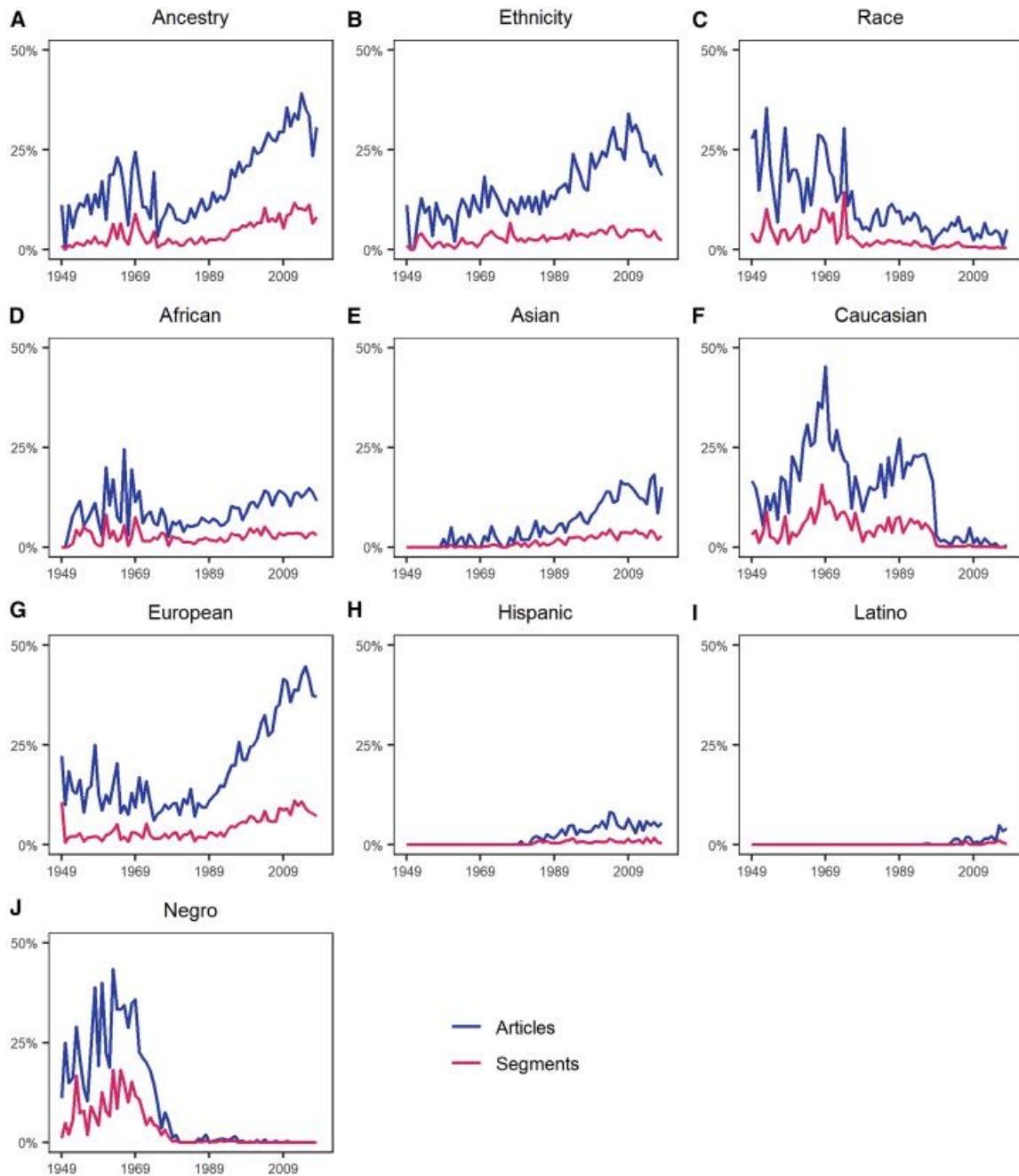


Figure 7: Use of population categories in the American Journal of Medical Genetics. Percentage of 200-token segments containing ‘ancestry’, ‘ethnicity’, ‘race’ and other population descriptors (A-J) Yearly percentage of 200-token segments (pink) an article (blue) containing ‘ancestry’ (A), ‘ethnicity’ (B), ‘race’ (C), ‘African’ (D), ‘Asian’ (E), ‘Caucasian’ (F), ‘European’ (G), ‘Hispanic’ (H), ‘Latina/o/x’ (I), and ‘Negro’ (J) from 1949

– 2018 Byeon et al., (2021) p. 2218. available under a [Creative Commons Attribution 4.0 International \(CC BY 4.0\)](https://creativecommons.org/licenses/by/4.0/) License

Figure 7 shows an increase in the term European and a decline in the use of Negro, not accompanied by a corresponding rise in the use of African or other related terms. This suggests that Negro and African may have referred to distinct concepts or populations, which is unlikely. Moreover Negro, historically used in the U.S. to describe African American populations, appears not to have been simply replaced with African but instead phased out without a clear successor term for the same group. Possibly revealing US dominance in science and the discourse of race in science. These multiple readings are important to note as they bring light to the limitations of quantitative analysis, which, by design, cannot fully capture the meaning, context, or intent behind the use of specific terms. These historical mappings of human population labels are helpful to consider the social and scientific shifts influencing the trends identified in the data.

For example, the study did not include terms such as *Black*, *White*, or Native American, citing difficulties with statistical significance and the complexity of these categories. This omission significantly limits its capacity to examine how racism may shape the use of population descriptors in scientific literature, reflecting the broader challenges of addressing race and racism directly in this field. This study also reveals late adoption of terms such as Latina/o/x and Hispanic as a research category, a further example that highlights how entire groups can be essentially absent from scientific research and discovery, limiting what we can know or understand about them, including the inequities they face due to this lack of inclusion (González Burchard et al., 2005; Montoya, 2011; Schulz et al., 2003).

Overall, the narrow range of terms available to the authors of this paper point to prevailing assumptions about which groups constitute the “major” categories of human populations. These assumptions, like the categories themselves, are historically contingent and subject to change over time. Gibbon, (2016) argues that genomic researchers are not outside the social construction of race and ethnicity.

While some scientists may argue that trends shown in this paper should move to removing any discrete, biologically fixed racial groupings (1000 Genomes Project Consortium et al., 2015; Collins & Mansoura, 2001; Yu et al., 2001; Zhao Zhongming et

al., 2000), others argue that including race in scientific studies can highlight inequality (Bustamante et al., 2011). The continued use of outdated and biologically suspect terms like “Caucasian”, however raises questions about awareness, privilege, and the scientific literacy to deal with these questions of inequality (Popejoy, 2021).

How scientists report race, ethnicity, and ancestry becomes additionally important because such work can be misinterpreted or used to support harmful narratives (Forzano et al., 2021; Novembre & Barton, 2018). Despite researchers issuing clear statements against the misuse of genetic findings, distorted interpretations continue to circulate in public discourse and are taken up by groups seeking to claim biological differences between races (Carlson, 2022; J. J. Lee et al., 2018; Racimo et al., 2018; Smart, 2022).

As race and ethnicity descriptors fail as accurate proxies for the biological classification of human populations (Hunt & Megyesi, 2008b; Kaufman & Cooper, 1995) allelic variation (differences in the versions of a gene found in a population), which is more closely tied to geography, can reveal more robust biological associations. However, flaws in sampling and study design, which all require description of groups may result in findings still fall short of truly representative findings (Fine et al., 2005; A. Fuentes et al., 2019). Though population labelling is governed to some extent by a researcher's individual intent, institutional, disciplinary, and regulatory environments within which researchers operate have a great influence. (Jacob, 2022; Sankar et al., 2015).

Simply eliminating racial terminology without addressing underlying biases could backfire; ignoring race entirely may allow structural racism and its effects to remain unexamined (G. Dawson, 2003; Harden, 2021). Rather than removing racial categories, many now advocate that a more critical understanding of race and its intersections with genomic science is necessary if health inequalities are to be addressed (Gurdasani et al., 2019; Mersha & Abebe, 2015; Tishkoff & Kidd, 2004). Grappling with these challenges is an ethical imperative for genomics, given the discipline's aspirations to improve health for all populations. As Hinterberger, (2012) argues attention should shift from population labels to the population concept itself, since even “individualised” genomic medicine depends on defining and acting on populations, however they are constructed.

How groups are constructed in genomic research is perhaps just as ethically consequential as the labels applied to them, since both processes shape how human variation is categorised and understood. In genomics, where statistical association is a primary tool of genetic ancestry inference (A. Lewis et al., 2022), such classificatory decisions take on amplified significance, shaping both the scientific record and the social meanings attached to genetic variation.

2.5 Constructing Groups in Genomic Research

Genomic researchers employ statistical methods to discover and contribute to the knowledge of genes and their function in human populations. While these methods can identify broad patterns, they are constrained by the diversity and quality of the reference datasets. When applied uncritically, such analyses can inadvertently reify socially constructed categories, giving the false impression of discrete population boundaries, or overemphasise differences while downplaying shared variation (Peterson et al., 2019) . The need to aggregate population data to create groups for comparison echoes earlier forms of racial grouping, although approached with greater awareness of the underlying complexity and sharedness of human populations, as illustrated in Figure 8.

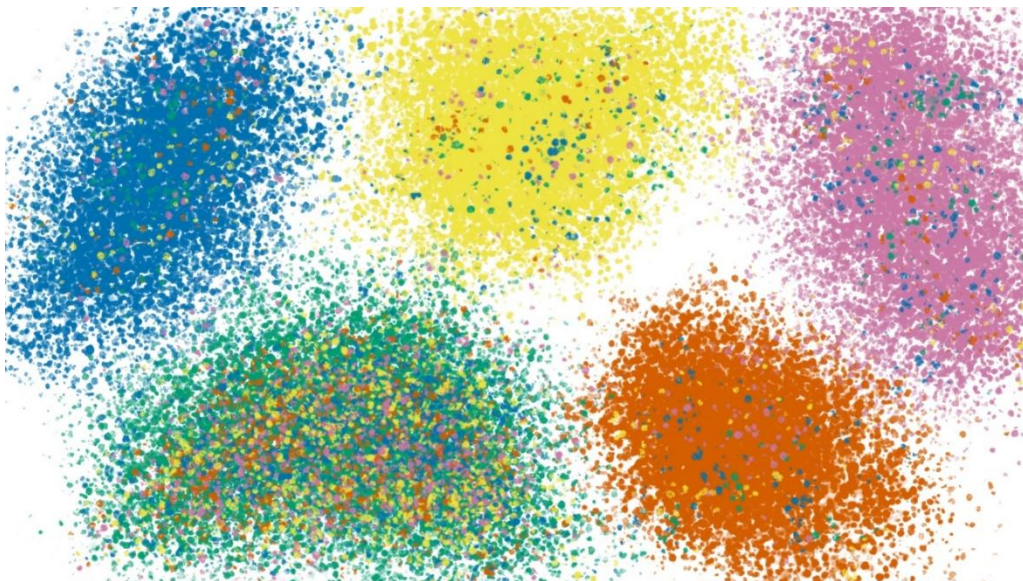


Figure 8: Clustering still requires drawing boundaries. Statistical methods reduce complex genetic ancestry data into clusters of similarity, the 5 colours chosen here reflect the genetic ancestry labels and their relation to Blumenbach's earlier discussed 5 racial groupings. Produced by the author.

2.5.1 Tools for Making Groups

Clustering of individuals depends not on the revelation of fixed, natural divisions but on the method used, the scale of analysis, and the sampling strategy (Rosenberg et al., 2002; Serre & Pääbo, 2004). Some have noted that while genetic ancestry in scientific discourse is perceived as having a stronger biological basis, it reflects only recent migration patterns (A. Lewis et al., 2022).

While statistical approaches to clustering human genetic data may show some correlations or associations between population groups, many studies report no significant variability that neatly distinguishes large populations (Belbin et al., 2018; A. Lewis et al., 2022). As the size and diversity of datasets increase, the continuous variation in previously apparent differences in traits such as intelligence, behaviour, or even physical characteristics increases in visibility.

Principal component analysis (PCA) is a statistical technique that reduces high-dimensional genetic data into a small number of components that capture the greatest variance, allowing researchers to visualise broad patterns of similarity among individuals or groups. In human genetics, PCA is commonly used to summarise genomic structure and to identify large scale gradients of ancestry (Patterson et al., 2006).

While these types of technique generalise patterns of genetic similarity, they struggle to capture the continuous and overlapping nature of human ancestry, which rarely aligns with discrete clusters. PCA's reliance on researcher decisions about which samples to include or exclude can introduce artificial boundaries, limiting its ability to represent the full complexity of human variation (Lawson et al., 2018).

Other tools used to trace ancestry, such as the ancestral recombination graph (ARG) do not require categorisation by population or contextualisation through geographical or cultural identity (Speidel et al., 2019). They reveal a fundamental contradiction between

scientific insight and scientific practice, however. ARG ancestry estimation still almost always imposes categories, defaulting to continental labels that limit both the accuracy and the explanatory power of the resulting data (Lewis et al., 2022). Mathieson & Scally, (2020) have similarly critiqued the limitations of current approaches, underscoring the inadequacy of existing models in capturing the true spectrum of human genetic ancestry.

2.5.2 Inconsistent Approaches

The inconsistent and often ad hoc application of population categories further complicates the picture (Bustamante et al., 2011; Hunt & Megyesi, 2008a; Yudell et al., 2016).

Popejoy and Fullerton, (2016) identified considerable heterogeneity in how populations were constructed across genomic projects. Panofsky and Bliss (2017) examine how geneticists label populations and identify eight classification systems based on racial, geographic, and ethnic logics. They find that these systems were often blended inconsistently, with growing use of quasi-racial “continental” terms over time. They suggest that even inconsistency in categorisation can be interpreted as a form of individual scientific autonomy. Nonetheless, many scholars support the use of more clearly defined and transparent categories as a safeguard against misuse and misinterpretation (Ali-Khan et al., 2011; Caulfield et al., 2009; Yudell et al., 2016).

In large-scale genomic initiatives such as the HapMap and 1000 Genomes projects, both designed to accelerate variant discovery and data sharing, differences in data filtering and inconsistent ethnicity categorisation left many variants uncatalogued and rendered datasets inadequate for certain populations (Buchanan et al., 2012).

The HGDP’s controversial legacy informed the design of the International HapMap Project, launched in 2002 as a separate initiative but one shaped by the earlier political fallout. HapMap aimed to improve representation by sampling multiple continental populations and investing in community engagement. Yet, while it included globally distributed groups such as the Yoruba, Han Chinese, Europeans, and Japanese, these categories were often abstracted from their broader historical and social contexts (Adsit-Morris et al., 2023). In *The Postgenomic Condition* (2017), Reardon critiques this as

“bounded” ethical engagement, addressing concerns about inclusion through targeted consultation and procedural safeguards, while avoiding the more difficult task of confronting how power relations shape genomic knowledge, including who defines population categories, to what ends, and with what consequences.

This bounded approach, together with methodological inconsistencies carried forward into projects like 1000 Genomes, has left unresolved the core challenge: how to pursue genomic progress without reproducing historical patterns of exclusion and exploitation. These legacies continue to inform current debates over diversity in large-scale initiatives such as the UK Biobank and Genomics England, where questions of who is included, how, and why remain as politically and ethically charged as ever.

2.5.3 Epistemic Approaches to Sampling

The Human Genome Project’s (HGP) Ethical, Legal, and Social Implications (ELSI) programme initially treated genomics as an unquestionable public good, focusing largely on technical concerns such as data privacy, while leaving deeper issues of equity unexamined (Adsit-Morris et al., 2023; Reardon, 2009).

Jenny Reardon’s *Race to the Finish* (2005) and later *The Postgenomic Condition* (2017) argue that genomics has never been independent of questions of race, identity, and power. Analysis of the HGDP reveals how even well-intentioned diversity initiatives could misstep: its plan to sample “isolated indigenous populations” provoked strong protests and was described by critics as a neo-colonial “vampire project” exploiting marginalised groups (Reardon, 2005; TallBear, 2013). These controversies exposed the field’s persistent difficulty in addressing the politics of sampling and representation. Genomics, far from operating within a post-racial (Bonilla-Silva, 2022) landscape in which race is assumed to have lost its social and political significance, necessarily embeds social decisions about how human differences are defined and valued. Reardon further argues that the HGDP framed diversity as something for science to preserve, a stance that held the authority of genomic institutions while avoiding more disruptive questions about how knowledge production itself.

These examples highlight that sampling strategies are never purely technical; they emerge from existing epistemic assumptions about who can be studied, who is available for extraction, and what counts as meaningful human difference.

Moreover, the structures of scientific practice often perpetuate exclusion through mechanisms such as “helicopter science” where research is conducted in and on marginalised communities without meaningful collaboration or credit for local scholars and the erasure of indigenous knowledge systems (Atutornu et al., 2022). These practices not only reproduce hierarchies in authorship and career advancement but also undermine the epistemic legitimacy of alternative ways of knowing.

These broader structures intersect with the scientific assumptions researchers bring into the laboratory long before sampling decisions are made. It has long been clear how genetics is entwined with understandings of identity and human difference, and that this is not a question of the ‘consequences’ of research, but of the decisions and assumptions made long before samples are even collected (Rabinow, 2002)

The ways in which scientists categorise, cluster, and analyse data remain grounded in cultural assumptions about human difference (Bliss, 2011). Educational experiences shape these assumptions long before researchers ever enter a laboratory. Donovan (2017) found that adolescents, though able to articulate race as a social construct, frequently reverted to biological explanations. Jamieson and Radick (2017) similarly showed that determinist thinking among university students was shaped by early curricular exposure to biological essentialism. If racial determinism becomes embedded at the educational stage, its legacy may influence how researchers pose questions, design studies, and interpret results. Research in science education suggests that these epistemic habits are cultivated through curricular framings that often reproduce hierarchical and Eurocentric models of knowledge (Gandolfi, 2021), while limiting opportunities for students to interrogate how scientific categories acquire authority (Allchin, 2020; Waight et al., 2022). The implications of these inherited frameworks may extend beyond scientific practice itself. As seen in Rabinow's (1996) account of biosociality, biological classifications can become the basis for new forms of social identity and collective belonging. In this sense, educationally inherited essentialist frameworks may shape not only how knowledge is produced, but also the social

meanings and identities that emerge around genomic understandings of difference (Rabinow, 2002).

The discussion so far has focused on race, ethnicity and genetic ancestry as mechanisms by which inequities are reproduced. However, these are no doubt part of a broader set of scientific practices; workforce diversity, unequal access to resources, and extractive research models that together sustain the structural hierarchies within which science operates.

Who is present within the research workforce shapes what is imagined as scientifically important and which populations are viewed as accessible, legitimate, or visible in genomic sampling.

Representation within the research workforce compounds these epistemic concerns. The underrepresentation of *Black*, Asian, and Minority Ethnic scientists in the UK is well documented. Of the 11.3% “non-white” scientists at Sanger just 0.7% identified as *Black*, compared to Cambridge’s 3.3% population baseline (S. Ahmed, personal communication, May 2022), mirroring national disparities (Adedokun et al., 2016; Bonham & Green, 2021; Higher Education Statistics Agency, 2021). Gibney (2022) reports that fewer than 1% of UK university professors are *Black*, and that *Black* students remain significantly less likely to progress to postgraduate research compared to their *White* peers. These disparities are common in scientific institutions, where leadership and senior research roles remain overwhelmingly *White*. Such demographic imbalances risk narrowing the perspectives that inform research agendas, leading to genomic science that primarily serves dominant populations (Gibney, 2022).

As Marks, (2012) notes, traditional science often defines validity as a statistical fit between theory and reality, a narrow epistemic frame that sidelines other forms of insight (Gandolfi, 2019). Alternative epistemologies could therefore challenge this reductionism. As Feyerabend, (1981) argued that what makes science distinctive is not its methods, but its epistemic structure, its ways of claiming and validating knowledge.

The challenge, then, is not simply to diversify participation in science, but to transform its epistemic foundations. These critiques suggest that sampling practices are not only technical procedures but reflections of deeper epistemic commitments, institutional power, and structural inequities.

A justice-oriented epistemology in genomics would centre pluralism in values, reflexivity in method, and accountability in practice. It would confront the dominance of European datasets and push for nuanced, context-specific descriptors that reflect social realities, not false biological taxonomies. As Broussard (2023) argues in the case of technology, injustice in science is often systemic and not incidental. When populations are mislabelled or omitted entirely, genomic research can reinforce hierarchies under the guise of neutrality.

The following section explores the question of scientific responsibility, considering what it might mean for genomic researchers to take seriously their social and moral obligations in the face of structural injustice.

2.6 Scientific Responsibility

The historical construction of race and the attendant hierarchies discussed in Sections 2.3.2 and 2.3.3 suggest an ethical and epistemic responsibility to critically examine the inherited structures that continue to shape contemporary scientific practice. This position aligns with broader calls for greater scientific responsibility and reflexivity (Forssén et al., 2011). Douglas & Branch, (2024) argue that the mid-twentieth-century ideal of science as value-free and detached from social responsibility is no longer tenable. In their view, the principle of "freedom from responsibility" must give way to one of "freedom with responsibility", echoing Panofsky & Bliss's, (2017) account of science's self-regulating authority and prompting a reconsideration of the social contract between science and society. Similarly, Graves et al., (2022) contend that scientists, as members of society, have responsibilities that extend beyond advancing disciplinary knowledge to advancing the broader needs of humankind. Meeting this challenge requires more than technical expertise. It demands an understanding of the social contexts within which genomic science operates and a willingness to interrogate how established methodologies, classificatory systems, and inherited assumptions may undermine, rather than enhance, equitable outcomes. If population is, as Hinterberger, (2012) argues through a Foucauldian lens, a problem of power and politics, then responsibility in genomics must include examining how populations are constituted across scientific, institutional,

funding, and ethical infrastructures, not merely how research findings are communicated.

While many scientific institutions explicitly articulate socially oriented missions (Genome Research Limited, 2022; The Francis Crick Institute, 2025; The Royal Society, 2025) , scientists themselves are often motivated by diverse and sometimes competing priorities, including career progression, publication metrics, disciplinary norms, intellectual curiosity, and public benefit (Besley & Nisbet, 2013; Rainie, 2015). As a result, social responsibility is frequently treated as distinct from the responsible conduct of research, which remains largely focused on procedural obligations such as producing reliable results, avoiding misconduct, sharing findings, and mentoring future researchers (Pimple, 2002; Weed & McKeown, 2003). Yet questions of scientific responsibility inevitably raise questions of justice. Determining what responsibilities scientists have, to whom they are owed, and how they should be enacted requires engagement with broader theories of social justice.

Social justice has been conceptualised through a range of theoretical traditions within the social sciences. Rawls & Kelly, (2001) theory of justice as fairness provides a foundational account of distributive justice, focusing on the equitable distribution of rights, opportunities, and resources through institutions that secure equal liberties, fair equality of opportunity, and arrangements that benefit the least advantaged. Fraser (2009) extends this framework by arguing that justice requires not only redistribution but also recognition and representation, drawing attention to the cultural and political dimensions of inequality. Honneth(1996) similarly foregrounds recognition as a condition for social participation and human flourishing, while Young, (1990) critiques what she terms the "distributive paradigm", arguing that justice must also address structural processes of oppression, marginalisation, powerlessness, and cultural domination. Extending these critiques, Mills (2005, 2017) argues that liberal theories of justice often abstract from the historical realities of colonialism, slavery, and racial domination, such that ideal theory can "ignore, marginalise, or trivialise the distinctive experience of the oppressed" (Mills, 2005p.168). Collectively, these approaches shift attention from the distribution of resources towards the operation of power within institutions, social structures, and systems of knowledge production.

These critiques are particularly relevant to genomic medicine. Advancing social justice requires more than individual integrity or procedural compliance; it demands engagement with the structural conditions that enable or constrain equitable outcomes. Young, (2002, 2011) work is particularly instructive in this regard. She argues that injustice emerges through both oppression, which constrains self-development, and domination, which constrains self-determination. Importantly, her social connection model of responsibility extends responsibility beyond individual culpability to encompass participation in structural processes that reproduce injustice. From this perspective, scientific responsibility cannot be understood solely as adherence to ethical procedures or professional norms. It also entails recognising and addressing the ways in which scientific practices may contribute to, sustain, or challenge broader structures of oppression and domination. Genomic research, despite its empirical and ostensibly objective orientation, is neither insulated from nor external to these processes. Researchers frequently recognise the tensions inherent in conducting socially contested research, balancing the potential to advance clinical knowledge against the possibility of reinforcing harmful narratives or inequalities (Forssén et al., 2011). Consequently, while many scientists explicitly reject racism and may understand their work as contributing to social progress, they often struggle to articulate what social justice requires in practice (Bliss, 2011; Ossorio & Duster, 2005). As Martschenko (2022) observes, researchers frequently express uncertainty about what social responsibility entails and whether it is possible to prevent the misuse or reinterpretation of scientific findings.

For these reasons, this thesis draws on Critical Race Theory (CRT) as its primary social justice framework. CRT places race and racism at the centre of analysis rather than treating them as secondary forms of disadvantage. It understands racism as ordinary, structural, and historically embedded within social institutions, law, policy, and knowledge production (Crenshaw, 1988; Delgado & Stefancic, 2017). This orientation is particularly relevant to genomic medicine, where race is formally rejected as a biological category yet continues to re-emerge through classificatory practices, population categories, research infrastructures, and scientific narratives. As an emancipatory framework, CRT is concerned not only with analysing race and racism but with exposing and challenging the structures that reproduce racial inequality.

The relationship between science and social responsibility remains politically contested and only partially defined (Glerup & Horst, 2014). This ambiguity enables researchers to

frame their role primarily as producers of knowledge while responsibility for downstream consequences is displaced onto other actors (Douglas & Branch, 2024; Forssén et al., 2011). However, scientific responsibility extends beyond the production of accurate knowledge to encompass accountability for the social consequences of classification. Within genomic medicine, decisions about how populations are defined, categorised, and represented are not merely technical choices but interventions with implications for equity, inclusion, and the distribution of benefits and harms. Scientific responsibility therefore becomes inseparable from questions of social justice, requiring attention not only to methodological validity but also to the social, political, and historical consequences of classification practices and the ways scientific knowledge may reproduce or challenge existing forms of racial inequality. Building on the concerns about epistemic formation discussed in Section 2.5.3 (Gandolfi, 2021; Waight et al., 2022), the remainder of this section examines two interconnected domains of scientific responsibility: science's role in shaping public understandings of human difference, and the pursuit of data justice and adequate representation as conditions for equitable genomic futures (Panofsky & Bliss, 2017).

2.6.1 Contribution to Public Understandings of Human Difference

Science has long shaped how societies understand human difference, from its historical role in codifying race as biological to its influence on contemporary ideas about identity and belonging. These narratives circulate far beyond laboratories; they appear in classrooms, policy debates, public health messaging, and popular media, becoming part of the everyday “common sense” through which people interpret human variation (El-Haj, 2007; Nelson, 2016; Panofsky, 2014; Wailoo et al., 2012). Yet the authority of scientific narratives can obscure the political choices they contain. TallBear (2013), for example, shows how genomic framings of indigeneity echo older colonial logics, while Nelson, (2008, 2016) illustrates how *Black* communities negotiate and reinterpret genetics, revealing science as socially and culturally embedded rather than neutral.

How scientists conceptualise human difference often diverges from how the public or research participants understand it. Researchers may rely on broad labels such as European or South Asian for analytical convenience, whereas individuals hold far more

nuanced or fluid identities (Ridgeway et al., 2019; Schwartz-Marín & Wade, 2015; Weller, 2010). Debates persist over the reliability of self-reported ancestry: some highlight inconsistency as a limitation (Shim et al., 2014), while others argue that paired with genomic data, self-identification offers important insight into social and environmental influences on health (Lorusso & Bacchini, 2015; Mersha & Abebe, 2015). Genetic similarity does not map neatly onto social identity, yet both shape how human variation is understood.



Figure 9. How the public views ancestry. The discrete and largely continental groupings reflect how public understandings often lack access to the discontinuous and overlapping patterns seen in genomic data. Produced by the author.

For some public, fixed and racialised ideas of human groups persist, reinforced in part by direct-to-consumer ancestry testing, which presents identity as pie chart percentages rather than as the complex, continuous patterns recognised by population genetics (Nelson, 2018; Phelan et al., 2014; Roberts, 2011). Public perceptions, though, with lived experience of fluid groups, then treat ancestry as discrete and largely continental, as illustrated in Figure 9. Categories circulate between scientific and public spheres, creating feedback loops that shape both interpretation and the incentives of genomic research (Panofsky & Bliss, 2017).

Outram et al. (2018) show that many people, particularly those they describe as non-*White* and lower socioeconomic groups in the United States, continue to view race as relevant to explaining social inequality, suggesting a broader scepticism toward claims that society is “post-racial.” Similarly, Schwartz-Marín & Wade (2015) demonstrate how ideas of nation, race, and belonging shape public interpretations of genetic ancestry, revealing persistent tensions between visible physical traits and the abstract claims of genomic science. At the same time, ancestry testing can highlight the fluidity of identity, prompting individuals to acknowledge mixed heritage and move away from rigid racial frameworks (Aspinall, 2018; Atkin et al., 2022).

Together, these dynamics illustrate how scientific practice, public interpretation, and commercial genomics continually interact to shape understandings of human difference. Even as genomics rejects the notion of biologically discrete races, the categories used, the language chosen, and the ways results are communicated can reproduce older racial logics in new forms. These patterns have real consequences for whose genomes are studied, whose health needs are prioritised, and whose perspectives shape scientific knowledge.

If scientific narratives shape how human difference is understood, then the responsibility extends beyond communication to the practices through which data is gathered, and people are recruited into research, since these processes crystallise conceptual assumptions into scientific fact. The next section examines how decisions about sampling, inclusion, and representation are shaped by historical and structural forces, and why confronting them is essential to producing equitable genomic science.

2.6.2 Ethical Governance, Consent, and Structural Responsibility

In genomics, the prominence of informed consent as an ethical touchstone reflects procedural orientation. Consent is a widely recognised and easily communicated concept, one that can be addressed through clear policies, forms, and institutional oversight. It offers an appealingly concrete response to ethical concerns, allowing researchers to demonstrate accountability while leaving deeper structural critiques untouched.

Informed consent rests on the principle that freely given agreement lends moral legitimacy to actions that might otherwise be unacceptable (Boulton & Parker, 2007). Within biomedicine, it is widely regarded as the primary mechanism for protecting individual rights and welfare. Yet, as Manson & O'Neill (2007) argue, informed consent cannot serve as an ethical panacea. It fails when individuals cannot fully grasp the information, when transactions are flawed, or when excessive demands for “fully explicit” consent make it impractical.

For example, numerous ethical concerns have been raised about the handling of genetic data from marginalised populations, including the documented mistreatment of Uyghur and other ethnic minorities in China, as well as Roma communities in Europe (Lipphardt et al., 2021; Normile, 2021). Without situating consent within a wider framework of justice, relying on it risks offering reassurance without resolving the underlying inequities in research practice (Manson & O'Neill, 2007).

Informed consent can be procedural and formulaic, but does not have to be and, in fact, often is not in practice (Boulton & Parker, 2007). Recognising this requires attention not only to policy and formal guidelines but also to the moral and epistemic life of laboratories, research projects, and collaborations. It is precisely this gap between procedural compliance and the lived moral dimensions of scientific work that Parker (2012) urges us to examine. Drawing on Kleinman's critique of bioethics, Parker argues that conventional frameworks, grounded in abstract principles, often fail to capture the everyday moral reasoning through which researchers navigate their practices. A key contribution of Parker's work to the ethics of genomics is his distinction between the moral and the ethical. Ethical procedures refer to the formal mechanisms that govern research, while moral judgment concerns more profound questions of what is right, fair, or just. As a result, practices such as consent forms or data-governance protocols can be followed meticulously even when the underlying moral issues remain unexamined. A study may, for example, classify participants into population groups in line with institutional guidelines yet give little attention to whether those categories reproduce social inequalities. Presented as neutral or routine, such categories are historically contingent, shaped by legacies of racialisation, colonial science, and institutional standardisation. They become ethical objects only when these inherited norms are made visible and open to critique.

Parker's framing highlights a risk: harmful or exclusionary practices can persist precisely because they are embedded in what is considered ordinary or even virtuous within scientific culture. The history of eugenics, once framed as a moral good, shows how scientific authority can legitimise unethical action. In contemporary genomics, similar risks arise when institutional norms and methodological conventions go unexamined.

Social science research has shown that the most consequential ethical issues in genomics often arise not in formal consent processes, but in the everyday routines such as where categories are made, applied, and normalised. Although these studies vary widely, they share a focus on the classificatory habits, institutional norms, and historical inheritances that procedural ethics cannot reach.

The following section examines these social structures within science, exploring how racialised thinking can become embedded in routine practice and shape genomic research and its applications.

2.7 Studying Social Structures in Science

This work builds on a wider Science and Technology Studies (STS), tradition of laboratory and biomedical ethnography, including studies of scientific practice by scholars such as Cetina(1999), Latour & Woolgar, (1986), Mol (2002) and Waldby and Mitchell (2006), which demonstrate how scientific facts, classifications, and forms of value emerge through social, material, and institutional processes rather than being simply discovered.

A range of theoretical approaches have been used to examine scientific practice, including Actor-Network Theory(Latour, 2005), co-productionist approaches (Sheila Jasanoff, 2004), Foucauldian analyses of knowledge and power (Foucault, 1980; Rose, 2007) and broader social constructivist accounts of scientific knowledge (Cetina, 1999; Latour & Woolgar, 1986). Each offer valuable insights into the production of scientific facts and institutions. For this thesis, critical Realism (Bhaskar, 1975) was adopted (see 3.1 rationale for the theoretical framework). This orientation is particularly important for a study concerned not only with how genomic classifications are enacted, but also with the

broader institutional and structural conditions through which they emerge and acquire authority.

Within STS a substantial body of ethnographic scholarship has examined how genetic and genomic knowledge is produced in practice (Martschenko, 2022; M'Charek, 2005; Rabinow, 2002). Rather than treating scientific categories as neutral reflections of biological reality, these studies investigate how classifications of population, ancestry, race, and disease are assembled through everyday laboratory work, institutional routines, technological infrastructures, and broader social and political contexts.

Together, they provide an important methodological and conceptual foundation for understanding the social life of genomic knowledge. Their work resonates with Bowker and Star's (2000) influential argument that classification systems are never merely technical tools but social and political infrastructures that shape what becomes visible, measurable, and actionable. This insight is particularly important for studies of genomics, where categories of ancestry, population, and difference increasingly inform research, policy, and clinical practice.

Montoya's *Making the Mexican Diabetic* (2011) exemplifies this approach through sustained ethnographic engagement across clinics, laboratories, and public health infrastructures in Mexico and the United States. Following the trajectory of "Mexican-American" blood samples, Montoya demonstrates how national and racial categories are produced, accrue authority, and travel across institutional contexts. This work exposes the limits of procedural fixes, such as consent, when broader governance frameworks and public health agendas drive the classification process.

Fullwiley (2011) traces how funding priorities, publication norms, and laboratory routines render race a convenient, if problematic, proxy. She employs multi-sited ethnographic fieldwork across the United States, France, and Senegal of researchers in sickle cell and related genetic work. Her work illuminates the gap between declared principles and daily methods, showing how race persists as a moral routine rather than an object of sustained ethical interrogation.

The findings of Fujimura and Rajagopalan (2011) underscore that analytic decisions can proxy Race, sedimenting categories into practice. They combine ethnographic observation and interviews with scientists conducting GWAS on human variation and complex disease. They identify efforts to replace racial labels with "genetic ancestry"

while detailing the socio-material and technical work and software that underpin these shifts.

Martschenko, (2022) adopts a different vantage point, using semi-structured interviews with socio-genomics researchers working on intelligence and educational attainment. Her analysis foregrounds scientists' own understandings of "social responsibility," revealing tensions between pursuing scientifically and socially contested research, the perceived potential benefits, and the difficulties of public engagement. Importantly, for this thesis the study surfaces uncertainty among researchers about what social responsibility entails in practice, highlighting the absence of shared frameworks for addressing the deeper structural and historical dimensions of their work.

The studies reviewed here demonstrate the value of ethnographic approaches for examining the ethical and epistemic life of genomics beyond the limits of procedural ethics. Ethnographic approaches make visible the tacit norms, institutional pressures, and everyday routines through which population categories are produced and normalised. They therefore provide a valuable means of examining how classificatory practices become embedded within scientific institutions and wider systems of governance.

Other scholars have pushed this inquiry into adjacent terrains. Reardon, (2009, 2017) interrogates the institutionalisation of ethics in genomics, particularly through ELSI programmes, showing how formal governance can contain ethical debate within procedural boundaries that sidestep colonial histories and the political economy of genomic knowledge. By insisting on collective rights and relational accountability TallBear (2013) reframes the ethics of genomics from the standpoint of Indigenous sovereignty, challenging autonomy-based consent models. M'Charek (2005) demonstrates how race is not simply discovered within genetic research but materially enacted through laboratory practices, technologies, databases, and interpretive decisions. Her work is particularly influential in showing how racial categories persist not because they are biologically self-evident but because they are continually assembled through scientific practice.

Collectively, this body of work has been highly successful in demonstrating how genomic classifications are produced and stabilised. By tracing the social and material life of categories, these studies make visible the points at which technical problem-solving and

ethical judgement intersect. While many of these studies address questions of inequality and categorisation, relatively few engage in sustained dialogue between Critical Race Theory and ethnographic analyses of contemporary genomic practice, particularly within the context of UK genomic medicine.

Several theoretical traditions have examined the production and consequences of racial categories, including racial formation theory (Omi & Winant, 2014) and postcolonial theory (Haddour, 2019; Partner & Foot, 2013) offer important insights into the historical production of racial categories and the enduring legacies of colonialism. Racial formation theory is particularly valuable for understanding how institutions participate in the creation and transformation of racial meanings over time. Critical Race Theory was selected because it provides a more explicit framework for examining the relationship between racialisation, structural racism, and institutional power (Delgado & Stefancic, 2017; Rollock & Gillborn, 2011). As discussed in Section 3.3, this orientation is especially relevant to genomic science, where contemporary classificatory practices remain connected to longer histories of racial categorisation while raising questions of accountability, responsibility, and reparative justice.

Like Fullwiley(2011), M'Charek (2005), Montoya, (2011)) and (Fujimura and Rajagopalan, (2011) this thesis examines how categories of human difference are produced and negotiated within scientific practice. However, it extends this literature by focusing on contemporary genomic medicine in the UK, a context shaped by distinctive histories, institutions, and commitments to diversity in genomic research. By bringing STS scholarship on classification into dialogue with Critical Race Theory, the thesis examines genomic categorisation not only as a technical and epistemic practice but also as a site through which broader questions of inequality, accountability, and justice are negotiated.

2.8 Gaps in Literature

There is a growing body of literature examining the use of race and ethnicity as proxies for genetic ancestry in genomic research. However, much of it lacks critical engagement with how these concepts are operationalised across different institutional and cultural

contexts. Notable contributions such as guidance from the National Heart, Lung, and Blood Institute's Trans-Omics for Precision Medicine (TOPMed) programme (Khan et al., 2022) and recommendations for improving inclusivity in pre-clinical research by Zaaier & Capes-Davis (2021) offer pragmatic interventions. Yet these often assume universal applicability and overlook the socio-political conditions required for their implementation. Sustaining such recommendations requires sustained effort and coordinated collaboration, yet scholarship on the practical challenges of building meaningful partnerships in a global genomic research environment marked by entrenched power asymmetries, disciplinary silos, and uneven research infrastructures remains limited.

Although the debate on human population descriptors is recognised as global (Popejoy et al., 2018) the literature remains dominated by Euro-American perspectives. The marginal visibility of contributions from scholars and contexts outside North America and Western Europe reflects a narrow epistemic geography, constraining the field's capacity to account for diverse experiences and knowledge systems. This is particularly significant given the UK's ambitions to lead in genomic science. Despite this positioning, there has been relatively little critical reflection on the UK's colonial legacies and their continuing influence on how race, ethnicity, and ancestry are conceptualised and used in genomic research.

The UK's distinctive history, shaped by colonial taxonomies and postcolonial legacies, means that, while race and ethnicity are socially constructed, they remain materially consequential in structuring inclusion, classification, and authority within research.

As Reardon (2017) and Parker (2012) both argue, in different but intersecting ways, the ethical challenges of genetics are embedded in institutional norms and epistemic cultures. Building on these perspectives, the literature reveals several critical gaps:

- A lack of empirical work on how UK genomic researchers operationalise race and ancestry in practice.
- Limited engagement with the historical and political contexts that shape these practices in the UK.
- Few studies that move beyond procedural ethics to examine how moral routines in science may reproduce or challenge structural inequities.

Discussed in the introduction, race and ethnicity in the UK are not neutral descriptors. Scholars such as Benjamin (2019); Nelson (2018); Reardon (2017) and TallBear (2013) urge genomic research to centre racial justice, include scholars of colour and Indigenous experts, and critically reimagine what scientific responsibility entails. Parker (2012) reminds us that ethics is not separate from the moral routines of science.

These arguments provide the conceptual and ethical impetus to engage with the gaps outlined above and, in doing so, establish the foundation for the empirical analysis that follows, examining how researchers navigate these issues in real-world laboratory and institutional settings.

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3 Methodology

Recognising race, ethnicity, and genetic ancestry as historically contingent yet materially consequential categories also requires methods that can make their operations visible within everyday scientific practice. For me, this is not only an intellectual task but also an ethical one. As a researcher whose own history is bound up with the legacies of empire, slavery, and colonial science, I cannot separate my scholarship from the question of integrity: whose perspectives are centred, whose voices are excluded, and how my own presence shapes what is seen.

Ethnography provides one way of attending to these dynamics, enabling me to trace how concepts of race and ancestry are embedded in the moral routines of genomics while also making my own positionality accountable. Accountability also means opening yourself up to scrutiny, and for this, public and patient engagement was equally vital, both as a check on my interpretations and as a resource to ground them in the lived concerns of those for whom genomics is not abstract but personal. These commitments shaped the design of this study, from its early scoping through to analysis and dissemination.

The following chapter, therefore, sets out the project's methodological framework, as shown in Figure 10. I begin by outlining the theoretical perspectives that underpin this approach, critical realism and critical race theory, before turning to the pilot work, research context, data collection, analytic strategies, and ethical practices that structured the study. In making this shift, I move from establishing *why* race and ancestry matter in genomic science to explaining *how* I investigated their significance in practice.

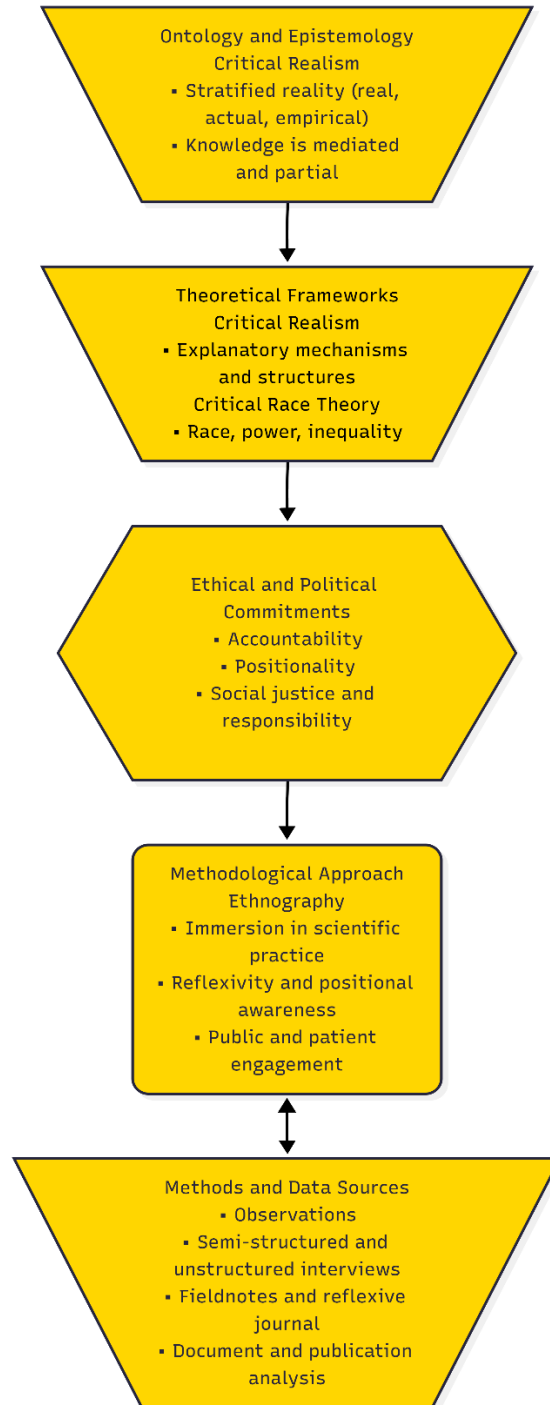


Figure 10. Conceptual flow of ontology, theory, and methods.

The figure provides an overview of how ontological and epistemological assumptions inform the theoretical frameworks, ethical commitments, and methodological approach adopted in this study, leading to the selection of ethnographic methods and data sources. Produced by the author.

3.1 Rationale for Theoretical Framework

As discussed in the previous chapter, the use of race, ethnicity, and ancestry categories is complex, contested, and at times deeply problematic (Birney et al., 2021; Dobson & Arjunan, 2021; Hardcastle et al., 2022). Some scholars have argued that terms such as race are so biologically inadequate and socially harmful that they should be abandoned altogether (Timmer, 2023).

Yet the erasure or replacement of such terms with seemingly less offensive alternatives, pursued in the name of accuracy, can obscure science's historical and ongoing role in producing inequities. Even when unintended, these practices risk reproducing racist ideologies (Graves et al., 2022; Popejoy, 2021; Saini, 2020). Attempts to 'do away' with the problem swiftly and painlessly may, in fact, limit scientists' capacity to recognise, engage with, and address the injustices that can emerge from their work.

This tension highlights the need for approaches that interrogate the social frameworks underpinning scientific practice and grapple with race's nuanced and contradictory operations. Critical Realism (CR) and Critical Race Theory (CRT), which I turn to in the following chapters, provide such frameworks. Through these perspectives, I outline the methodological tools and ethical commitments that guide this study, showing how they make visible the entanglements of science, race, and power while also opening space for more accountable and just forms of research.

3.2 Critical Realism as a Framework for Race and Ancestry

Critical Realism (CR) provides a way of holding together the tensions between structure, agency, and context that surfaced in both my early research encounters and the broader debates from the literature outlined above. Where scientific categories of race and ancestry risk flattening complexity or erasing social dynamics, CR insists that beneath what is observed lie real but often unacknowledged structures shaping experience. It allows attention to the ways individuals exercise agency within these constraints, without reducing their actions to mere reflections of social forces. In this sense, CR offers a

productive foundation for developing methods that seek to make visible the entanglements of science, race, and power, while also recognising the everyday negotiations of those most affected.

3.2.1 Ontology, Epistemology, and the Three Domains

Roy Bhaskar's (2013) discussion of realist theory, commonly described as Critical Realism (CR), he distinguishes between the 'real' and the 'observable' world. The observable world is constructed from perspectives and experiences and can be measured. The 'real' world, however, exists independently of human theories, perceptions, or constructions. What makes this realism *critical* is its recognition that our access to reality is always mediated by social and cognitive frameworks; we never encounter it directly, but only through the categories, concepts, and assumptions available to us. Critical realists, therefore, argue that while unobservable structures in the 'real' world generate the patterns and events we perceive, the ways we interpret those patterns are inseparable from the frameworks through which we come to know them. Categories such as race or ancestry, for example, do not simply reflect reality but actively shape how it is apprehended. The task of research could be argued to be both to investigate the structures that underpin observable events and to remain reflexive about the interpretive lenses through which those structures are made visible.

Critical Realism separates ontology from epistemology, giving each a thorough account. Ontologically, it asserts that reality exists independently of our perceptions, while epistemologically it recognises that our knowledge of that reality is always mediated, partial, and open to revision. In doing so, Critical Realism outlines three ontological strata (Collier, 1994); see Figure 11).

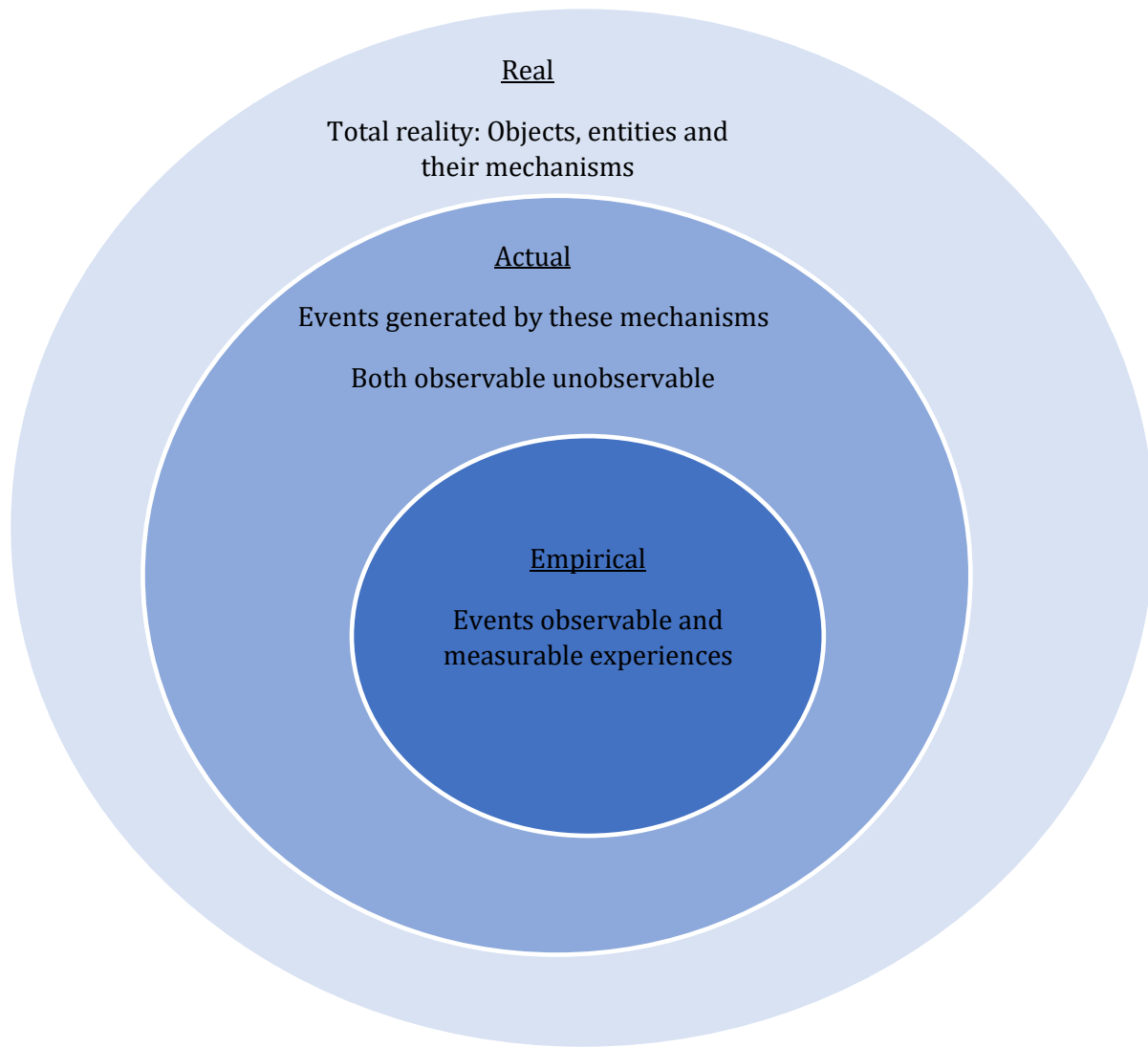


Figure 11: The vector model of critical realism. Produced by the author. *The three domains of Critical Realism (adapted from (Bhaskar, 2013):*

- **The Real:** underlying mechanisms, structures, and entities that exist independently of human observation (though often unobservable).
- **The Actual:** events generated by these mechanisms, whether observed or not.
- **The Empirical:** the subset of events experienced or measured through observation.

3.2.2 Applications and Implications

This stratified ontology allows us to distinguish clearly between what appears, what happens, and what truly underpins those occurrences. Researchers such as Bhaskar (2013) and Lawani, (2020) have highlighted CR's capacity to generate thought-provoking, insightful research. By acknowledging these layers, Critical Realism supports a nuanced analysis of how power relations, cultural norms, and material conditions shape both research practices and outcomes. Importantly, it also offers a framework for developing interventions that address inequities and foster transformative change, most notably in fields such as social work and community psychology, where CR has been explicitly mobilised to guide emancipatory and justice-oriented practice (Botha, 2025; S. Peter et al., 2025).

While Bhaskar's framework describes a logical flow from the real to the actual and into the empirical, the relationship is not unidirectional. Once empirical categories and measurements are institutionalised, they begin to shape practices, policies, and forms of knowledge that generate new actual events and, over time, become sedimented into the real structures of social life. In the context of genomics, categories such as race, ethnicity, or ancestry are not simply neutral descriptors but empirical tools that direct research, organise care, and delimit participation. Their use can therefore reproduce inequalities even as they claim to measure biological reality. For this study, the value of Critical Realism lies in making visible both the generative mechanisms that underlie events and the ways in which empirical practices feed back into those mechanisms. It opens space to ask whether, and how, commitments to social justice might be woven into empirical work, even in discovery science, so that categories are not only instruments of measurement but also sites of ethical responsibility.

In what follows, I turn to Critical Race Theory (CRT), which complements this realist orientation by interrogating how categories such as race and ancestry are themselves produced, contested, and mobilised within systems of power.

3.3 Critical Race Theory

Critical Realism (CR) holds the capacity to suggest strategies for addressing social problems, and the socio-historical context of this study prevents any claim to neutrality around race. Critical Race Theory (CRT), therefore, offers the tools to understand and challenge the social problem of racial inequality. CRT extends this insight by showing how empirical practices, such as the use of racial or ancestry categories, do not simply record reality but actively participate in shaping social outcomes, embedding inequity, or opening possibilities for justice. In this way, CRT offers a framework that links explanatory depth to the pursuit of social change (Borrell, 2018; Tsosie, 2005).

Critical race theory (CRT) regards race as socially constructed, embedded, and nuanced. CRT theorists, therefore, seek to understand and challenge discrimination against persons of colour that has historically served the interests of the dominant *White* population (Delgado & Stefancic, 2017; Rollock & Gillborn, 2011).

3.3.1 Origins and Expansion

In the 1980s, a collective of legal scholars of colour began to formalise a body of work that theorised the systemic and often obscured operations of racism. This framework redirected attention away from overt individual acts and behaviours, towards the structural and institutional dimensions through which racism persist (Bell, 1995; Crenshaw, 1988, 2011).

Since, CRT has expanded to fields of enquiry beyond the American legal system, addressing the rights of specific marginalised groups through its concern with social justice. Racial scholars argue that racism influences rates of morbidity, mortality, and overall well-being (Borrell, 2018; Tsosie, 2005). This expansion has generated distinct bodies of literature and priorities, but across them CRT retains a set of shared core principles that emerged from its legal foundations and continue to anchor its application in new domains.

3.3.2 Core Principles of CRT

CRT is often summarised through six interrelated principles (see Figure 12), each offering a lens for understanding how race and racism operate within social life and scientific practice.

First, *Race is ordinary*; embedded in everyday life, this ordinariness, to the point of invisibility, is part of its power: racism does not only appear in spectacular acts of prejudice, but is sedimented into institutions, practices, and knowledge systems. As demonstrated earlier in this thesis, genetics has historically been a key site where ideas of race were embedded and naturalised, often with enduring effects on racialised and minoritised people (Fullwiley, 2008; Graves et al., 2022; Rutherford, 2021b). Seeing racism as ordinary underscores CRT's insistence that it is systemic rather than exceptional, requiring methods that can make it visible even in the most routine spaces of science.

The next CRT pillar *race is socially constructed* also aligns with, but extends beyond, the current consensus. Race is understood not as a biological essence but as a product of social thought, political projects, and relational positioning (Delgado & Stefancic, 2017). Yet what people mean by "socially constructed" is not uniform: it varies across countries, cultural and ethnic communities, professional disciplines, and personal experiences (Popejoy et al., 2020). In genomics this multiplicity is especially consequential. As Birney et al. (2021) and Hardcastle et al. (2022) argue, the categories deployed by scientists may appear neutral, but they are shaped by contested histories and carry diverse interpretations that influence how they are applied and understood.

Interest convergence is another key pillar, which suggests that racism ultimately serves the interests of a dominant *White* majority, and that progress toward racial equality tends to occur only when it also benefits that group (Bell, 1995). In the context of precision medicine, this principle resonates strongly. Increased data diversity holds not only potential for more equitable outcomes for racially minoritised and underserved populations but also brings advancements that benefit the broader population. However, there are legitimate ethical concerns that such efforts may continue to offer disproportionate advantages to populations who are already well represented in

genomics, such as those of European descent (Bentley et al., 2017; Popejoy & Fullerton, 2016) .

In the pillar of *differential racialisation* CRT highlights how groups are racialised and minoritised differently over time, depending on political, cultural, and economic needs. For instance, behavioural traits have historically been constructed in racialised terms to justify inequality, while groups such as *White* Irish in the UK, once minoritised, came to be absorbed into the *White* majority through social and political change (Edwards, 2018; Garner, 2003). Similarly, individuals of “mixed ancestry” may align with a *White* majority identity in some contexts and not in others, demonstrating the contingent and shifting nature of racial stratification (Hickman, 1998). Echoing debates in genomics about the unstable meanings of ancestry and race.

CRT also insists on the *importance of minority voices* as a key principle. Perspectives from minoritised groups, it asserts, bring knowledge of exclusions, silences, and practices that dominant groups often fail to recognise (Delgado & Stefancic, 2017). This commitment is both conceptual and methodological, requiring reflexivity about whose voices are centred and how they are incorporated into analysis. Within this study, it offers a framework for acknowledging how positionality shapes the interpretation of genomic practices and institutions.

Finally, *intersectionality* or anti-essentialism makes clear that no one has a unitary identity (Crenshaw, 2017). Inequalities arise not only from race but from the intersections of race with gender, class, sexuality, and other axes of identity. CRT therefore resists essentialising accounts of race that flatten these complexities. Although critics argue that intersectionality risks diluting focus or undermining other struggles for justice (Borrell, 2018; Fernander, 2022), CRT explicitly positions itself as a framework capable of holding multiple dimensions of inequality together. This resonates with scholars in science and medicine who emphasise the interwoven ways that categories of race, ancestry, and social difference are mobilised in genomics, shaping both scientific knowledge and its social consequences (Fullwiley, 2008; Hardcastle et al., 2022; Popejoy, 2021).

This relationship between CRT and genomic practice is illustrated in Figure 12, which maps the six core principles of Critical Race Theory onto the analytical concerns developed throughout this thesis.

Critical Race Theory Principles and Their Application in This Thesis

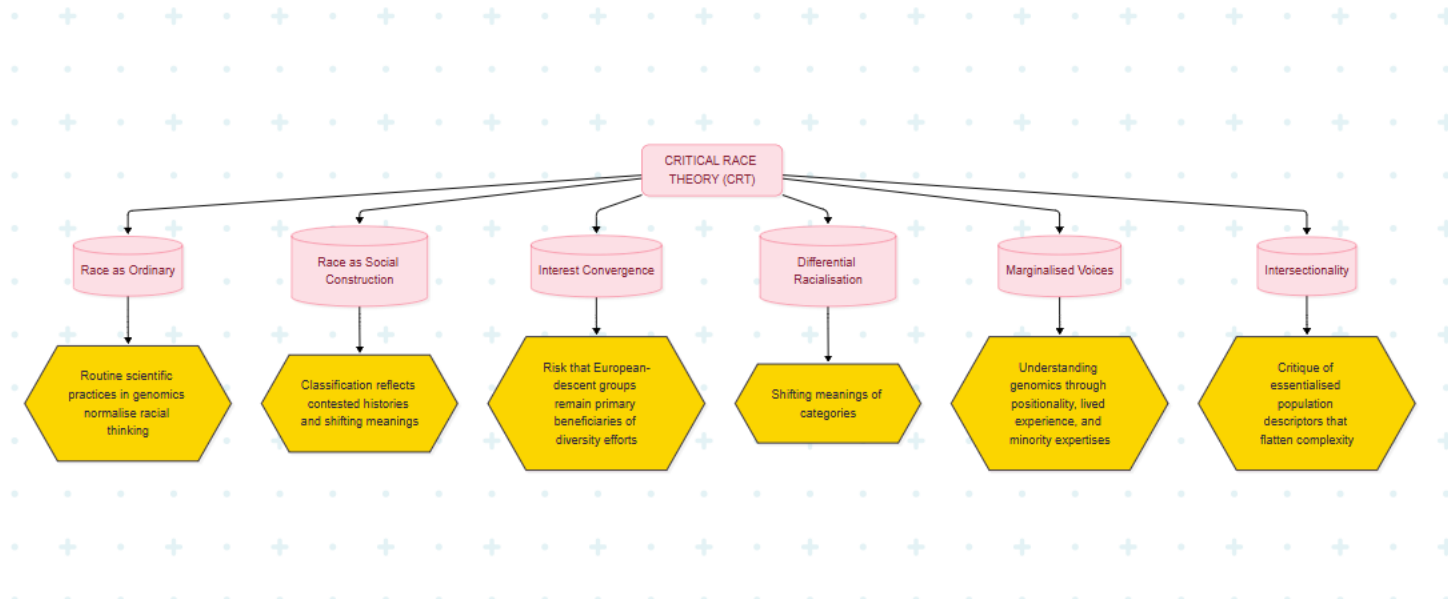


Figure 12. Critical Race Theory principles and their application in this thesis. This conceptual map shows the six core principles of CRT on a single level, with corresponding applications to genomic research placed beneath each principle. It illustrates how CRT helps illuminate the structural, historical, and epistemic forces that shape population descriptors, sampling decisions, data practices, and scientific responsibility. Produced by the author.

3.3.3 CRT, Realism, and Genomics

Although CRT has been widely applied, one criticism concerns the limited attention it pays to questions of realism (Anderson, 2021). From a realist perspective, domestic structural racism can be understood as a manifestation of recurring patterns generated by underlying economic and social structures (B. Carter, 2002; Nayak, 2006). Similar arguments have been advanced in cultural and postcolonial scholarship, where Gilroy (2013) demonstrates how race and class are entangled in sustaining systemic inequalities in Britain. In contrast, more idealist approaches argue that racism is primarily a function of language, symbols, and psychology (Delgado & Stefancic, 2017). While these positions are often presented as oppositional, I do not interpret them as mutually exclusive. Rather, I understand CRT as a set of tools that can illuminate the actual and empirical domains of genomic research within the broader stratified ontology described by Critical Realism.

Complex social structures therefore underlie the conceptualisation and embodiment of race and racial ideas, as suggested by CRT, and these can be interrogated through the ontologies of CR. Together, they emphasise the importance of precise enquiry into the economic, social, and linguistic constructions of race and ethnicity in genetics across the genomic ecosystem. This study leverages the strengths of both frameworks: CR provides explanatory depth on the layered structures that shape genomic research, while CRT highlights the racialised dynamics and inequalities embedded within it. Used together, they enable an approach that is at once analytical and normative, seeking not only to describe but also to challenge the injustices that persist in this field (Delgado & Stefancic, 2017).

3.3.4 Toward Methodological Application

In public health, CRT has already been employed to clarify racial biases woven into scientific knowledge and to inform policy interventions (Ford & Airhihenbuwa, 2018). This orientation underpins the framework's capacity to inform this study's research questions, which are themselves oriented toward equity and accountability. Moreover, CRT's stance against positivism, combined with CR's alternative to both positivism and post-positivism Bhaskar (2013) provides a philosophical foundation for the methodological approach adopted in this work. In this context, ethnographic inquiry, which I will discuss in more detail, offers a particularly suitable means of engaging with the practices, categories, and interactions that constitute race and ancestry in genomics.

3.4 Ethnographic Orientation

Ethnography, broadly defined, is the study of people and cultures through sustained engagement in their natural settings, using observation, participation, and conversation to capture the meanings and practices that shape social life (Hammersley & Atkinson, 2019). Rather than generating data in controlled conditions, ethnographers immerse themselves in the everyday activities of a community or institution and produce what Geertz described as “thick description” (1973, p. 3) of its norms, values, and dynamics. In contemporary practice, ethnography often combines observation with interviews, document analysis, and field notes, creating holistic accounts of how people live, work, and interpret their experiences (Savage, 2000).

While ethnography is often associated with constructivist or interpretivist epistemologies, it can also be productively situated within a critical realist framework. Critical realism acknowledges the existence of real social structures and mechanisms that shape experiences, while recognising that these are always mediated through interpretation (Bhaskar, 2013; Porter, 1993). Ethnography, with its attention to lived practices and situated meanings, offers a means of accessing the ways such structures are enacted and reproduced in everyday settings. As Banfield (2019) argues, critical realism provides ethnographers with a philosophical grounding that balances the search for causal explanations with sensitivity to context and meaning. In this study, ethnography is used in precisely this way: to illuminate how broader institutional and social mechanisms around race, ancestry, and genomics are expressed through the daily practices, discourses, and decisions of researchers.

The method, however, has a contested history. Its origins in anthropology are entangled with colonialism, where Western researchers often studied non-Western societies in ways that reinforced hierarchies of knowledge and power (Clifford et al., 1986). Recognising this legacy is important, particularly in fields marked by inequities.

In this study, ethnography is redeployed to examine a powerful scientific institution rather than an “othered” population. This shift destabilises ethnography’s colonial genealogy by directing its lens toward the structures and practices of genomic science itself. In line with calls for “studying up” (Nader, 1972) the focus here is on elites, institutions, and knowledge producers whose categories and practices carry significant social consequences.

As discussed in Section 2.7, STS scholars have examined scientific practice through a variety of analytical traditions, including laboratory studies, Actor-Network Theory, co-productionist approaches, and biopolitical analyses. While these perspectives inform the broader intellectual context of this research, the present study adopts an ethnographic approach centred on the everyday practices and

interpretations of genomic researchers, reflecting its focus on classification, meaning making, and the negotiation of human difference in scientific work.

3.4.1 Ethnography in the Study of Genomics

Ethnography is particularly well suited to the study of social structures in action. As Young (2010) notes, attention to the ways structures are enacted offers a means of apprehending society as a whole. Ethnography, grounded in participant observation and extended engagement, provides tools to capture these dynamics in situ, cultivating an in-depth understanding of complex institutional and cultural practices (Goodson & Vassar, 2011; Reeves et al., 2008; Van Maanen, 2011). Its defining feature is immersion: the ethnographer gains insight by inhabiting the same spaces, rhythms, and conversations as participants, enabling comparative judgements about what is particular to a setting and what may be more widely shared (Flick, 2022; Hammersley & Atkinson, 2019).

Ethnography is not limited to observing behaviour; it also attends to the discursive practices through which institutions and communities make sense of themselves. Fieldnotes, interviews, and documents become resources for analysing how categories are articulated, justified, and contested in everyday talk and formal communication (Emerson et al., 2011; Poole, 1990; Rinaldo & Guhin, 2022). This attention to discourse is particularly relevant in the genomic sciences, where language is central to how concepts such as “race,” “ancestry,” or “diversity” are stabilised and gain authority.

Questions of validity and reliability are rightfully questioned in ethnographic work (Hammersley, 2018; Hegelund, 2005; Mantzoukas, 2012). Sampling often proceeds opportunistically, by availability, referral, or institutional pathways, and the reader encounters the site through the ethnographer's interpretive lens. Yet these features do not negate rigour. Small (2009) argues that purposive case selection, when guided by iterative engagement with data, produces findings that are highly valid for answering context-specific research questions. Reliability can be enhanced through transparency: being explicit about positionality, access, and analytic decisions strengthens credibility, while digital and collaborative practices expand possibilities for documenting and sharing interpretive processes (Murphy et al., 2021; Savage, 2000). Triangulation further addresses limitations by drawing on multiple data sources, methods, and settings to cross-check interpretations (Alhojailan, 2012; Jonsen & Jehn, 2009; Marshall & Rossman, 2014; Reeves et al., 2008). Situated on the Genome Campus, this study benefited from both data triangulation (through varied sources of institutional knowledge) and methodological triangulation (qualitative immersion with potential quantitative checks), enhancing the validity of its conclusions (Flick, 2018, 2022; Fusch et al., 2018; Heale & Forbes, 2013).

My positioning as a genetic counsellor reinforced the appropriateness of ethnography in this research. Genetic counselling practice is grounded in Rogerian principles of empathy, listening, and person-centred communication (Thorne & Sanders, 2012). These same skills are valuable in ethnographic encounters, where the goal is to create a space where participants can speak openly about their experiences, including sensitive and challenging topics such as race, ethnicity, and ancestry. Similar arguments have been made about the role of nurses and other healthcare workers in ethnographic research, where clinical communication skills strengthen rapport and data quality (Cruz & Higginbottom, 2013; Reeves et al., 2013; Savage, 2000). This alignment of professional training with methodological demands enhanced the credibility of my role as an ethnographer, while also requiring reflexivity about how my dual positioning, as clinician and researcher, shaped both access and interpretation.

Semi-structured interviews formed a natural extension of this approach. Like counselling practice, they balance openness with gentle structure, allowing participants to introduce meanings in their own terms while ensuring that key areas of inquiry are addressed (Denzin & Lincoln, 2018; Kvale & Brinkmann, 2009). This format was particularly appropriate for eliciting in-depth reflections on sensitive and contested concepts such as race, ancestry, and diversity, where rigidity could foreclose nuance, but entirely unstructured conversations risked producing gaps across accounts. Semi-structured interviews therefore supported both the *depth* required for understanding lived and institutional experiences, and the *comparability* necessary for analysing patterns across multiple conversations (Bryman, 2016; Rinaldo & Guhin, 2022).

Ethnography has also been used productively in related areas of genetics and science. In genetic counselling, it has illuminated how knowledge and values are co-produced between clinicians and patients (Cowley, 2016; Outram et al., 2022). Montoya's (2011) and Fullwiley's (2011) work examining diabetes and sickle cell, respectively, illustrate how racial categories are constructed, applied, and revised in scientific practice, often producing contradictions and inequities when race is treated as a discrete descriptor. Their detailed analyses show ethnography's capacity to uncover the institutional drivers and cultural assumptions underpinning genomic "knowledge." More directly relevant, Fujimura and Rajagopalan (2011) studied biomedical scientists working on human genetic variation and common complex diseases, tracing how researchers alternately relied on genetic ancestry or racial categories in their work. Their findings highlight both the methodological ingenuity and the institutional constraints through which these categories are mobilised, as well as the broader political and social landscapes that shape them.

In this study, ethnography was well-suited to interrogate the categories, practices, and institutional cultures of genomics, while the literature review and public engagement strands provided supporting data and perspectives. Examples of critical realist ethnography demonstrate how the method can sit

between constructivist and realist positions, revealing how lived practices reflect underlying social mechanisms while remaining attentive to meaning and context (Porter, 1993). This capacity to move between the observable and the structural made ethnography complementary to a critical realist framework in this project.

At the same time, ethnography has long been a productive site for the development of Critical Race Theory in education and the social sciences, offering narrative and situated accounts that expose racial inequality and the limits of dominant epistemologies (L. Parker, 2015; Solorzano & Yosso, 2002). Critical race ethnography demonstrates how lived experience and storytelling can challenge structures of exclusion while generating new insights into the operations of power.

3.5 Defining and Approaching the Research Question

To clarify the research questions, I undertook two pilot studies. Here, I briefly summarise their contribution to the main study design. The themes that emerged from this process are central to the discussions in Chapters 4–7, where they are examined in greater detail.

3.5.1 Pilot Study One

In early campus meetings and scoping discussions, two senior genomic researchers at Sanger were repeatedly identified as key figures. Both were highly visible in faculty-wide seminars and discussions on equality, diversity, and equity, and their scientific groups were actively addressing challenges around diversity in genomic datasets.

Given the depth of interaction with these researchers, I proposed formalising interviews with them as a pilot study. Approval was granted by the Faculty of Education Research Ethics Committee (REC) and the Higher Degrees Office of the University of Cambridge. Consent materials were prepared in collaboration with my supervisor (Appendix A and B).

Two semi-structured interviews were conducted via Zoom to explore the project's context and rationale. The interviews sought to incorporate researcher perspectives into the doctoral design, while also identifying additional scientists who might contribute to the main study. Recordings were manually transcribed and thematically analysed using a hand-coding approach (Morales et al., 2018).

3.5.2 Pilot Study Two

Following Pilot Study One and an initial review of the literature, the rationale for the research questions became clearer. However, gaps remained, particularly regarding the global genomic ecosystem and the ways in which human population categories had evolved and been operationalised in the UK. The more recent history of these debates, especially from the early 1990s at the Sanger, was largely absent from the literature.

To address this, I conducted a second pilot study to investigate the UK context more directly. This included an interview with a scientist and clinical geneticist who had worked closely with the Sanger Institute in the 1990s, informal interviews with two archivists (one attached to the research group of a pilot one participant, and another working more broadly across the archives), and a conversation with a science historian specialising in genetics, race, and ethnicity. Data were gathered through audio recordings, field notes, emails, and meeting minutes. Some of which formed the Introduction and literature review section of the study.

The decision to pursue ethnography was shaped by my early experiences on the Wellcome Sanger Campus. As captured in the excerpt from a journal entry below, moving between my clinical role as a genetic counsellor in London hospitals and the secluded, highly resourced environment of the Sanger Institute made visible the distance between genomics as a research enterprise and the diverse publics it seeks to serve.

.....
Journal entry, 10.6.22:

The campus is still supporting hybrid working so I don't come here many days in the week. The journey to campus represents a change of state for me. Every stop on the train is a reminder that I am further away from the noise, colours, and chaos of South London. To reach the campus, I jump on a shuttle bus and arrive on a sprawling site where space, in contrast to London, seems underutilised, and chaos isn't welcome. I am not sure I will ever feel like I am part of this place, but I am fascinated by it and every conversation I have. I feel uneasy at times. I wonder if something is beneath the still surface, or perhaps the feeling is my experience of perceiving myself as 'matter out of place'.

.....

The questions that emerged were shaped as much by the insights of scientists, archivists, and historians as by my perspective as a genetic counsellor and a Black woman entering this space. This reflexive positioning is consistent with the philosophical frameworks of Critical Realism and Critical Race Theory introduced earlier. It is therefore necessary to reflect on how position and grounding informed the design of this study, both methodologically and ethically.

3.6 Study Design

I designed the study as an ethnographic investigation that drew on multiple sources of data to address the research questions. The primary sources were observations and semi-structured interviews with genomic researchers across the research contexts. The ethnographic analysis also incorporated documentary evidence, including photographs, written material, study reports, and published research articles. In addition, participants willingly shared materials not otherwise publicly available, such as meeting agendas, draft documents, and conference presentations, which were treated as research data. The research was conducted in the UK across several geographical locations. Genomic researcher data was collected primarily at Sanger and the offices of Wellcome Connecting Science, both located on the Genome Campus where I was based. Additional observations were carried out at related sites, including the University of Cambridge (Kavli Centre for Ethics, Science and the Public). For public audience observations and interviews, sites included the Wellcome Collection in London and the Black Cultural Archives, as well as events at the Sanger in Hinxton.

Table 2 provides an overview of the study design, illustrating how each method of data collection related to the research questions.

Table 2: Overview of Study Design, Data and Analysis

Research aim	Primary data sites	Types of data	Analytic approach
1. Examine how race, ethnicity, and ancestry are conceptualised, operationalised, and contested within genomics	Wellcome Sanger Institute; institutional publications	Semi-structured and unstructured interviews with scientists; observations; fieldnotes; published genomic papers	Thematic analysis of interviews; ethnographic interpretation; discourse-informed analysis of population descriptors
2. Analyse the institutional, methodological, and structural conditions under which these categories are maintained or challenged	Wellcome Sanger Institute; Genome Campus; wider genomic collaborations	Interviews with senior and junior scientists; observations of meetings and routines; institutional documents; publications	Ethnographic analysis; institutional analysis; systems-oriented mapping of structures, roles, and constraints
3. Explore the broader ethical and political implications of classification in genomic science	Wellcome Sanger Institute; public engagement sites (Wellcome Collection, Black Cultural Archives, Beale's High School)	Interviews; observations; fieldnotes; reflexive journal; public-facing discussions	Reflexive thematic analysis; ethical interpretation informed by Critical Race Theory and Critical Realism
4. Develop a framework through which social justice can be embedded into genomic research practice	Cross-cutting across all sites	Synthesised findings from interviews, observations, publications, and reflexive materials	Integrative analysis; theoretical synthesis; framework development grounded in empirical findings

Alongside ethnographic observation and interviews, I undertook a narrative review of the literature published by Sanger during the fellowship period. Scientific papers are genomics' primary mode of communication, representing what can be publicly acknowledged, formalised, and legitimised through peer review. This dual focus reflects the critical realist principle that the empirical does not simply capture reality but is mediated through categories and practices that shape what can be observed. Papers extend the institutional conversation beyond my immediate access and provide a record of what is considered acceptable discourse.

The process of acculturation on-site provided both access and constraints. My positionality as a genetic counsellor familiar with the culture of genomics but simultaneously an outsider critically examining its structures shaped the kinds of data I could generate. Negotiating access required sensitivity to institutional hierarchies and an awareness of how my identity and presence influenced what was disclosed. Problems of access are well-documented in the ethnographic literature (Hammersley & Atkinson, 2019; Reeves et al., 2013; Savage, 2000), and my experience aligns with these accounts, including the selective willingness of participants to engage and uneven exposure to institutional conversations

3.7 Public Engagement as a Parallel Stream

Public engagement was positioned as a necessary parallel stream alongside the ethnographic approach and was integrated throughout the study. These activities acted as both regulator and interlocutor, creating intentional spaces in which the project could be challenged on the potential distributive implications of its findings and ensuring that the research remained grounded in social justice concerns. Public engagement thus served as a “third commentary,” supplementing the perspectives of scientists and the literature with those of publics who are often positioned as the subjects, but not the authors, of genomic discourse.

3.7.1 Position and Grounding

Position and grounding played a crucial role in shaping the design of this study, both as methodological foundations and as integral aspects of its ethical framework. Relationships with the public and wider communities, established before and during the project, were equally important, fostering trust and ensuring accountability to those the study sought to represent.

Although CRT emphasises the role of researcher voice and narrative, investigating racism entailed confronting a phenomenon deeply entangled with my own lived experience, past and present. Researching race while adapting to the new physical and institutional environment of the Genome Campus was at times unsettling. As Rattay et al. (2025) introduce, such work can be uncomfortable for researchers, with implications for both interpretation and well-being. In response, I worked with my supervisor to implement a package of support, including mentorship from a Black woman scholar specialising in racial justice in education and counselling from a Black woman therapist.

The societal voice was also present throughout the project. Building on my background in public engagement, I participated in public-facing events on genetics and ethics, facilitated deliberations with patient and public groups, and collaborated on community projects throughout the fellowship (Appendix C). These experiences affirmed the high level of public interest in how genomics might respond to racial health inequalities.

To mitigate potential bias arising from my positionality and immersion among scientists, I also engaged with artistic and critical perspectives. Specifically, I collaborated with the artists Larry Achiampong and David Blandy on the Genetic Automata project, commissioned by the Wellcome Collection and Wellcome Connecting Science (Blandy & Achiampong, 2022). This series interrogates race, racism, and ancestry in science through film and public dialogue. During the fellowship, I hosted an open conversation with the artists, scientists, and staff at the Genome Campus in September 2022, and participated in a panel at the Museums Association Conference in November 2022 on tackling racism in science. These events overlapped with my ethnographic fieldwork, creating opportunities for cross-fertilisation between artistic critique, scientific practice, and community engagement.

My position and grounding thus informed a study design that is critically engaged and ethically situated. The intersection of professional expertise, personal identity, and community engagement afforded both insider and outsider perspectives, enabling me to navigate professional and cultural contexts with reflexivity and sensitivity. In what follows, I build upon these foundations to describe the design of the main study, explaining how the clarified research questions, philosophical commitments, and pilot work converged to shape an ethnographic inquiry into genomics research.

3.8 Data Collection Site

Wellcome Sanger Institute

The primary site of data collection was the Wellcome Sanger Institute in Hinxton, Cambridgeshire. Founded by the Wellcome Trust in 1992 and named after biochemist Frederick Sanger, the Institute gained global recognition for sequencing around one-third of the human genome in the Human Genome Project, the largest contribution of any single centre (Rogers, 2025). It is now one of the world's major hubs for large-scale genomic discovery, combining high-throughput sequencing with population-scale analysis.

After the Human Genome Project, Sanger expanded into sequencing diverse genomes, including pathogens, parasites, cancer cells, and model organisms, creating shared resources that shaped scientific understandings of human variation and disease (Genome Research Limited, 2022; Wellcome Sanger Institute, 2020). Its research spans six programmes: Cancer, Ageing and Somatic Mutation; Cellular Genomics; Human Genetics; Parasites and Microbes; Tree of Life; and Generative and Synthetic Genomics, all supported by extensive sequencing and data infrastructures.

The Genome Campus also hosts the European Bioinformatics Institute (EMBL-EBI), forming an integrated environment for laboratory science, computational biology and large-scale data management. As a research setting, Sanger provided a key vantage point for examining how population descriptors, scientific practice and responsibility intersect, given its reliance on global genomic datasets and categories that hold both scientific and social significance.

3.8.1 Participants

'Scientists'

Participants from Sanger included both senior and junior scientists engaged in genomic research, particularly in human genetics and cell-based studies where considerations of race, ethnicity, and ancestry arise in relation to datasets.

I conducted primary interviews with six senior researchers, all of whom were heads of research groups. Heads of group were selected because of their strategic influence over research questions, funding applications, and publications. Accordingly, the inclusion criteria focused on participants who could provide insight into the broader genomic ecosystem and research culture at Sanger.

I did not collect demographic characteristics (e.g. ethnicity, age, gender) due to the small sample size and recruitment approach. However, issues of identity and self-identification were explored as part of

the semi-structured interviews with primary participants, in line with the research objectives concerning race, ethnicity, and ancestry.

I conducted secondary interviews with a further six scientists. In addition, multiple informal discussions and observations with scientists took place in both individual and group settings.

‘Public’

I selected participants based on their interest in genomics and the ethical issues surrounding race, ethnicity, and ancestry. Demographic data (e.g. age, gender, ethnicity) were not systematically collected due to the small sample size and recruitment approach. However, recruitment sites were chosen specifically for their engagement with Black, Asian, and other groups underrepresented in genomic research. All participants spoke English, and no communication needs were declared, although one participant disclosed additional learning needs. The ability to communicate in English was not set as an exclusion criterion. Although public audiences ranged from 20 to 80 people, four individuals volunteered to participate in a formal interview across the three sites.

The three sites’ participants were recruited from and were connected to the *Genetic Automata* film series, these were:

- **Black Cultural Archives (BCA):** Adult attendees (18+) of *Genetic Automata: Black Resistance to Scientific Racism*, an event that combined an exhibition of the film series with archival examples of Black resistance to scientific racism.
- **Beale’s High School:** Young people (15–16 years old) enrolled in *Genetic Automata: Project Intergen*, a creative workshop series that used the films to foster intergenerational conversations around science and ethics.
- **Wellcome Collection:** Adult attendees (18+) of *Passing Down*, a panel discussion linked to the *Genetic Automata* exhibition, featuring the artists Larry Achiampong and David Blandy.

Although public data from these sites did not directly contribute to the thesis's findings, encounters there shaped my positionality and provided an essential point of critical accountability.

3.8.2 Recruitment

I employed snowballing and opportunistic approaches to recruit scientists. Following the pilot phase, in which a smaller number of scientists were engaged, participants recommended other colleagues who met the inclusion criteria and were open to being interviewed. Some of these scientists also extended

invitations to their meetings, open days, and laboratory spaces, which provided opportunities for further interviews and observations. In addition, on-campus events relating to data diversity, ethics in science, and the *Genetic Automata* film series offered further opportunities for scientists to self-select into informal conversations on the research topics.

Young people at Beale's Academy participated voluntarily in alignment with their school's core curriculum. Attendees at the Wellcome Collection gallery event and the Black Cultural Archives also self-selected to participate. At each public event, opportunities for individual interviews were offered. Interested participants were contacted directly via email (or, in the case of young people, through a responsible adult) and interview times were arranged at their convenience. Interviews were offered both in person and virtually.

3.9 Data Collection

Interviews

All scientists who volunteered for formal interviews were provided with participant information sheets and consent forms via email (Appendix B1 and B2). Once consent was obtained, between two and four interview dates were scheduled, spaced at least one month apart. At least one interview was conducted in the participant's usual working environment. Initial interviews were semi-structured, providing a degree of consistency across participants, while subsequent interviews were open and unstructured, allowing earlier conversations to shape later discussions (Appendix D).

Public participants who volunteered at events were also provided with information sheets and consent/assent forms (Appendix B4 – B9). Public interviews were conducted within three months of the exhibition observation, at a time chosen by the participant. These interviews were semi-structured and designed to explore non-scientists' attitudes toward the use of population descriptors in genomics, as well as to reflect on emerging themes from the scientists' data.

Interviews were conducted either in person or online via Zoom, each lasting 45–60 minutes. In-person interviews were recorded using the researcher's iPad; online interviews were recorded via Zoom. In both cases, transcripts were auto-generated (Zoom or Turboscribe™) and then cross-checked against the audio file for accuracy. Once verified, all files were downloaded to a password-protected external drive, and copies were deleted from Zoom, Turboscribe™, and recording devices.

Observations

Consent for observations was sought in writing from the relevant event organiser, team leader, or faculty management. An invitation to observe was treated as tacit consent, with organisers asked to remind participants of the presence of an observer.

At Sanger, observations took place during team meetings following invitations from heads of programme, with notification of the observer's presence always made explicit. These included listening to routine conversations about the practicalities of research and interactions among scientists across different career stages. Less specific observations were also made in communal areas such as dining spaces, the library, and meeting rooms, as well as at open campus-wide seminars, workshops, and meetings with both internal and external researchers. Online webinars conducted via Zoom were recorded onto a laptop and saved on a password-protected external drive.

In relation to Beale's High School's Genetic Automata: Project Intergen, participation began with attendance at a workshop. Participants were informed that further observations would take place during a later visit to display their artworks at the Wellcome Collection, and that the opportunity to volunteer for individual interviews would be offered during this initial workshop. At the exhibition, a notice was displayed allowing participants to decline observation. Engagement took the form of conversations with participants about their artworks and the Genetic Automata display. Recordings made in public spaces were used for reflection and destroyed following analysis.

At the Black Cultural Archives and Wellcome Collection Passing Down events, notices were displayed to notify participants of the observation and provide the opportunity to decline. As with other sites, recordings made in public spaces were used for reflective purposes only and destroyed after analysis.

Field notes

Fieldnotes were an important complement to interviews and observations, providing additional context and supporting interpretation. They captured details of who was involved (e.g., a junior scientist, but never names or identifying information), verbatim or semi-verbatim quotations, and non-verbal aspects of interaction such as setting, tone, and body language. They also noted the tools and materials used (for example, the categories applied in classifying data). Notes were handwritten and anonymised either during writing or immediately afterwards.

Fieldnotes were taken in a variety of settings: while sitting in shared spaces at Sanger, during community events, while moving through exhibitions, and after panel discussions. In these contexts, they often centred on personal reflections and impressions rather than direct accounts of conversation. Conversations were never covertly listened to, and no identifying information was recorded. When photos or recordings of public spaces were taken, these were stored securely and used only for reflective purposes.

Alongside these fieldnotes, I also kept a reflexive journal throughout much of the research. This journal recorded thoughts, feelings, and reflections on my own well-being, and became a tool for self-regulation and accountability. It served as an autoethnographic resource, offering insight into how my position and perspective shaped the research, and supporting interpretation when analysing

interviews and observations some time after the events themselves. In this way, both fieldnotes and the reflexive journal operated not only as records of data but also as aids to reflexivity and critical engagement with the research process.

Publications review

To complement the ethnographic and interview data, I undertook a targeted review of institutional publications. The purpose of this review was to trace how human population categories and related ethical considerations appeared in the scientific outputs most visible within the Sanger community.

I adopted a discourse-analytic lens to examine how population descriptors are used and discussed within published genomic papers. This approach treats the publications as ethnographic texts, attending to what is said, by whom, and what is implied or left unsaid.

The weekly research bulletin provided the basis for the review. Circulated internally across the Institute, the bulletin compiles recent publications and preprints associated with Sanger and its partner organisations. As such, it represents more than a record of outputs: it provides a window into the scientific discourse circulating within the Institute, encompassing both the work that staff produce and the research judged to be relevant to the community.

This source is particularly valuable because of the position Sanger researchers occupy within global consortia and collaborations. Staff are frequently one of many contributors to large-scale projects, working with datasets collected across diverse populations, geographies, and research contexts.

The bulletin therefore captures a wide cross-section of genomic science, extending well beyond a narrow disciplinary or institutional focus. In this sense, it offered a robust starting point for exploring how categories of race, ethnicity, and ancestry are invoked within contemporary genomics.

- Sources and Search Process

Papers are identified and compiled by the Sanger Library team, who maintain comprehensive saved searches across multiple bibliographic databases. The library process for circulation of publications can be found in Appendix E.

- Selection of Eligible Papers

Between 9 May 2023 and 26 January 2024, the weekly bulletin circulated 920 papers across 22 bulletins. From this, 148 papers were identified as eligible for the review.

Eligibility was determined with reference to the scope of the study and the ethnographic fieldwork. Specifically, papers were included if they:

- Related to the work of group leaders and lead scientists I had interviewed.

- Reported genomic research making use of human population datasets.
- Represented clinical, computational, wet-lab, or dry-lab studies.

Exclusions were also clearly defined. Papers from the Tree of Life programme, which focuses on biodiversity research, were not included, as they did not address human population categories.

A small number of case studies involving very few patients were retained (n), as were a limited set of papers that addressed patient experience or ethical, legal and social implications (ELSI) in genomics (n). These subsets, while minor in number, were considered relevant for understanding how human population categories and ethical issues were articulated in scientific outputs.

By foregrounding publications linked to both my interview participants and human population data, this selection strategy ensured the review was closely aligned with the study's broader aim: to interrogate how categories of race, ethnicity, and ancestry circulate within genomic science and what ethical implications accompany their use. This approach also follows established practice of using scientific texts as a site for discourse analysis (Kerr, 2000; Myers, 2022).

- Review Method

Papers judged to be in scope were then entered into an Excel spreadsheet for systematic recording and analysis.

For each eligible paper, the following fields were completed:

- Publication date
- Author position (Sanger staff or external)
- Area of interest
- Datasets or collections used
- Source of data
- Population categories identified in methods
- Population categories identified in conclusions or discussions
- Other mentions of population categories (e.g., location or terminology)
- Summary of findings and ethical considerations

Where papers referred to race, ethnicity, or ancestry without specifying the labels used, supplementary material or the direct dataset sources were reviewed to clarify which human population categories were being referenced.

To minimise omissions, a secondary search was conducted within the eligible papers for common terms that might indicate relevant categorisations or ethical reflections. These included: population,

ethnicity, ancestry, cohort, nationals, individuals, adult, children, men, women, diversity, ethics, ethical, fairness, diverse, demographics, metadata.

This structured approach enabled the review to do more than catalogue publications: it created a basis for analysing how population categories and ethical considerations are mobilised within scientific discourse. In this way, the document review formed part of the ethnographic analysis and helped address the study's central questions about the social and ethical framing of genomics.

The interviews, observations, fieldnotes, and publications review provided a multi-layered empirical base for understanding how population categories and ethical reasoning emerge in genomic practice. These methods also supplied the material necessary for the subsequent interpretive analysis and systems-level mapping.

3.10 Analysis

The analytic strategy used in this study brought together a set of interpretive, reflexive, and situated qualitative approaches that were suited to examining how race, ancestry, and genomic practice acquire meaning across scientific, institutional, and public settings. Rather than relying on a single methodological tradition, the analysis was shaped by the demands of each data source and by the epistemological orientation that underpinned the project. The following analysis outlines the interpretive procedures used across textual, ethnographic, interview, and organisational materials, and shows how meaning was generated through immersion, reflexivity, and sustained engagement with the genomic field.

Coding began with interview transcripts, which were read repeatedly and coded inductively within Atlas.ti. Initial codes captured recurring discussions of race, ancestry, diversity, inclusion, responsibility, governance, and classification. Through iterative discussion with supervisors and the wider research group, these codes were refined and grouped into broader thematic categories. These emergent themes subsequently informed the analysis of other data sources, including fieldnotes, institutional publications, and systems mapping materials, allowing patterns identified in one dataset to be explored, challenged, or elaborated through others. While the coding process remained inductive, it was also informed by the study's theoretical interests, particularly questions concerning racialisation, institutional practice, and the production of scientific categories.

The analysis combined formal qualitative techniques with reflexive practices drawn from both ethnography and genetic counselling. Reflective models such as Gibbs' reflective cycle (1988) as cited in Brown (2019) provided a structured way to interrogate non-verbal communication, emotions,

and researcher positionality. Reflective techniques from genetic counselling practice were used to deepen interpretation and maintain critical self-awareness.

All interview transcripts, fieldnotes, documents, and images were managed in Atlas.ti. Interview data were examined using an emergent thematic analysis approach (Cruz & Higginbottom, 2013) in which coding developed inductively and was refined through ongoing engagement with the material and regular discussion with supervisors and advisors (Reeves et al., 2008). Rather than following a predefined coding manual, patterns were identified through repeated readings and interpretive attention to how participants framed population categories, addressed equity and inclusion, and negotiated ethical issues within genomic practice.

Fieldnotes were integral to the analysis. Although not formally coded, they were read reflexively and revisited throughout the study. They provided contextual detail, captured interactional dynamics, and situated themes emerging from interviews and publications. Reflexive journals and memos complemented this process by documenting the researcher's own positioning and responses, ensuring transparency in how interpretation was shaped by lived experience within the research setting.

For the review of institutional publications, eligible papers were scanned for population descriptors through keyword searches such as population, ethnicity, ancestry, cohort, and diversity. These terms were traced back into the full text to recover their meaning and contextual use. Where relevant, additional papers using the same terms were reviewed to identify recurring themes. The process drew on discourse-analytic sensibilities by attending to the ways scientific texts mobilised categories and ethical claims Kerr (2000); Myers (2022); Soini and Birkeland (2014).

Although this study does not constitute a full CBPR approach (Israel et al., 2005) it draws on CBPR principles of recognising participants as knowledge contributors, foregrounding dialogue and voice, and integrating engagement into the analytic process through the use of intentional public spaces as described in 3.8.3 and Appendix C. Finally, across interviews, observations, and publications, I examined references to institutions, partners, and collaborators. This analytic strand sought to map the social and organisational ecosystem of genomics. Conceptually, this aligns with institutional ethnography (Smith, 2005) and social network approaches (Crossley, 2010), in which attention to named actors and organisations provides insight into the structuring of a research environment. This mapping informed the development of a visual systems representation that traced the connections, flows, and influences shaping genomic practice. Ethnographic immersion strengthened this process by grounding the map in observed routines, relationships, and institutional dynamics rather than abstract categories.

Critical Race Theory informed the analysis by sensitising attention to processes of racialisation, structural inequality, institutional power, and the ways in which seemingly neutral classificatory practices may reproduce broader social hierarchies. Rather than functioning as a fixed coding framework, CRT shaped the interpretation of themes and guided questions about how race and ancestry were understood, justified, and operationalised within genomic practice. Critical Realism informed the analysis in a different way. It supported movement between empirical observations, such as interviews, publications, and fieldnotes, and consideration of the wider institutional and historical mechanisms that might generate those observations. This orientation encouraged interpretation not only of what participants said and did, but also of the structural conditions through which particular classificatory practices became possible and acquired authority.

Together, these strategies ensured that the analysis moved across individual experiences, textual framings, and institutional relationships, while remaining grounded in critical race semiotics (Lawrence & Hylton, 2022). This analytic movement also informed the structure of Chapters 5-7, which progress from discourse and meaning-making, to institutional practice, and finally to the wider genomic ecosystem within which classifications acquire authority and social consequence. Rather than organising findings solely by data source or thematic code, the chapters were designed to move across different levels of genomic practice identified through the analysis. Chapter 5 examines the language, meanings, and classificatory practices evident in institutional publications and researcher accounts, establishing the discursive environment within which genomic work takes place. Chapter 6 builds on these findings by exploring how questions of equity, responsibility, collaboration, and institutional commitment are negotiated in practice. Finally, Chapter 7 broadens the analytic lens to consider the wider genomic ecosystem within which these activities occur, drawing on systems mapping and ethnographic observation to examine relationships, influence, and institutional power. This progression reflects a critical realist concern with understanding how observable practices relate to wider organisational and structural conditions, while also reflecting a CRT-informed interest in how racialisation and inequality become embedded and reproduced across interconnected institutional contexts.

3.11 Ethics and Informed Consent

This study was conducted in accordance with the ethical principles outlined by the British Educational Research Association (BERA, 2018) and received approval from the Faculty of Education Ethics Committee, University of Cambridge.

My research involved exploring how scientists and publics engage with categories of race, ethnicity, and ancestry in genomics. These topics are deeply entangled with histories of inequality and contemporary experiences of exclusion, and I recognised from the outset that questions could be

challenging or uncomfortable for participants. Ethical engagement therefore required more than procedural compliance: it required a commitment to reciprocity, reflexivity, and care.

Informed and Ongoing Consent

Consent was understood not as a single event but as a dynamic, ongoing process. Participants were provided with clear information about the study, its aims, and their rights, both in writing and verbally, before any interview or observation. They were reminded that participation was voluntary and that they could withdraw at any time without consequence. For interviews, consent was confirmed both in signed forms and again at the start of each recording. For observations, explicit permission was sought from organisers, and participants were notified of the researcher's presence through verbal reminders or notices.

For young people under 18, consent and assent were obtained through a two-step process: responsible adults (teachers or youth coordinators) facilitated parental consent, and young participants themselves confirmed their willingness to take part. This approach emphasised agency as well as protection.

Anonymity and Confidentiality

Protecting confidentiality was particularly important given the relatively small and specialised community of senior genomic scientists at Sanger, where identities could be inferred even if names were removed. To mitigate this risk, data were anonymised in transcripts and analysis, and identifying details (such as names on slides or institutional logos) were redacted before sharing beyond the supervisory team. However, I was transparent with participants that complete anonymity could not be guaranteed, and this was explicitly discussed during the consent process.

Participants were also given the option to relinquish anonymity if they wished to be named in outputs or acknowledged for their contribution. This was important in recognising participants not only as research subjects but also as potential co-producers of knowledge.

Care and Wellbeing

Given the sensitive nature of the research, attention was paid to participant well-being. I acknowledged that discussions of race and ancestry in science could provoke discomfort, particularly for those whose personal or professional identities were implicated. All participants were offered information about support resources, and I ensured that debriefing and ongoing communication were possible. Although these offers were not taken up, their availability was an important part of the ethical stance of the project.

Researcher Reflexivity

Finally, ethics in this study extended to me as a researcher. I drew on reflexive practices from genetic counselling and ethnography, including maintaining a reflective journal and engaging in supervision and mentoring. These practices provided accountability, supported my well-being, and helped ensure that ethical principles were enacted not only procedurally but also relationally throughout the project.

3.12 Limitations

Although positionality was intentionally embedded in the methodological approach, it inevitably shaped interpretation. Bias and prejudice are always present in qualitative research (Hegelund, 2005; Morse et al., 2002). Iterative conversations with my principal and second PhD supervisors, mentor, and the wider research group provided space for reflexive discussion about how my race, background, and experiences influenced the analysis. This reflexive structure, described to participants in the consent process, was central to ensuring accountability.

The study was also shaped by the scale and focus of the ethnographic fieldwork. Participant observation, purposive selection, and semi-structured interviews necessarily limited the sample size in a time-limited project. However, the ethnographic approach provided the depth and contextual richness that quantitative methods may not capture (Goodson & Vassar, 2011). Focusing on scientists based at the Wellcome Genome Campus reflected both accessibility and the aim of producing specific policy-relevant insights within the UK context. This narrow focus limits generalisability (Savage, 2000) but immersion in the field enabled me to seek out voices beyond immediate networks, reducing the risk of over-reliance on snowball sampling and allowing alternative or less prominent perspectives to be included.

A further limitation lies in the scope of the research. By centring on genomic researchers working primarily with large quantitative datasets, the study does not fully address how population descriptors are used across all areas of genomics. Nonetheless, the methodological framework developed here is transferable. Its replication in other institutional or disciplinary contexts could support the development of more locally responsive frameworks for equity in genomics, as opposed to generic solutions which have often proved inadequate.

Challenges specific to ethnography were also encountered. Immersion in the research setting demanded careful negotiation between objectivity and subjectivity, as close engagement risked influencing interpretation (Hammersley & Atkinson, 2019). Building trust required sustained investment of time, patience, and cultural sensitivity (Fetterman, 2019). Bringing a critical social science lens into a genomic science environment also carried risks of being perceived as overly exposing or insufficiently respectful of disciplinary norms. While there is substantial ELSI (ethical,

legal and social implications) work in genomics, it often remains siloed, complicating cross-disciplinary engagement.

Ethical concerns further shaped the study. Ethnographic work involves recording sensitive aspects of participants' professional lives, raising issues of consent, privacy, and context-specific vulnerabilities (Denzin & Lincoln, 2018). Limited resources, time pressures, and the unpredictability of fieldwork added to these challenges (Emerson et al., 2011). Together, these factors underscore the resilience and ethical commitment required to undertake ethnography in scientific settings.

The review of institutional publications also had limitations. While it provided a valuable overview of how social justice questions intersect with data categorisation in genomics at a major UK institution, it risked bias towards literature that aligned with my emerging interpretations. Reproducibility was limited by the context-specific and purposive nature of the review, and the dataset may not be exhaustive. To mitigate this, I clearly documented the review process, remained transparent about its intentional focus, and regularly discussed findings with my advisor.

Finally, while data saturation was achieved through a sufficient number of formal interviews, complemented by extensive informal interviews and over six months of participant observation (DiStefano & Yang, 2024; Staller, 2021) the study remains bounded by its institutional and temporal context.

4 Race, Ethnicity and Ancestry on the Page: Sanger Publications May 2023 Jan 2024

4.1 Introduction

Studies investigating the language around human population descriptors have shown that race and ethnicity are used inconsistently and in conceptually complex ways in genetics (see section 2.5.2). This chapter examines how these issues appear in publications involving Sanger scientists, focusing on papers circulated in the internal weekly research bulletin between May 2023 and January 2024. Treating these outputs as part of the ethnographic field, the chapter explores how population descriptors are constructed, justified, and problematised within a UK genomic research environment.

The bulletin collated work across Sanger's programmes (see section 3.8.1), spanning discovery science, methods development, and translational applications. Studies range from Harrison et al., (2023) who provided an example of clinically oriented research through a randomised trial evaluating ruxolitinib for polycythaemia vera, while Audain et al. (2023) demonstrated large-scale gene discovery in rare disease by analysing extensive exome datasets for congenital heart disease. Other studies, such as Liao et al. (2023) in the Human Pangenome Project and Lotfollahi et al. (2023) on deep-learning integration of single-cell data, illustrate the Institute's participation in large international consortia and its reliance on shared population datasets.

A discourse-analytic lens is used to read these publications as texts, attending to the language through which categories of race, ethnicity, and ancestry are produced, circulated, and rendered legitimate. This involves examining what is said about population groups, how often and by whom, as well as what remains implicit or unsaid. The chapter first analyses how population descriptors are constructed and harmonised across studies, then traces how these practices embed hierarchies of race, ancestry, and Europeanness, and finally considers how ethical frameworks in the papers' address, or fail to address, these dynamics.

4.2 The Construction of Populations

This section examines how human data is categorised across different areas of genomic research, focusing on how groups are conceptualised, constructed, and labelled within studies involving Sanger scientists. It considers the purposes these groupings serve, the methods through which they are

generated, and the interpretive and ethical challenges they present. In doing so, it explores how statistical approaches, data sources, and harmonisation practices contribute to the visibility and reproduction of concepts of race, ethnicity, and ancestry within genomic science.

The purpose of creating groups in genomic research should be restated. Across genomics, where the goal is to understand genes and their functions, comparison is fundamental. Differences in gene frequency, structure, or function are typically assessed between defined populations or relative to a reference dataset, approaches that enable researchers to identify variation and infer biological meaning (Jobling et al., 2013; Lewontin, 1972).

Population descriptors and descent-oriented classifications are often treated as proxies for shared genetic origin; for example, the label “European” is used with the implicit assumption of recent common ancestry. Although this review focuses on race, ethnicity, and genetic ancestry (Table 1), descent-oriented groups are not always necessary. Even when used, the meaning or salience of any given population label can shift across time, place, and scientific context (Dauda et al., 2023; Popejoy et al., 2020).

Conceptual frameworks such as *genetic ancestry*, *genetic similarity*, and the socially constructed categories of *race* and *ethnicity* were present, often in overlapping or conflated ways. This overlap reinforced the need to interrogate not just which terms were used, but how and in what sense they are being used.

A label on its own did not reveal whether a study is invoking *race*, *ethnicity*, or *genetic ancestry* as its underlying population construct. For example, the label “White” might be used to indicate a social category (race), a cultural identity (ethnicity), or a cluster of shared genetic variation (genetic ancestry). Distinguishing among these uses required close attention to textual and methodological cues, what was being compared, how similarity was inferred, and which assumptions were made explicit or left unsaid.

Harmonisation and the Reproduction of Labels

In this section, I consider harmonisation of data and how it can push choices about categories, especially for population descriptors.

Across the 147 papers in my sample, population grouping was far from routine in either method or justification. Of the 84 studies that named a dataset or collection, 66 also reported some form of population categorisation.

It was clear that the dimension of genetic ancestry was most often referred to in the reviewed papers and used across a range of global datasets. Evidence that the group label was used to capture genetic ancestry was inferred both from definitions of how continental labels are commonly used and from examples where they are associated with the term “ancestry” or “genetic ancestry” (Qu et al., 2023; Rasooly et al., 2023; Rhie et al., 2023; Stefanucci et al., 2023). Genetic ancestry was used as a representation of the DNA inherited from shared ancestors (Kang et al., 2023).

The most common label by far to indicate an ancestry group was the use of continental descriptors; European, African, East Asian, South Asian, West Asian, Oceanian, and American and sometimes these were referred to as “major” or “superpopulations” as shown in Figure 13

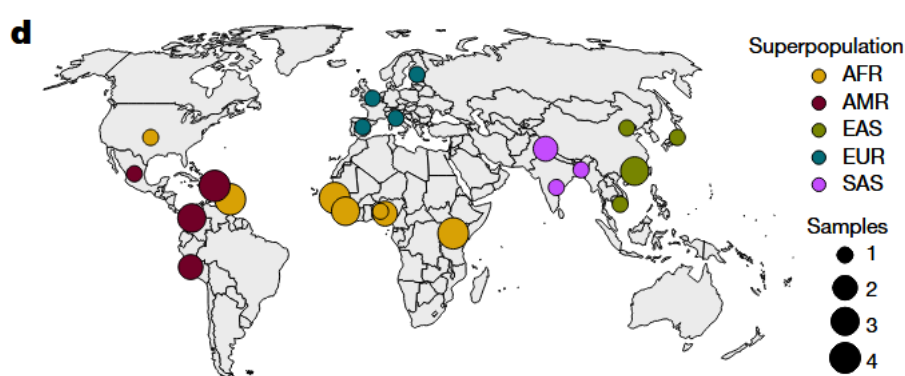


Figure 13 Illustration representing the geographical location and sample size according to continental groups of 1000 genome data (Hallast et al., 2023 p. 356). With permission from Springer Nature.

(Chu et al., 2023; Chundru et al., 2023; Hallast et al., 2023; Henry et al., 2023; Qu et al., 2023; Ravn et al., 2023; Rhie et al., 2023; Sikkema et al., 2023; Wang et al., 2023).

“Among the 1,008 individuals, 194 had African ancestry originating from the African continent (excluding African ancestry in the SW USA (ASW) and African Caribbean in Barbados (ACB))” (Hallast et al., 2023 p.6).

The frequent use of geographic and continental group labels reflects the findings of other studies that have shown their routine use in genetic studies to demonstrate the idea of genetic ancestry in population groups (Panofsky & Bliss, 2017).

In several cases, the definitions of categories were only available in supplementary materials, making the categorisation work effectively invisible in the main text. As a result, superpopulations from

resources such as the 1000 Genomes Project were invoked routinely and rarely problematised (Briest et al., 2023; Hallast et al., 2023; Liao et al., 2023; Meng et al., 2024).

In at least 43 of those papers, the labels appear to have been reproduced directly from the source dataset, such as the UK Biobank, the Office for National Statistics (ONS), or the 1000 Genomes Project, rather than constructed or justified within the study itself (Chen et al., 2023; Kentistou et al., 2023; Wall et al., 2023). Most often, these studies carried categories forward with limited local rationale.

Wang et al., (2023) defined ancestry groups by relying on the classifications already assigned within the contributing GWAS cohorts, which typically combined genotype principal component analysis with cohort-level ancestry labels. They then conducted fixed-effect meta-analyses within these predefined groups before combining the ancestry-specific results. Figure 14 illustrates how these inherited categories structure the comparison of risk variants, since the analyses depend on broad continental groupings rather than clusters generated anew for the study.

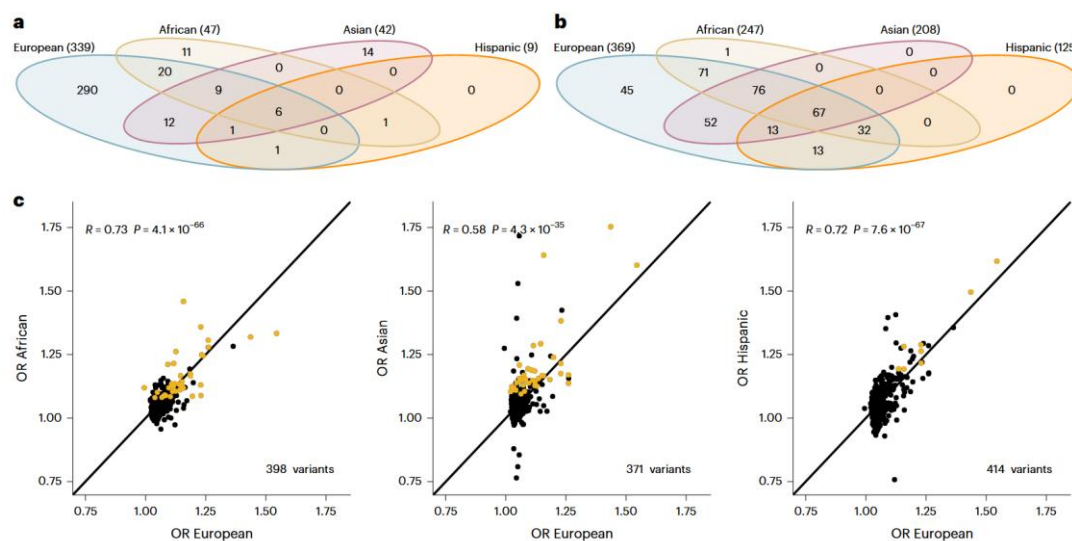


Figure 14 A and B Overlap of ancestry-specific prostate cancer risk variants and C Effect size comparisons across ancestry groups (Wang et al., 2023 p. 2067). With permission from Springer Nature.

Although the specific labels showed some variation, they recurrently used the same geopolitical labels to describe populations or datasets. Even when the source datasets or studies were mentioned, the

ancestral grouping and labels that may have been part of an original dataset were not always listed (see Figure 15) (Chung et al., 2023; Lord et al., 2024).

Characteristics	Men, N = 746,386			Women, N = 896,112		
	Mean (SD) or n (%)	No. (%) of persons with value	Median (IQR) of measures per person	Mean (SD) or n (%)	No. (%) of persons with value	Median (IQR) of measures per person
Earliest age during follow-up, years	50.4 (12.7)	746,836 (100)	-	51.3 (13.6)	896,112 (100)	-
History of diabetes [§]	21,936 (3.0)	746,836 (100)	-	20,662 (3.0)	896,112 (100)	-
Blood pressure-lowering medication prescriptions [^]	119,230 (16.0)	746,836 (100)	-	203,394 (22.7)	896,112 (100)	-
Current smoker, n (%) [#]	192,509 (25.8)	438,638 (59)	3 (2–6)	176,719 (19.7)	424,268 (47)	4 (2–7)
Systolic blood pressure mm Hg, mean (SD) [#]	136.7 (18.5)	699,229 (93)	4 (2–11)	131.6 (20.3)	870,075 (97)	6 (3–14)
Total cholesterol mmol/litre, mean (SD) [#]	5.4 (1.0)	532,620 (71)	2 (1–5)	5.6 (1.0)	624,189 (70)	2 (1–5)
HDL cholesterol mmol/litre, mean (SD) [#]	1.3 (0.4)	490,003 (66)	2 (1–4)	1.6 (0.4)	573,842 (64)	2 (1–4)

* Included 1,642,498 individuals from Clinical Practice Research Datalink, Hospital Episode Statistics, and the Office for National Statistics, England, United Kingdom, 2004–2019, aged 40–85 years, without prevalent CVD and statin treatment before study entry, and had least 1 measurement on any of systolic blood pressure, total cholesterol, HDL cholesterol, or smoking status between their study entry and study exit dates.

[§]Defined as ever having recorded diagnosis of diabetes before the study entry

[^]Defined as ever having recorded use of blood-pressure lowering medications before the study entry

[#]Proportion or mean (standard deviation) of the first ever measurement

<https://doi.org/10.1371/journal.pone.0292240.t001>

Figure 15 : Image of table of characteristics UK biobank data was used in the study ancestry groups are not routinely listed. (Chung et al., 2023 p. 6). Available under **CC BY 4.0** licence.

Many studies re-use existing population labels due to pressure to harmonise data across projects and biobanks. For example, Sikkema et al. (2023), in their review ‘An Integrated Cell Atlas of the Lung in Health and Disease’, align their population descriptors with public datasets to enable comparison (Figure 16), illustrating how harmonisation can perpetuate inherited categories rather than invite reconsideration.

M	N	O
tissue_dissociation_protocol	harmonized_ethnicity	selfreported_ethnicity_as_collected
Dispase + collagenase	european	white
Dispase + collagenase	european	white
Dispase + collagenase	european	white
Dispase + collagenase	european	white
Dispase + collagenase	european	white
Dispase + collagenase	european	white
Dispase + collagenase	european	white
Dispase + collagenase	european	white
Dispase + collagenase	admixed american	latino
Dispase + collagenase	european	white
Dispase + collagenase	european	white
Dispase + collagenase	european	white
Dispase + collagenase	european	white
Dispase + collagenase	european	white
Dispase + collagenase	european	white
Dispase + collagenase	european	white
Dispase + collagenase	european	white
Dispase + collagenase	european	white

Figure 16 An overview of the samples included in the Human Lung Cell Atlas (HLCA), donor metadata and harmonisation. (Sikkema et al., 2023 p. Supplementary table 2). Available under **CC BY 4.0** licence.

More broadly in large collaborations, harmonisation enables data sharing, reproducibility, and comparability, yet this technical uniformity imposes subtle constraints (Gonçalves & Musen, 2019).

Within these papers, although harmonisation is seldom named, the uniformity of population labels across projects reflects this dynamic: the drive to maintain shareable data reproduces a narrow descriptive vocabulary even as analytical methods evolve.

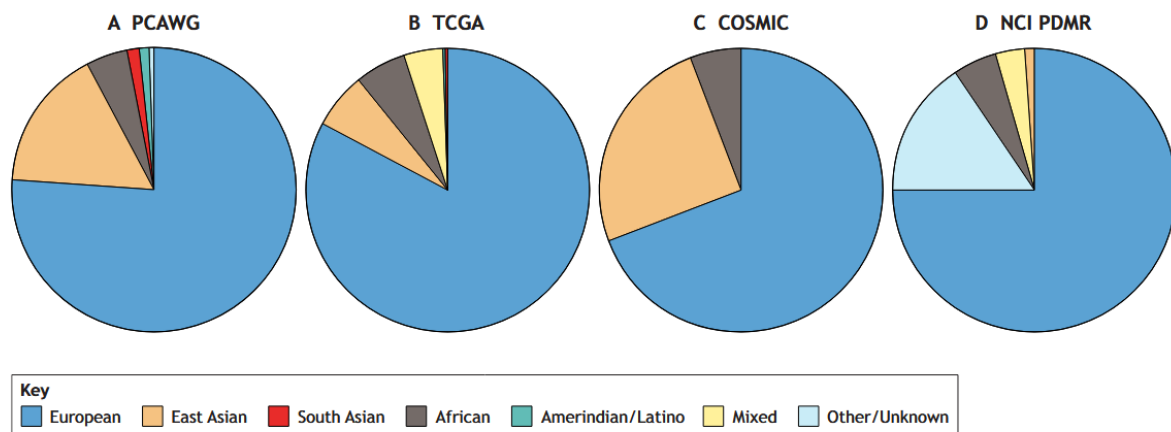
Groups described as sharing genetic ancestry were most often generated through some type of genetic inference, although detailed methods for defining an ancestry group were not always included, particularly we can assume because not all studies generated the datasets themselves (Foreman et al., 2023; Jindal et al., 2023; Karim et al., 2023; J. J.-K. Lee et al., 2023). When labels were generated or harmonised within a study rather than inherited, the dimensionality and naming were inconsistent.

Groups defined using similar quantitative procedures, such as PCA-based ancestry inference or genetic similarity thresholds, might be named by geography in one paper (Chundru et al., 2023) by ethnicity in another, sometimes within the same project (Meng et al., 2024). We also see superpopulation labels (EUR, AFR, EAS, SAS, AMR) used as if self-explanatory, without parallel discussion of how these terms were aligned or validated for the specific dataset at hand (Qu et al., 2023; Ravn et al., 2023; Rhie et al., 2023). Even when alignment appears tidy at the continental scale, it is seldom tested at finer granularities, sub-continental, diasporic, or admixed identities, where the correlation between self-report and computation may weaken or change.

4.2.1 Alternative Labelling

Here, I examine the population groups used and, in particular, how word choices signal the constructed nature of these categories. By comparing terms such as “labelled” and “European-like,” I show that although authors sometimes gesture towards the provisional character of these groupings, they continue to rely on inherited continental and subpopulation labels.

Some alternative labels were used to capture ancestry dimensions that may suggest a value held about the methods for determining the groups. One paper (Molina-Aguilar & Robles-Espinoza, 2023) referred to groups at one point underneath an image depicting lack of representation of samples from non-European ancestry in large public repositories as ‘Labelled’ by their non-genetic continental term (Figure 17).



Disease Models & Mechanisms

Fig. 1. There is a lack of representation of samples from non-European ancestry in large public repositories. (A–D) The pie charts depict samples in each collection by ancestry in the PanCancer Analysis of Whole Genomes (PCAWG; A), The Cancer Genome Atlas (TCGA; B), the Catalogue of Somatic Mutations in Cancer (COSMIC) Cell Lines Project (C) and the National Cancer Institute (NCI) Patient-Derived Models Repository (PDMR) (D). (A) The PCAWG contains 2583 high-quality samples, including 1968 (76.2%) labelled as European, 416 (16.1%) labelled as East Asian, 124 (4.8%) labelled as African, 35 (1.4%) labelled as South Asian, 26 (1%) labelled as Admixed American/Amerindian/Latino and 14 (0.5%) labelled as Unknown (Campbell et al., 2020). (B) TCGA, as analysed by Carrot-Zhang et al. (2020), contains 10,678 samples, including 8836 (82.7%) labelled as European, 669 (6.3%) labelled as East Asian, 651 (6%) labelled as African, 41 (0.4%) labelled as Native/Latin American, 27 (0.3%) labelled as South Asian and 454 (4.3%) labelled as having at least 20% admixed descent. (C) The COSMIC Cell Lines Project contains 1007 samples, including 308 with ancestry labelled by COSMIC and 699 with ancestry labelled by Kessler et al. (2019), together including 697 (69.2%) labelled as European, 253 (25.1%) labelled as East Asian, 56 (5.6%) labelled as African and 1 (<0.1%) labelled as South Asian. (D) The NCI PDMR contains 979 samples, with 734 (75%) labelled as European, 155 (15.8%) labelled as 'not applicable', 49 (5%) labelled as West African, 31 (3.2%) labelled as Mixed (all ancestries, including Native and Latin American <80%), 9 (<1%) labelled as East Asian and 1 (<1%) labelled as Native/Latin American.

Figure 17 Example of image showing the use of ancestry as a label rather than an essential quality. (Molina-Aguilar & Robles-Espinoza, 2023 p. 3). Available under **CC BY 4.0** licence.

This paper focused primarily on the lack of ancestral diversity in cancer research data rather than on a specific genomic analysis. The authors may therefore have been intentionally drawing attention to the limitations of continental labels for representing the full diversity of datasets. However, because the prefix “labelled” was confined to an image description and not used elsewhere in the text, it reads less as a consistent terminological shift and more as a commentary on the act of labelling itself. This choice subtly points to the constructed nature of these groupings. It highlights the decision to assign a label rather than revealing an inherent truth about biological difference. The wording calls attention to the scientists’ agency in creating not only the dimensional categories but also the linguistic labels used to define them.

Interestingly, Sergouniotis et al. (2024), who used UK Biobank data to explore genetic predictors of retinal morphology, adopted a slightly different linguistic strategy. They appended the suffix “-like” to ancestry terms, creating expressions such as “European-like ancestries,” “African-like ancestries,” and “Asian-like populations.” This phrasing suggests a degree of approximation rather than categorical certainty. For instance, they wrote:

“After applying standard genetic and OCT quality control filters (Patel, 2016; Currant, 2021), we defined a subset of the UK Biobank population that (i) can be considered genetically well-

ixed (i.e. includes participants that were assigned by genotype principal component analysis (PCA) to a cluster with subjects of mostly European-like ancestries).” (Sergouniotis et al., 2024 Pg 5)

Although the authors did not explicitly justify this terminology, the use of “-like” aligns with the methodological description of how clusters were defined. It acknowledges that ancestry groups are not homogenous and that clustering reflects degrees of genetic similarity rather than absolute categorical boundaries.

This approach implicitly recognises that ancestry categories are analytic constructs, shaped by computational thresholds rather than discrete biological divisions. Yet, in both Molina-Aguilar & Robles-Espinoza (2023) and Sergouniotis et al. (2024), the continued use of continental descriptors reinforces the prevailing convention in genomics: to divide populations along broad, geo-political lines even while acknowledging their constructedness.

Within these continental ancestry categories, subpopulations were often defined in more granular terms that incorporated sociocultural, non-genetic, or biological dimensions. Examples include classifications such as Brahmin, African-American, British Asian, Ashkenazi Jewish, Gambian in West Africa, and Finnish (Chu et al., 2023; Chundru et al., 2023; DeGorter et al., 2023; Liao et al., 2023; Rasooly et al., 2023; Thygesen et al., 2023).

However, in most cases, these secondary labels originated from the conventions of the source datasets, suggesting that researchers rarely departed from inherited classificatory systems when defining group identities within genomic analyses.

4.2.2 Statistical Approaches

Next, I reflect on how researchers define comparison groups in human genomic studies and how different classificatory choices shape the kinds of variation that become visible.

Although genetic ancestry is frequently discussed as a dimension, what is often demonstrated through genetic inference methods is genetic similarity (quantifiable genetic overlap between individuals, populations, or species (Dauda et al., 2023)). It is conceptually linked to ancestry because those who share recent ancestry tend to have greater areas of similarity within their genomes. Although genetic similarity can be measured directly, ancestry can only ever be inferred. It relies on assumptions about reference populations and historical time frames, meaning it inevitably reflects the choices researchers make about how to construct a group.

Statistical tools are used to cluster complex genomes using averaged measures of similarity, producing apparent structure within what is, in reality, continuous variation. They rely on quantitative inference to define ancestry-based groupings (Rouskas et al., 2023) and to find genetic similarity in pre-labelled samples such as Rasooly et al. (2023) who conduct a large-scale meta-analysis of heart failure GWAS using control data from individuals of European ancestry.

These tools make invisible variation legible and analysable, but at the cost of reifying fluid genetic relationships into discrete units for comparison.

Not all datasets in human genetics use descent-oriented groups. Some research instead employs alternative criteria such as disease phenotype, age, sex, or healthcare utilisation to establish comparison sets (Ferreira et al., 2023; Ismail et al., 2023). These examples show that such classifications may be unnecessary and demonstrate approaches that capture variation beyond ancestry-based framing. In contrast, many studies adopt specific descent-oriented groupings without explaining their purpose or analytical value, and these group labels often play little role in the results or conclusions (Jiang et al., 2023; Kentistou et al., 2023).

Methods such as principal component analysis (PCA), ADMIXTURE, and logistic regression models are employed to infer genetic similarity among individuals (Chundru et al., 2023; Rasooly et al., 2023; Wall et al., 2023). These methods cluster individuals according to shared allelic patterns, generating statistical proximities that are often interpreted as indicators of ancestry. PCA, for instance, reduces genetic variation across loci into a few dimensions, allowing samples to be positioned according to relative similarity as discussed in section 2.5.1 (Novembre & Barton, 2018). Yet at the same time these approaches can blur boundaries, producing overlapping clusters that reflect continuous variation rather than the discrete groups implied by the labels applied to them. As shown by figure 18 the PCA plots of array and RNAseq data using the authors' selected SNP sets do not show discrete separation of the broad self-reported ethnicity groups they have chosen to label as clusters. Instead, individuals assigned the same label are widely dispersed, and groups overlap substantially across the axes. This pattern underscores the limited correspondence between the inferred genetic structure and the categorical boundaries implied by the labels.

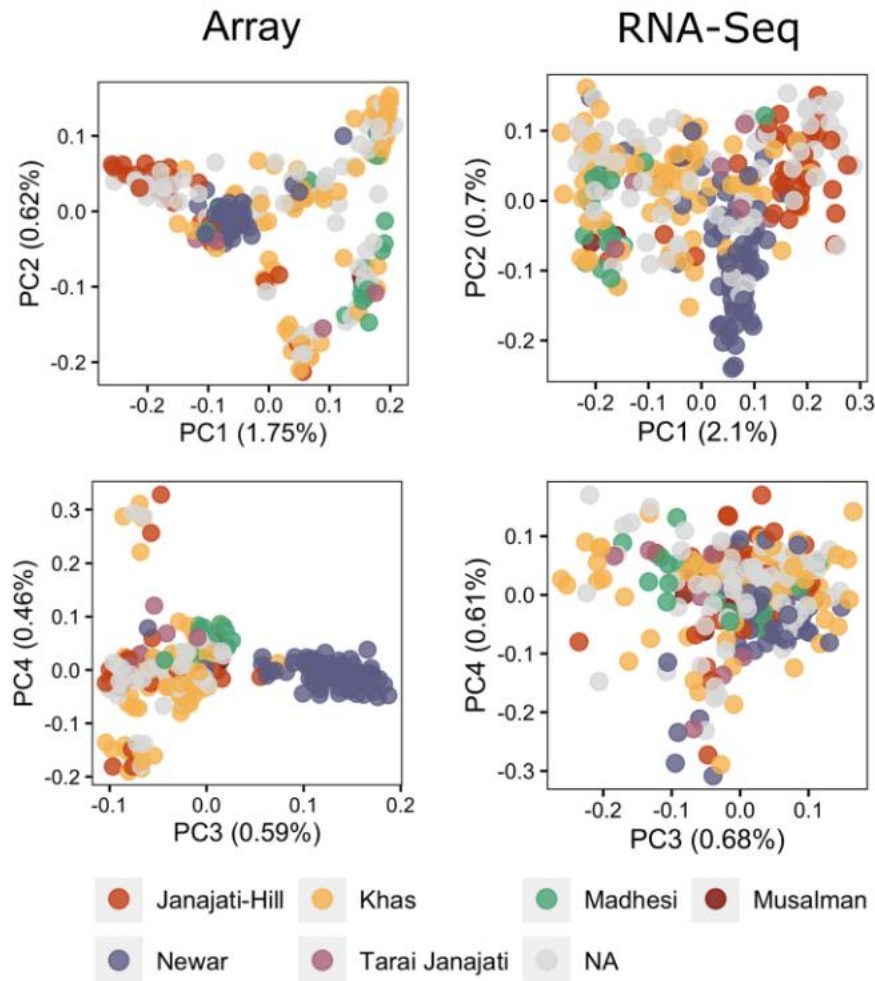
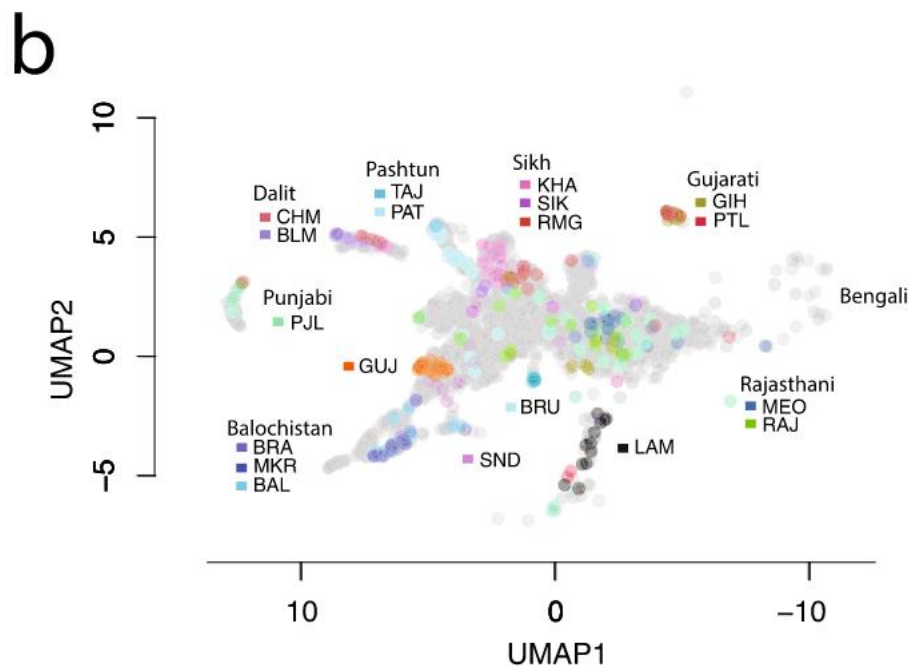


Figure 18 PCA plots of Population structure. Consistency with self-reported ethnicities is claimed by the overlapping points suggest blurred boundaries. (Fachrul et al. 2023 p. 5). Available under **CC BY 4.0** licence.

ADMIXTURE estimates, for each genome, the proportions derived from different inferred ancestral components. Researchers use these ancestry profiles to compare groups, examine admixture histories, and evaluate how self-reported or assigned labels correspond to genetic patterns see Figure 19) (Lawson et al., 2018).

UMAP visualises these relationships in high-dimensional genetic data into a visual map that preserves local relationships between individuals. Researchers use these maps to detect fine-scale substructure and to see how samples with known backgrounds group within larger clusters as, shown in Figure 20.

3).Available under **CC BY 4.0** licence.



metropolitan nature of the recruitment centre (Wall et al., 2023 p.3). Available under **CC BY 4.0** licence.

When applied to pre-labelled datasets such as the 1000 Genomes or UK Biobank references, these approaches reproduce the initial labelling conventions. This introduces a circularity in which computational outputs appear to validate categories that were socially or historically defined rather than genetically grounded (Dauda et al., 2023). This example illustrates a broader methodological dependence on legacy datasets that define what counts as an ancestry group. The result is a self-reinforcing logic in which labels derived from earlier classifications guide new analyses, which in turn reproduce the same categorical boundaries Chundru et al., 2023 write:

“We recognise that these genetically inferred ancestry groups do not capture the full genetic diversity of human populations. The classifications were based on genetic similarity to individuals in the 1000 Genomes and HGDP reference datasets, inferred from PCA.” (Chundru et al., 2023 p. 4)

While this makes no explicit difference to the methods they return to it as a limitation of their study. Despite their findings due to understudied populations and sample sizes their power for gene discovery is reduced.

This example illustrates a broader methodological dependence on legacy datasets that define what counts as an ancestry group. The result is a self-reinforcing logic in which labels derived from earlier classifications guide new analyses, which in turn reproduce the same categorical boundaries. As seen in studies by Senkin et al., (2023) and Malawsky et al. (2023), the drive for harmonisation across datasets maintains the appearance of coherence even as the underlying definitions diverge.

4.2.3 Identity, Inference and the Power to Classify

Following the previous discussion of computational approaches to inferring ancestry, this section turns to how population groups are constructed when researchers combine these inferred categories with self-reported identities. Rather than revisiting the mechanics of inference, the focus here is on how self-defined and algorithmically defined labels interact, how each is used in analysis, and how the authority to classify shifts between participants and data-driven systems.

When primary data are collected directly from participants, self-reported and self-identified categories serve as the mechanism for defining population groups. These labels, often chosen from predefined lists, are used to capture ancestry, ethnicity, or other social descriptors (Aung et al., 2023; Sergouniotis et al., 2024). The terms 'self-reported' and 'self-identified' were used interchangeably in these papers. Their common usage and comparative lack of critique or limitation saw them both

treated as valid indicators of population structure. Often with statements about how they are comparable with ancestry that was inferred computationally (Fachrul et al., 2023).

As Malawsky et al. (2023) comment shows:

“A UMAP with 4 PCs identified two large clusters which largely corresponded to individuals who self-reported as being of Pakistani or Bangladeshi origin.” (P.e3)

On the one hand, participant identity becomes validation of algorithmic categorisation. Yet, on the other hand, there is omission of how apparent alignment is driven by the social and historical structuring of identity, not biological essentialism. Factors such as migration histories, language, and census regimes contribute to these alignments (A. Lewis et al., 2022).

Sikkema et al., (2023) acknowledge that ethnicity metadata were based on self-reports and harmonised across datasets, noting that while useful, these measures imperfectly capture the diversity within human populations. They write:

“Ethnicity metadata were based on self-reports and harmonized across datasets, which is an imperfect approach to representing the diversity of the atlas. SNP-based inference of genetic ancestry constitutes a more objective and therefore preferable approach...” (Sikkema et al., 2023 p.17)

This perspective situates self-reported ethnicity as both limited and necessary. It acknowledges that categories are subjective, yet argues they provide meaningful context for genetic association. Wider scholarship has discussed this interaction between identity and science as dialogic: participants interpret and reshape their own understandings of race and ancestry through engagement with genetic studies (Condit et al., 1998; Smart, 2022).

This assumption reflects a persistent tension in genomics: the need to balance participants' own understandings of identity with researchers' statistical needs for population stratification. The combining of self-reported and computationally inferred categories to create composite groupings also reflects a pragmatic effort to align social identifiers with genetic data for analytical consistency.

However, such hybridisation can obscure rather than clarify the conceptual basis of population labels. For example Meng et al. (2024), in their GWAS study of major depression, describe using PCA assignments for most individuals to ancestry groups by matching them to reference populations, while simultaneously relying on self-report for “Hispanic or Latin American” groups.

In studies where genetic inference and self-identification are jointly used, individuals whose genotypes do not align neatly with reference clusters may be reclassified to fit the closest

computationally defined group (McLaren et al., 2023). At times, this misalignment also resulted in removal (Aung et al., 2023; Meng et al., 2024).

This process demonstrates the inherent instability of ancestry categories. It transforms subjective social identifiers into data points that can be adjusted or overwritten to maintain statistical coherence. The analytical precision of computational methods thus absorbs a foundation of socially contingent and historically inherited categories.

The interplay of methodological necessity, inherited classification systems, and harmonisation pressures produces a language of population that is both technical and social. In the following section, the analysis turns to the values embedded in these classifications, exploring how assumptions about hierarchy, typicality, and whiteness are reproduced through the language and practices of genomic population science.

4.3 Hierarchies and Power in Genomic Categorisation

In this section, I examine how genomic research papers embed racial hierarchies through routine labelling and measurement. I show how race, ethnicity, and ancestry are repeatedly conflated in both language and practice, including cases where “ancestry” substitutes for race as a putatively neutral proxy. I then trace how Europeanness and whiteness become co-constituted as methodological baselines, while “diversity” is articulated as what lies outside that norm. Drawing on examples of “non-European” and related formulations, I reveal how classification choices shape what becomes visible in analysis and what is rendered invisible in these papers.

4.3.1 Embedded Racial Hierarchies and Conflation of Categories

This section examines how race, ethnicity, and ancestry are routinely conflated in genomic research linguistically, procedurally, and analytically. I show how “ancestry” then functions as an ethically acceptable proxy for race, naturalising social categories as biological facts.

This review found several examples of the use of *race* labels, but with some variation in how they were applied and understood. Numerous papers used racial terms such as “White” and “Black” to describe populations (Aung et al., 2023; Jamrozy et al., 2023; Lagou et al., 2023; M. Peter et al., 2023; Sikkema et al., 2023; Taylor et al., 2024). Fewer studies use the word *race* itself; however, when race was used it was operationalised together in data collection. For example, Aung et al.

(2023) discuss *race* and *ethnicity* together as part of self-reporting as shown in Figure 21 taken from a table of study cohort characteristics:

Race and ethnicity, No. (%) ^a
Asian or Asian British
Black or Black British
Chinese
White
Mixed race
Other ethnic group

Figure 21 Section of table of race and ethnicity as study cohort characteristics. The image demonstrates how *race* and *ethnicity* are treated as, interchangeable demographic variables (Aung et al., 2023 p. 812). Available under **CC BY 4.0** licence.

Participants assigned themselves to these available options. This reinforces researcher-defined categories, which often appear when self-report is claimed. In providing only predetermined options, researchers have power over who is included and how identities become part of the study.

This conflation is both conceptual and procedural: it reflects underlying assumptions about what constitutes meaningful human difference and reproduces them through research design. In this way, classification practices not only record but also help sustain the analytical blurring that underpins racial categorisation in science.

In contrast, Qu et al.,(2023) demonstrates an even more complex conflation between *ethnicity* and *ancestry*, simultaneously describing data as “multi-ethnic,” “trans-ethnic,” and “ancestry-specific.” For example:

“Five sets of SNP weights were generated based on the respective LD patterns from the following populations, African American (AA), Hispanic American (HAMR), East Asian (ASN), Northern European (EUR) and all of the above populations (i.e. trans-ethnic)”(Qu et al., 2023 p. 3)

“We show that in the absence of a well-powered trans-ethnic GWAS ... trans-ethnic scores can be generated from European ancestry GWAS using Bayesian approaches.”(Qu et al., 2023 p. 2)

Here, *ethnicity* and *ancestry* are used interchangeably to describe genomic datasets, allowing ancestry to function as a proxy for both *race* and *ethnicity*. This reflects a second form of conflation: a

linguistic and methodological substitution in which “ancestry” becomes an ethically acceptable stand-in for social categories that are inaccurate or outdated. Thus, even when *race* is excluded by design, its logic remains embedded in the conceptual and analytical framework of population description.

While certain European countries refrain from collecting *ethnicity* data, invoking GDPR interpretations or broader commitments to removing racial classifications from public discourse (Simon, 2025), only one study in this review explicitly advanced this rationale. Here Riba et al.,(2023) state:

“The information collected in the Patient domain (demographics) encompasses a list of items suitable for all countries. We did not include items that cannot be collected in some EU member states (e.g., ethnicity, for legal reasons).” (Riba et al., 2023 p. 9).

Given that the collaborations represented in these studies were international, this limitation is significant. Ethnicity data collection is restricted in France and Germany, where legal and historical frameworks discourage the categorisation of populations by *race* or *ethnicity* (Simon, 2025). Of the publications reviewed, France appeared in 22 author affiliations and Germany in 31, suggesting that a considerable proportion of studies were led or co-authored from countries where such data cannot be collected. This highlights how national ethical standpoints vary, yet are negotiated largely invisibly within international research collaborations.

The absence of consensus about when and how ethnicity data are statistically useful also reinforces conceptual confusion. If banning the collection of *ethnicity* data is perceived as a strategy to prevent racial harm, *race* and *ethnicity* become conflated in both intent and effect. The assumption that not naming *ethnicity* is synonymous with preventing racism reflects an unresolved tension between the ethical management of difference and the scientific need for demographic accuracy.

Hageman et al.,(2023) illustrate how race and ethnicity are treated as interchangeable, both conceptually and through their conflation with social determinants. They write:

“Race/ethnicity may also improve risk prediction but was not included in the current study because of the close correlation between race/ethnicity and social economic determinants” (Hageman et al., 2023 p. 1712).

This phrasing treats “race/ethnicity” as a single unit and positions it as a correlate of socioeconomic factors. It suggests two perspectives: first, that race and ethnicity are not themselves social factors but are associated with them, and second, that social determinants are viewed as confounders in risk prediction. In either case, the choice to exclude these variables shows how conceptual and procedural uncertainty around race and ethnicity shapes what becomes visible or invisible in analysis.

Most studies listed both *ancestry* and *ethnicity* descriptors within their metadata and participant descriptions for example, “European ancestry,” “African ancestry,” or “Asian ethnicity” without

clarifying any conceptual distinction. Throughout, there is an implicit assumption that these terms have intrinsic, widely understood meanings requiring no further explanation, much like “height” or “sex” (Aung et al., 2023; Jamrozy et al., 2023; Lagou et al., 2023).

Similarly, Sikkema et al. (2023) explicitly harmonise self-reported racial and ethnic data with genetic superpopulation categories:

“Self-reported ethnicity categories were then harmonized with the superpopulation categories as follows: Black was categorized as African, White as European, and Latino as admixed American...” (Sikkema et al., 2023 p. 1581)

This literal relabelling assumes that race and ethnicity correspond directly to shared genetic ancestry, effectively translating social identity into biology. The suggestion that *ethnicity-specific thresholds* could improve analytical accuracy (Chung et al., 2023) reinforces this conflation, treating social classification as a proxy for biological difference.

While Sikkema et al. (2023) offer the most literal example, earlier cases demonstrate that this is not an isolated practice. Across genomic studies, *race*, *ethnicity*, and *ancestry* are used interchangeably in ways that naturalise social categories as biological facts. Given the evident conflation, blurring, and inconsistency, this assumption seems to rest on an uncritical confidence in the shared meaning of these terms.

Some papers even refer to “multi-ancestry” or “ancestry-specific” analyses while simultaneously categorising data using social or ethnic terms such as “Hispanic/Latino” or “South Asian” (Qu et al., 2023; Sikkema et al., 2023). This recurrent blending of terms demonstrates that ethnicity and ancestry become operationally intertwined, suggesting that avoiding racial terminology or substituting it with ostensibly genetic population descriptors, functions more as a methodological workaround than a genuine conceptual resolution.

The substitution also reflects assumptions about the legitimacy of *ancestry* as a concept. While ancestry can be statistically inferred, such inference presumes that what is being inferred exists as an underlying, essential reality. Yet individuals do not physically carry the DNA of all their ancestral lines, and different computational methods infer *ancestry* in different ways meaning that ancestry is, in practice, constructed through the application of computational methods. Figure 22 presents an illustration of the UK Biobank retinal imaging dataset. The image visually depicts how genetic and optical coherence tomography (OCT) quality filters narrowed the cohort to those deemed genetically “well-mixed” and with sufficiently high-resolution retinal images.

UK Biobank data quality control

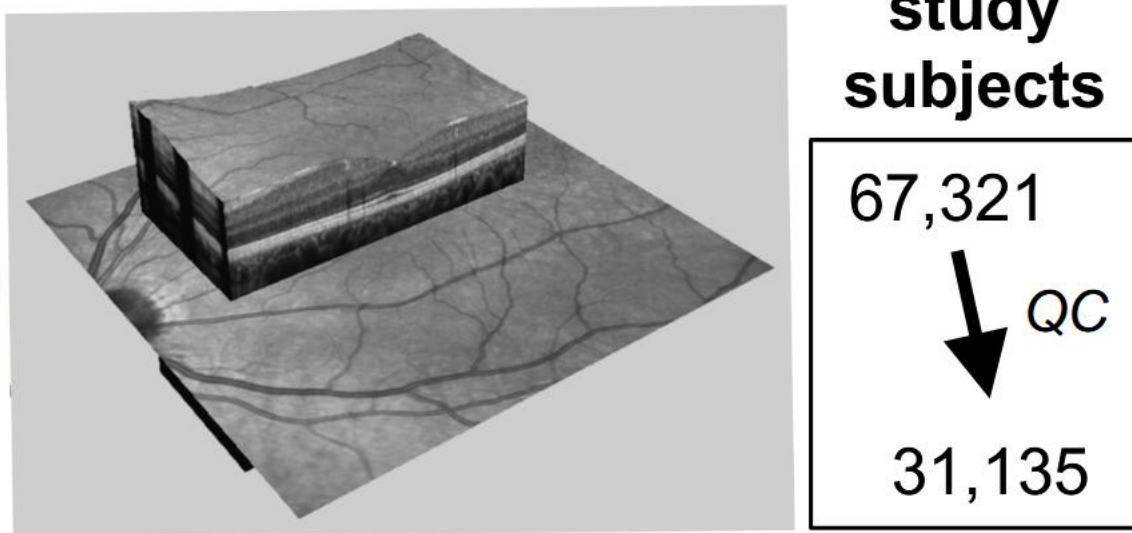


Figure 22 Quality Control. The creation of the cluster of participants with European-like features, from Sergouniotis et al. (2024) p. 129. Available under [CC BY 4.0](#) licence.

The straight edges of the stacked retinal blocks evoke the straightening performed by the quality control process itself, where irregular human variation is reshaped to meet the analytical requirements of the study. Genetic ancestry appears only as a criterion for filtering, not as a dimension of inquiry, yet it determines whose data become legible enough to be visualised. The image therefore captures not simply a dataset being refined, but a population being constructed to fit the contours of the methods.

This belief in the capacity of these methods to reveal something authentic or pre-existing lends them an aura of scientific and ethical legitimacy, as though *statistical power*, not in the technical sense, but as an instrument of scientific authority could uncover the true nature of human difference. It is within this aura that certain populations become the measure of accuracy and legitimacy.

4.3.2 Admixture and the Problem of Purity

The preceding discussion has shown how the authority of genomic classification continues to structure language and meaning in science. This same power is manifest in the operation of

admixture and the computational tools designed to measure it. They embody the tension between recognising human interconnectedness and re-inscribing boundaries in the name of analytical rigour.

Across the Sanger-affiliated publications, approaches related to admixture and ancestry inference were common. Twenty referred conceptually to “*mixed*” or “*admixed*” populations, and many employed computational methods such as ADMIXTURE, STRUCTURE, or comparable ancestry-deconvolution tools to make an analytic sense of the population structure in such groups. These tools are standard not only in population genetics and population genomics but also in the genetic epidemiology of disease, where they are used to adjust for or interpret population stratification. Admixture-based reasoning has therefore become routine across these domains, providing a basis for examining both its technical applications and its conceptual implications.

Many of the Sanger-affiliated papers reviewed explicitly addressed cohort diversity, using terms like *mixed*, *admixed*, *multi-ethnic*, or *diverse* especially to denote inclusion of non-European populations. A smaller number of papers used “*admixed*” or “*mixed ancestry*” as descriptive categories when defining sample inclusion, interpreting results, or framing population-level findings. Such instances show that the computational language of admixture frequently extends into conceptual and classificatory practice, shaping how diversity, relatedness, and belonging are discussed within the literature. For example Qu et al. (2023) states:

“The Brazilian population in the ELSA-Brasil cohort is three-way admixed. We therefore compared the transethnic score in this population to each of the three founders: American, African and European” (Qu et al., 2023 p. 6)

This phrasing illustrates how individuals and populations are conceived as both *mixed* and yet still divisible into constituent ancestral parts. The description of Brazil as “*three-way admixed*” simultaneously acknowledges historical interconnection and reinstates a tripartite model of origins. In such cases, mixture does not dissolve categorical boundaries but reaffirms them: difference is retained as a measurable proportion. The language of admixture thus enables a form of multiplicity that remains tethered to discrete ancestral groups, preserving the logic of separation within the very act of describing combination.

These tensions are also visible in the visual language of admixture. Standard admixture plots present individuals as composites of predefined ancestral components, turning continuous histories of gene flow into discretised colour blocks. As Figure 23 illustrates, these images give the appearance of tidy ancestral partitions even when the underlying data are far more complex.

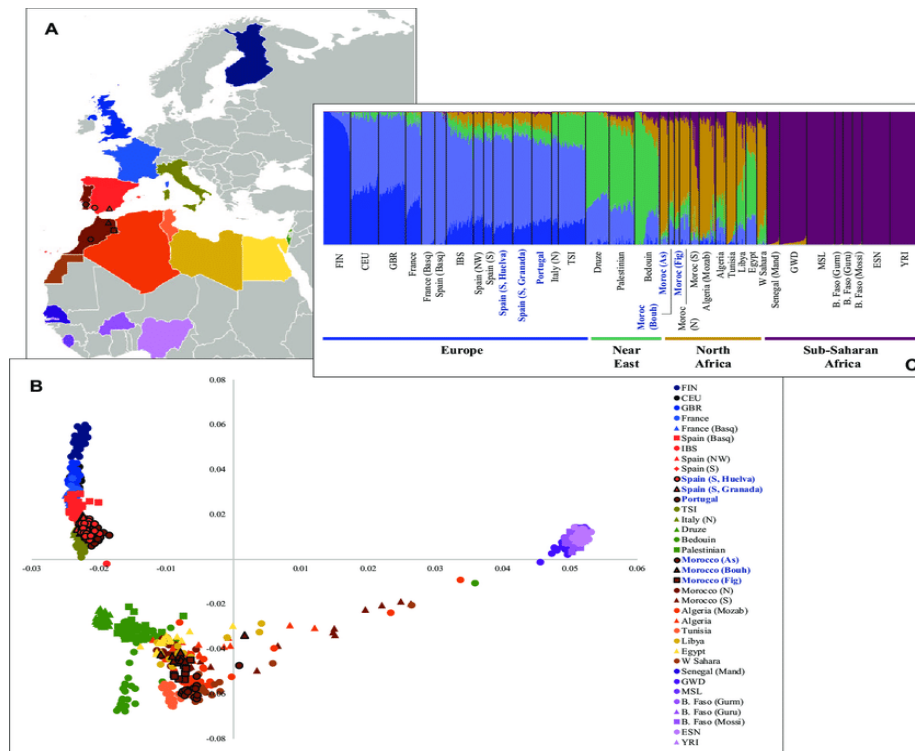


Figure 23 Overview of the genetic structure and global ancestry inference. (A) Countries represented in the database (B) Principal components analysis (PCA). (C) ADMIXTURE from Hernández et al. (2020) P. 1044. Available under [CC BY 4.0](#) licence.

In population genomics study admixture was integral to the scientific approach. As an analytic output, the figure demonstrates how population structure is statistically inferred; as a visual artefact, it shows how those inferences are stabilised into discrete and legible ancestral components. Another example was the CCR5Δ32 evolution study, which used local ancestry analysis around the CCR5 locus to infer a European origin and to interpret its distribution in Latin American and other populations (Ravn et al., 2023). Inclusion decisions were based on coverage and filtering, but admixture itself was not framed as a problem to be removed it was central to the evolutionary narrative they sought to construct.

In contrast, despite the frequency with which admixture appears as a descriptive category in GWAS and medical genomics, relatively few studies treat admixed genetic data as an object of analysis in its own right. The level of genetic diversity within study samples was frequently treated as a problem to be managed rather than a feature to be explored (Chen et al., 2023; Chundru et al., 2023; Henry et al., 2023; Meng et al., 2024). Population structure typically had to be controlled for through principal component analysis or by stratified analyses within broad ancestry groups, as in the HPRC pangenome (Liao et al., 2023) and muscle-strength GWAS (Huang et al., 2023).

Heterogeneity, likewise, was at times treated as a technical problem. As in Figure 24, taken from Fachrul et al. (2023) the PCA plot contains grey points labelled “NA”, representing participants who were not placed into any predefined ethnicity categories; their muted colouring marks them as present

yet unrecognised, included in the visual field but excluded from the labelled groups. The aggregation of “mixed,” “minority,” or “Other” categories, as in Jamrozy et al. (2023a), also illustrates how those who do not fit neatly into boundaries, whether racial or ancestry categories, are rendered methodologically invisible.

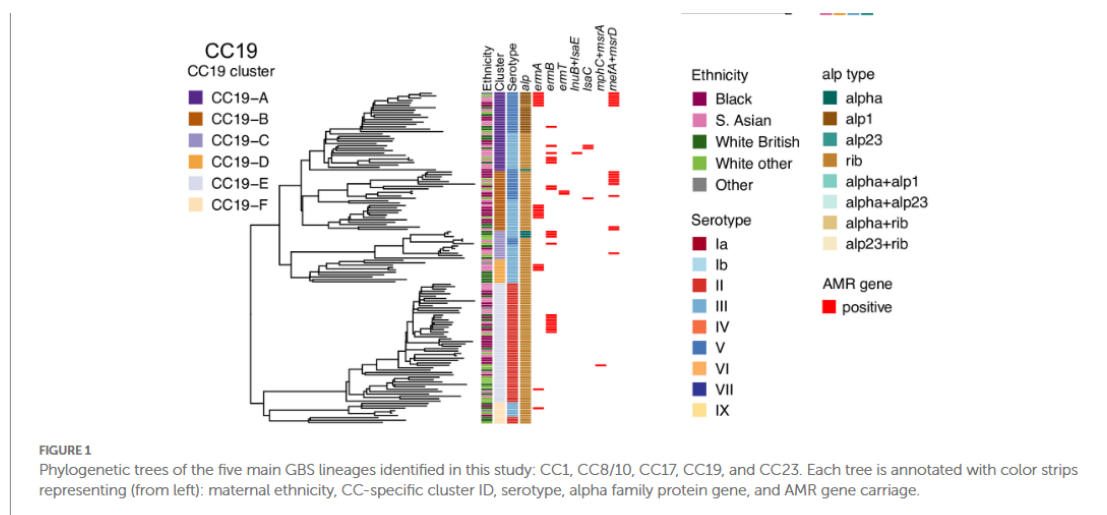


Figure 24 The grouping of minority categories and aggregation as “Other” . (Jamrozy et al, 2023 p.5). Available under [CC BY 4.0](#) licence.

Linguistically, *Other* marks a form of conceptual expulsion: any blurring of categorical boundaries results in erasure, as individuals lose both ancestral and ethnic identifiers. This means that admixture is not only a genetic descriptor but also a disciplinary mechanism, it delineates who belongs within the frame of analysis and who falls outside it.

Significantly, labels such as *Admixed American* or *Mixed American/Latino/Amerindian* lacked clear or consistent definitions (Honigberg et al., 2023; Lord et al., 2024) erasing the complexity of lived ancestry and to computational manageability. When individuals do not conform to pre-existing categories, they are either reclassified or removed entirely (Aung et al., 2023; Faquih et al., 2023; Warriar et al., 2023). As Aung states:

“We also excluded participants who lacked information on race or ethnicity ... those who selected ‘do not know’ or ‘prefer not to answer’ were excluded from the study sample.” (Aung et al., 2023 p. 809)

This example shows how inability or refusal to align with predefined labels results in exclusion, reinforcing rigid boundaries reminiscent of early racial taxonomies. At the same time, the exclusion of admixture is increasingly recognised as a source of inequity, one that limits discovery and perpetuates bias in genomic resources as Li states:

“There is a strong need for more diverse and deeper human multiomic data, which will generate novel insights as a complement to other pre-clinical models for drug discovery and development.” (Li et al., 2023)

As in many large GWAS, analysis also restricts genetic relatedness by excluding close relatives, ensuring that only one individual from each family network contributed to the final dataset (Aung et al., 2023; Honigberg et al., 2023). While framed as a routine quality control step, relatedness filtering disproportionately affects populations characterised by dense kinship networks, small community size, or historically constrained sampling frames. As TallBear (2013) and (Fullwiley (2011) have shown, such assumptions embed particular social models of population into genomic practice. The result is that populations may be excluded for being too similar as readily as for being too different. These exclusions are statistically motivated, but they nonetheless carry social consequences, particularly for marginalised and underrepresented groups whose demographic structures depart from this ideal. However, exclusion is not the only available statistical response to relatedness and population structure. In their GWAS study Meng et al (2024) use GENESIS, a statistical method to adjust for population structure and relatedness, within the analysis. Researchers, therefore, have methodological choices available, yet the social consequences or ethics of technical choices were not discussed in any of the papers reviewed.

Admixture itself\ therefore, appeared in several guises: as a statistical parameter inferred through modelling (Chundru et al., 2023; Senkin et al., 2023), a confounder to be controlled or eliminated (McLaren et al., 2023), or a measure of diversity arrayed along a continuum (Sousa da Mota et al., 2023). In each case, it operated within a logic of containment; an effort to stabilise populations as internally coherent and externally distinct. When categories such as *native*, *indigenous*, or *reference* populations were contrasted with *admixed* ones, the implication was that only certain groups were sufficiently “*pure*” or homogeneous to serve as baselines. This framing reproduces a longstanding classificatory hierarchy, positioning mixture as deviation and purity as scientific norm.

The logic of purity, though often implicit, was made explicit in methodological statements that set inclusion and exclusion thresholds, where technical criteria served to reinforce moral and historical assumptions about difference. For example:

“Principal component analysis (PCA) was performed by EigenStrat software ensuring only European ancestry populations were included, and those with less than 80% European ancestry were excluded (n = 167)” (Jandu et al., 2023 p. 2)

Subpopulations such as Finnish or Ashkenazi Jewish were often singled out for their homogeneity (Chu et al., 2023; Toivonen et al., 2024), reinforcing the importance of purity and coherence in research methodology. This logic is common across genomic studies, where researchers justify

restricting analyses to a single ancestry group on the grounds that doing so reduces potential bias caused by population stratification, even while acknowledging that the resulting findings may not apply to populations of other ancestral groups. In each case, exclusion is framed as a technical necessity to avoid bias.

European datasets, due to their sheer size and coverage, dominate reference panels and thereby define the statistical centre of genetic similarity (Garwany et al., 2023; Julian et al., 2023; Verdier et al., 2023) and so exclusion due to size was regularly observed. Van der Vossen et al. (2023) note that participants *“of Ghanaian origin were excluded due to the low number of second-generation participants.”* And certain groups, particularly Hispanic/Latino populations, were described as admixed and consequently excluded for lacking statistical power (Meng et al., 2024).

The pursuit of purity in genomic research rests on a belief in bounded, coherent populations that can serve as reference points against which others are measured. The technical language of bias reduction and population stratification conceals a value system that equates homogeneity with validity and mixture with instability.

In these settings most employ admixture language rhetorically to mark diversity, inclusion, or global representation while the underlying methodologies seek to minimise its effects. Admixed individuals are often filtered out, down-weighted, or assigned to the nearest homogeneous cluster, reflecting a persistent methodological discomfort with hybridity. This asymmetry between the prevalence of admixture as a conceptual sign of diversity and the paucity of methods designed to model it directly, reveals how genomics continues to value purity and partition even as it celebrates mixture.

Diversity, however, is not rejected outright. It is embraced when it can be made orderly and compatible with the classificatory logic of genomics; transformed from a challenge to an attribute that can be managed, quantified, and contained. Within this framework, admixture is acceptable only when measured proportionally, not when it exposes the continuum that disrupts categorical order.

The authority of statistical reasoning thus feeds directly into the dominance of whiteness and Europeanness as the unmarked standards of truth in genomic research, the reference points against which all other populations are rendered comparable, derivative, or diverse.

4.4 Whiteness and European Dominance

This section examines how whiteness and Europeanness are reproduced as overlapping categories of legitimacy in genomic research. The first part considers how whiteness is constituted linguistically and symbolically in genomic discourse. The second examines how European dominance in datasets

sustains this logic through methodological practice, shaping both the ethical framing and scientific authority of diversity itself.

4.4.1 The Authority of Whiteness

The aura of scientific and ethical legitimacy attached to genomic methods, as though statistical power could uncover the true nature of human difference, extends directly into how populations are categorised and valued. Within this framework, the use of whiteness in genomic discourse moves beyond representation into a hierarchical logic in which European identity becomes synonymous with legitimacy, accuracy, and scientific neutrality.

While categories such as *White British* or *White European* appear to describe geographic or ancestry-based populations, the repeated pairing of racial and geopolitical terms blurs those boundaries and attaches whiteness to ideas of authentic or “true” European origin (Hanssen et al., 2023; Henry et al., 2023; Jandu et al., 2023; Lagou et al., 2023; Malawsky et al., 2023; M. Peter et al., 2023). Across the dataset, geopolitical pairing occurred far more often with *White* than with *Black* population labels. *White* appeared with national or continental qualifiers ten times, for example *White British*, *White European*, or *White American* (Aung et al., 2023; Fachrul et al., 2023; Jamrozy et al., 2023; Lagou et al., 2023; Malawsky et al., 2023; M. Peter et al., 2023; Taylor et al., 2024), while *Black* was geopolitically qualified only three times, as in *Black British*, *Black African*, or *Black Caribbean* (Jamrozy et al., 2023; Sikkema et al., 2023; Taylor et al., 2024). Combinations such as *White European* are commonplace, yet terms such as *Black European* or *Asian European* are entirely absent.

The label *White* was frequently attached to groups defined as having European ancestry (Akbari et al., 2023; Henry et al., 2023; Jandu et al., 2023; Kentistou et al., 2023; Malawsky et al., 2023; Rasooly et al., 2023; Reay et al., 2024; Sikkema et al., 2023; Warrier et al., 2023). *White British* was used as a specific regional identifier (Jamrozy et al., 2023), yet no examples of *White African* or *White Caribbean* appeared. By contrast, *Black British*, *British Pakistani*, and *Asian British* were used more often (Aung et al., 2023; Malawsky et al., 2023; M. Peter et al., 2023). This linguistic hierarchy reinforces whiteness as presumed and therefore unmarked, while racialised identities must be explicitly named and geographically qualified. Whiteness thus functions as the invisible standard of comparison against which all other groups are measured (Reardon, 2017; Roberts, 2011).

these naming practices show how whiteness operates as a linguistic norm in genomic classification, rendering European identity natural, neutral, and authoritative. This symbolic authority is not confined

to language alone but is reinforced through the material dominance of European datasets and reference populations, which is examined in the following section.

4.4.2 European Dominance

European dominance in genomic data mirrors and extends this linguistic hierarchy. It is sustained not only by historical asymmetries in data collection but by a methodological logic that treats European ancestry as the scientific baseline. Across the reviewed literature, bias towards European datasets is routinely framed as a technical condition, while simultaneously functioning as an ethical justification for the very dominance it describes.

Jandu et al. (2023) make this logic explicit, stating that “studies should also increase recruitment from genetic ancestry groups other than *White* Europeans.” The phrasing acknowledges imbalance while positioning *White European* data as the default against which all other ancestries are defined. This framing recurs in references to “European dataset bias” and calls for more “diverse” samples (Ip et al., 2023; Jamrozy et al., 2023; West et al., 2023). Diversity is invoked without conceptual grounding, reinforcing White European populations as the implicit standard from which difference is measured.

This dominance is frequently described as both a practical and moral dilemma. Several studies note their reliance on European populations (Honigberg et al., 2023; Larsson et al., 2023; Qu et al., 2023; Stefanucci et al., 2023). For example, Akbari et al. (2023) write:

“Because participants in the INTERVAL cohort are overwhelmingly of European ancestry, we have not been able to study genetic variation extraneous to the European ancestry population.”(Akbari et al., 2023 p.10)

Such acknowledgements operate as ethical disclosures that paradoxically normalise exclusion, positioning non-European representation as desirable yet ultimately supplementary.

In other cases, the methodological virtue of whiteness is made explicit. Henry et al., (2023) justify their reliance on European-only data by stating that:

“to obtain a more accurate estimate, we used the European ancestry meta-analysis summary statistics with pre-computed tagging files derived from 2,000 *White* British individuals.”

Here, European ancestry is equated with quality and control, and whiteness functions as a measure of methodological purity. A similar logic appears when non-European participants are excluded on the grounds that they introduce undesirable variability. As Krzak et al. (2023) state:

“Patients of non-European ancestries were also excluded to reduce confounding,” (p. 20)

Reinforcing the assumption that analytical coherence depends on European homogeneity. Adlam et al. (2023) likewise justify the absence of non-European data by citing socio-economic limitations in recruitment, while concluding that greater ancestral diversity would enhance replication power. Diversity is thus idealised as a theoretical good while operationalised as an analytical risk. Rouskas et al. (2023) in their analysis of Greek genomic data, note that the GTEx (GTEx Consortium, 2020) dataset is dominated by individuals of Central and Northern European descent, with only small numbers of African-American and Asian-American donors. As discussed in sections 1.2 and 3.3.2, this reflects a stratification within Europe itself, in which a narrower, Northern and US-centred form of Europeanness functions as the normative reference, positioning other European populations as peripheral within the category of whiteness.

Together, these rhetorical strategies construct what may be understood as a moral economy of bias. Underrepresentation is acknowledged as an ethical concern, while homogeneity is framed as a practical necessity. European datasets retain their status as both scientifically robust and ethically defensible. Even where diversity is foregrounded (Costanzo et al., 2023; Duckett et al., 2023; T. Liu et al., 2023). European dominance remains the reference point that validates progress, transforming bias into evidence of moral awareness rather than a prompt for methodological reform.

DeGorter et al. (2023) exemplify this dynamic in their discussion of African genomic data, starting the abstract to their paper with:

“Mapping the functional human genome is often limited to European-descendent population samples. To aid in overcoming this limitation...” (p. 2)

The formulation positions Europeanness as the origin of completeness and non-Europeanness as corrective. Ethical gestures towards inclusion thus remain dependent on the continued centring of Europe as the point of comparison.

Through these examples, ethical discourse itself becomes a mechanism for reproducing hierarchy. To acknowledge bias is to reaffirm European authority as the foundation from which correction must begin. Calls for diversity emerge within, rather than against, the conceptual grammar of whiteness, ensuring that European, and by extension *White*, data continue to function as the standard of scientific truth.

4.4.3 Producing Non-Whiteness

This section examines how the concept of non-whiteness is positioned within genomic discourse. If whiteness and Europeanness operate as unmarked standards of legitimacy, then non-whiteness appears as their inverse: necessary for scientific completeness, yet always articulated in relation to a dominant European frame. Inclusion is therefore framed simultaneously as an ethical and scientific imperative, while remaining tethered to European comparison.

Across the dataset, this positioning is visible in the language used to mark difference. References to non-European populations appear far more frequently than inverse formulations. The term non-European occurs 24 times, compared with eight mentions of non-Finnish, five of non-*White*, and only three of non-African. By contrast, terms such as non-Asian are entirely absent. This asymmetry underscores how European datasets function as the normative centre of genomic comparison, while other populations are named primarily in relation to that norm as categories of exception or diversity. Whiteness and Europeanness are never marked as deviation, signalling their status as the unexamined baseline.

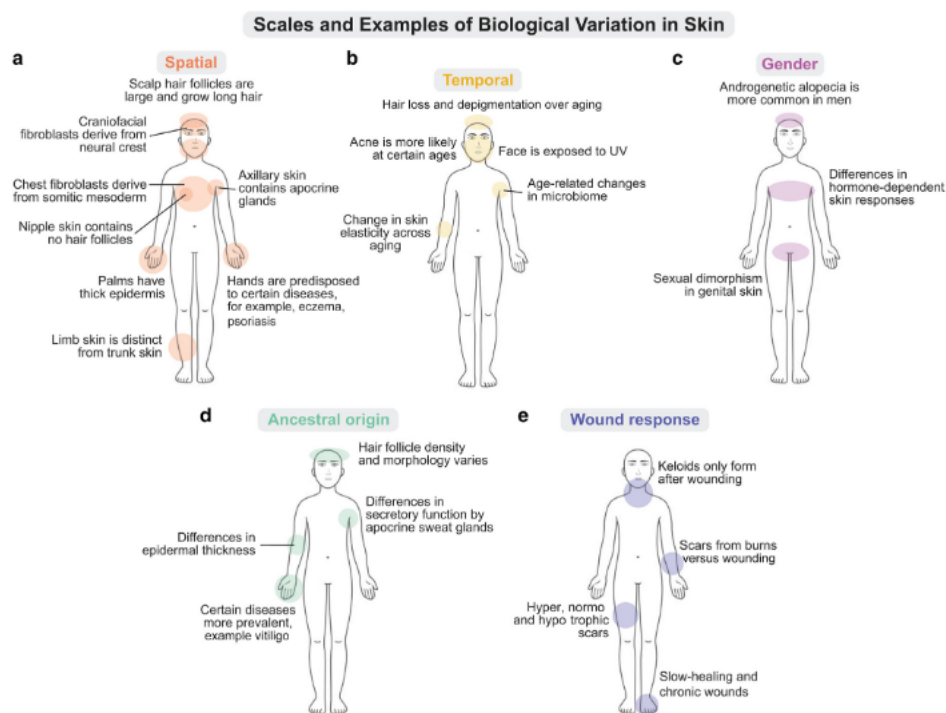


Figure 25 explaining the importance of biological variation in skin. (Almet et al., 2023 p.1668).

Available under **CC BY 4.0** licence.

Figure 25 illustrates how ancestry is naturalised as a biological axis of variation. Taken from a study aiming to build a consensus Human Skin Cell Atlas (Almet et al., 2023), the figure situates ancestral origin alongside spatial, temporal, gender, and wound-response factors as an equivalent biological scale. The authors later argue that a comprehensive atlas requires data from genetically diverse ancestries, explicitly naming “African, Asian, Hispanic, and Middle Eastern populations” as underrepresented. While framed as a call for inclusion, this formulation positions these ancestries as completeness-enhancing additions rather than as foundational to the definition of variation itself, leaving the normative reference population implied.

A similar logic appears in analytic practice. Duckett et al. (2023), in their study of delayed growth and puberty, report that excluding participants of “non-European ancestry” “did not change the overall frequency of LoF MC3R carriers” and conclude that this “overrepresentation is not confounded by ancestry differences.” (p. 1586). This conclusion is enabled by the overwhelming dominance of European ancestry within the cohort, leaving insufficient variation to meaningfully test such effects. Non-European ancestry is treated as a potential source of confounding to be checked and set aside, while European ancestry remains the unmarked baseline against which stability is assessed.

Related patterns appear in large biobank analyses. Carss et al. (2025) frame diversity as a means of enhanced characterisation, noting that “within- and cross-ancestry genomic associations” yield novel insights, while Barclay et al., (2023) acknowledge more directly that UK Biobank participants are not representative of the UK population Ip et al. (2023), examining lifestyle factors and colorectal cancer risk, explicitly identify the underrepresentation of non-*White* participants in UK Biobank, naming “non-white” as a category of absence. In doing so, whiteness is rendered visible as an ethnic category only through its absence, reinforcing its implicit dominance even as generalisability is questioned.

The technical consequences of this asymmetry are made visible in Figure 26, taken from Sousa da Mota et al. (2023):

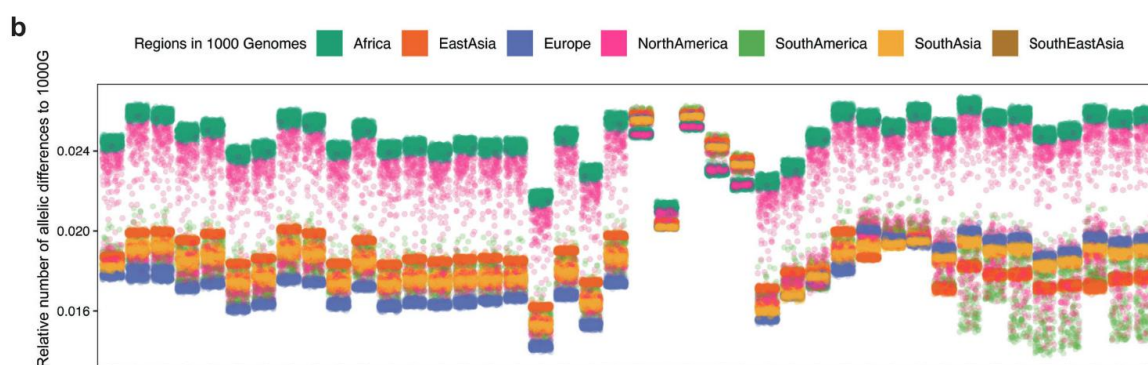


Figure 26 The imputation error rate (NRD) for 42 ancient genomes. (Sousa da Mota et al. 2023 p.4). Available under [CC BY 4.0](#) licence.

The figure depicts non-reference discordance (NRD), an imputation error rate measuring how often non-reference genotypes are incorrectly predicted across 42 ancient genomes. Genomes of African ancestry consistently show higher NRD values, reflecting lower imputation accuracy. The authors explicitly attribute this pattern to bias in reference datasets, which are overwhelmingly European, and caution that their population labels “Africa”, “Europe”, “Western”, and “Asia” are “vague” and “not fully descriptive.” Even with this reflexivity, the visualisation positions non-African as a conceptual category rather than a descriptive one, marking distance from a European reference point that functions as the standard of accuracy. The figure thus demonstrates how reference bias is not only linguistic but infrastructural, shaping error, reliability, and analytical confidence.

Together, these patterns construct non-whiteness as difference-as-deficiency. Populations described as non-European or non-*White* are rendered ethically necessary yet analytically secondary, valued for their contribution to diversity rather than for their capacity to redefine the centre of genomic knowledge. Terms such as underrepresented and understudied frame absence as a moral problem, while leaving intact the classificatory structure that produces it (DeGorter et al., 2023; Ismail et al., 2023; McLaren et al., 2023). Efforts to increase diversity are frequently framed in terms of scale rather than a redistribution of epistemic authority, equating representation with quantity while leaving underlying categories unchanged (Jain et al., 2023; Kang et al., 2023).

In this way, non-whiteness is incorporated into genomic discourse as a supplement to an already privileged dataset. Difference is acknowledged, measured, and managed, but rarely allowed to unsettle the authority of European reference populations. The following section examines how these linguistic and technical hierarchies are reinforced through the continued circulation of colonial terminology, historical narratives, and classificatory practices.

4.4.4 Colonialism, Slavery, and Hierarchy

The authority of genomic classification does not only reproduce contemporary hierarchies but also carries the linguistic and conceptual residues of colonial thought. Terms and categories that once sustained racial science continue to appear, sometimes uncritically, in the vocabulary of modern genomics. This section considers how the persistence of this language sustains the conceptual link

between racial difference and scientific legitimacy, binding present-day research to older imperial grammars of knowledge and value.

Several papers use the term *Caucasian* (Almet et al., 2023; Duckett et al., 2023; van der Vossen et al., 2023; Yu et al., 2023). Examples are given in text and graphic form as shown by Almet et al., (2023) in Figure 27. John et al. (2023), a cancer genomics study investigating spatial heterogeneity in multiple using multi-region and single-cell sequencing, describes its patient group as “31 *Caucasians*,” without clarifying whether this refers to race, ethnicity, or ancestry. Yu et al. (2023) use the term interchangeably with *European*, reinforcing a racialised equivalence.

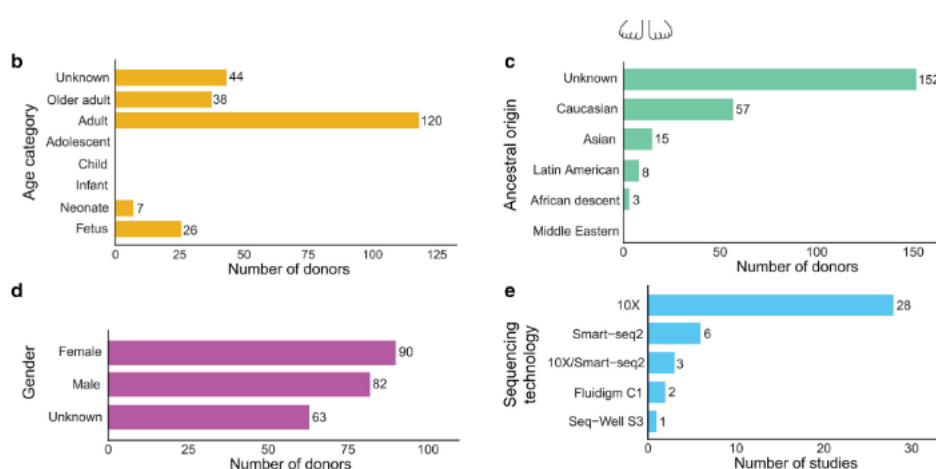


Figure 27. Caucasian as an ancestral origin. (Almet et al., 2023 p.1670). Available under **CC BY 4.0** licence.

Caucasian has long been associated with pale skin and *European ancestry*, particularly in U.S. contexts where it remains a racial descriptor synonymous with *White* (Comstock et al., 2023; Morning, 2011; Shamambo & Henry, 2022). Although the term’s nineteenth-century origins in racial anthropology are well established (see chapter 2), it continues to appear in genomic studies as a synonym for European ancestry. In this usage, the scientific neutrality of ancestry disguises its colonial inheritance. To describe participants as *Caucasian* is to invoke a category invented to hierarchise human variation while claiming the authority of objective measurement. The term translates an ideology of race into the language of genetics, allowing a colonial vocabulary of difference to persist under the guise of biological description.

Although van der Vossen et al.(2023) refer to Turkish participants as having Mediterranean, Middle Eastern, and Caucasian ancestry, it remains unclear whether Caucasian is used here in the narrow geographic sense of ancestry linked to the Caucasus region, or as a broader synonym for European-adjacent or West Eurasian populations. In contemporary genomics, the latter usage is far more

common, despite its recognised imprecision (John et al., 2023; Morning, 2011; Yudell et al., 2016). This ambiguity itself is instructive: it illustrates how a term with explicit origins in racial hierarchy continues to circulate in scientific discourse under the ostensibly neutral language of ancestry, without methodological clarification.

Although these examples represent a minority, the uses of *Caucasian* in John et al. (2023) and Yu et al. (2023) also demonstrate the persistence of imprecise and racialised terminology that equates whiteness with European biological coherence ((Popejoy, 2021); as discussed in sections 1.3.1., 2.2.2 and 2.4).

The use of *Oriental*, like *Caucasian*, is an explicitly colonial term long rejected as inaccurate and racist (Yu Hsi Lee, 2016). Colonial language also appears explicitly in (Jamrozy et al., 2023), whose study on group B streptococcus in pregnant women categorised participants as “Mixed,” “*Oriental*,” and “Other”:

“The remaining women (n = 23, 4%) represented minority groups composed of Mixed, Oriental, and other ethnicities, which were grouped here as Other.”

(Jamrozy et al., 2023, p. 6).

Its appearance, even inadvertently, sustains a classificatory logic in which whiteness is normative and superior, while identities outside of that classification are residual, peripheral, and indistinct.

Van der Vossen et al. (2023) further reveal how the legacies of slavery and colonialism remain unevenly invoked. Their study compares *ethnic origin groups* to explore environmental and acculturative effects on the gut microbiome, yet slavery is mentioned only in relation to the Surinamese population:

“African Surinamese (‘Creoles’ descending from former slaves, mixed Western African and Dutch ancestries), South Asian Surinamese (‘Hindustanis’ descending from workers arriving to Suriname post-slavery, South-Asian/Indian ancestry).”

(van der Vossen et al., 2023, p. 3).

By contrast, Moroccan and Turkish participants are described solely through migration and acculturation.

In Figure 28 visual language mirrors this logic. *European*, *Turkish*, and *Moroccan* identities are represented as nationally coherent and culturally fluid, while *Surinamese* identity is divided into separate ancestral components, *African* and *South Asian*. Only some populations are granted neutrality under the national flag; others become legible only through explicit lineage. The use of flags collapses nationality, ethnicity, and ancestry into a single national symbolism. In doing so, the image

reproduces a logic of classification that relies on social census categories rather than genetic measures, embedding the social construct of race within genomic representation.

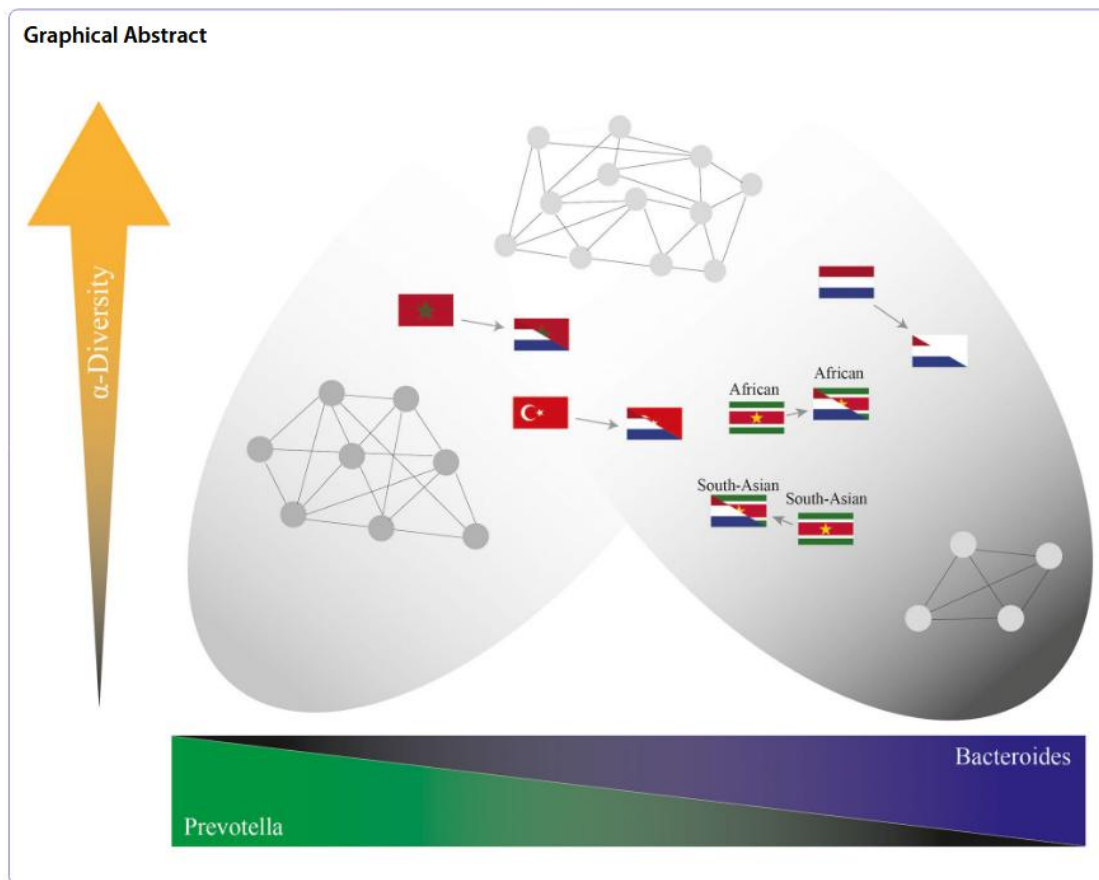


Figure 28. How microbiome shifts correspond with the westernisation of diet and urban environment. From van der Vossen et al. (2023) p. 2. Available under [CC BY 4.0](https://creativecommons.org/licenses/by/4.0/) licence.

This selective historicisation reproduces a racialised asymmetry: some groups are anchored to colonial histories of enslavement and subjugation, while others are framed as culturally mobile but historically neutral. Such uneven contextualisation exposes how *ethnicity* functions as a flexible category absorbing racialised meanings where colonial legacies are invoked, yet remaining depoliticised elsewhere.

This asymmetry extends beyond visual rhetoric. In Qu et al. (2023), in the same sentence, they say European ancestry, while other groups are presented simply “African Americans”, “Hispanics”, and “Latinos”. In fact, the word ancestry is only used alongside the European group throughout. This distinction delegitimises some groups by denying them the epistemic status of having an “ancestry,” while *European* identity retains scientific depth and continuity. Broader ethnic or racial labels such as *Hispanic*, *Latin American*, or *Asian* are used interchangeably, flattening their complex geopolitical and cultural distinctions.

The language and visual rhetoric of genomics continue to reproduce colonial hierarchies under the guise of scientific description. Terms such as *Caucasian* and *Oriental*, selective invocations of slavery, and the conflation of nationality with ancestry sustain a classificatory order in which whiteness remains the unspoken reference point of biological coherence. The authority of genomic representation thus rests on boundaries drawn long before the advent of molecular genetics, translating historical categories of race and empire into contemporary scientific frameworks.

These continuities do not end with terminology. They extend into how diversity, inclusion, and representation are conceptualised across genomic discourse. The following section examines how the same colonial logics are rearticulated through the moral language of equity and global participation, showing how the pursuit of diversity itself often remains problematic.

taken together, these analyses point towards the ethical frameworks through which genomic practices are governed, justified, and made accountable. Ethical oversight in genomics aims to promote fairness, equity, and respect for participants, yet the ways in which ethics is defined and enacted often mirror the same hierarchies that organise scientific reasoning itself. Examining how ethics is articulated in recent Sanger-affiliated publications reveals how the moral language of inclusion and diversity becomes institutionalised through procedural norms, while deeper questions of justice, representation, and historical responsibility remain largely unexamined.

4.5 Ethical Frameworks in Practice

The preceding analysis has shown how colonial hierarchies and classificatory logics persist within genomic science, shaping both language and methodological authority. This section turns to the ethical frameworks through which these practices are governed and legitimised. Across the Sanger-affiliated publications reviewed, ethics is addressed overwhelmingly through procedural mechanisms, most informed consent, institutional approval, and compliance with established guidelines. While these frameworks aim to ensure accountability and protect participants, they rarely engage with the ethical implications of population categorisation, differential representation, or historical inequities. As a result, the moral language of equity and inclusion is institutionalised mainly through compliance and documentation, while broader questions of justice, power, and responsibility remain outside the ethical frame.

In practice, ethics statements focused almost exclusively on informed consent, data protection, and institutional approval, typically following journal-mandated formats such as the International Committee of Medical Journal Editors Recommendations (ICMJE, 2025). Ethical compliance was treated as a matter of procedural sufficiency, with little attention to how population categories are

constructed or how exclusion and underrepresentation might constitute ethical issues in their own right.

4.5.1 Beyond Procedure: The Missing Ethics of Representation

There were limited papers to explore that engaged with ethics as a broader practice of equity and responsibility, as described in 3.5.1 these papers were not chosen by the staff themselves and so may not represent a generalised approach. Nonetheless, the campus library aims to capture the entire publication output across the Institute. Molina-Aguilar and Robles-Espinoza (2023) call for increased representation of non-European populations in cancer research, while Sharaf et al. (2023) highlight how linguistic and infrastructural barriers constrain participation in African biodiversity genomics. Işık et al. 2023 and Samuel, (2023) explore environmental and educational ethics, arguing for frameworks that integrate sustainability and social responsibility into research design. Others, such as Bernier et al. (2023) and Welter et al. (2023), focus on data stewardship and FAIR principles, demonstrating how accountability is framed through technical reproducibility rather than distributive justice.

Efforts to promote equity within institutions also featured in papers by Bradke et al. 2023 and Williams et al. (2023), who describe initiatives to build inclusive research cultures and training programs. At Sanger, postdoctoral fellowships that address the underrepresentation of Black heritage researchers and the Behavioural Competency Framework illustrate how organisational policies aim to formalise inclusion. However, these measures remain unevenly implemented and often symbolic.

Engagement with participants and communities, a key theme in ethical practice, also reflects this asymmetry. Several studies describe meaningful attempts to address sensitive issues through collaboration (Malawsky et al., 2023; Senkin et al., 2023), while others cite the lack of public or patient involvement as a limitation (Jindal et al., 2023).

The conflation of race, ethnicity, and ancestry, the dominance of European data, and the legacies of colonial extraction remain largely outside formal ethical scrutiny, even as they continue to shape the moral landscape of genomics.

4.5.2 Final Thoughts

Across this analysis, the review of Sanger-affiliated publications has shown how genomic science continues to grapple with the entanglement of technical practice, historical legacy, and ethical responsibility. Focusing on Sanger as a place-based site makes it possible to trace how institutional

location, people, and published literature intersect, but the patterns identified here are not unique to Sanger. Rather, they reflect wider dynamics in genomics that were introduced in the literature review, including the conflation of race, ethnicity, and ancestry, the dominance of European reference populations, and the persistence of classificatory hierarchies rooted in colonial histories. The pursuit of objectivity in genomics, therefore, remains inseparable from the social and political contexts in which knowledge is produced.

Across the publications reviewed, ethical reflection is largely procedural, centred on consent, governance, and regulatory compliance, while deeper questions of justice, representation, and historical responsibility are rarely addressed. Although the moral vocabulary of equity and inclusion has become embedded within genomic research, it is often enacted symbolically rather than transformatively, reproducing existing hierarchies of legitimacy rather than unsettling them. Taken together, these findings suggest that contemporary genomics has expanded its ethical language without fundamentally reconfiguring the structures through which authority, normality, and difference are defined.

The next chapter follows these patterns of discourse into lived experience. By turning to interviews with scientists at Sanger, the analysis examines how institutional narratives are interpreted, negotiated, and sometimes resisted in practice. This shift from published texts to personal accounts makes visible the human dimension of genomics, showing how ideas of fairness, responsibility, and diversity are encountered as everyday challenges within scientific work, rather than solely as abstract ethical commitments.

5 Power and the Making of Populations: Scientific Practice and Meaning at the Sanger Institute

5.1 Introduction

In this chapter, I examine how power operates within the scientific, institutional and interpersonal life of Sanger. Drawing on ten months of ethnographic fieldwork that combined in-depth interviews with scientists (conducted both in person and over Zoom see figure 30 *The digital interview*), digital and in-person participant observation, attendance at seminars and meetings and sustained immersion on the campus (figure 29 auditorium entrance), I explore how authority, hierarchy and institutional identity shape the ways in which human populations are conceptualised, categorised and managed as data.



Figure 29. Entrance to the Campus Auditorium.

The main venue for Institute away days, town halls and international scientific meetings, illustrating one of the physical sites of institutional life. Photograph taken by the author, 25 Oct 2023.



Figure 30. Digital interview screenshot. A Zoom recording interface, illustrating the use of online interviews as part of the digital ethnographic dataset. Captured by the author, 3 November 2023.

Power at Sanger is both structural and relational: it is held, felt, exercised, and at times resisted. Scientists describe moments of both control and constraint, where influence over research direction, collaboration, and communication is distributed unevenly across positions, disciplines, and global contexts.

Rather than treating power as a fixed possession, this analysis approaches it as something enacted through scientific practice. This analysis builds on the understanding of power developed in Chapter 2, where classification, knowledge production, and institutional authority were shown to operate as mutually constitutive rather than separate domains.

Power appears in the ability to define problems and decide which questions matter; in the authority to name, group, and communicate difference; and in the institutional privilege that grants access to data, funding, and legitimacy. At Sanger, positionality, whether as a senior leader, early-career researcher, or global collaborator, determines not only who can act but what can be known and said.

The following three themes trace how this dynamic unfolds through different aspects of genomic science.

- **Power in Communication** examines how authority, hierarchy, and scientific language sustain the persistence of racialised ideas, even when scientists believe they have moved beyond them.

- **Power in Methods** explores, methodological choices methodological choices and research aims becomes a moral and political process that shapes equity in genomic outcomes.
- **Power Beyond Data** (addressed in chapter 6) considers how inclusion, diversity, and institutional responsibility are defined and enacted within Sanger's ecosystem, revealing how global influence intersects with local culture.

Across these themes, power is not simply a matter of domination or control but of *production*: it determines which truths are made credible, which people and populations are rendered visible, and how science itself becomes a site where inequality is both recognised and remade.

5.2 Power in Communication: Silence, Avoidance, and the Language of Racial Ideas

This section examines communication as a site of power in genomic science, showing how race remains a contested and often uncomfortable presence in everyday scientific practice. Among scientists at Sanger, these tensions were navigated not only through what was said, but through avoidance, substitution, and silence. Ethnicity frequently functioned as a more acceptable alternative to race, yet remained an imprecise proxy for genetic ancestry, social identity, or environmental exposure. Attention to these communicative practices reveals how institutional and disciplinary norms shape what can be articulated explicitly and what must remain implicit.

Reflecting on the range of labels that emerged during interviews and observational work (Table 4), it was striking how rarely such terms were clarified or questioned, despite their acknowledged ambiguity. Labels associated with race and ethnicity were more readily recognised as problematic, while geographically framed descriptors of ancestry were often treated as self-evident. As shown in the preceding discourse analysis, shifts in terminology can confer legitimacy on categories that might otherwise appear contentious. Setting these labels alongside the dimensions they were used to describe makes visible the overlaps and tensions between race, ethnicity, geography, and genetic ancestry.

Table 4. Labels used by participants to describe population differences across domains

Dimension	Example of labels used
Geo-political	African, African American, Asian, Brazilian, Finnish, Lower Middle-income Country/Global North/Global South, Bangladesh, Indian. Iranian, Polish, Western, Japanese
Race	African American, Asian, White British/Irish,
Ethnicity	African American, Asian, Black ancestry, Caucasian, Caribbean. Jewish, Japanese
Genetic similarity	African/Africa Ancestry, European/ European Ancestry, Japanese
Health demographic	Underrepresented, Understudied, HDSS,
Genetic Ancestry	Jewish Ancestry, South Asian Ancestry, White European Ancestry

Across interviews, race and ethnicity were frequently used interchangeably and regarded as sensitive or risky, something to be carefully managed or avoided. Although race was widely acknowledged as a social construct, it remained present through categories such as ancestry, ethnicity, and location. Scientists expressed awareness of the historical and ethical concerns surrounding race yet often framed genetic diversity in ways that sidestepped direct engagement. This avoidance was reinforced by disciplinary conventions, institutional pressures, and concerns about public interpretation. Race was thus simultaneously disavowed and reproduced.

Mapping labels and dimensions in this way provides a foundation for the analysis that follows. Three subthemes are examined in turn: first, how race and ethnicity are ejected or avoided in everyday interaction; second, how certain terms come to be treated as accurate or acceptable; and finally, how racial ideas persist despite sustained efforts to distance scientific practice from them. Together, these subthemes show how race cannot be fully excluded from genomic communication, even when its explicit invocation is constrained.

5.2.1 Avoidance and Rejection

The first of these themes revealed how race and ethnicity were often understood less as dimensions of human variation and more as terms that evoke discomfort and unintended associations.

Reluctance to engage with race was often expressed through concern about how others might respond, rather than through an outright denial of its relevance.

Harmful ideas

Even mentioning race was felt to risk legitimising divisive or harmful ideas, reinforcing a tendency to keep such discussions at a distance. As this participant pondered:

“Genomics research in that it's extremely powerful, it's liberating, it's empowering, but with it comes responsibility and understanding of what the repercussions and consequences could be. One of those was how do we report variations and heterogeneity, between individuals collectively, and not sort of somehow add fuel to the concept of, you know, racial entities ”

Group leader, Developmental Biology.

This question captured a central dilemma: the need to describe biological variation and heterogeneity without invoking the problematic language of race. The description of genomics as “powerful”, “liberating”, and “empowering” is striking. These terms frame scientific participation as a source of authority and freedom, positioning science as a space that confers legitimacy in contrast to a less controlled non-scientific world. The satisfaction expressed here signals an awareness of power, even as responsibility is emphasised as its ethical counterweight. This concern echoed examples discussed in 4.3.1, where ethnicity data collection was rejected outright in certain national contexts. Here, the entanglement extends further: even apparently neutral technical terms were treated as capable of carrying racial meaning. In this sense, the language of biological difference is never entirely free from the possibility of being read through race, no matter how carefully it is framed.

“You know,in the past. science is used to solve.. like create differences between people that are actually meaningless. You know, in my opinion, race is one of them. Science is [was] used to kind of like say, some people are superior, inferior”

Group Leader, Developmental Biology.

This participant often referred back to historical understandings of race that accompanied their anxiety. This anxiety also surfaced in public engagements with the history of scientific racism beyond the research setting (Figure 31). Another scientist also drew on historical frames in their thinking and judgment, as this participant notes:

“I mean historically.. very sinister kind of historically. They're very sinister kind of categorisations of race, right? Like, I mean, Nazis, and But there what does it? What does it mean when someone says that nowadays? I think they are basically referring to skin colour, aren't they? That's so coarse and doesn't really mean very much.”

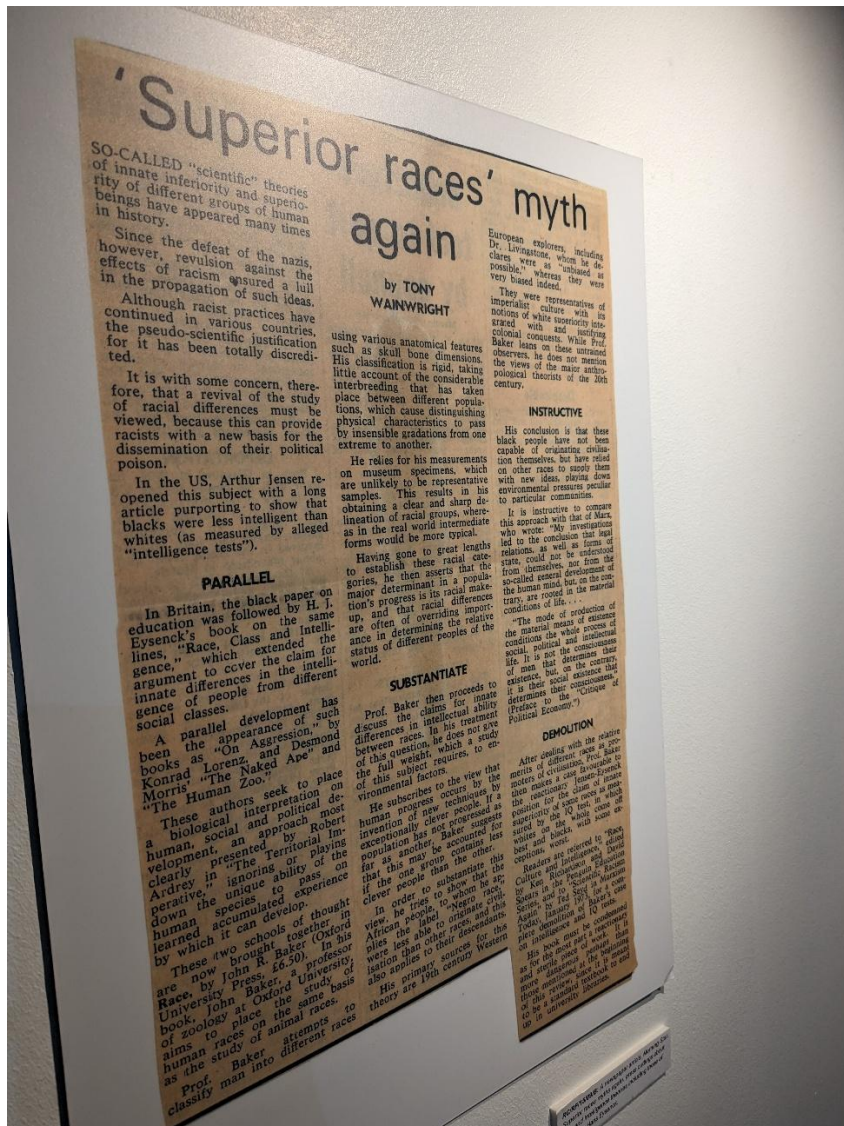


Figure 31. Newspaper cutting displayed at the Black Cultural Archives as part of the exhibition *Challenging the Narrative: Black Resistance to Scientific Racism* (Appendix C). The cutting critiques the historical and ongoing harms of scientific approaches to race and the revival of racialised scientific studies. Photograph taken by the author, 25 January 2024. Permission given by Black Cultural Archives.

This discomfort often shifted onto ethnicity, particularly when ethnicity appeared as a proxy substitute for race. For some, ethnicity was experienced as imprecise and vaguely opportunistic:

"[Ethnicity]—that's a hard one, because it's a term that I think has sort of become trendy nowadays. But also, I don't really know what it means. It's because of the bad association with race, I think—that's another one that's come up that people are using."

Group Leader, Discovery Science.

Yet other participants defended ethnicity as biologically meaningful:

" Personally, I think there's it's very important to represent different ethnicities in your data sets because biology differs a lot. You know, between different ethnicities"

Postdoctoral Researcher, Discovery Science.

And in some cases, the preference for ethnicity was explicitly enforced through hierarchy, with race removed from documents through senior authority:

"I was explicitly discouraged to use the word race. And then my supervisor, basically changed everything to ethnicity. And to me, as a non-English speaker, I honestly don't quite understand the implication of these race word, but I'm just instructed not to use it"

Senior scientist, Bioinformatics.

Even where the historical and political contours of the terminology were not fully understood, senior positions could exercise power to remove race from texts and replace it with a supposedly safer alternative. What emerged across accounts was a recurring tension between what was meaningful and what was permissible. Scientists were not only debating terminology; they were negotiating when and how race or ethnicity could be "welcome" within scientific reasoning, especially in relation to evidence, justification, and harm. As one Group Leader in Developmental Biology reflected:

"Is ethnicity needed? How does it interact with other social determinants of health? Is it always a surrogate for SES? What is the evidence that this analysis is meaningful—is there any evidence? And what's the evidence that this has actually contributed more harm, directly or indirectly, to a particular ethnic group? 'Cause I think that is really needed then that would give you more impetus and persuasive evidence of whether this should continue or whether we're just wasting our time."

Group Leader, Developmental Biology.

Their uncertainty about meaning also extended to questioning practical value, asking:

"Where has this really been valuable in highlighting something that has made a big difference to that ethnic group?"

Group Leader, Developmental Biology.

While race was met with immediate resistance by all, ethnicity remained an uneasy but more acceptable alternative for some. This created a paradox: race was too politically and socially charged to engage with directly, yet ethnicity, vague for some and meaningful for others, also proved inadequate. Together, these ambivalences contributed to avoiding race altogether, while allowing ethnicity to surface only under certain conditions, especially when it could be framed as scientifically neutral or methodologically necessary. This unease translated into everyday practices of avoidance, where discussions of race were redirected, softened, or silenced within the institutional life of the lab.

Race and Gender

In discussions around race and ethnicity, comparisons with gender and sex frequently emerged when participants were asked why race was seen as problematic. These comparisons often highlighted perceived similarities such as discrimination, microaggressions, and the historical underrepresentation of women in research, but they also revealed how gender provided a more comfortable framework for acknowledging inequality. In doing so, participants could recognise exclusion while avoiding the deeper discomfort associated with race, and they sometimes reinforced the idea that discomfort around race and diversity primarily affects racialised and minoritised groups, rather than implicating institutional structure.

“let me give you an example and this is what switched it on in my brain like basically if we stop tracking say male/female gender pay gap how many people are employed are Male or Female, because we just say sex doesn't matter everybody is an individual we are just hiring whatever, paying whatever and everyone is the same then you are losing any ability to track any inequalities that are attached to those labels that's the argument to this coarse grain analysis.”

Group Leader, Discovery Science.

This participant had described ethnicity as a “crude” measure, yet when challenged on the implications of discarding categories entirely, she turned to gender as an analogy. The example centred on the ability to track inequality, rather than on any inherent scientific value in the category itself.

“ Sometimes, you know, you hear something, but because it's so alien to your own experience, you cannot even begin to understand it, let alone process it in any way. So, you need to almost emotionally advance to be able to recognise it and I think that's even the same for kind of like gender issues.”

Group Leader, Discovery Science.

“Yeah, gender issues, you need to be more emotionally advanced to kind of understand gender issues otherwise, you could be, you know, right smack in the middle of a gender inequity issue and it will just pass you by.”

Group Leader, Discovery Science.

The frequency with which sex and gender were invoked suggests that race and ethnicity were often understood through comparison to gender, both framed as sites of inequity and underrepresentation. Yet these were the only parallels participants drew; other axes of inequality linked to genetics and health, such as disability, were notably absent. The prevalence of gender discourse, especially compared to the brevity with which race was often curtailed, indicates how race remained a more complex and uncomfortable topic to dwell on.

“It’s not just around ancestry, you know, we talk about—yanno—historically... Actually, I think we used to talk about gender quite a lot, I think, actually, just because scientists don’t like using the word sex. Okay? But yeah, yeah, there’s a—there’s a biological sex, so there’s a genetic sex, and that’s male or female. And I guess other situations—XY, and what have you—but you can be... you could be X... your sex can be XX female, and your gender can be male. I mean, that’s entirely fine, just as it’s entirely fine to be ancestrally, erm, you know, to identify as... your ancestry genome might be one thing, and then actually, kind of culturally, your—this—your ethnicity could be different to your ancestry, is what I’m saying.”

Group Leader, Discovery Science.

Here again, gender and sex are used as a model to explain ethnicity and ancestry, while race remains absent from the discussion. The reflection that scientists prefer gender because they “don’t like using the word sex” echoes earlier tendencies to favour ethnicity over race. In both cases, the more acceptable term is used to smooth over conceptual and political discomfort, producing a slippage between accuracy and acceptability.

“the same with sex and gender ..so basically, now in the publications you shouldn't use gender, you should use sex. Because obviously, the implication of different genders right? And the sex assigned at birth. So yeah I think here, similar , I'm trying to understand them...the implications”

Postdoctoral Researcher, Discovery Science.

The discomfort was tangible. It surfaced in hesitations, self-corrections, and visible searching for the ‘right’ word, producing moments of insecurity and trepidation around how to speak at all. This unease extended beyond interviews: during fieldwork, emails relating to this research were repeatedly flagged

or blocked because of the use of the word 'race', underscoring how the term itself had become institutionally prohibited.

Power and Prohibition

The widely accepted position that race has no place in genetics, as observed in the broader literature, was also reflected in participants' responses. One noticeable way of avoiding engagement with race was through its prohibition or the perception that it should not be discussed. As a result, key issues such as racism and the role of race in systemic inequality remained largely unexamined.

"I try to avoid using the word race...full. Stop "

Programme Lead, Discovery Science

I asked why and then after a long pause they remarked:

"I find it's a word well which is quite evocative and has a kind of erm it's. It's a word which is difficult to define. And I think many people use it [in] different ways....there's a way in which I might want to use the word race that's not the same as the way somebody who might be listening to what I'm saying wants to use race. So I try to stick to ethnicity and ancestry"

Programme Lead, Discovery science.

As in earlier examples, the authority and institutional power attached to role appeared to shape how discussions of race could be directed or contained. In a previous case, a younger scientist's work had been rewritten by a more senior colleague to remove references to race altogether. Here too, seniority enables control over what becomes sayable.

"I feel as scientists we are almost adding to the idea—the notion that there is some biological basis to these ethnicities."

Group Leader, Developmental Biology.

Despite questioning meaning, this participant acknowledged their participation in that very process of meaning-making: the biological framing of ethnicities. The statement was delivered with frustration, as something they felt they should not be contributing to, yet recognised as a recurring outcome of scientific practice.

Some participants also dismissed the categories as obviously flawed, but that dismissal could itself function as a way of narrowing discussion:

"It's just, I think that what they are trying to capture, you know, those ethnicity, or whatever boxes blah blah blah, you know, is that they're actually not trying to capture your ethnicity at

all. But it's with socio-economic kind categorisation, which is a completely different thing. And then that gets back to like income, or your parents' income, or your... and then they don't want to ask that, you know."

Group Leader, Discovery Science.

Like this group leader's "blah blah blah", this next scientist's reflections exemplify a particular distancing that carried a mix of amusement and dismissal, mocking the notion of categories as if their redundancy were self-evident. As they spoke, there was a physicality to this rejection: gestures that seemed to invite complicity. The familiarity of this exchange may also have been shaped by symmetries in positionality and having encountered one another in institutional spaces concerned with race and equity. These shared social locations seemed to blur the line between collegial rapport and epistemic alignment, creating a sense of collusion that both connected and unsettled.

"Even when you are trying to see the analysis by categories if you plot differences between the ethnicities it gives more credence to the idea that there are things that determine or form the basis of racial groupings. When you look at it the initial groupings have nothing to do with that they just had some daft measurements of pretty superficial anthropomorphic type measurements."

Group Leader, Developmental Biology.

By mocking the historic "daft" measurements used to justify race, the speaker positions themselves as intellectually above such practices, yet the derision also enables a form of detachment that absolves them from deeper engagement with how contemporary forms of classification continue to operate. The laughter and dismissive gestures neutralising the political charge of the topic, transforming critique into performance. Such distancing is emblematic of what I repeatedly observed: avoidance that went beyond discomfort, a deliberate narrowing of discussion that restricted its depth and scope.

The avoidance of racial discourse limited the possibility of sustained engagement, diverting conversation toward broader institutional or logistical concerns such as recruitment, participation, or global economic disparities. Yet these issues are inseparable from the racialised assumptions underpinning ideas about data quality and representativeness. As discussed in Section 2.4, such reluctance to engage aligns with wider critiques of race avoidance in scientific discourse (Fullwiley, 2007; Ossorio & Duster, 2005) and reveals how the presumed neutrality of genetic research can obscure the structural inequalities that sustain its infrastructures (Roberts, 2011). In this context, ideas of "accuracy" and "acceptability" emerge as modes through which categorisation can be discussed without discomfort, framed as technical challenges rather than ethical or political questions. What is produced through this discourse is not merely methodological caution, but a form of epistemic evasion

that sustains the illusion of scientific objectivity while leaving the social and historical dimensions of classification unexamined.

5.2.2 Accuracy and Acceptance

If race and ethnicity are frequently rejected or avoided, the question that follows is what kinds of categories are instead treated as accurate, acceptable, and legitimate within genomic practice. This section examines how terms such as ancestry come to be privileged as scientifically neutral, and how claims to accuracy function as a way of resolving ethical discomfort without abandoning classification altogether.

Ancestry as accuracy

Scientists consistently expressed a preference for terms such as ancestry, framing them as more accurate and scientifically appropriate than race. These terms were described as objective and semiotically uncomplicated, even though the methods used to produce them, particularly clustering techniques, rely on subjective and contextual decisions about how difference is constructed. Geopolitical labels were routinely assigned to genetic clusters, reinforcing the impression that such groupings possess a real and stable biological existence, despite concurrent acknowledgements of their constructed nature.

“So because so I think my understanding for most of these is very pra..practical. It's it's about composition of genome differences or the ability to compare but but usually as a very genomic sense, and I think a lot of that is is, is detached from social constructs around race or ethnicity. It's just, you know, the basis you have in your genome.”

Group Leader, Infectious Biology.

Here, group construction is framed as a practical necessity, grounded in comparison rather than social meaning. Race and ethnicity are positioned as external to the data itself, while genomic difference is treated as something that simply exists and can be measured.

“I don't think we would use race in general. I don't think it's a very accurate formulation. I would imagine we would mainly refer to genetic ancestry and the reason for that is that people can have different perception of who they identify themselves with in terms of continents, nationalities and so on but that isn't as important from a drug discovery perspective as what your genetic background is.”

Senior Scientist, Translational Informatics.

Race is rejected here not because it is ethically troubling, but because it is framed as scientifically imprecise. Identification is characterised as a perception, while ancestry is positioned as a more reliable inference. This distinction allows ancestry to appear neutral, even though it too is produced through interpretation and modelling.

The authority to label

Distancing of who can be classified who does the classification becomes more explicit as this scientist shifts to addressing a hypothetical “you”:

“So we can infer your genetic ancestry based on the DNA. We can plot it, we can give you some kind of coordinates artificially inferred, and we can provide you with a label based on seeing where your point is mapping on that plot, with what other populations it’s mapping with, right? And it is tricky because also you get a lot of admixture so it’s absolutely smearing and blurring in all sorts of directions. So you’re somewhere on the spectrum of that, but that is important information, so we would be using genetic ancestry predominantly, and that then has a label associated with some kind of population or a larger group.”

Senior Scientist, Translational Informatics.

The use of “you” raises questions about who is being positioned as the object of classification. While admixture is acknowledged as complicating inference, the method remains privileged. The data must fit the analytic framework, and the assignment of a label is treated as both necessary and improving accuracy relative to race.

“And so when I think about genetic ancestry, it’s about where your kind of genome sits in that three dimensional space. That’s a very different thing from ethnicity.”

Group Leader, Discovery Science.

Genetic ancestry is presented as multidimensional and sophisticated, contrasted with ethnicity as comparatively flat. Notably, this distinction rests on belief rather than on how ancestry is actually constructed. Although ethnicity is rejected as imprecise, it remains conceptually tethered to ancestry as a way of explaining difference.

Scepticism towards self-reported ethnicity further reinforced this hierarchy. Self-identification was frequently portrayed as unreliable, particularly when set against genetically inferred classifications.

“People may raise concerns when you cluster people like based on their like genotypes...and it’s not actually reflecting. Well, the self-reported ethnicity is often very inaccurate in terms of like how their genetics actually speaks...but for scientific research, of course you don’t.... it’s

not possible to rely on self-reported ethnicity because that would cause a lot of trouble in the analysis.”

Senior scientist, Bioinformatics.

Here, the scientist positions themselves apart from the public, acknowledging that self-identification matters socially while insisting it cannot be trusted analytically. Ethnicity is framed as meaningful only insofar as it can be corrected by genetic data.

Others expressed discomfort with the assumption that ethnic groupings have a biological basis at all:

“It’s this idea that there is some sort of biological basis by these groupings and they want to find out. When I feel that it is probably slightly flawed and there are other ways of looking at the data.”

Group Leader, Developmental Biology.

The interviewee’s laughter underscored the perceived absurdity of searching for biological meaning in ethnic categories, even as such categories continued to structure analysis.

“I just I just assume that you know, there’s so only much you can trust on self-reported ethnicity when you don’t have the genotype data. So the genotype is real with that you can infer this ethnicity. But of course I don’t have enough expertise to comment on this.”

Postdoctoral Scientist, Discovery Science.

Among less senior researchers, uncertainty and deference were more explicit. Genotype was described as “real”, while ethnicity became something inferred, revised, or corrected. Although ethnicity was framed as unreliable, it was not discarded entirely. Its value shifted depending on how and by whom it was produced.

Across seniority levels, hesitancy and expressions of limited expertise revealed a shared uncertainty about how best to classify human variation. Yet this uncertainty did not disrupt the underlying assumption that genetic analysis could reveal a more accurate version of identity than self-report. Ethnicity was treated as having an objective existence that could be accessed through genetic data, even as participants acknowledged its instability.

Although ethnicity was widely viewed as more acceptable than race, it remained an imprecise and contested category, particularly when self-reported. The recurring appeal to genetic ancestry reflects a persistent faith in the objectivity of genomic methods, despite the socially constructed nature of the populations they produce. Avoiding race, then, does not eliminate its influence. Instead, it is

reconfigured through the language of accuracy, inference, and methodological necessity, a transformation that becomes more explicit in the following section.

5.2.3 Inescapability

As shown in sections 5.2.1 and 5.2.2, scientists at Sanger actively rejected race as an analytic category, favouring ancestry, ethnicity, or geography as more acceptable alternatives. However, here I show how these interviews also demonstrate that such substitutions do not remove racial logic from genomic practice. Instead, race reappears through the dominance of European reference categories, the moral pressure of classificatory “boxes”, and the embodied language scientists use to situate themselves within genetic space. In this sense, race is not eliminated but reconfigured.

European ancestry as norm

A central site of this persistence was the dominance of data labelled *European ancestry*. Interviewees described this imbalance not simply as a historical legacy, but as actively shaping discovery and translation. As one participant observed:

“The vast majority of the data that exists is from predominantly white European populations. And that really skews the discovery, drug discovery.”

Group Leader, Translational Informatics.

Another acknowledged incremental change while emphasising continued concentration:

“The ancestral diversity of our genome association is improving, but still very, very heavily focused on European ancestries.”

Programme Lead, Discovery Science.

In these accounts, *European ancestry* functioned as both a methodological reference category and a social descriptor. It was frequently used interchangeably with *white*, while populations outside Europe were more often discussed through broader ethnic terms:

“If an atlas is 70% white or 70% European, what about the rest of the ethnicities?”

Postdoctoral Researcher, Discovery Science.

This slippage shows how scientific terminology remains entangled with everyday racial language, even when race is explicitly rejected as an analytic category.

Being put in boxes

These tensions were experienced personally. Scientists described frustration with limited or imprecise options for self-identification:

“It doesn’t kind of describe it... my dad is [from two countries], my mum is [another country]... ancestors were from the British Isles.”

Programme Lead, Discovery Science.

and

“The only thing I can choose from those forms is like a white background.”

Postdoctoral Researcher, Discovery Science.

Comparisons across national contexts further highlighted the instability of these categories:

“In Malaysia, you are either Indian, Malay, Chinese. Here you have White, European, Black... What do those things mean?”

Group Leader, Developmental Biology

For some, even resisting classification felt morally fraught, although resistance itself is powerful:

“Am I being obstructive... as a white middle-class person to not go along with the system?”

Programme Lead, Discovery Science.

Despite recognising the inadequacy of these categories, participants sometimes returned to discrete racial constructs when describing genetic variation. One scientist described their own genome as composed of locatable parts:

“Small segments which are very, very African... but across my entire genome, it’s firmly and centrally in the white European, white British cloud.”

Programme Lead, Discovery Science.

The language here is telling. Genomes are imagined as divisible into segments, while European ancestry is described as central and stable. By contrast, “very, very African” appears peripheral and exceptional. This way of thinking underpins a methodological logic that permits the exclusion of some parts and the prioritisation of others, whether for their similarity or, their relative size (section 4.3). Human data thus becomes a collation of neatly defined ancestral fragments rather than a blend of multidimensional information shaped by both social and biological histories. This asymmetry recurs in population labelling, where European ancestry uniquely carries the additional marker of whiteness.

Relatedly, one senior scientist described themselves using explicitly racial terminology, “Either white or Caucasian male, heterosexual”.

The use of *Caucasian* reflects the persistence of outdated racial language within scientific spaces, made more consequential by seniority and institutional authority. As shown in Section 4.4.4, the term carries a long colonial genealogy that continues to be rearticulated in contemporary genomic discourse under the guise of scientific neutrality.

Demands of Description

The responsibility to describe human difference clearly and accurately sits uneasily alongside inherited linguistic habits. For early-career researchers in particular, the pressure to follow established norms, satisfy reviewers, and keep work legible to both scientific and public audiences makes racialised and ethnic labels difficult to avoid, even when they are critiqued.

Public engagement and language choices therefore become sites where these tensions are most visible:

“It is sometimes necessary to have public engagement as part of the research, including the choices of words for writing up the manuscript, which is part of the cycle of research, and follow-up activities such as results presentation in conferences, Twitter, LinkedIn, interviews etc. For these purposes, the choices of words to describe genetic similarity could be different.”

Senior Scientist, Bioinformatics.

Similarly, another participant described how wider engagement requires shifting language when communicating with non-specialist audiences:

“We will describe the populations and their geographical areas. But it will be centred in genetic ancestries. When we engage with a wider group, we’ll have to change the language to be more accessible.”

Senior Researcher, Discovery Science.

Language is presented here as something that must be made accessible, positioning public communication as a translation problem rather than a substantive rethinking of classification. Yet the emphasis on what remains “centred” also implies stability and hierarchy, and the move to geography does not escape ancestry-based belonging. In practice, this ‘best-fit’ logic sorts people into categories that appear objective but remain shaped by social histories of who is assumed to belong where.

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Vignette - feelings about race outside the field - 11 December 2023

A public participant engaged through parallel public-facing work articulated a markedly different relationship to race than that expressed by most scientists interviewed. Rather than treating race as an awkward or risky category to be managed, she framed it as a historically contingent construct whose harms arise not from its invocation alone, but from the failure to confront the social and political systems that sustain inequality. While she rejected racial labels for herself, describing herself foremost as human, this position reflected critical awareness of how race has been mobilised through science, law, and policy to naturalise hierarchy.

In contrast to scientists who positioned race as scientifically imprecise or publicly dangerous, her critique was historically expansive and ethically explicit. She situated genomic discourse within longer trajectories of slavery, colonial extraction, and racial science, arguing that the central problem lies in separating biological explanation from social conditions such as poverty, discrimination, and power.

This perspective complicates the apparent convergence between public and scientific critiques of race. While both reject race as a biological essence, they do so from different positions: scientific discomfort often seeks distance from history, whereas this participant insisted that ethical responsibility requires sustained engagement with it.

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Participants repeatedly described public engagement as a point where these tensions surface most clearly, requiring shifts in language across audiences. Yet such shifts did not undo the underlying classificatory logic. Rather, they reinforced distinctions between authoritative scientific description and its translation for broader publics.

These socio-cultural dimensions of classification were not always clearly understood, particularly among early-career scientists, who often felt they lacked guidance on what to record, report, or avoid. This uncertainty left many vulnerable to reproducing outdated racial categories:

“I don’t know, did you follow the lung cell atlas kind of tweet? So basically, the human cell atlas community kind of reconstructed the lung cell atlas, and then there was a subset of donors where there was recording of very crude things, like white, black, like very crude. And then there was a whole Twitter thing about, like, this is obviously too simplistic, but is it good to even record the data or not? .”

Programme Lead, Discovery Science.

The next participant linked this directly to the absence of shared norms:

“I’ve never seen race as a term, which I think is a good thing, it’s not there. We usually see ethnicity, and, as I said, the only column I showed you was self-reported ethnicity. But even when you publish a paper, there are no guidelines on what should be included, at which terms, and yeah, I think it’s still very ambiguous.”

Postdoctoral Researcher, Discovery Science.

For those who already viewed ethnicity labels as flawed or meaningless, this feeling of an absence of guidance served only to entrench their use, exposing how critique and compliance can coexist within the same practice:

“You submit some work, and the reviewer will ask, did you look at ethnicity? Then you have to do it, and it becomes this vicious cycle that is reinforced. And there’s no group of people or organisation or learned society that says we need to stop this. It’s not coming from the President of the Genetic Society.”

Group Leader, Genomics.

Taken together, these accounts show how classification becomes less an act of belief than an infrastructural demand, sustained through habit, hierarchy, and the continual requirement to describe.

Across these accounts, racial ideas persist not as explicit analytic commitments, but as the language through which authority, precision, and belonging are routinely expressed. Efforts to move beyond race do not eliminate its effects; instead, race is reconfigured into the vocabularies of ancestry, population, and data. As authority shape what can be said and known about human difference, genomic research risks continuing to privilege certain forms of knowledge and certain groups of people, sustaining long-standing inequities under the guise of neutrality. In this way, genomic communication continues to reproduce racial hierarchy.

5.3 Power in Methods: Constructing Knowledge and Difference

The making of groups is far from neutral; it is a powerful act that structures both knowledge and equity within genomic research. Scientific categorisation influences not only the analytical design of research but also its ethical and social consequences (2.5). This second theme shows that while grouping individuals is necessary for analytical clarity, it carries significant risks: flattening the complexities of human diversity, embedding inclusion and exclusion into research practice, raising questions of accountability, and complicating efforts to achieve genuine data diversity. Underlying these processes

are values, biases, and assumptions that shape how knowledge is produced and whose lives it represents.

5.3.1 Flattening diversity

As discussed throughout this thesis, human populations are profoundly complex, shaped by genetic, cultural, and environmental variation. While this multidimensionality is widely acknowledged, the ways diversity is operationalised in genomic research often flatten that complexity. In practice, diversity must be rendered analytically tractable. This necessary reductionism (Sarkar, 1998) enables comparison, but it also reshapes what diversity is understood to be.

Participants described ancestry inference as an exercise in approximation rather than precision. Genomes were situated within multidimensional space, yet ultimately reduced to summary positions that facilitate analysis rather than reflect lived or biological complexity. As one programme lead explained:

“We use tools like principal component analysis to assign a genetic ancestry... you can place someone in a multidimensional space and say, this person’s genome is here. Of course, each individual will have small sections that look very African or very European, but you don’t capture that *precisely*. *What you get is an overall genome ancestry average, which is what we often need.*”

Programme Lead, Discovery Science.

What is produced here is not a representation of genomic complexity but an operational simplification. The data are compressed to fit methodological frameworks, rather than methods being designed to accommodate complexity. Diversity becomes an average, and variation is managed rather than explored.

Frustration with this narrow framing of diversity was especially pronounced in infectious disease research, where environmental and social conditions are inseparable from biology:

“I am frustrated about the definition of diversity... because in truth it just fails. Diversity isn’t just ancestry. It’s diet, environment, disease exposure, age, poverty. Suddenly, that feels like the real world.”

Group Leader, Infectious Disease Biology.

Here, the laboratory is described as producing a partial and simulated world, constrained by tools that struggle to accommodate complexity. Reduction becomes not just a methodological necessity but an epistemic limit.

These pressures extend into data labelling and harmonisation. Sensitivities around ancestry annotation were widely acknowledged, yet simplification remained unavoidable:

“There has to be a certain level of simplification and harmonisation so that datasets can speak to each other, but you have to make sure you’re not misrepresenting something important.”

Group Leader, Translational Informatics.

The scientific drive for comparability continues to shape how diversity is operationalised, particularly through the construction of homogeneity within datasets. One group leader outlines the methodological rationale:

“That allows us to do these kinds of homogeneous comparisons between cases and controls... we correct for variation within the sampling, and define our case controls from within it.”

Group Leader, Discovery Science.

Homogeneity here is not discovered but produced. While such practices enable statistical precision, they also obscure the heterogeneity they claim to manage, sustaining the impression of stable and bounded groups.

Participants were acutely aware of the limitations of continental categories, even as they continued to rely on them:

“It’s insane to label Polish and Finnish people under the same umbrella. A geneticist knows that. But you still need that branching.”

Group Leader, Translational Informatics.

This awareness captures a central tension. Scientists recognise that these categories are insufficient, yet they remain necessary for producing usable data. As Gitelman (2013) argues, the ideal of pure data is an impossibility. Genomic science depends on simplification and exclusion to function.

As one participant reflected:

“You have to have a comparator. That’s how science works. It’s reductionist.”

Group Leader, Infectious Disease Biology.

Reduction enables discovery, but it also delimits what can be known. Through practices of averaging, harmonisation, and comparison, diversity is transformed into manageable difference. In this process, absence becomes institutionalised. Those least represented in genomic datasets are often those most exposed to structural inequality, excluded not by accident, but by the very mechanisms that claim to make human difference scientifically knowable.

5.3.2 Inclusivity and Exclusivity

The creation and application of categories in genomic research affect who is included in or excluded from studies, shaping data accessibility, analytic comparability, and downstream applicability. Group labels organise recruitment and participation, but also govern how data are stored, searchable, shared, and analysed. Participants repeatedly returned to the consequences of categorisation, particularly how participants experience these divisions, and how “low numbers” become a routine justification for exclusion.

A Programme Lead described how small ancestry-labelled groups can be treated as a confounder to be removed, framing this as a historically established solution:

“There are certain labels that we don’t have in numbers to be able to include... when we were doing one of our very first genome-wide association studies... there were some Jewish ancestry individuals in the cases, and we didn’t have any Jewish ancestry individuals in the controls... so we had to remove the small number of Jewish individuals from the study...”

Programme Lead, Discovery Science.

They then emphasised that representativeness is negotiated as a matter of “appropriate number”, rather than a fixed threshold, and positioned newer methods as expanding what is possible, without removing the underlying constraint:

“Once you start to get to having... an equivalent number... there are methods you can use... there is less need now to say, okay, I’m very tightly controlling genetic ancestry here... but still... if you have a very small number of individuals with a very divergent ancestry... you are going to run into problems, so you may need to remove them.”

Programme Lead, Discovery Science.

What is striking here is how quickly the analytic problem is located in the data rather than in the research design or the wider infrastructural conditions that produced the imbalance. “Divergent ancestry” becomes undesirable, and exclusion is normalised as methodological hygiene. That logic also helps explain why minority populations remain missing from the kinds of studies that require large, balanced case-control architectures.

A less senior scientist described the same practice more bluntly, reinforcing how exclusion can become routine when ancestry-labelled counts are low:

“Often you have... a few like 10 individuals who are not European. And then basically just exclude those people because you just simply don't have that power...”

Senior Scientist, Bioinformatics.

They gestured towards alternatives, but each requires additional resources, access, or time, for example, modelling structure, or drawing on external biobank data “to expand your research”. In other words, inclusion is technically imaginable, but practically costly.

A further example makes the dependence on aligned categories and sample sizes explicit. The same Programme Lead explained how, when controls include an ancestry-labelled group that cases do not, the controls are reduced to match:

“When you get... the UK Biobank data... [it] had more individuals of South Asian ancestry... and then we didn't have any [disease] cases of South Asian ancestry, so we would remove the South Asian ancestry individuals from our control set...”

Programme Lead, Discovery Science.

This shows how interoperability is not just about phenotype harmonisation, but also about the numerical distribution of people across categories. Categories become the conditions of comparability, and comparability becomes the condition of inclusion.

Another participant linked these questions to national histories and to different logics for defining who “counts” in a cohort:

“Brazilian populations are so mixed for historical reasons... within the study, we are only asking that... parents and grandparents were born in the country... we have other studies... where... ancestry is self-reported... but... we have a group of scientists who will analyse the sequencing data to establish global ancestry and local ancestry.”

Senior Scientist, Global Health.

Two inclusion logics sit side by side here. One is bureaucratic and geopolitical (country of birth across generations). The other is classificatory and technical (self-report and then genetic inference). Both can widen a cohort in one sense, while still funnelling participants back into ancestry structures that determine analytical eligibility. The power lies not only in who is recruited, but in who remains intelligible to the analysis once “ancestry” is “established”.

Participants also described how metadata labelling and harmonisation reproduce a hierarchy of epistemic control. As one Group Leader put it, teams inherit the annotations of those who produced the datasets:

“We really rely on the data sets that are already out there, and how they are annotated by the primary teams... what the most invested people have decided is the right way of calling the data... And we adopt this... How do we... harmonise the phenotypic data...? Because the phenotypic data could be slightly different... Then you have it in white Europeans.”

Group Leader, Translational Informatics.

The phrase “most invested people” is doing quiet work here. It points to where infrastructural authority sits: consortia, centres, and teams able to define taxonomies, set standards, and decide what becomes interoperable. To adopt their labels is to inherit their assumptions about how populations and diseases should be delineated. Exclusion therefore occurs well before analysis, embedded in the upstream politics of annotation.

That politics also surfaced through analogy: the same kinds of classificatory disputes that structure cell annotation were offered as a model for population grouping:

“There’s huge community... annotating cells... some people will feel passionate about blending some cells together or not. And that’s totally the same for humans... we can blend some people together... there’s a historical sensitivity. And then there’s also pure data... and whether it makes sense to merge people together or not.”

Group Leader, Developmental Biology.

Alongside merging, participants also described how small-population estimates can be subordinated to larger-population estimates, where “accuracy” is equated with scale:

“We compare... a GWAS from a small population against... a bigger population... [if] no evidence of... heterogeneity... we will use the genetic estimate from the big population because it comes from a big population. It's more accurate.”

Programme Lead, Discovery Science

This sits uneasily with the earlier acknowledgement that small groups are often removed because power is low. If low power makes the difference hard to detect, “no heterogeneity” may partly be an artefact of underrepresentation. The move from exclusion to generalisability is therefore not straightforward.

Some participants did describe attempts to quantify the epistemic cost of exclusion directly:

“[They are] leading a project whereby you... have different ancestry groups... and then saying, if I took one of the data sets out... does that change my conclusion?... you post hoc work out... whether having a non-diverse sample affects the conclusions...”

Group Leader, Developmental Biology.

This reframes diversity as a sensitivity test, making visible how unstable conclusions can become when representativeness is depleted.

Even proposals to bypass labels by returning to “raw data” were described as institutionally difficult, distributed, and slow:

“If we get the raw data... you have to go through data access committee... 20 different sources... But it's not impossible... eventually you will get the data... and then... map... the diversity.”

Programme Lead, Discovery Science.

The appeal to “raw” sequence data can sound like a route to neutrality, but it also risks obscuring what your wider thesis emphasises: data are never raw in any meaningful sense, and access itself is structured through power, ownership, and institutional hierarchies.

Participants sometimes named the affective discomfort produced by these limits to scientific control:

“It is scary... we are getting to a different point where the facts are not sufficient, and that is scary for a reductionist basic scientist.”

Group Leader, Infectious Biology.

Yet alongside discomfort sat a strong claim that ethnicity has biological importance in practice, particularly in clinical translation where European-dominant resources limit generalisability:

“For genetic studies ethnicity has been a very, very important aspect... most of the data are generated from European people, and whether they can be generalised... is very questionable.”

Senior Scientist, Bioinformatics.

And, crucially, inclusion was sometimes justified through benefits to the already-dominant groups:

“If you want to find the potential causal variants... European people... have the same... linkage disequilibrium... including people from different ethnicities... helps... identify causal variants... [and] also benefit European people themselves.”

Senior Scientist, Bioinformatics.

Europeanness operates as the unmarked baseline (see section 4.4.3) and how methodological discomfort with hybridity and structure produces exclusion (see section 4.3.2). Read through that lens, inclusion is not only an ethical gesture but also an instrumental strategy that can consolidate value for those already best served by existing infrastructures.

Taken together, these accounts show that inclusion and exclusion are not incidental. They arise at the intersection of methodological conventions, infrastructural constraints, and epistemic authority. What looks like a technical limitation (low numbers, harmonisation barriers, access committees) is also a social and institutional mechanism that determines whose diversity is valued, and whose absence becomes statistically permissible. This returns us to the ethical stakes of categorisation and to the moral work done when populations are made comparable by being reduced.

5.3.3 Responsibility and accountability

This section explores how scientists at Sanger negotiated the responsibilities and accountabilities that come with creating and using human population categories. Across interviews, researchers spoke about the risks of misrepresentation, the pressures to demonstrate inclusivity and the practical limits of what they felt able to control. Responsibility appeared dispersed and uneven, often acknowledged

yet rarely claimed. What follows are conversations shaped as much by avoidance and uncertainty as by confidence in scientific method.

Curiosity seemed to be a strong driver for why researchers enter science, often discovery scientists described their work partly as the pursuit of knowledge for its own sake.

“I mean, there is that curiosity and that sort of why, like a child, right, why why why why why, so that does have to, that is a kind of driving force for a lot of people I think, and certainly for me that has always been really the major reason I go into research and it is just curiosity and wanting to know the answers”

Programme Lead, Discovery Science.

The metaphor of childlike curiosity and innocence is casually assigned to scientists, portraying them as motivated only by the desire to understand a new world. .

“Have I always wanted to do this? No, I think for a long time I had no idea what I wanted to do, I just kind of was driven by curiosity.”

Group leader, Translational Informatics

Even where the point being made is that curiosity precedes a fully formed research direction, this sometimes flippant framing invokes an image of scientific intention that obscures the power and purpose underpinning research trajectories

“I think it's naive, you know, no one does anything they're not interested in, they just don't do it.”

Group Leader, Infectious Biology.

In one exchange, after I asked what harm might arise from working with under-represented samples, the scientist leaned back, frowning slightly as though uncertain how to approach the question. I repeated it, trying to move the conversation beyond the usual moral position that everyone should be included simply because inclusion is assumed to be ethically self-evident. Given that several participants had already raised concerns about the sensitivities of using certain language and the risk of misrepresentation, I had not expected the question to provoke discomfort. She resisted reframing her stance and instead replied:

"it's more about the good that we can do I would say it the more positive way is it's less about the harm of having a sample that's under-representative and more about the benefit that we gain for better representation, yeah? I guess that's how I would put it the obligation, basically."

Programme Lead, Discovery Science.

Another scientist described how exclusion is rarely foregrounded during analysis:

"if you're working with data that you have in your hands, unless you're probably directly working with comparing it, I don't think you often think about that its impact will be limiting, right? I think you're just more focused on like, you know oh, it is going to help someone, right? You're not thinking who, it's not going to help and what are the reasons for that?"

Group Leader, Translational Informatics.

The use of inference models enables researchers to work with existing data without directly imposing labels. Yet, as discussed earlier, summarising findings within frameworks such as genetic ancestry or ethnic categories still exposes research to misinterpretation. Many participants expressed concern that their work might be co-opted for harmful purposes, although the wider culture of science does not consistently encourage deep engagement with these risks.

One senior scientist in his attempts to explain distinctions between sex, gender, ancestry and ethnicity states:

"You could be X. Your sex can be XX female. and your gender can be male, I mean, that is entirely fine, just as it is entirely fine to be ancestrally erm you know, to identify as, if if ancestry genome might be one thing, and then actually, kind of culturally, you this, your ethnicity could be different to your ancestry is what I am saying, in much the same way that your gender can be different to your sex. ...when I was a PhD student and post doc, we would kind of use those words interchangeably but now I try, because old habits die hard, I try to use them differently and encourage people in my team to use those ancestry and ethnicity sex gender to mean very different things. ...You can have two questions one to ask their sex and one to ask their gender but actually even doing things like that is quite erm it is quite sensitive for some people ... I am a geneticist so you know kind of confronted with it. 'Yes, but what does the genes say' ? Most people, perhaps, are not. They are like, well, my gender is this, and that is the most important thing."

Programme Lead, Discovery Science.

Hearing this, I noticed the uneasy laugh, a small but telling moment. It revealed a tension beneath his seemingly clear explanation. He used gender and sex as an analogy, which helped him frame ancestry as similarly flexible, but it also moved the discussion away from the deeper sensitivities surrounding how ancestry is defined, measured and imposed. The responsibility of creating categories was present but unspoken. Scientists construct the classifications that participants must later inhabit, and even the simplest demographic question can feel intrusive or loaded. His explanation shifted the sensitivity towards the public, while his own position remained grounded in

what genetics requires. The laugh exposed the gap between scientific neatness and lived experience and showed how classification is negotiated, softened and occasionally sidestepped.

A recurring theme in discussions of genomic equity was the idea that increasing diversity in datasets is the ultimate solution. While this is an admirable goal, it remains challenging given the complexity of human variation and the structural barriers to collecting comprehensive data. Efforts to improve diversity often focus on either maximising existing datasets or encouraging participation in underrepresented groups. These offer practical short-term benefits but cannot fully address the deep biases within genomics (Fatumo et al., 2022).

One senior scientist described their sense of reassurance about the generalisability of common variants:

"So I'm kind of reassured now that we're getting to large enough sample sizes that I can rule out for the vast majority of the loci that there are these differences in effect, size, because we'd have found them, I think, for some, there are kind of modest differences in effect, size. And I think ultimately we'll, you know, because standard errors get smaller. You obviously get more power to define tiny differences, we will find the differences, but they will be relatively small. And certainly, if you were trying to use these genetic insights to give you genetic findings to give you insight into the disease biology, accepting that, there's a kind of very much imperfect correlation between an effect size which variants and the relevance of a particular piece of disease biology. Then, you know, I still, I still think, if you learn about IBD biology in one population, then, based on that, you develop a drug. Then that drug stands a really good chance of working."

Programme Lead, Discovery Science.

Here, they argued that a drug developed in one population is likely to work widely. This reflects an expectation that scientific findings should be generalisable. I questioned this confidence given the simultaneous emphasis on diversity and the insistence that population differences matter. If differences across groups are minor, why does diversity dominate the discourse?

They later offered an important distinction:

"Now, I should say. All of this, I just said, applies to common variants that we are getting out of GWAS. I am not so confident that the same will be true of rare variants. Rare variants typically are younger than more common variants, which means that they are more likely to be population specific, because they are more likely to have mutated recently and therefore be present in a smaller number of populations. And therefore you know that that problem I was talking about earlier about. Oh, I am looking at population A, so I can see the variants, which a

specific population B is more likely to be thing more of a problem for rare variants than it is for common variants. So that is an issue, especially because we believe that rare variants, if they are associated with disease, are likely to have larger effect sizes. So I think we are at a certain point with our understanding around common variants where yes, there are differences erm but very likely a discovery in one population, which is kind of like an awesome, exciting ten out of ten discovery, is probably a kind of awesome eight or nine out of ten discovery for the other population. I do not know how true that would be for rare variants. I think that the jury is still out on that one"

Programme Lead, Discovery Science.

The metaphor of the jury still being out was illuminating. It acknowledged incomplete evidence and suspended judgement, revealing how uncertainty becomes normalised in scientific practice. It also raised the question of who is responsible for deciding when evidence is sufficient. In research, judgment is dispersed across reviewers, committees, funders, replication efforts and public or regulatory scrutiny. When the jury is still out, responsibility is unclear, yet the consequences of uncertainty are felt unevenly, often by populations not captured in datasets.

Researchers acknowledged that data gaps will always exist. Genetic findings apply most reliably to the populations represented in a study, yet genomics often aspires to universal relevance. As one infectious disease biologist explained:

"And so if you really want to understand all aspects of disease, then you have to understand all people And so it's not just a fascination with underrepresented populations. It's just an acceptance that we're not normal. And then and suddenly, on top of that, you say, well, actually, we're really benefitting from this data and other people need to"

Group Leader, Infectious Biology.

Across interviews, the categorisation of human groups emerged as a contested and ethically charged process. Scientists drew on concepts of descent, ancestry and ethnicity in ways that often reflected hierarchies of acceptability. Ethical principles such as maleficence, beneficence, autonomy and justice appeared intermittently but rarely operated as a coherent framework. Instead, they were mobilised unevenly and shaped by the immediate practical demands of research. Yet these conversations also revealed that scientists were not simply carried along by structural norms. Many demonstrated an active awareness that their own decisions, language choices and methodological habits played a role in shaping how categories were constructed and used.

Some participants reflected that these discussions expanded their awareness and opened possibilities within their own practices:

"It's made me rethink how we focus on what type of research we do. And I think we touched briefly on the idea of having concepts around social justice at the forefront, and then the specific biological areas that would meet that..."

Group Leader, Developmental Biology.

This reflection suggested that scientists were not entirely passive within the constraints of genomic methodology. Instead, they recognised points where they could intervene, however subtly, in research direction, lab culture or the framing of questions.

By having these conversations intentionally, one participant reflected on the effect they were already having on them:

"You're just resurfacing certain concepts by talking to you... one starts to ask themselves: Is this something in my research I could do?... Because this could have an impact on my research."

Group Leader, Translational Informatics.

These moments showed that agency was often perceived as quiet and incremental rather than structural or dramatic. It surfaced in the ability to notice when a concept or ethical concern reappeared, to adjust a methodological choice, or to steer their group gently toward more reflexive practices. Scientists described these possibilities as small openings rather than sweeping reforms, yet they were openings that mattered. They revealed that responsibility was not only distributed and contested but also something researchers could actively shape in the everyday choices of their work.

However, alongside this sense of individual agency sat a much more sobering account of how accountability was understood within scientific culture itself. Here, participants shifted from the possibilities within their own practice to the broader structures that shaped what scientists were permitted, expected or allowed to do. One participant remarked:

"Science does not incentivise supervisors positive or negative behaviour. Science has been a free for all. Its only now things have changed."

Group Leader, Developmental Biology.

I leaned forward (Figure 32) as they continued, aware of my own sympathy for their description.

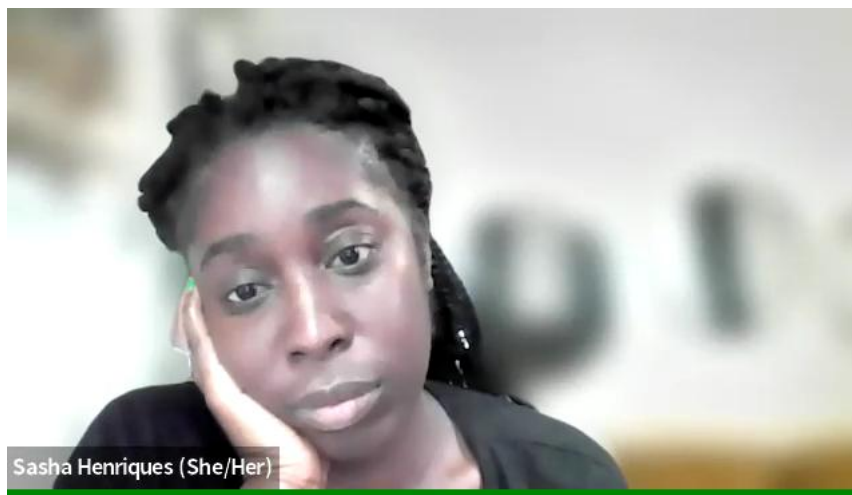


Figure 32. Still from a digital interview during a discussion of institutional accountability and the cultural norms that can facilitate uncollaborative behaviour. Captured by the author, 30 Oct 23.

"For a long time science has allowed this anything goes approach as they just wanted people to be creative. Even now funders do not really penalise anyone."

Group Leader, Developmental Biology.

The interviewee leaned back in her chair, running her fingers repeatedly through her hair as she reflected on what she understood as 'bad' behaviour within the Institute. Her account did not present this behaviour as overtly malicious, but rather as a by-product of the underlying values that shape scientific success. She described a culture in which reward structures privilege individual accomplishment over collective practice, creating conditions that work against sharing, collaboration and mutual benefit. In her view, these norms form the substrate upon which uncollaborative, or ethically questionable practices can take root, particularly in the absence of clear mechanisms for accountability.

Their reflections described a system where accountability has been historically diffuse. Creativity was prioritised over oversight, and correction often entered only slowly. Many described progressing through careers without confronting the ethical dimensions of their work until they were compelled to. Listening to them felt like hearing a sad but familiar story, marked not by individual failings but by structural dynamics.

These accounts also reinforced an earlier observation: many participants felt they lacked clear guidance on how to navigate these responsibilities. Only one participant provided the name of any formal guidance. Expectations associated with ethics review, data access committees, and journals were instead encountered primarily as requirements to be met to get studies approved or published and were often experienced as impositions rather than sources of guidance (explored further in

Chapter 6). In this sense, the apparent shifting of responsibility suggests less a complete absence of guidance than a limited engagement with existing resources, shaped by how that guidance is encountered in practice. In this context, their sense of agency was both real and constrained, emerging in small decisions precisely because larger frameworks for accountability remained uncertain.

Across this section, responsibility and accountability emerged as central yet unsettled features of genomic research. The creation of population categories enables comparison and analysis but also simplifies diversity, introduces inclusion and exclusion, raises ethical questions about representation and reinforces structural inequalities. Scientists acknowledged these tensions while often distancing themselves from their consequences. These methodological practices show how power operates through research methods, shaping both scientific findings and the meanings assigned to human difference.

In the next section, I turn to the final operation of power I observed at Sanger: the moments when conversations moved beyond categories and data, revealing how the institution positions itself in relation to access, equity and its wider social responsibilities.

6 Power Beyond Data: Access, Equity and Institutional Responsibility

For many scientists, questions of equity, diversity and inclusion were not simply abstract values but conditions that shaped what counted as accurate, progressive and impactful research. At the same time, they described working within systems where access to resources, infrastructure and decision-making power was unevenly distributed. Power beyond the data therefore appeared in who could set priorities, who could participate in collaborations and whose needs were taken into account when designing studies.

This theme focuses on three interconnected subthemes that show how power in genomic science extends beyond questions of data representation to encompass challenging practical dynamics governing funding, access, collaboration, and the institutional structures that organise research.

6.1 A Complex Challenge

Diversity in genomic data is perceived as both a challenge and a necessity, as demonstrated throughout this research. The bias in the data, skewed towards European data, is understood to limit the applicability of genomic findings and to slow scientific progress. This was discussed in interviews and more widely across Sanger, as captured in this moment in Figure 33.



Figure 33. Image from a Sanger hybrid in-person and virtual discussion fireside chat featuring senior institute leaders. (Matthew Hurles (R) is Director of the Wellcome Sanger Institute, and Ewan Birney (L) is the Interim Executive Director of the European Molecular Biology Laboratory (EMBL). Here, their appearance reflects their institutional roles rather than participation in this research. This moment in the session highlighted their expressed need for much stronger research cohorts across Africa, including well-supported birth and adult cohorts and sustained investment in local infrastructure, and emphasised that such efforts must be led within Africa rather than managed remotely. The image illustrates the broader institutional context. Captured by the author. 12 December 2023.

6.1.1 Funding

The complexity of achieving more representative data came through strongly in the interviews, alongside participants emphasising its importance

"I think people think about it [ancestral bias in datasets] a lot but do not have enough time to do anything, or perhaps the word is worry. People worry about it a lot but actually do not have the opportunities perhaps. But I do think this is not just a researcher problem, this is a whole structural research problem. And so I think even broadening that conversation and highlighting the areas and the discussions that people are having is key to solving this. So actually I think that is why I was so keen to mention funders. I think they really have a role in this"

Group Leader, Infectious Biology.

Funding emerged as a significant aspect of these participants' research experience (figure 34). The complexity of it was often remarked on, as in the moment above, where this group leader was 'keen' to emphasise how funding facilitates or constrains efforts to improve diversity in datasets. Another participant described how funding shapes available resources and influences how studies involving understudied populations are conducted.

"There's a lot of infrastructure costs that are needed to support patient recruitment, especially if you want to get accurate phenotype data. And oftentimes, in the size of a traditional grant, you don't have enough space to fund those elements."

Programme Lead, Discovery Scientist

This highlights how funding constraints narrow scientific priorities, sidelining recruitment strategies that would otherwise be methodologically preferable. This challenge was particularly pronounced when obtaining data from historically underserved and marginalised populations.



Figure 34 Funding display at away day. Photograph showing a screen celebrating a research group’s funding, taken during an open day where I was invited to observe one participant’s work in greater depth. The image highlights the resources that sustain their research and the broader investment shaping work at Sanger. Captured by the author. 4 October 2023.

While engagement with communities was positioned as an expected element of the research process, particularly in translational work, this participant highlighted how the absence of targeted funding produces a structural gap that restricts engagement with underserved communities:

“All funders are asking now for some kind of public engagement, which is good. But to engage with the public in a way that the public will engage back in a meaningful way we need to have funds to do that”

Group leader, Translational Informatics.

Scientists expressed a preference for diverse data to be accessed and processed at its source, and these conversations were clear in how that also required much more funding, infrastructure and time, for example:

"I kid you not, it is four times more expensive than it is here... you know, this investment into looking at diversifying the datasets is going to just take much longer because you need to invest in building blocks to create that environment and infrastructure to be able to generate the data."

Group leader, Translational Informatics.

The lack of funding intersected with broader equity concerns, including collaboration and capacity building, which were seen as essential for ethical and effective research in global contexts. Researchers often viewed the pursuit of representative datasets as constrained not by scientific will but by the financial and institutional priorities that shape what research is possible as discussed here:

"So, do funders understand, know, in more detail than us, the way we should be behaving, and what we should be doing? the answer to that, in my opinion, is no, categorically not"

Group Leader, Infectious Biology.

And this programme lead, in addition to echoing the well-understood reasoning for appropriate levels of funding, introduces the need for funding designs that address the complexities of working across geo-political regions.

"So if we are now going to go and try and get, you know, samples from other ancestries, engage with local clinicians in non-European ancestry populations, quite rightly, they are going to need some funding, and traditional fund design ain't used to this."

Programme Lead, Discovery Science.

Although this scientist was clear about the limitations of traditional funding models, they did not view all funding as short-sighted. Instead, they pointed to a shift among some funders, including those seen as non-traditional and even historically established organisations, towards more intentional approaches to addressing inequity. One such example was described as follows:

"A few years back, we were aware of the inequalities, and how much of the studies were driven by Europeans for European populations and the Chan Zuckerberg Initiative launched a call to look at different ancestries" Senior Scientist, Global Health.

This perception of change was echoed by a programme lead, who framed it as part of a broader movement among major funders to prioritise diversification in human genetics:

"I think what I was saying about this movement towards wanting to diversify human genetics. I think, actually, some funders like Chan Zuckerberg initiative, for example. And increasingly like Wellcome and NIH, you know they are buying into that"

Programme Lead, Discovery Science.

However, regardless of whether funders were characterised as traditional or non-traditional, they were often described as operating within inflexible structures. In reflecting on the additional time required for research in African and Asian contexts, particularly in comparison with the timelines built into standard funding cycles, this participant noted:

“But if you ask a grant funder, ‘Look, we need two years,’ they say, ‘No, no, we’re only funding 12 months.’ You can’t deliver a grant in that country on that topic in 12 months. It’s impossible to do.”

Group Leader, Infectious Biology.

These comments point to a structural tension between stated commitments to equity and the temporal constraints imposed by funding regimes. While funders may articulate support for inclusive or global research agendas, fixed timelines function as a limiting condition that shapes what research can realistically be undertaken, and where.

This pressure was explicitly linked to questions of power and responsibility, particularly in relation to well-resourced institutions and their funders:

“But the science has to happen really quickly.... I feel a little bit like part of the problem of Sanger is perhaps Wellcome. Because you know they can. Who holds the power and the money. Passivity is an action in itself; by being passive, they have enabled it.”

Group Leader, Developmental Biology.

In focusing on Wellcome as Sanger’s principal funder, participants were clear that Sanger is not simply acting at a funder’s command. Rather, frustration was directed at what was perceived as funder passivity. Funders were understood to hold significant power to shape research priorities, particularly in relation to data diversity, but to be failing to exercise that power in ways that would enable structural change. This sense of constraint was often expressed through irritation, defiance, or open critique, even as participants acknowledged the authority funders hold.

“I’m going to knock the funders again.... I mean, these are my views. So I genuinely don’t care. I honestly don’t. Because I think it’s important, really important. I think what you’re doing is important.”

Group Leader, Infectious Biology.

This willingness to speak critically reflects a tension between dependence and resistance. Although funders were widely recognised as powerful actors within the research ecosystem, senior scientists

often positioned themselves as ethically justified in challenging funding decisions when they were seen to limit progress on equity and inclusion.

6.1.2 Social and Ethical Concerns

Across interviews, participants consistently expressed that questions of ethnic, cultural and ancestry diversity and equity mattered to them, even where they felt under-equipped to address them fully. For some, this took the form of explicit moral commitment; for others, it was framed as a recognition of the need for further understanding, guidance, or expertise in navigating the ethical and social dimensions of genomic research.

“If we wanted to look at or focus on the populations that have been, or are, in significant disparity, we would need to focus the collection sites there, and that’s not what we’re doing. I’m also not entirely sure how we would achieve that without someone from the social sciences leading recruitment; do you see what I mean?”

Senior Scientist, Global Health.

They recognised that the limitation lay not only in the funding framework but also in the expertise needed to engage ethically and meaningfully with communities. Ethics also features in the considerations raised by the next participant, who states:

“it’s an interesting one. Because in a way. what do you do? If you have a data set that, let’s say, has been unethically collected. But it’s there now, it’s extremely valuable. And it’s filling in the gap that’s there, right? And do you deliberately decide not to use it? That’s almost like, you know, that’s even worse.”

Group leader, Translational Informatics.

Here, ethics is framed not as a single moral rule but as a set of distributed concerns across the research ecosystem. Ethical value is located in funding priorities, dataset governance, relationships with communities, and procedural constraints such as consent and access, positioning ethics as both infrastructural and practical in shaping what research can proceed.

“For me it is like you don’t tell your children they can do anything and jump on the settee or something. You have to set some sort of ground rules. But these ground rules don’t seem to exist in science... So where are the rules? By country by institution. Ethical practice is governed by who... which policies override each other and which issues are seen as ‘bad’ enough to be punishable?”

Group Leader, Developmental Biology.

The metaphor of scientists as children in need of boundaries positions responsibility in a notably passive way, suggesting that ethical conduct in part depends on someone else defining the rules.

However, this regulatory and procedural nature of the research environment wasn't always seen as fit for purpose as this group leader expressed below:

"If you are writing a grant, you cannot engage with the community prior to getting the grant, it is unethical. You cannot get the permissions for it. You are constrained by a system that enforces what seems to cover all of the bases, but in itself stops you doing what would probably be the ideal situation."

Group Leader, Infectious Biology.

A further tension emerged between the moral imperative of inclusion and the procedural ethics that shape how, and whether, that inclusion can be realised. As one Programme Lead explained, ambitions to represent humanity sit uneasily alongside the limits imposed by ethical governance structures:

"There was a discussion about whether there should be a piece on definitions, but it's tricky for a number of reasons. So one thing is that, on the one hand, there's this desire for the [programme] to represent the whole of humanity. On the other hand, when you get samples, often there's no record or very little record, depending on how the ethics is structured, of what you can record about the sample donor."

Programme Lead, Discovery Science A.

Accessing genetic data that some believed had not already been shaped by inaccurate categorisation or analysis (raw) was described by some as technically possible but procedurally burdensome. The same participant outlined how managed-access systems shape everyday research practice:

"The other way to get information about genetics is from the sequence, but a lot of samples are managed access rather than open access and so there's a whole application process that very few people go through, frankly, because it just takes time. So they just use the processed data rather than the underlying genetic data, because it's such a process to dig into it, such a palaver."

Programme Lead, Discovery Science A.

Describing the process as a "palaver" signals how procedural ethics, can come to be experienced as an obstacle to scientific work. One response to these constraints was to move away from human tissue altogether, reframing research design to avoid regulatory entanglements:

“And we try and move to cell free, you know, to cell models, and and so on, where there's no, none of those constraints. So we think about that. We think about the sample types and we're constrained by things like the Human Tissue Act... and the problems that happened at Alder Hey and the ethics around storing tissue.”

Group Leader, Infectious Biology.

Here, donors are effectively separated from data, allowing research to proceed with fewer ethical and bureaucratic obligations.

6.1.3 Open Source and Equal Benefits

Yet, as another senior figure noted, these governance mechanisms do not only protect participants; they also safeguard datasets themselves as valuable assets:

“So it's often as protection [restricting access to samples/data], protection of the participants themselves, but then there's also protection of assets, having a diverse or unique population genetic level datasets of value.”

Director, Informatics Science.

This framing is revealing. Genetically unique and historically understudied populations are recognised here as valuable assets. This raises questions about who can access that value and who ultimately benefits from it. While procedural ethics and data access controls are often experienced by researchers as hindrances, they can also operate as mechanisms of stewardship. Such processes create the conditions for retributive benefit, where communities whose data have long been extracted without return may have greater leverage over how value is generated, shared, and translated into outcomes.

Sanger's approach to data sharing was framed as positive and in contrast to the regulatory and bureaucratic obstacles they often observed elsewhere:

“Our resources are sort of open source and we have to be able to share that data so it's slightly different for us in that we would like to be able to analyse and then make that publicly available.”

Director, Informatics Science.

Open data sharing is a fundamental aspect of the research culture in genomics institutions, notably Sanger. It helped establish and enforce the Bermuda Principles, which required the open, rapid

release of sequence data into the public domain. However, inequities can be exacerbated by open data access. Researchers in well-resourced settings can more readily navigate open-access databases, while those in lower-resource environments often lack the necessary infrastructure or training to engage with these resources fully. This scientist reflected on a campus event that I had personally helped organise with Wellcome Connecting Science's education team (Fig. 35 – 38).

“ I had never thought that open access would actually hinder some parts of scientific development. My whole view had been that open science is simply what it is, that you cannot do anything differentactually it is not like that, It has different impacts because of the power. It is not a balancing power.”

Senior Scientist, Global Health.

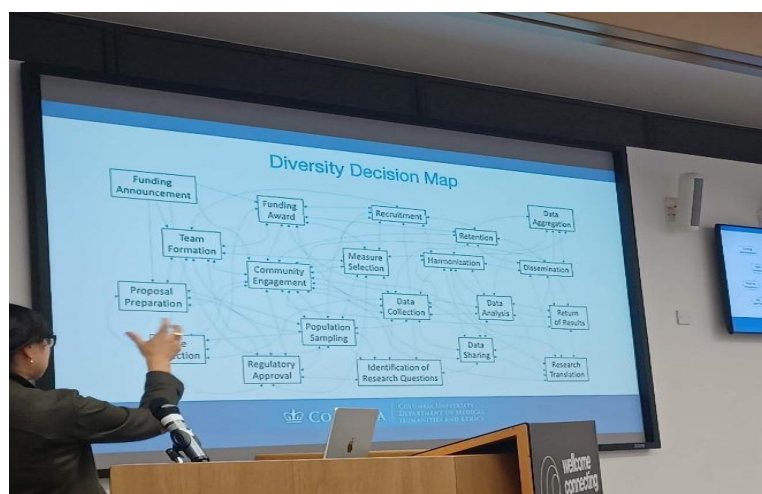


Figure 35



Figure 36



Figure 37

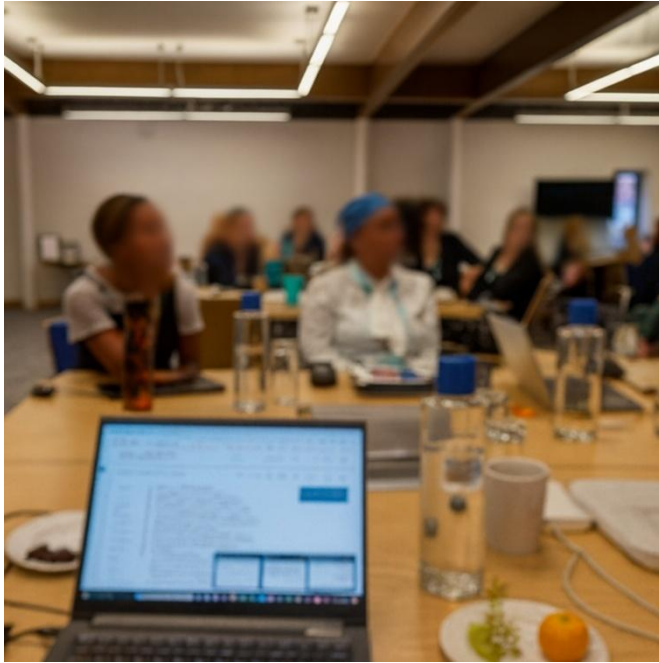


Figure 38

Figures 35–38.

Figure 35. A presentation slide illustrating the complex regulatory and methodological challenges involved in obtaining and conducting research with diverse data.

Figure 36. A senior representative from a major institution discussing the benefits of ancestrally diverse data and the implications of its absence.

Figure 37. A researcher discussing the challenges of uneven access to data that is nominally open source.

Figure 38. An international and interdisciplinary group of scientists at a workshop focused on global data practices. The workshop brought together Sanger staff and international researchers and practitioners in genomics to discuss equity in data practices, and I contributed to its design. The application process was structured to ensure global, disciplinary, and gender representation.

Photographs taken by the author, Wellcome Sanger Institute Conference Centre, 20 September 2023.

This disparity reinforces existing imbalances in scientific knowledge production and participation.

“it's not everybody who has access to the data and it's not everybody who's enabled because the way that the data is actually provided .. it's only for the limited few, who have like high computing resources and the ability to code. The vast majority of people don't have access to it.”

Group Leader, Developmental Biology.

When acknowledged, this was framed as a fundamental obstacle to achieving equitable research outcomes for underserved populations. Access to samples and data from these communities, and their reciprocal ability to generate meaningful uses and outcomes from that data, remains uneven and biased.

“if the benefit of this expensive project is to be equitable then we need to provide the data the insights the knowledge outputs in a fair and equitable way in a portal and that's something I would say we really need to work on”

Programme Lead, Discovery Science.

The bias in European datasets, therefore, represents more than a lack of diversity in samples. It reflects wider inequalities in data access, shaped by the uneven distribution of resources and technological power. The question of how to make data accessible was raised by this group leader:

“We need to generate lots of data and we need to provide the data, but how do you provide the data so that everybody has equal opportunities and access to it?”

Group Leader, Developmental Biology.

Several participants recognised this, with one in particular emphasising the need for capacity building to address this constellation of challenges:

“We are trying to increase the representativity [sic] of Latin American populations in genetic studies we are trying really to build capacity , by training people that is only the very, very beginning of trying to have a more equal representation of all populations in genetic studies, and how the genetic studies themselves will impact society.”

Group Leader, Translational Informatics.

Despite these challenges, there was optimism. One scientist stated of their work that they “can’t be more excited about this” (Senior Scientist, Translational Informatics), while another described translatability across ancestries as the “dream”

Group Leader, Translational Informatics.

This hope extended to imagining a future in which the complexity of obtaining samples would be less burdensome, although issues of categorisation at this stage remained largely unexamined. There is an implicit assumption that the ways data are operationalised, and the exclusions and inequities that follow, are primarily a consequence of insufficient data.

“Okay the fullness of time we hope that all population everyone's sequence in the world and we can we can do well power these irrespective of your ancestry but while we have this disparity can we use the [data] in an informed way the data that we have the small amount of data that we have in these non-European ancestry groups to bridge that gap.”

Programme Lead, Discovery Science B.

Similarly, future access to NHS data was described as “promising” in its potential to:

“provide, at least for the ancestry diversity within the UK, near full coverage. Obviously, in a global sense, you won’t get good coverage, but for all the different groups that are common in the UK, you would get good representation.”

Programme Lead, Discovery Science B.

When data are sourced elsewhere, the problem seemed to be resolved, framing diversity primarily as a question of representation. This positions healthcare as an intermediary layer between researchers and patients, privileging clinical and genetic data while marginalising environmental and social dimensions.

Across this section, diversity in genomic data emerged as a shared priority that was nevertheless constrained by funding structures, ethical governance, data access regimes, and uneven global capacity. Participants consistently framed these barriers as structural rather than individual, producing a situation in which commitments to equity coexist with practices that reproduce exclusion. The next section turns to collaboration and teams, examining how scientists attempt to navigate these constraints through partnerships, capacity building, and working across disciplinary and geopolitical boundaries.

6.2 Teams and Collaboration

This section examines how collaboration operates as a site where diversity, power, and responsibility are negotiated within genomic science. It shows how teams and consortia are shaped by uneven resources, infrastructures, and histories, and how these conditions influence whose knowledge is valued, how trust is built, and what forms of equity are imagined as achievable.

6.2.1 Diversity for Discovery

In their accounts, diversity carried two overlapping meanings: first, as a characteristic of the scientific workforce shaping team culture, interpersonal dynamics, and decision making, and second, as a property of the data generated through the Institute's collaborations and global research networks. These understandings frequently intersected, with participants suggesting that who is included in scientific work influences what becomes visible in research, how questions are framed, and which forms of absence are recognised. This quote discusses this relationship:

“one cannot think about because you're just sitting in your closed little bubble and somebody who's coming just from a different background, like, Oh, you should have looked at, you know, X, because obviously X always comes up with Y, and you're like, I never thought about it because it never showed up in my little bubble.”

Group Leader, Translational Informatics.

Participants emphasised the importance of encountering varied ways of thinking and recognised the value of different personal attributes and experiential backgrounds within research teams. Here, the metaphor of a “closed little bubble” signals a narrowed and insulated framing of scientific practice. Cultural experiences were understood as making some issues visible while rendering others less perceptible, illustrating how different backgrounds can disrupt habitual assumptions and open new lines of enquiry.

A similar emphasis on collaboration and difference appeared in reflections on interdisciplinarity and team science:

“And I also think that the idea that you have this lone genius, eureka moment, is rubbish, because you know, what can one person do, build a garden shed? If you want to build a skyscraper, you cannot have one person building a skyscraper. So the more ideas you have and can bounce about, having a critical mass, the faster you can get things done”

Group Leader, Developmental Biology.

The dismissive tone towards the trope of the solitary eureka moment positions such imaginaries as “rubbish”, contrasting modest individual achievements with ambitious collective undertakings. Impressive outcomes were framed as requiring cohesive teams, and diversity within those teams was frequently linked to progress that benefits society.

“the collaborative agreement that we have between industry and the academic institutes, ...we can all come together in and develop things together. It is sort of contributing to humanity and humankind in supporting everyone.”

Director, Informatics Science.

How “humanity” benefits from collaborative working also featured in discussions about data diversity. Participants emphasised that different experiences shape what researchers notice or overlook, and therefore how absence is identified.

Scientists highlighted the international composition of the Sanger workforce and expressed confidence in celebrating this aspect of diversity. Comments like this were not infrequent:

“It’s very international especially at the PhD and post doc level... people come here for training from all over the world.”

Programme Lead, Discovery Science.

Workforce diversity was framed as both intrinsically valuable and instrumentally beneficial, contributing to a richer intellectual environment and more reflexive scientific practice.

“you often forget what [data] is not there unless you actively think, whereas the person who has no access to it is far more likely to see it as you know [missing], I think that’s one, the kind of like the lived experience and things that you take for granted, I guess.”

Group Leader, Developmental Biology.

Those differently positioned in relation to resources and infrastructures were seen as more likely to notice the absence. Missingness was framed as something overlooked rather than deliberately excluded, suggesting that ‘active’ thinking are what these issues require, a different orientation to standard scientific practice.

This need for international perspectives was also linked to perceived limits of Eurocentric education. One scientist reflected on how restricted educational exposure could explain, though not fully excuse, a blind spot around social justice:

“I have heard comments that almost imply there is a true lack of understanding of the world beyond the Western world. Maybe that’s because of how education is in this country. You

don't really learn about the countries, you don't learn about the cultures. Even if you do learn other languages, it is very much European, Northern European languages. So, there isn't a lot of opportunity to really understand. That may be it .. it's lack of awareness."

Group Leader, Developmental Biology.

The hesitation in this account suggests an attempt to make sense of limitation without treating it as a complete justification. Framing blind spots as lack of awareness allows difficult questions to be softened. Although participants spoke about data diversity in terms of increasing ancestral representation, ancestry itself remained complex and underexamined. Overlooking one domain could be presented as a technical omission, while the causal significance of the other was quietly set aside.

6.2.2 Collaboration

In large collaborations and data initiatives, diversity was described as most consequential when it actively shaped the work rather than simply being present. Within consortia, representation was framed as particularly important for driving discovery at scale. An example of this is exemplified in Figure 39.



Figure 39. Image from a Sanger blog post celebrating a consortium's work. A group of scientists in facemasks look up towards the camera, visually marking the scale and collective spirit of the Atlas of Variant Effects Alliance. The photograph foregrounds both the diversity of the community involved and the collaborative labour required to generate knowledge at this magnitude.<https://sangerinstitute.blog/2024/01/16/nine-billion-and-counting-mapping-the-human-variants-and-their-effects-on-health-and-disease/> 16 Jan 24

However, as this Group Leader explained, such diversity requires deliberate investment and intentional design:

“If you want to do it properly, you really need to kind of invest and get the right people to do it....You just need a lot of representations, and perspectives of people who are from those countries that have been affected by, you know, like they should be part of the panel or part of the kind of like group, you know.”

Group Leader, Developmental Biology.

The repeated “you know” signals assumed shared understanding. This participant identified as not *White* and often spoke explicitly about this, which may have shaped the shared ground they assumed we held, which didn't require them to finish their points. They were suggesting that those who are marginalised, underrepresented or from countries that were exploited were needed in designing better research. Their view was that intentional representation mattered to them across the broader ecosystem and that it in fact brought an absent perspective.

Although international teams were widely regarded as necessary for high-quality genomic science, participants also emphasised that both forms of equity and practical complexities were important. As discussed here by a group leader explaining a developing consortia:

“this is going to be about, imparting expertise and knowledge... identifying what issues and organs are relevant... where there is local generation of analysis, or even a funding engagement project, which is focused on equal partnership. So talking about how do we work as a consortium? What makes an equitable partnership?”

Group Leader, Developmental Biology.

These reflections on the need for equity are connected with broader concerns in global collaborations, where disparities in funding and infrastructure acutely shape what research is possible.

"I can relate to that from our work... Running work in developed countries is so much easier, and your return on investment is much faster. ...if you're working in other countries. you will

have to work for two years to figure out how you're going to collect the samples. You know, there's no infrastructure. ...you don't have access to the technologies that you have in European countries."

Group Leader, Translational Informatics.

Participants contrasted European and non-European settings through the familiar developed-underdeveloped divide, emphasising the additional labour required before research could even begin. Collaboration was as such described as demanding of intention and investment.

"It's 10 times cheaper to sequence here [in the UK]. So what we have is that we have a strategy whereby high value, low number sequencing is done in there [in a lower resourced country]... Where you need thousands of samples and it is going to take years ... we can afford to do here."

Group Leader, Infectious Biology.

In describing equitable approaches, practical solutions like this one were given, and participants also highlighted that unevenness in collaboration was not one that could easily be addressed, as this group leader described:

"..by definition you are always keeping one person ahead because people are already ahead and pushing forward and everyone else is just trying to catch up and might be 10 years behind. In partnerships, people's careers, ages, and capacities are all very different, so it gets a bit blurred."

Group Leader, Infectious Biology.

This description positions an ecosystem with leadership distributed across and within institutions, revealing the power dynamics that operate between them. It suggests an acceptance that some gaps may be too large to ever achieve the equity initially imagined as an aim of these initiatives. Even so, there remained a sense of hope that, while benefits might be uneven, they would at least be available to all.

"...everyone can benefit. Regardless of where you are, you start to get people benefiting concurrently."

Group Leader, Infectious Biology.

This more utopian framing, however, again sidestepped deeper questions of inequity and the scale of the disparities involved.

6.2.3 Trust

Despite these sidesteps, disparities and unevenness have real-world consequences for the relationships that can be built between researchers working in vastly different economic and political contexts. Building and sustaining these relationships as an independent strand of collaboration emerged as work that researchers were engaged in, either explicitly or implicitly. As this senior scientist explains:

“yeah. I don't know if you are aware but Sanger is doing a piece of work on the relationships between researchers in Sanger with researchers in low and medium income countries and I think that is a step forward”

Senior Scientist, Global Health.

Where relationships were being built across unequal economic and political contexts, trust became a central component of collaboration. In these settings, European research, including work led by Sanger, was itself something to be tested, and this scrutiny shaped how relationships could be developed and sustained. A reflection from one of my conversations with a Sanger PhD student, supervised by one of my participants, allowed me to explore the dynamics of trust within these relationships.

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Vignette – Mistrust and Justice – 2 March 2024

It was a bright Friday morning, and I met a bubbly scientist in a café in the centre of Cambridge. They were a PhD student supervised by one of my participants, who had suggested that I speak with them about their experiences of working with understudied communities. The suggestion had been made to help me better understand why these communities might lack trust. This young scientist was open and enthusiastic about tackling these issues. They expressed frustration and a sense of disillusionment with how engagement with communities was often carried out. Their own experience had been something they had pushed for, yet the framework of support around them was not as well developed as it needed to be. I could immediately see why their supervisor had spoken about them with such positivity. The mistrust they described was of course rooted in history, yet speaking with this scientist made clear that current engagement with historically excluded populations remains far from sufficient to address these injustices. It prompted me to reflect on how historical justice had been

positioned in my interviews. Almost all the scientists referred to it as something belonging to the past, something they were not responsible for, and something that simply created a challenge to be overcome. How current inequalities continue to reinforce feelings of exploitation and isolation was rarely, if ever, part of the conversation. In fact, harm, particularly around racial and ethnic categories, was often framed as the product of 'outdated' views. I am not convinced that the observable inequity supports the idea that such harms are outdated or no longer reproduced through research practices and the unequal distribution of their benefits.

.....

One scientist's experience at a meeting with global collaborators revealed how organisational reputation was understood, particularly in contrast to the benefits that collaboration can bring

"I actually really learnt about how strong emotions were about this type of work, you know, whether people were adequately represented, who was benefiting from it, you know, what was the role of those people, the strong views about helicopter science and so on."

Group Leader, Developmental Biology.

The scientist's account shows how striking the emotional weight of the discussion was for them. What they encountered was not a casual critique but a clear articulation of how power operates in global genomics. The reference to helicopter science underscored this. In the context of a data powerhouse like Sanger, the term evokes fears that knowledge, samples and value flow in one direction while local communities gain little meaningful participation or long-term benefit. For this scientist, the openness of collaborators made visible the depth of these worries and the broader landscape of mistrust that Sanger becomes associated with, regardless of individual intentions.

This mistrust of European research and researchers extended to the non-science communities they sought to engage with :

"it comes back to this idea of community, it's actually suspicion...No one trust any medics and ... they don't trust us. If you come around saying you want a vaccine, they're gonna say, no, we didn't have it because of trust, and it's not just understanding it's lots of things"

Group Leader, Infectious Biology.

The researcher positions themselves as part of the 'us' that communities do not trust. This 'us' reflects a layered distance between scientific and non-scientific worlds, and an identification with medicine as an inherently benevolent enterprise. Within this framing, healthcare is imagined as something that

should be offered and accepted, while community refusal appears as a barrier, revealing a tension between assumptions of benefit and with the assuredness of “they’re gonna say no”, the legitimacy of rejection.

6.2.4 Choosing a Team

These interpersonal dynamics were also visible in how research teams were formed and selected. The process of building collaborations rested not only on formal expertise or institutional alignment, but on interpersonal relationships.

“First of all, I met the person who runs it... we spent a few days together just talking about ideas... As a result of that, we wrote a grant...”

Senior Scientist, Global Health.

Such accounts point to the influence held by senior figures, whose capacity to form teams and steer consortia extends beyond data generation. They also highlight how collaboration can appear relatively effortless for those with established networks, while relying on time, proximity, and the informal labour of relationship-building.

“The researchers in these countries are truly part of this project... they feel like they have ownership..that we’re coming to help, support, and facilitate, but they feel like they are driving this project.”

Group Leader, Translational Informatics.

While leadership here was framed through partnership and facilitation, it was rarely articulated elsewhere as redress for past extraction; it often felt more like benevolence. As the desire to share leadership sat alongside an absence of, and at times avoidance of, restitution, collaboration felt similar to the pursuit of data diversity itself. It tended more towards the imperative to drive discovery than for justice.

Across these accounts, collaboration emerged as a site where diversity, trust, and power intersect. While participants expressed strong commitments to equitable partnership, responsibility for enabling such change was often deferred beyond individual teams and between institutions. The final section of this chapter, therefore, turns to institutional commitment, examining how responsibility for equity concerns are taken up, limited, or displaced at the organisational level.

6.3 Institutional Commitment

This section examines how responsibility was understood and enacted within the Institute. Drawing on staff reflections, it considers how power, leadership and institutional culture shaped the capacity to address questions of equity. These accounts reveal both the limits of individual influence and the broader structures that govern what becomes possible within the organisation.

6.3.1 Aims and Recognition

Across interviews, responsibility was rarely framed as a neutral or straightforward process. Staff situated it within the practical realities of working in a hierarchical scientific organisation, where authority, recognition and the distribution of labour shaped what individuals could influence. The nature of Sanger's work and its scientific capacity were central to participants' understanding of the Institute's position. Scientific discovery was described as integral to both individual and institutional identity, and participants articulated a strong sense of epistemic authority, as reflected in the following comment:

“One of my principal tasks and goals in life is new scientific insight, regardless of whether people think it's useful or not, as long as it's true and it's valid.”

Group Leader, Discovery Science.

Other participants reinforced this by contrasting academic freedom with the constraints of industry, using this comparison to underline Sanger's ability to pursue curiosity-driven research. As one Group Leader noted:

“I do not know how industry thinks about it... I am sure that one of the primary drivers is that they need to make money... whereas a pure academic environment has the liberty to be far more curiosity driven.”

Group Leader, Translational Informatics.

This freedom, combined with Sanger's early and unprecedented sequencing capacity, established the Institute's global standing from the outset. These advantages provided the momentum for an organisation that continues to position itself, and be positioned by others, as world-class and operating at a scale that exceeds most genomic centres worldwide (Fig 40 Wellcome Sanger Institute, 2020).

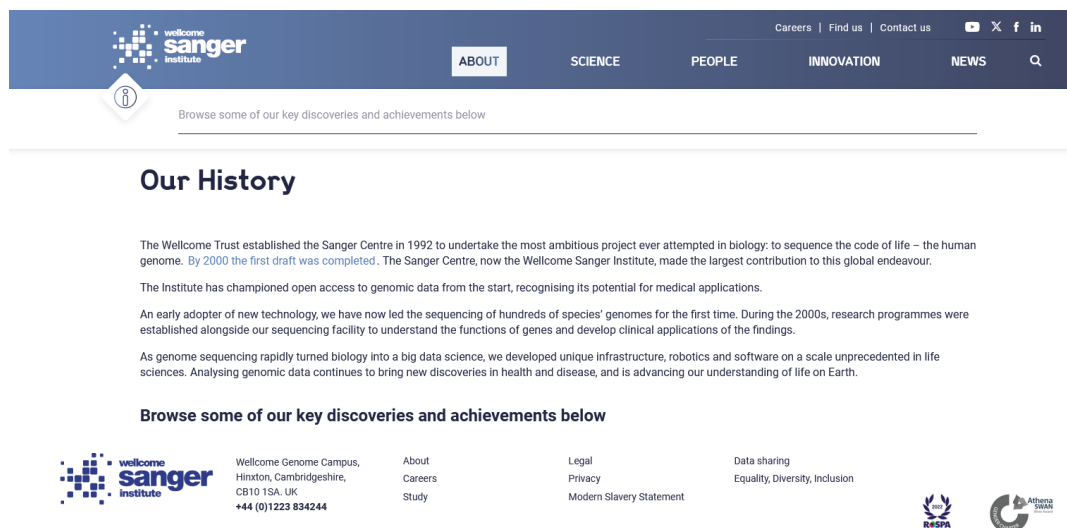


Figure 40: Extract from the Wellcome Sanger Institute ‘About Us’ webpage. Outlining its origins and achievements while presenting the institute as a dominant actor in genomics and positioning itself at the forefront of the field’s development. <https://www.sanger.ac.uk/about/our-history/> 17 October 25.

The impact of Sanger's sequencing capacity was universally understood as this group leader recalled:

“The capacity to sequence is a great example of how the power balance changed... we had resources... most other places, both in the UK and worldwide, did not have the same opportunities. ”

Group Leader, Infectious Biology.

Given Sanger's international reach, it operates in contexts where trust, reputation, and the legacy of historical inequalities continue to matter. As seen in 6.2 Sanger's interactions with global partners, collaborators and participants encountered the wider reputational landscape surrounding Sanger. As explored in 2.3, Sanger, as part of British and global genomic initiatives, is connected to racialised hierarchies and exclusion within science that may influence its standing.

The earlier reflections from the Group Leader in Infectious Biology, for instance, hinted at sensitivities rooted in longer histories of inequality that influence how scientific work is received internationally. These unspoken histories form part of the backdrop against which Sanger is perceived, especially in regions with legacies of colonialism and extractive research. Other reflections pointed even more directly to these concerns, such as the following account, which foregrounded the sensitivities involved in working in postcolonial contexts

" And you know, one of the sensitivities is like, what's gonna happen to the data? ...There's so much sensitivity in that continent of, you know, Europeans coming in, colonising, taking stuff to benefit themselves, right? And I am—it's a really interesting experience for me, because it's not part of my history. So I feel like I'm more objective, maybe, to this, but I am very sensitive to understanding the sentiments, to make sure that we are not going to do something similar."

Group leader, Translational Informatics.

Data concerns were understood to be tied to the historical injustices of extraction and exploitation carried out by European powers. Even though this scientist sees themselves as outside that history, they recognise its legacy. Interestingly, they also feel they are somehow more objective because they are neither from the countries that were colonised nor from those that carried out colonisation. It raises the question of which form of objectivity is being claimed. Is it the objectivity of not feeling the sensitivities involved, or the objectivity of not having been responsible?

Even if their countries of origin are connected to this legacy of empire, scientists are still unlikely to have a direct relationship to the gains of colonisation. However, through their association with Sanger, working in a well-resourced and politically stable setting, all these participants share a privilege connected to these histories. By raising these points, they attempt to place themselves outside this privilege or complicity, even while continuing to benefit from their position at Sanger.

One Group Leader reflected on how geopolitical tensions shape the reception of scientific work, though their comments focused on their research's immediate impact rather than its deeper causes. These tensions are rooted in global inequalities produced by historical injustices and uneven resource distribution (Epo et al., 2024):

"It is a blocker, right, so geopolitical stigmatising of disease and geopolitics, closing borders and preventing access is the biggest problem we have."

Group Leader, Infectious Biology

This scientist frames the pursuit of science as something that should ideally transcend geopolitical borders, positioning these barriers as external obstructions to scientific progress. Their account presents these borders solely as constraints, rather than as boundaries shaped by longer histories that continue to influence how science is conducted. They noted that even deciding which diseases become subjects of research is shaped by cultural differences, such as stigma that can influence what work is considered possible or desirable. Their description of borders and restricted access as the 'biggest problems' also shows how keenly they feel the weight of state and organisational power in their everyday practice. It is the research centre 'we', rather than the individual scientist, that becomes

constrained and positioned as an outsider. These limits arise not just from personal scientific aims but from wider political alignments and the geographical position of a UK genomics research centre.

The reputation of the Institute itself also emerged as a concern for some participants, who recognised how internal experiences feed directly into how Sanger is spoken about beyond its walls. One Programme Lead in Discovery Science reflected on the frustrations of PhD students who felt overlooked, an incident I had also heard of from the PhD perspective in student retreat in Bath Figure 41. They noted how such experiences travel with people once they leave and contribute to the Institute's external image:

"... now we only have people who have seen us and lived with us for some time, and then there is the Institute's reputation externally as people leave and go on to say things."

Programme Lead, Discovery Science.



Figure 41. Materials from a creative session during the PhD retreat.

The retreat was an informal moments in which PhD students expressed concerns and reflections that were less visible in formal scientific environments. Photograph taken by the author. 12 Aug 2022.

This privilege was often acknowledged, however, with scientists describing Sanger as a place of considerable resources that enables work and carries influence.

“It’s a little bubble. It’s so like almost out of reality in the sense that what we live here in terms of access to everything is so different from most parts of, you know, surely Latin America, Africa, parts of Asia and so on.”

Senior Scientist, Global Health.

The participant’s description of Sanger as “a little bubble” signals an awareness of the distance between the Institute’s well-resourced environment and the material realities of many regions where its research partners and sample contributors live. Framing everyday life at Sanger as detached from “reality” highlights how abundance can obscure structural inequalities and dull sensitivity to the conditions shaping global health research.

Across interviews, several participants used metaphors such as a “bubble” (6.2) or “fishbowl” (7.2.3) to describe their position, conveying insulation, fragility, and distance from the outside world. While participants used these images to reflect both privilege and disconnection, my analytic framing of this position as a delicate dream captures how institutional protection and funding sustain this insulation while also rendering it precarious. Reliance on a small number of powerful funders reinforces this fragility, shaping what can be questioned or acted upon and making commitments to inclusion feel provisional rather than secure. As this quote states:

“it’s also the opportunity that we have at Sanger and partly because of the amazing people who are here and the amazing funding that is here, the amazing technologies”

Programme Lead, Discovery Science.

This insulation matters because Sanger’s position as a UK scientific institution is historically entangled with uneven global power relations and past injustices. Occupying such a protected space requires active recognition of these dynamics, particularly if privilege is not to limit responsibility or weaken commitments to equitable practice.

Although Sanger was seen as a place of technical, financial, and human-resource wealth, for some, this institutional privilege was accompanied by a sense of responsibility.

"I have a really unique opportunity to give back in some way, and so I think that's an additional motivator for me to feel that. You know, we can make a change, and we can make a change in a at the level that's, you know, we can potentially impact the way they will really think about drug discovery and treating patients in those countries"

Group leader, Translational Informatics.

The use of the phrase “give back” evokes the language of charitable work, yet it also points to an unspoken sentiment about the gains that come with being at Sanger, in Europe, and in proximity to whiteness. These gains persist even when the taking that enabled them is historical rather than ongoing, and this history forms the backdrop against which such sentiments are expressed as this group leader experienced:

“I spent quite a lot of time talking to people because I guess I was not a white person, they were more open, yeah, talking about it in a way that was quite insightful.”

Group Leader, Developmental Biology.

It is also notable that this participant positioned themselves outside whiteness in a way similar to the earlier speaker who saw themselves as separate from European history. In both cases, this distancing is treated as significant to their insight and access to a social awareness they think others, who were not *White* or Western European, may have. This framing distinguishes whiteness from simply identifying as *White* and instead casts whiteness and European in terms of what they continue to represent throughout this thesis: a form of power and dominance embedded in data practices and institutional presence.

This alignment between whiteness and institutional power was also echoed in material I encountered at the Genetic Automata exhibition, which pointed to how scientific origin stories, from evolutionary narratives to anatomical typologies, have been framed through European defaults that continue to structure global hierarchies of knowledge (Figure 42). These positions bring considerable advantages to those working within them, yet as these quotes show they can be experienced by underserved communities as benefits gained at their expense.

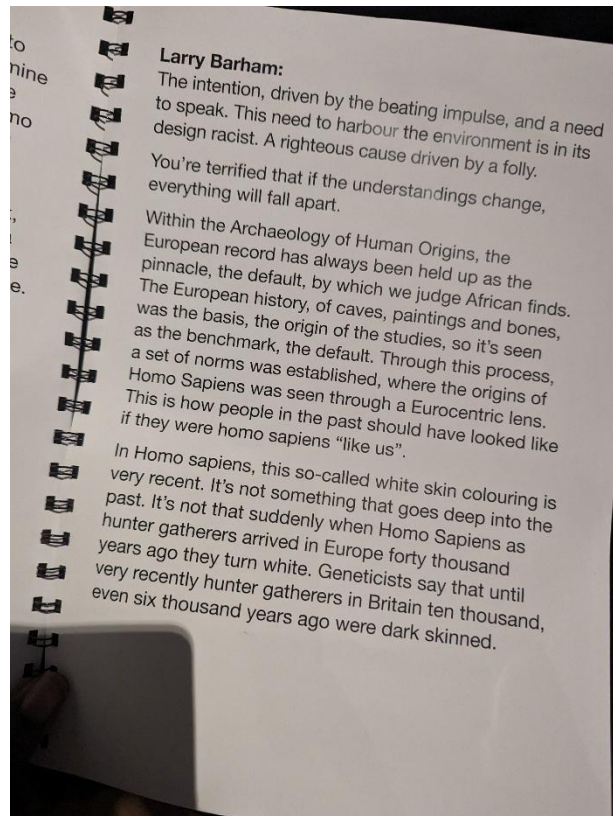


Figure 42. Photograph from the Genetic Automata exhibition, a public art installation exploring race, identity and scientific classification. The exhibition visited frequently alongside my fieldwork, acted as a parallel interpretive space that helped contextualise and reflect on themes emerging from the ethnography. Photograph taken by the author, 9 Feb 24. Permission given by Wellcome Collection.

6.3.2 Valuing Social Context

Social and historical awareness was frequently described as something external to the routine experience of working at Sanger, rather than as a form of expertise embedded within scientific practice. While many researchers recognised the importance of understanding past harms and histories, this knowledge was often positioned as optional, personal, or acquired elsewhere, rather than institutionally cultivated.

"I think it would have been super important for me to kind of understand historically the relationships that did not work in other countries, how much of this helicopter science impact, impacted other groups, universities, people I think that would have been quite useful for me. If I have had all my life in Sanger, that would have been like transformative because working in the UK is already very different"

Senior Scientist, Global Health.

Although this scientist felt that their own experience had made them attuned to the social context of research in other countries, they recognised that such awareness does not arise automatically from being at Sanger. In some ways, the environment of the centre the ‘bubble’ can obscure a fuller understanding of these dynamics, rather than make them more apparent.

This moment sharpens the question of what expertise is needed to work responsibly across such divides and how scientists understand their role in addressing the political and ethical terrain that accompanies global data work. As such this participant reflects on the responsibilities and the skills required to navigate them:

“I feel as though unless this type of work actually has people with experience, and who can comment on that in a more meaningful way, it’s actually very difficult to do. It’s a bit like, you know, it’s very easy to see how a cell biologist is not going to be able to undertake genomics experiments necessarily. But when it comes to these things, we don’t seem to recognise it as actually requiring specialists and experts. You know, you can’t become an overnight sociologist.”

Group Leader, Developmental Biology.

This scientist’s reflections were revealing of the hubris and dismissiveness that can surface as others have recognised when contemplating the place of social and historical understanding within these disciplines (Adsit-Morris et al., 2023). They treated knowledge, however, as a specialised skill, placing social expertise on the same level as technical competence, yet the platforming of social science in settings like Sanger is often not equal, highlighting the gap between what is needed and what is formally valued. Social science is often seen, as has been shown, useful in its support of scientific endeavour (recruitment, improved data diversity) rather than in its own right (Keston et al., 2025).

In a later interview, the same group leader drew attention again to how personal attributes, including having a few “brown” or “black” faces present, are far from sufficient for building trust or addressing deeper structural concerns. For them, this pointed to a wider issue about how the value of ethics and social justice is embedded within the organisation, as the reflection below shows:

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Vignette – Data diversity on the Agenda – May 2023

The Diversity in Data workshop at Sanger brought together senior staff to consider how diversity, representativeness, and generalisability should shape the centre’s scientific work. A task of a strategy given to them by their main funder, the Wellcome Trust. Although the agenda involved concepts

deeply entangled with social and historical concerns, the discussion remained firmly technical. Surrounded by scientific and policy staff, I was the only person in the room with social science research training, and the absence of that perspective felt pronounced.

This contrast hit me especially sharply because elsewhere in the Institute I had often been invited into team meetings where researchers actively sought my insight. In those settings, my work was treated as necessary and overdue, and I was sometimes spoken to as though I held answers or authority. That sense of welcome made the atmosphere of this senior-led initiative stand out even more.

As my role shifted into that of an ethnographer, I made sure to explain that I was now studying the organisation rather than contributing to its strategy. Ahead of the workshop, I asked that my position be communicated and requested permission to take observational notes. Only afterwards did I learn, indirectly, that a separate email had been sent questioning the governance of my study and denying that permission. I had not been copied into the exchange.

At the workshop, I introduced myself for transparency and made clear that I would not be collecting data. Even so, I was questioned about my ethics process in front of the group. The tone felt corrective, as though I had not understood the basics of research approval. A few weeks later, I was formally asked to revise and resubmit my protocol for Sanger, despite already having full university approval. In that moment, my project seemed to be treated not as legitimate research but as something risky or intrusive.

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The work on representation continued after this meeting, and a strategy was later published without involvement from the social science research and dialogue team of which I was part. Despite repeated claims from interviewees that social science expertise was valued and urgently needed, its absence from the formal process was striking. This encounter made visible how social science could be welcomed in everyday conversations, yet marginalised or treated with suspicion in the spaces where strategic decisions were made.

These senior scientists all saw this historical knowledge of social injustice as important. One group leader shared their feelings after a conversation with participants at a conference regarding their strong mistrust of institutions like Sanger due to exploitation and data governance concerns:

"I think it's not. It's not discussed enough or there's no real awareness of it...every Genome Centre probably has had a role in some of these things, because that's how historically it

was...I think it's also not necessarily fair to kind of like put a lens that judges previous things with modern expectations and values. What society felt was appropriate changes. But it's important to understand that those things happen, ."

Group Leader, Developmental Biology.

The protection of the institution despite this is clear statement of liability, reintroduces the distance seen earlier with the participant who positioned themselves outside historical injustice. Here the 'blame' is transferred from the institution itself, calling for its judgement to be 'fair'. Similar arguments appeared in the 2.2.2 when discussing legacies of scientific racism, where scholars such as Blumenbach were often excused on the basis that their views were acceptable at the time. Although the logic of this position is understandable, it nevertheless does little to confront the embedded ideas, practices and datasets that continue to shape genomic research.

This produced a recurring tension. Participants often described this knowledge as personally important, yet expressed reluctance to see it as something that should be mandatory or institutionally provided.

"Me, personally need to know. Whether it's necessary? It depends. I mean, it's not necessary to know the historical derivation of those rules, if you like, but to understand often increases compliance. Yeah. So I think it's really about understanding that, making it more likely to think and to comply. So it's not necessary, strictly, but I think it's certainly desirable that people understand the derivation."

Group Leader, Infectious Biology.

These ruminations were common, and I often held silence, creating space for these almost contradictory justifications to surface. The image reflects how this pause was also enacted bodily, with my hand held near my mouth (Figure. 43).

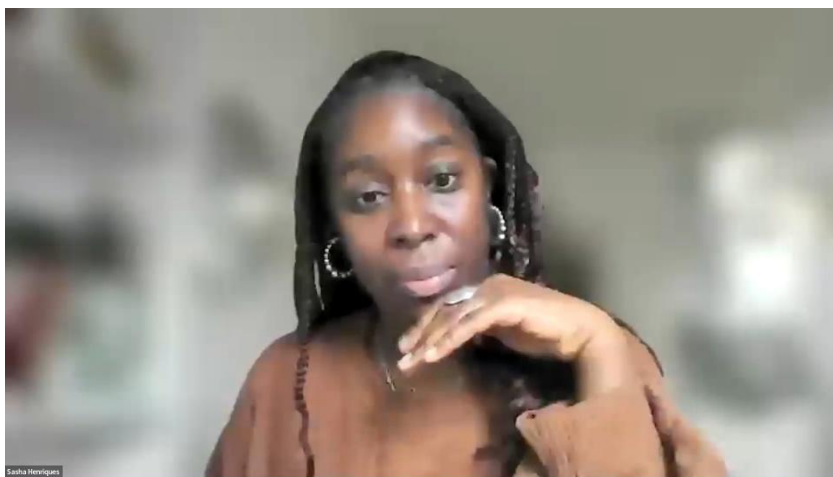


Figure 43. Holding silence. Screenshot from a Zoom interview with a Group Leader in Infectious Biology, taken at the point in the conversation when they discussed the importance of historical understanding within scientific practice. Captured by the author, 20 Dec 23

A similar dynamic appeared here. The researcher acknowledged the value of historical awareness yet simultaneously positioned such understanding as beyond what most scientists could reasonably be expected to acquire. Instead, they suggested commitment to fairness should emerge from interpersonal decency rather than structured learning or institutional provision:

“I think it helps a lot. Yeah, you know. On the other hand, I wouldn't you know, we can't all be like historians. History is a huge field and I'd like to think that just you know one human being meets another human being listens to what that that person has to say so, and then kind of so takes it from there”

Programme Lead, Discovery Science A.

In marking what scientists are not expected to be, these accounts expose the historically contingent construction of scientific identity, and raise the possibility that its boundaries could be drawn differently.

6.3.3 Boundaries of Responsibility

Within these constrained boundaries, responsibility for engaging with ethics and historical context was rarely claimed collectively or institutionally. When I did ask who should carry responsibility for engaging with the historical and social dimensions of scientific practice, participants frequently located it with less senior researchers. The reasons given tended to centre on the idea that junior staff had

more time or that, because they were carrying out the technical work, they were better placed to think about these issues. Senior leaders were described as being focused on strategy rather than the day-to-day contexts where questions about history, ethics or representation emerge.

Though these histories sit beneath much of the work, they were absent from any formal training after the undergraduate level and, even then, appeared inconsistently. Speaking to scientists across the Campus, it became clear that exposure to these topics was patchy.

“No, we're not trained for that at all, I would say. We're not trained for that, like we're not trained in that. I mean, I wasn't, at least, obviously, I was an undergraduate 30 years ago, might have changed now. But it's not in, not in any level of training. Not at any point. Is it part of it”

Programme Lead, Discovery Science A.

Although this was a senior scientist, her limited awareness of what early-career researchers encounter in terms of training on these issues was not far removed from my own experiences discussing them with PhD peers. Some PhD students who had trained partly overseas described a different experience. One student, for example, spoke about her time in the United States, where subjects such as eugenics and scientific racism had featured more explicitly in her scientific education. These issues surfaced more openly in self-initiated or extracurricular contexts than in the everyday scientific training offered at Sanger, a pattern illustrated in the following reflection:

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Vignette – Scientific Racism in Cambridge – Nov 2022

In November 2022, I received an email from fellow PhD students at the Wellcome Genome Campus expressing concern about a new academic appointment at the University of Cambridge and a blogpost he had made. The appointee, Nathan Cofnas, is a philosopher whose work argues that biological differences between human populations (often discussed in terms of “race”) merit continued investigation (Echaniz-Furuta, 2022). He contends that restricting research on group differences in traits like intelligence stifles ‘free inquiry.’ Cofnas has publicly denied being a proponent of racism (Cofnas, 2022), yet many of his supporters are drawn from circles associated with White nationalist views (Hood, 2022).

The episode surrounding statements made publicly by Nathan Cofnas placed students in an uneasy position, caught between their scientific identities and their institutional affiliations. During this period, several PhD students sought guidance on how to respond, including whether to issue a public letter as university students and as researchers based at Sanger. The advice they received was cautious.

They were discouraged from speaking publicly in connection with Sanger, on the grounds that the Institute was not formally implicated because Cofnas was not an employee. This position, while institutionally understandable, left the students feeling unsupported in their frustration and uncertain about how to voice their concerns. Some turned to the Research and Dialogue team within Wellcome Connecting Science, where I was based, in search of acknowledgement and direction. Although Wellcome Connecting Science issued a clear statement addressing scientific racism, no equivalent statement was released through Sanger's senior communications channels. The wider context remained unsettled, as the University of Cambridge continued to grapple with the issue, initially restricting Cofnas' supervisory duties and later upholding his fellowship following an internal review, accompanied by a formal statement issued in October 2025 (University of Cambridge, 2025).

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The presence of visible controversies involving institutions with global reach and influence serves as a reminder of the need for socially responsible genomic research that takes historical context seriously. Yet within Sanger, the burden of engaging with these difficult historical and social questions repeatedly fell to early-career researchers, rather than being carried collectively or taken up at the leadership level. As shown by this participant's response to my suggestion of ELSI training for the researchers:

“So yeah, it is a wonderful thing to have, but I do not know how easy it would be to achieve. But, you know, it does not need to be faculty, maybe scientists who work on these projects, PhD students, technicians, postdocs, you know, would they be interested in having these conversations?”

Senior Scientist, Translation informatics

Just Good Science (Figure 44) was a programme run by the Engagement and Society team at Wellcome Connecting Science. It provided a space for people across the Campus to explore the everyday ethics of genomics, focusing on the practical, technical and epistemological questions that surfaced in routine work but often sat outside formal ethics discussions. By drawing attention to these day-to-day issues, the project encouraged researchers and other professionals to reflect on the social and ethical implications of their work. The meetings were open to everyone, and participants could join as and when they wished, contributing whether they held clear views or were still thinking things through



Figure 44. Blocks used in the Just Good Science initiative, photographed in Hinxton Hall at Wellcome Sanger Institute. Positioned within a building historically funded through wealth accumulated by the East India Company, the image underscores how present-day conversations about ethics and fairness occur within spaces shaped by longer histories of colonial extraction. Photograph taken by the author, 17 Jan 24.

I attended Just Good Science on several occasions, including facilitating a session with scientists on the use of race, ethnicity and ancestry in their work. Another session I joined later in my data collection focused on collaboration between the ‘Global South and Global North’. It centred on how positive aspects of collaboration with Sanger might be replicated and how to ensure that ‘knowledge goes back’ to partners in less-well-resourced settings, including capacity building and more equitable research agendas. These meetings were engaging, honest and, over time, became a kind of community for me on campus. I often saw other PhD students and postdoctoral researchers there, many already interested in these conversations, just as senior leaders had predicted. The same faces reappeared in other venues where ethical and equity concerns emerged, from the Cofnas situation described earlier to campus discussions on race and equity, where senior leadership presence was confined to a very small, consistently recurring group. This dynamic was also noted in my interviews:

“Classic example, a lot of activities are attended by people who are an ethnic minority. But then there's never any Senior leadership members who attend. You know how many of those activities go... and then it doesn't matter because their absence is not noted. Their absence is tolerated”.

Group leader, Developmental Biology.

Those who were willing to engage with questions of equity and ethics were generally junior researchers, yet their capacity to influence institutional direction was markedly limited. Enthusiasm at this level has constrained impact without explicit endorsement from senior figures. As one group leader observed, the ability to effect change is closely tied to positional power:

"If the people around you do not support it, and if you have not got enough, I think that is the most important. Because if you have like a head or a director who says, actually, I do not see that as a problem that is not a priority."

Group Leader, Developmental Biology.

This reflection highlights how responsibility for these issues is frequently placed on those with the least power to act, while meaningful change requires endorsement from those who set priorities and determine what counts as important within the organisation.

A similar family metaphor emerged in another account, but here with a different emphasis on institutional responsibility:

"The environments in which we raise scientists are quite important... I think we can also raise them in an environment where actually how you do science, how we talk about science, is as important as the science that you do."

Programme Lead, Discovery Science B.

In this framing, leadership is cast as a parental role within the Institute, with responsibility for shaping the immediate environments in which scientists are "raised" and for embedding the values that guide their day-to-day practice. While this reflection concerns the responsibilities of individual leaders, the following section shifts the focus to how such responsibilities are understood and distributed more broadly across the organisation.

These reflections raise broader questions about leadership: where authority is concentrated, how responsibility is assigned and whether those in senior positions take up the ethical commitments that staff identify as necessary. Scientists described leadership as a form of positional power that sets agendas, distributes resources and determines who is included in ethical or institutional conversations.

"I think it would be kind of naive to think the conversation [data diversity] that takes place in a room with, all the members of a research team, takes place in a senior leadership team. And

there's far more weight and decision making that happens as you go higher up. And you know [their own] topics then become higher priority ”

Group leader, Translational Informatics.

One Group Leader in Developmental Biology challenged what they perceived as a widespread belief that diversifying leadership is sufficient to shift institutional culture. They made this point through a comparison with government:

“What is the progress if non-white people are in leadership roles but have views and implement policies that are exactly like what a white person would, or even worse the far extremes, [not] naming names. But if you look at some government leaders, do you know what I mean?”

Group Leader, Developmental Biology.

This assumption echoes other simplified narratives of equity in genomics, such as the idea that increasing the diversity of datasets will, by itself, produce more equitable scientific outcomes. Here whiteness is again framed as structurally tied to the oppression of non-*White* groups, even when leaders are not at the “far extremes”. This challenges any assumption that diverse leadership naturally disrupts entrenched cultures, simultaneously drawing my attention to the longstanding absence of non-*White* voices within Sanger’s senior leadership teams.

Notably, this comparison to government and political life was not unique. Both this participant and others drew on political analogies when describing Sanger, suggesting that they viewed the Institute’s structure, distribution of authority and exercise of power as resembling a political system as much as a scientific one. When I asked this participant directly about the responsibility of social justice in leadership, they remarked:

"That's a hard question, does it always become a responsibility? I mean I've seen people get to the privilege position? And they just abuse it right? I mean, you think about politicians. You would wanna think that a lot of them are motivated and driven by the fact, they want to serve their community."

Group leader, Translational Informatics.

This participant, while also using political analogy to talk about power and privilege, reflected on how these positions must be used cautiously. They noted that privilege itself can become problematic, distracting from the ‘good’ intentions individuals may bring to leadership roles and complicating their efforts to act responsibly.

Leaders were often perceived as distant from the everyday contexts in which ethical tensions emerge, and demographic diversity alone was seen as insufficient to shift entrenched institutional norms. Comparisons with government and political life underscored this point, suggesting that scientists understood Sanger not only as a research institute but as an organisation whose internal dynamics resembled the broader political structures that shape scientific authority, privilege and accountability.

Across this theme, a structural condition was revealed as one shaping scientific practice, collaboration, and institutional responsibility. Efforts to diversify genomic datasets were constrained not only by methodological and logistical difficulties but also by the legacies of unequal funding systems, uneven research infrastructures and the labour required to build trust in contexts where scientific institutions from Europe have long been associated with extractive practices. These geopolitical landscapes continue to influence how research centres like Sanger are perceived and how their work is negotiated. Scientists described the need for a deeper understanding of these histories and the responsibilities they create, yet such awareness was inconsistently held and unevenly taken up across the Institute.

These accounts show that power extends far beyond the production of data: it is embedded in institutional histories, colonial legacies, collaborative infrastructures, and everyday hierarchies. This sets up the final chapter, which turns outward to the broader genomic ecosystem. By examining how researchers experience funding structures, capacity building, geopolitical constraint and global-regional scientific inequalities, the chapter asks how social justice can be meaningfully embedded in the field. Understanding these wider structures reveals both the barriers and the points of leverage that shape what scientists at different levels of seniority can do. In this sense, the genomic ecosystem becomes the terrain through which questions of responsibility, equity and power must be navigated, and where possibilities for more just scientific practice may emerge.

7 The Genomic Ecosystem of Sanger Scientists

7.1 Introduction

Walking through the Sanger Institute's landscaped campus for the first time, I was struck not only by the beauty and calm of its setting, nestled in the Cambridgeshire countryside, but by the subtle ways in which that very calmness signified something deeper about the Institute's place in the genomic ecosystem. As a PhD researcher openly identifying as a Black woman, whose interests extended beyond the scope of my immediate project to broader questions of inclusion and ethics in science, my time at Sanger was shaped by a complex set of experiences.

This chapter explores these experiences alongside those of other scientists I encountered at Sanger. It seeks to reflect critically on the institution's identity, its purpose, history, and relationships with societal and industrial actors while also acknowledging how contradictions around inclusion, openness, and impact emerged throughout the daily life of the campus. Drawing on participant quotes, informal observations, and personal reflection, I aim to paint a picture of what it means to live, train, and work in a place that is simultaneously cutting-edge and remote; internationally diverse yet often culturally homogeneous in its leadership; and committed to open science while enmeshed in the opaque dynamics of industry collaboration.

7.2 The Wellcome Sanger Institute: Mission, Space, and Institutional Identity

Sanger was established to spearhead the UK's contribution to the Human Genome Project, a global scientific endeavour of unprecedented scale. Its founding purpose was focused and specific:

"It was to deliver the Human Genome Project... their mission was to deliver the Human Genome project so consortium... it wasn't a research institute. It was a centre with 100 employees that were just staff."

Programme Lead, Discovery Science B.

This quote reflects not only the Institute's functional origins but also the structural intention behind its early formation.

From its beginnings as a technical hub, the Institute expanded its mission, moving from sequencing into broader realms of genomic science, including the study of gene function, disease association, and the development of new technologies. Yet, there remains a tension between this expansion and the clarity of its original mission.

“Maybe it feels like, what is it that we want to achieve? And how do we want to do it and... what are the values?”

Group Leader, Infectious disease.

These questions continue to haunt the Institute’s evolution. One of Sanger’s unique qualities is its singularity of focus.

“Sanger is completely different from other... departments... it’s all about... generating genomic resources... for the community and that’s what it is —nothing else“

Programme Lead, Discovery Science A.

This purity of purpose, shaped by Wellcome’s funding priorities, sets it apart from academic departments, even as its leadership describes it as an academic environment. The language of “nothing else” is especially revealing; it implies a clarity of mission but also a boundary, a delimitation of what counts as valuable work and, potentially, who gets to do it.

Sanger’s role as a training centre for global scientists was often praised, particularly among PhD and postdoctoral researchers (6.2). The focus of purpose made Sanger a focal point in global genomic.

The pride in this global reach is not misplaced: training at Sanger represents a form of scientific capital that can be exported to other research settings, multiplying its influence. However, this celebratory narrative sometimes masked inequalities within the workforce. Permanent staff, particularly in non-research roles, remained predominantly British, and diversity in terms of ethnicity and class, especially among British staff, was still seen as lagging

“To be honest, I still think we fall short on supporting people from low-income backgrounds.”

Senior Scientist, Global Health.

As someone from a minoritised background, I found myself reading between the lines of these conversations. While the campus was open and serene, its spatial and visual cues did not always suggest inclusivity.

“You know there is no other signage on campus other than that about Sanger.”

Staff Scientist, Human Genetics.

Coming from a clinical environment where walls were crowded with health information, recruitment notices for clinical trials, and posters celebrating staff and community groups, Sanger's minimalist signage stood out. Its physical environment with looming sculptures, vast reflective domes, and intricate wooden carvings depicting the evolution of sequencing, reflected the Institute's ambitions but also reinforced a sense of institutional singularity. The décor was not just aesthetic but ideological: speaking to a curated identity of prestige, order, and calm, far removed from the noisy, lived-in feeling of a hospital or university department.

The campus always felt calm: wide open spaces, manicured grounds, and scientists queuing in an orderly fashion for food at lunchtimes, engaged in composed conversation before returning to their labs housed in large, multi-storey buildings. This composed rhythm, while comfortable for many, added to the impression of being “in a bubble.” (*Director, Informatics*). In contrast, the Francis Crick Institute (The Crick), a large biomedical research institute based in London, is nestled within a vibrant urban neighbourhood and integrates science communication into its very entrance. Sanger's physical and symbolic isolation contributed to both its sense of prestige and its detachment from the messiness of the social world. Like Sanger, the Crick is designed around scale, infrastructure, and collaboration.

Figure 45 and 46 show Sanger and the Crick and their surrounding landscapes.



Figure 45. The Wellcome Genome Campus. <https://www.sanger.ac.uk/about/campus-connections/wellcome-genome-campus/> 19 Dec 25.

'Image removed due to copyright restrictions. The original artwork 'Aerial view of the Francis Crick Institute London can be viewed at <https://www.jasonhawkes.com/media/1912768b-cf6c-4de6-91c8-ef2fd484ff58-aerial-view-of-the-francis-crick-institute-london>

Figure 46. Aerial view of the Francis Crick Institute London

<https://www.jasonhawkes.com/media/1912768b-cf6c-4de6-91c8-ef2fd484ff58-aerial-view-of-the-francis-crick-institute-london> 19 Dec 25.

This feeling extended to how certain types of work were perceived. My own research often elicited puzzled responses, unusual work undertaken by the Institute. I became known as the one who '*does the ethics PhD*', an identifier that felt peripheral. The ethical and social dimensions of genomic research, while officially acknowledged, were not uniformly integrated into the Institute's mainstream activities.

Still, efforts to engage the public were visible. I participated in outreach days designed to demystify the campus for school groups and other visitors. These groups were brought in on coaches, minibuses, and shuttles mirroring the way staff themselves commuted in and moved through Sanger like a curated exhibit. A one-way glass in one of the labs could slide back to reveal the sterile, white-lit space within: a room dominated not by bustling scientists, but by quietly humming machines (figure 47). As one staff scientist noted on a tour of the campus, this didn't quite carry the "wow factor" it might have in previous years. The spectacle of genomics was being replaced by the quiet automation of its machinery.



Figure 47. The Open Lab at Sanger. Photograph taken by the author. 19 Dec 25.

Values around data, in particular, were a key distinguishing feature of genomic research at Sanger. Data acted as both an interface and a boundary: central to the Institute's function, yet not shared evenly or freely. As one researcher explained,

"The other challenges are...Well, working with pharmaceutical companies, they are not academic environments, so they do not share information freely and easily. So there are challenges in navigating the space..."

Senior Scientist, Translational Informatics.

Sanger positioned itself as a leader in open science, but the benefits of open-source models were not uniformly experienced as one group leader remarked.

"Open source models tend to serve institutions that already have access and power."

Group Leader, Infectious Disease.

In theory, open science promised ethical transparency and democratised access; in practice, it often reinforced existing inequalities.

Sanger, described by some as an academic department, resembled in many ways the polished campuses of private genomic firms I had visited elsewhere. A strong sense of identity came through in discussions but also in anecdotal conversations I had on campus. A campus carefully branded, from signage to sculpture, with minimal intrusion from outside narratives. This sense of coherence, while admirable in some ways, also served as a visual and spatial confirmation of privilege of being shielded from the frictions, inequities, and messiness that define many other scientific and social spaces.

Belonging, Visibility, and Identity on Campus

The team I was part of, which had a public engagement and ethical remit, was frequently referred to in vague terms, '*they do the ethics*,' I often heard when being introduced to someone new, as if our work were niche or ancillary. And yet, we were involved in the visible performance of openness, welcoming visitors, narrating the institution's mission, and offering glimpses behind the glass. These curated experiences, while well-intentioned, often raised questions about who the science was really for, and what kind of visibility was being offered.

The *Black* scientists' group on campus was in its second generation. It had originally formed as a network specifically for *Black* scientists, but according to one computational scientist, amid internal concerns that the group might be too exclusionary, it became broader, the Race Equality Network. Even within this expanded framework, many of the members, predominantly African-born scientists,

still expressed a need for a space to share experiences of migration, race, and professional life. While the Race Equality Network remained, and the informal *Black* scientists' group emerged, socialising around WhatsApp, lunches, and potlucks. I was invited to both groups during my time and observed tensions between the institutional framing of itself and these organic, unofficial affinity spaces. Often, perspectives that never surfaced during my interviews with senior staff.

For example, I attended a panel discussion on International Women's Day where a question from the floor asked about the availability of *safe spaces* at Sanger. A Black woman scientist spoke candidly about the limitations of these spaces and how race continued to impact how minoritised scientists expressed themselves. Later, she shared personal stories of microaggressions, stories that echoed those I was hearing from others during my fieldwork.

From the outset of my project, it was scientists who were women who engaged with me most frequently and openly on EDI topics. At Race Equality Network events, they were often the dominant voices, both in the audience and on the panels. Two senior women were regularly called upon to speak or chair, a sign of both their leadership and the gendered labour in holding space for equity work. "*The usual suspects*," one Group Leader, (Developmental Biology) remarked, while EDI was increasingly part of the institutional discourse, its practice was uneven, and power remained concentrated in leadership, highlighting the limited change in who held authority and made decisions.

As the chapter unfolds, I turn to a series of visualisations to further illustrate the genomic ecosystem as experienced by the students and researchers at Sanger. These images help capture the complexity of a system in which ethical considerations are inextricably intertwined with scientific outputs. They reveal the spaces where values align and where they clash and offer novel ways to reflect on how scientists can be supported when their work lives within such nuanced ethical terrain.

7.3 Mapping the Genomic Ecosystem: A Sanger Perspective

This section presents visualisations that offer a situated and strategic view of Sanger's position within a complex and evolving genomic research ecosystem a

s experience by those within in. Each visual draws on data generated through ethnographic fieldwork, interviews with researchers, and attendance at relevant public and private genomic events. They provide a means of making visible the relationships, dependencies, and infrastructures that sustain, enable, and constrain genomic research at scale.

These visual tools are not intended to function as static depictions or universal frameworks. Rather, they reflect the contingent, embedded, and interpretive nature of system-mapping itself. They are

partial but revealing, offering a lens through which to trace flows of data, funding, influence, and ethical oversight across the many institutional boundaries that make up this ecosystem. Most importantly, they help surface the themes of power, autonomy, and alignment that will be explored in greater depth in the following section, which is part of the complex way that Sanger works.

“When people move from, you know, doing everything over here to everything over there, but actually science is in the middle. It's always that sort of middle ground where you work together to achieve a goal. There are some things that definitely you can only do there, or you can only do here, practically speaking, and because that's the right thing to do. But for most large questions, it's about how you work together, and then it's around, as you say, that opportunity for benefit.”

Group Leader, Discovery Science B.

“...we are a consortium that works with pharmaceutical companies to work closely with them to understand sort of the limitations from their side in the context of translating genetics to drug discovery and also from our side...to understand how we can impact the decision-making based on the science that we do in here.”

Senior Scientist, Translational Informatics.

These quotations, taken from interviews with researchers at the Sanger articulate the lived experience of working within a deeply interdependent ecosystem. They underscore the Institute's self-understanding as neither fully autonomous nor embedded in a hierarchy of external actors, but rather situated in a “middle ground” where value is co-produced through collaboration, negotiation, and shared purpose.

The ecosystem table (Table 5) sets out the institutional architecture that supports genomic research as a system. The examples provided are grounded in empirical accounts from scientists, and reflect the kinds of partnerships, dependencies, and obligations that routinely shape research activity. They do not aim to be exhaustive, but identify the major categories of organisations such as funders, policy bodies, healthcare providers, and research infrastructure. Understanding these actors and their roles is crucial to the further discussions in this chapter.

Table 6 describes the interfaces through which they interact. These interfaces include mechanisms such as data-sharing agreements, regulatory processes, funding allocations, collaborative platforms, and translational pathways.

(Table 5: Ecosystem table of actors, institutions as experience and described by Scientists at Wellcome Sanger Institute)

Type	Example	Description	Cluster
Patients	Patients with inflammatory bowel disease	Individuals participating in studies and donating samples/data to research.	Clinical interface
National Healthcare Organisation	UK NHS	The public health system in the UK delivers care, manages records and partners on research.	Clinical Interface
National Healthcare Body	NHS England	The policy and strategic body overseeing NHS services in England, including research partnerships.	Clinical Interface
Healthcare Provider	Addenbrooke's Hospital	Clinical institutions providing care and collaborating on medical research and innovation.	Clinical Interface
Clinicians	Network of Gastroenterologists	Healthcare professionals involved in clinical care and/or research.	Clinical Interface
Private Funder	GSK (also Industry Stakeholder)	Organisations funding research either for strategic/mission-aligned outcomes or returns on investment.	Funders and Strategy
Industry Stakeholder	Johnson & Johnson, GSK, Genentech	Companies that apply research outputs commercially (e.g. drugs, diagnostics, devices).	Funders and Strategy
Public Funder	NIH, UKRI Medical Research Council	Government-funded bodies that distribute public money to support research and health innovation.	Funders and Strategy
Philanthropic Funder	Bill & Melinda Gates Foundation, Wellcome, Chan Zuckerberg Initiative	Nonprofit funders aiming to address global health and societal challenges via targeted grants.	Funders and Strategy
Publishers	Springer Nature, Cell Press, AAAS, Elsevier, BMJ	Organisations disseminating peer-reviewed knowledge to research and professional audiences.	Genomic Communication and Culture
Social Media	Twitter (X)	Informal channel for real-time scientific communication and community engagement.	Genomic Communication and Culture
Conferences	ESHG, BSGM, ASHG	A formal channel for real-time scientific communication and exchange of ideas, networking, and collaboration across academia, healthcare, technology, and research.	Genomic Communication and Culture
UK Government	UK Government	The central political authority providing the legislative, policy and financial framework for science/health.	Governance and Policy

UK Government Department	DSIT	Departments that translate UK policy into research priorities and oversee regulatory processes.	Governance and Policy
Non-UK Government	Brazilian Government, US Government, Zambian Government	National entities involved in governance and funding for health/science in their respective countries.	Governance and Policy
Non-UK Government Department	US HHS, Zambian CDC	National-level agencies delivering health/research strategy, funding, and regulation outside the UK.	Governance and Policy
Independent Professional Organisation	British Society for Genetic Medicine, Black in Data UK, Black in Computational Biology Global Consortium	Membership bodies shaping policy, standards, equity, and representation in research and health.	Governance and Policy
Ethics Board	Internal Review Board/Research Ethics Committee	Oversees the ethical conduct of research, especially with human subjects.	Governance and Policy
Ethics Committee	UK HRA RECs, Joint Committee on Genomics in Medicine (JCGM)	Formal review bodies ensuring research is ethically and legally sound.	Governance and Policy
Research Institute	Sanger, EMBL-EBI, Genomics Unit, Center for Genomic Regulation, NIDDK, Broad Institute, NHGRI	Conduct advanced biomedical/genomics research and often develop new tools, datasets, or methodologies.	Research and Data Infrastructure
Academic Institution	University of Cambridge, Newcastle University	Higher education institutions where foundational and applied research is carried out alongside teaching.	Research and Data Infrastructure
Research Initiative	Our Future Health, Genomics England, Open Targets	Large-scale programmes that aim to collect data or drive specific scientific outcomes through collaboration.	Research and Data Infrastructure
Data Institute	European Data Science Institute (EDSI)	Organisations that develop infrastructure and expertise for managing and analysing big health/genomic data.	Research and Data Infrastructure
Research Institute (biomedical)	NIHR Cambridge Biomedical Research Centre	Bridge between lab research and clinical practice; often co-located with hospitals and universities.	Research and Data Infrastructure
Research Institute (International Health)	International Centre for Diarrhoeal Disease Research (icddr,b)	Research-focused organisations working in low-resource or global health contexts.	Research and Data Infrastructure
Academic Department	Newcastle FMS, Cambridge Department of Medical Genetics	Subdivisions within universities where research and teaching in a given field occur.	Research and Data Infrastructure

Bioresource	IBD Bioresource, UK Biobank	Collections of human biological samples linked to health/demographic data for research use.	Research and Data Infrastructure
Data Resource	UK Biobank, GWAS Catalog, EGA	Curated datasets used to support discovery and replication in genetics, medicine, and public health.	Research and Data Infrastructure
Research Group	Cellgen (Sanger)	Investigator-led units focused on specific research themes or methods.	Research and Data Infrastructure
Research Consortia	Human Cell Atlas Consortium	Large, often international networks aligned around shared goals and data-sharing.	Research and Data Infrastructure
Data Initiative	HDR UK, European Genomic Data Infrastructure (GDI)	Initiatives focused on national/international health data infrastructure and use.	Research and Data Infrastructure
Data Access Committee	Sanger DAC, UK Biobank Access Committee, NIH dbGaP	Gatekeepers that review and approve access to sensitive health/genomic data.	Research and Data Infrastructure

Table 6 Interfaces of Interaction Among Actors in the Genomic Research Ecosystem

Cluster	Types	Interfaces
Clinical	Patients; National Healthcare Organisation; National Healthcare Body; Healthcare Provider; Clinicians	samples; data; healthcare; patient data; clinical trials; guidelines; policy; strategy; publications/knowledge; research; drug development; investment
Funders and strategy	Private Funder; Industry Stakeholder; Public Funder; Philanthropic Funder	funding; investment; technology; commercialisation; grants; policy
Genomic communication and culture	Publishers; Social Media; Conferences	journal papers; books; articles; information; data; views; presentations; posters; research
Governance and policy	UK Government; UK Government Department; Non-UK Government; Non-UK Government Department; Independent Professional Organisation; Ethics Board; Ethics Committee	governance; policy; regulation; guidance; oversight; funding
Research and data infrastructure	Research Institute; Academic Institution; Research Initiative; Data Institute; Research Institute (biomedical); Research Institute (international health); Academic Department; Bioresource; Data Resource; Research Group; Research Consortia; Data Initiative; Data Access Committee	data; research; collaboration; training; knowledge/skills; education; data resources; technology; analysis; samples; publications/knowledge; global health initiatives; governance; policy; regulation

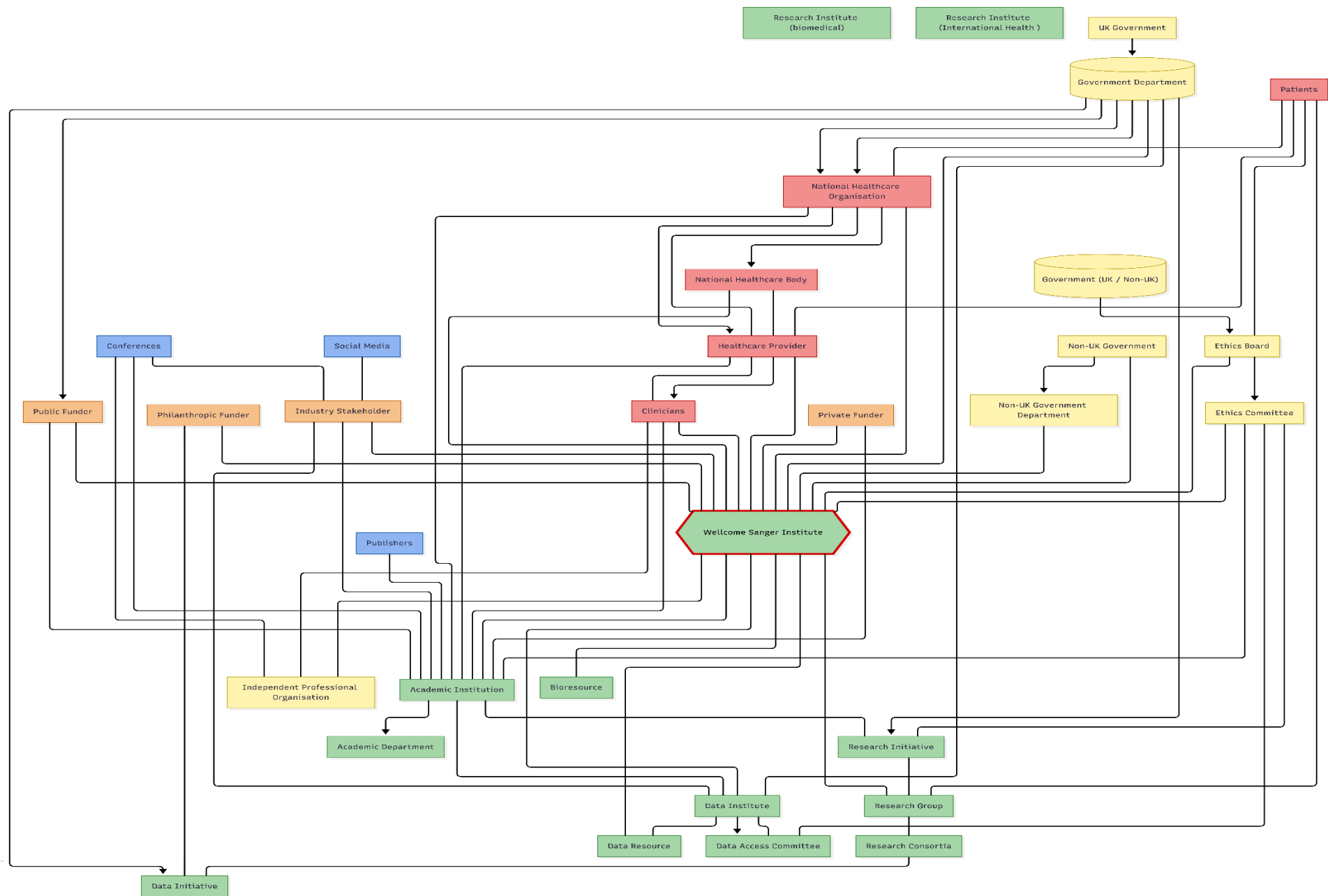
These tables outline the genomic research ecosystem surrounding Sanger. Table 5 identifies the main actors, clinical, institutional, and infrastructural, while Table 6 illustrates the interfaces that connect them through data sharing, governance, funding, and collaboration. Viewed jointly, they reveal a dynamic, interdependent system where roles overlap and boundaries blur. This ecosystem operates through negotiated relationships that enable coordination, innovation, and accountability. These tables therefore set the foundation for the next section, which visualises these connections in motion through a system map of flows, interactions, and relationships across the genomic landscape.

7.3.1 System Map: The Genomic Ecosystem in Motion

The system map (Figure 48) provides a graphical and conceptual representation of the direct and indirect relationships that define the genomic ecosystem surrounding Sanger. The map uses single connectors to trace the multidirectional flows of data, funding, expertise, regulation, governance, and clinical influence, offering a spatial language for understanding how knowledge is co-produced, translated, and circulated across institutional boundaries. Importantly, these connectors do not all represent the same kind of relationship: some indicate material or interactional exchange, while others denote normative and regulatory obligations.

The map was developed as an analytic tool to interpret qualitative data generated through interviews, participant observation, and documentary analysis. Using iterative coding, I traced how researchers, clinicians, institutional actors, and governance bodies described their relationships, dependencies, and points of influence within the genomic ecosystem. The diagram does not claim to reproduce the system as it exists in a formal or organisational sense. Rather, its structure reflects how participants themselves articulated the world they work within, including which actors were described as central, peripheral, constraining, or enabling. The hierarchical layout is therefore inferred from patterns in participants' accounts and my ethnographic observations, particularly the ways in which authority, responsibility, and decision-making were consistently described as flowing through certain institutions and committees more than others.

Figure 48: Sanger's Ecosystem Map. Analytic map of the genomic ecosystem surrounding the Wellcome Sanger Institute, based on qualitative coding of interviews and observations. Produced by the ..



Sanger sits at the core of the map (Figure 49). Although connected with other similar institutes, these appear just beyond the frame for clarity; in practice, they form part of the broader cluster of research life surrounding Sanger.

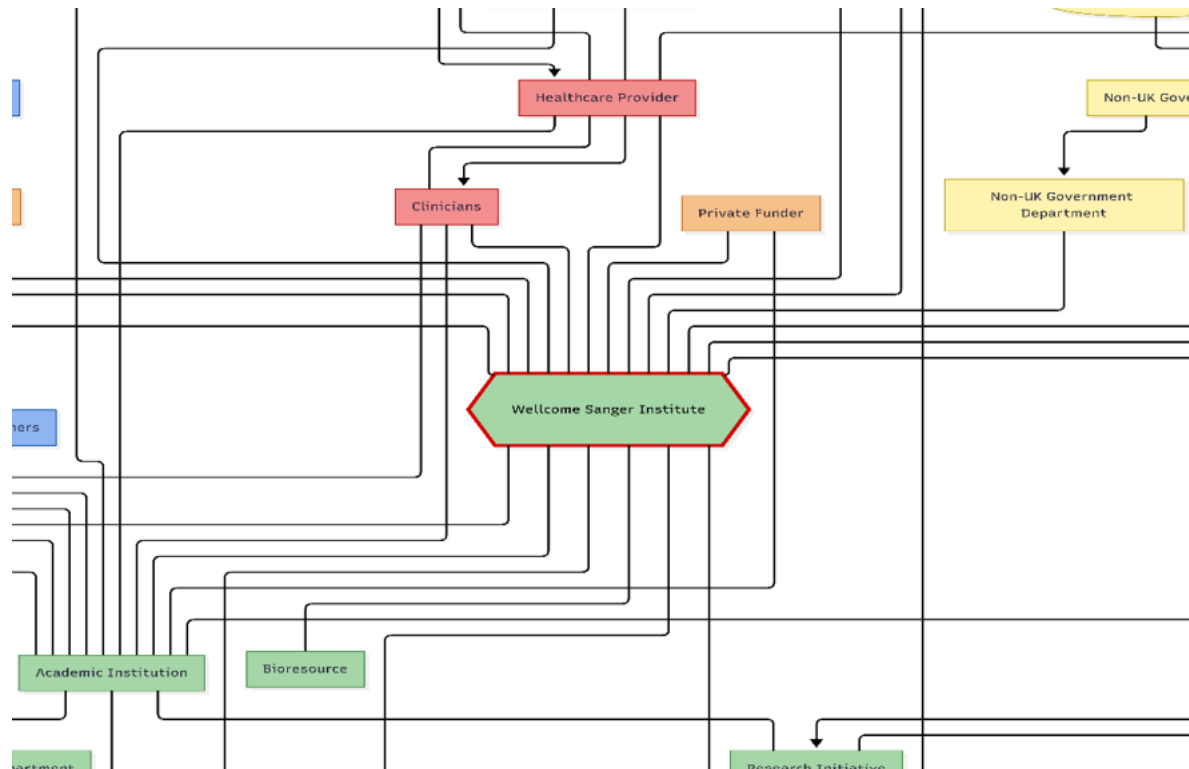


Figure 49

Within this cluster, scientific practice is rooted in the generation, analysis, and interpretation of genomic data, as discussed in the previous chapter. The map illustrates how these practices are sustained through networks of collaboration with both public and private partners.

Communication flows through this system in multiple ways, via publications (both professional and public, including social media), scientific conferences, training exchanges, and the mobility of individual researchers between institutions (Figure 50).

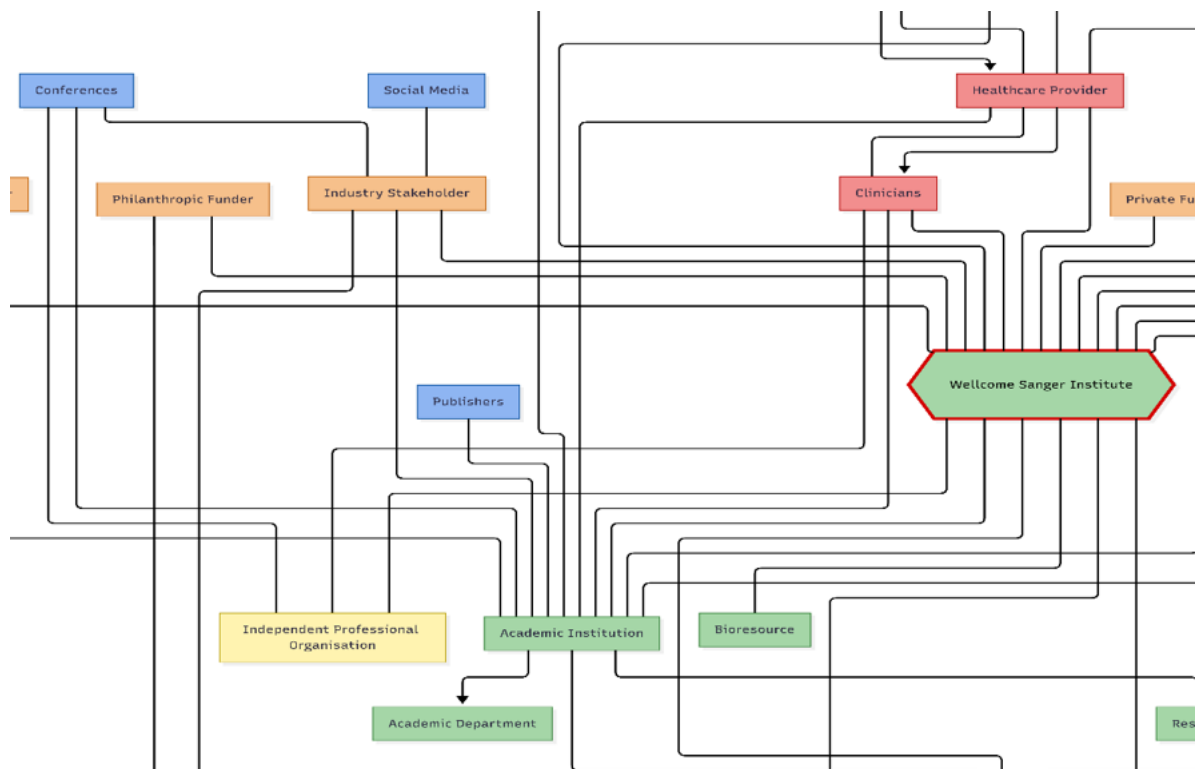


Figure 50

This communicative ecosystem, represented by the blue boxes, enables the diffusion not only of scientific knowledge but also of shared norms, values, and expectations regarding what constitutes good research.

“I mean, not global, but we need to communicate our work to other people in order to have it sort of recognised, or to share the information, share the knowledge, share the insights. And ultimately, that's what you need to do to get your PhD, to get your postdoc, to get your job. Once you've got your faculty position, you need to keep doing that to maintain your job.”

Programme Lead, Discovery Science A.

Conferences are key sites of value negotiation, where topics are not only shared but also platformed, deliberately positioned as worthy of attention. Scientists are encouraged to participate not merely to exchange findings but to advance their careers, as the following examples show. During one interview, a young scientist knocked on the door to speak with my interviewee; when the scientist left, and our conversation resumed, the participant remarked that this interruption perfectly illustrated a point she was coincidentally making, that effective communication and presentation are essential skills. While another participant organised regular group meetings, some online and some in an open, light-filled room above

the café, where early-career researchers practised presentations and received feedback from peers and more senior researchers.

Conferences also represented a site of commercial engagement. Private and industry stakeholders were highly visible, with researchers frequently approached by sales representatives promoting the latest assays or sequencing technologies. For many discovery scientists, relationships with industry were more direct and familiar than those with patients or the wider public. This familiarity carried an implicit understanding of industry's motivations:

“When you work with the private sector, they are rarely driven by equity, right? It's mainly for-profit organisations. So I think more often than not it is about that.”

Senior Scientist, Translational Informatics.

Top-down relationships in the system map, shown by directional arrows, illustrate belonging and ownership, highlighting how influence is understood to operate across layers of the ecosystem. The diagram is broadly hierarchical, with governance and funding positioned above the research environment. Professional bodies provide guidance and policy, but participants did not describe them as prescriptive or dominant. Instead, they are experienced as part of the culture of interaction between researchers, clinicians, and industry partners.

“I think having that senior buy-in has certainly had an influence—it influences the funding. It may not be obvious, but the funding comes from government and philanthropic sources. Wellcome is philanthropic, UKRI would be government, NIH, you know, and so on.”

Programme Lead, Discovery Science.

Governance and regulatory authority are enacted through government departments, legal frameworks, and professional bodies, indicated by colour yellow in the figure. Despite their formal authority, these structures often appeared distant from the day-to-day reality of laboratory work. Scientists perceived their operations as organic, practice-based, and voluntary, perhaps reflecting the layered nature of accountability rather than an absence of constraint. Misconduct, for instance, was viewed as something that would need to escalate to before prompting formal intervention.

Funding occupies a central, intermediary position between the clinical interface and research. The system depends on these streams to function, whether channelled directly into research institutes or mediated through public healthcare systems.

“And it's a challenge to get funding. For scientists in Asia, Africa, Latin America, it's much harder than for us here. CZI has made a first step, though there were criticisms of those funding mechanisms. Now Wellcome is—I'm actually meeting someone from Wellcome on Wednesday—interested in [Africa funding]. So, that will change things. The funding landscape is shifting.”

Programme Lead, Discovery Science B.

The clinical interface emerges as a vital site of exchange, providing both the biological material and data that drive genomic research and the route through which outcomes return to patient care. It thus acts as both an input and an output channel, shaping the ecosystem's translational capacity.

“We'll always go back and check that... often we aren't the consent holder; the consent holders are the people we're collaborating with. So that link with the UK and international network of [clinicians] is absolutely vital for my team.”

Programme Lead, Discovery Science A.

“We collaborate with clinicians and with bioresources that collect—they enrol patients and get samples in. Then we generate the GWAS or exome sequence data on tens of thousands of patients, which we ourselves haven't collected.”

Programme Lead, Discovery Science.

Yet this translation is often experienced as slow and indirect, contributing to a sense of separation between scientific intention (“science for society”) and tangible impact. While researchers express a strong commitment to public benefit, the path from discovery to application remains long and fragmented, shaped by multiple layers of policy, infrastructure, and context. The direct connection between patients and ethics committees in the map reflects not a relationship of interaction or access, but a relationship of institutional obligation, in which ethics review bodies are constituted to act in the interests of research participants, often in their absence.

Ideas, meanwhile, circulate far more freely than outcomes. Conceptual frameworks, population categories, and scientific narratives move rapidly through the system, while practical benefits advance more slowly, constrained by procedural complexity. The ecosystem feels culturally cohesive yet institutionally diverse, bound by shared scientific norms expressed through publications, public discourse, and networks that bridge organisations and disciplines.

Governance structures are often perceived as sitting apart from daily scientific life Figure 51, yet their influence is profound. Through legislation, professional codes, and ethics oversight, they define both the limits and methods of research, exerting subtle but far-reaching control.

“So, you're caught between funders’ objectives, initiatives, strategy, science, and to some extent, personal interests. I think it's naive to believe people do things they're not interested in. They just don't.”

Group Leader, Discovery Science.

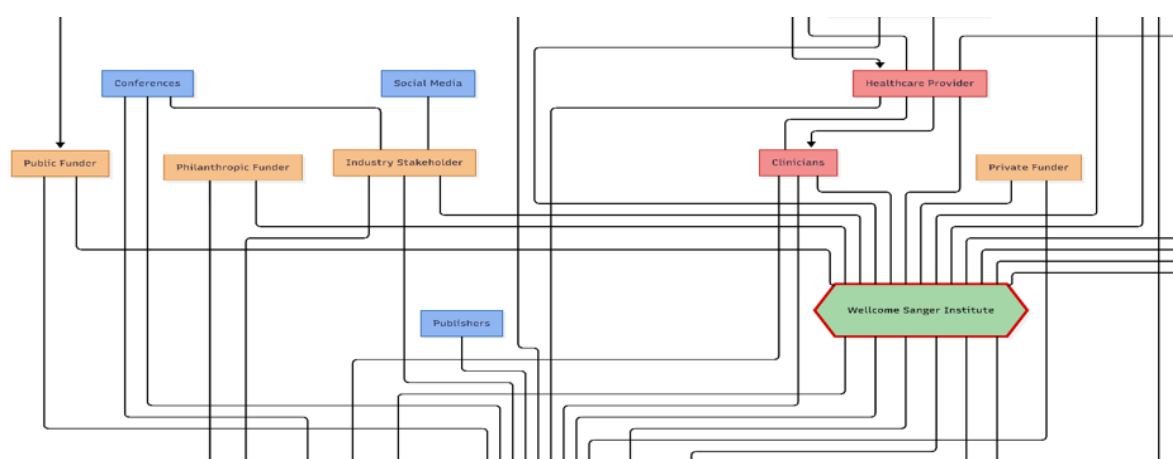


Figure 51

Funding, too, was described as an essential but strategically detached (Figure 51) force shaping priorities from a distance rather than through everyday engagement.

“In funding organisations, you have officers on the ground who have some influence on calls and programmes, but they need buy-in from their bosses. It's like anything in life.”

Programme Lead, Discovery Science A.

“The actual physical location of sequencing becomes a practical solution to delivering the grant. We focus on capacity building, sharing knowledge, and working together. The funders cared only about where sequencing was done, not the bigger picture.”

Programme Lead, Discovery Science B.

The clinical interface, then, acts as both source and sink, providing the biological materials and data that fuel discovery while absorbing research outputs back into clinical and policy

settings. Through this cyclical exchange between laboratory and clinic, the genomic ecosystem sustains itself, continually translating between knowledge and care.

“If we do something that severs that collaborative network, then we've got no access to patients. We've got no samples. We've got nothing. So we wouldn't want to do anything that in any way caused any ethical issues for the consent holders. We wouldn't want to do anything for the patients that was unethical with their data.”

Programme lead, Discovery science B.

Within this interpretive framework, patients appear at the periphery of the diagram (figure 52) yet remain symbolically elevated through direct normative connections to clinical providers and ethics boards.

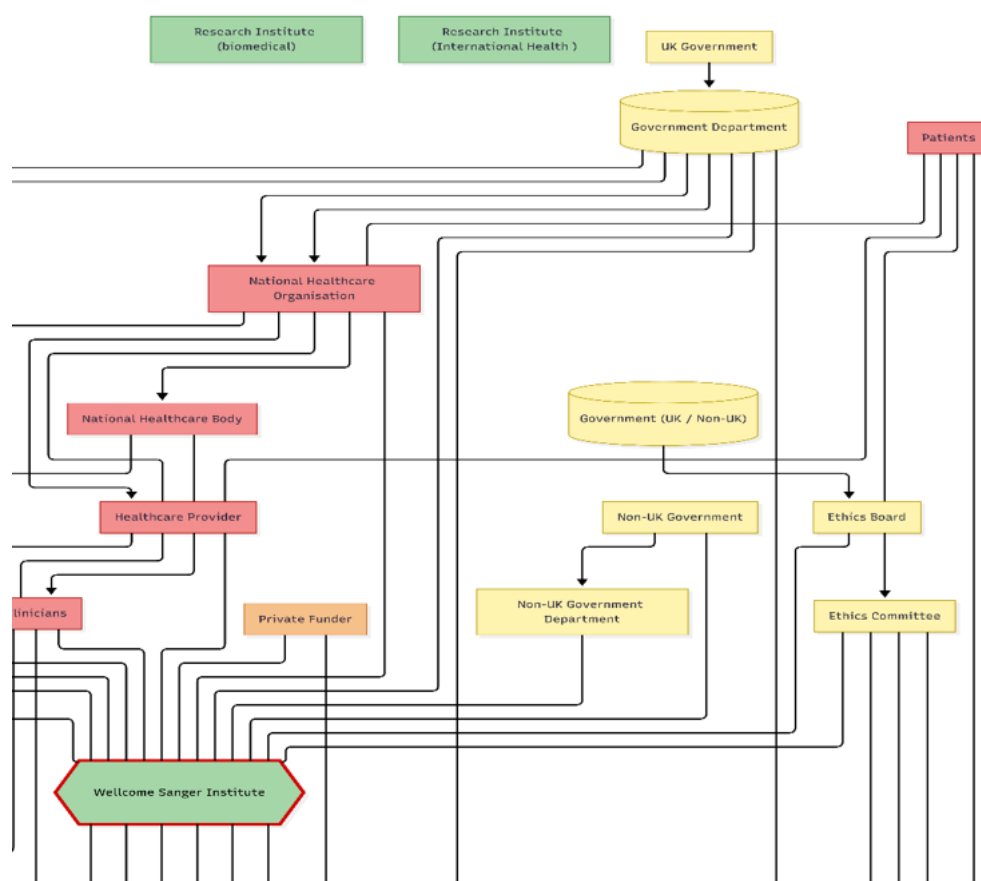


Figure 52

This positioning reflects a recurring duality in the empirical material. Patients were frequently described as essential to the legitimacy, justification, and ethical grounding of genomic

research, yet were rarely represented as active participants in the core workings of research design, data governance, or classificatory decision making. Their peripheral placement is therefore not a claim about importance, but an analytic reflection of limited inclusion in everyday institutional processes. This tension is telling of the epistemological gap identified across the thesis: those whose data and bodies underpin genomic research are central to its moral claims, yet marginal to its knowledge-making practices. It is within this gap that questions of power and justice for this group emerge most sharply, revealing how protection, representation, and exclusion can coexist within the same governance structures.

These visualisations provide a framework for understanding genomic research not as a series of isolated endeavours, but as a system of structured interdependence. Sanger does not merely operate within this ecosystem; it actively shapes and is shaped by it. Funding flows, policy structures, clinical partnerships, and regulatory bodies all play a role in defining what research is possible, permissible, and prioritised.

The system is neither fully open nor closed it is porous, contested, and subject to continual renegotiation. While certain structures (such as governance or funding) exert power from a distance, others (like the clinical interface or collaborative consortia) are sites of active participation and co-production. These dynamics set the stage for the next section, which will explore how themes of power and autonomy emerge from this complex and layered system.

7.4 Acts of Power and Autonomy: Visualising Institutional Influence

This section reflects on how power and autonomy operate across institutional domains at Sanger, using a series of diagrams to support an ethnographically grounded narrative. Drawing on in-depth fieldwork, these visuals do not aim to produce definitive or generalisable claims; rather, they offer an interpretive lens on the relational dynamics of institutional influence and independence.

The section how capacities such as accessing data, training researchers, and acting ethically are configured in terms of both the influence the Institute holds (power) and the degree to which it can act independently (autonomy). It then considers how Sanger navigates these domains in response to, or in tension with, broader commitments to social justice.

As the ecosystem analysis showed, interdependence is central: other institutions and actors hold their own power and autonomy. There are situations where others are directly affected

by Sanger's exercise of power or autonomy, and others where Sanger is shaped by theirs. From themes raised in the previous chapter, several acts of power and autonomy recurred across interviews. Their frequency and placement made them stand out as significant, both for the influence they exert and for how they are themselves influenced especially when read alongside the connections and authority mapped in the ecosystem.

7.4.1 Power and Autonomy in the Domains of Institutional Activity

This section examines how institutional power and autonomy are configured across Sanger's domains of activity, and how these configurations shift in practice. Building on the ecosystem analysis, I use fieldwork-informed diagrams to offer an interpretive account of how influence and independence are experienced, negotiated, and constrained. It maps power and autonomy across eight key types of activity revealed through the system map of Sanger, showing where authority is concentrated and where discretion is limited.

Across these eight domains, Sanger's positioning is uneven: high power does not always coincide with high autonomy, and divergence often signals sites of negotiation, dependency, or indirect influence. Where the two dimensions align, the Institute appears embedded and confident; where they part company, agency becomes more contingent and relational.

Domain 1: Funding issues Research (Power: Low | Autonomy: High)

Sanger holds influence over internal funding allocations, yet its external power in this domain is comparatively limited by the priorities and decisions of major funders. While the Institute remains a powerful scientific actor and exercises some influence beyond its boundaries, its capacity to shape the overall funding landscape is weaker relative to the degree of autonomy it holds over internal resource distribution. From interviews, it was clear that although Sanger can decide how to allocate resources strategically, these choices operate within structural dependencies that constrain what can be funded, for how long, and at what scale.

“We're very fortunate that the Chan Zuckerberg Initiative is a foundation that does seem to have some awareness and invests in that. Because, you know, for science, what you need is the backing from the funders.”

Programme Lead, Discovery Science B.

“So what it means is that what we have to—or are funded to—do, or supposed to do, and nothing else. That comes from the funding intention from Wellcome, which puts a hundred million pounds a year to the head of the director—one person.”

Programme Lead, Discovery Science A.

Domain 2: Prioritising Agendas (Power: Medium | Autonomy: Medium)

In this domain, power and autonomy are closely interdependent. Sanger’s ability to prioritise research issues has grown alongside its technical capacity, scientific standing, and collaborative networks. Interview accounts emphasised that shifts in institutional focus emerged through interpersonal decisions and evolving scientific opportunity, rather than through linear direction from funders.

Sanger neither fully determines research priorities nor simply responds to external agendas. The medium scores reflect this reciprocal configuration, in which influence is real but relational, and autonomy is meaningful yet shaped by wider scientific, funding, and policy environments.

Through publications, conferences, and training, Sanger exerts wide influence on scientific thinking. I observed that while researchers enjoy freedom to develop and communicate new ideas, broader intellectual control remains mediated by journals, funders, and the hierarchies of global science. Training emerged as a key vector of influence: a subtle, ongoing transmission of epistemic norms that shape how genomic science is done worldwide

“We try to harness the [research] community,.. It’s about helping answer scientific questions through our tools and data.”

Director, Informatics.

While Sanger’s reach and impact are a source of pride, assessing them was not straightforward.

“Assessing our impact is tricky, but we know we have thousands of users of our platform and genetics portal.”

Director, Informatics.

Domain 3: Data Sharing (Power: High | Autonomy: High)

Sanger’s leadership in genomic data infrastructure gives it strong influence over how genomic data are generated, curated, and made accessible. From its early role in shaping

open data norms, , open sharing has been embedded as an institutional culture rather than experienced as an external imposition. During my fieldwork, open data was consistently described as both a moral commitment and a strategic orientation, reflecting an internalised governance logic rather than compliance with external mandate.

Power and autonomy therefore align in this domain. While Sanger operates within global data-sharing expectations, it retains significant discretion over how these principles are enacted in practice, granting it unusual influence over a shared global resource.

“Sanger is completely different from other institutes or universities... historically it’s been all about generating genomic resources—genome projects—for the community, and that’s what it is—nothing else.”

Programme Lead, Discovery Science A.

“We used to say it was like working in a fishbowl—clear glass—part of our mandate was to make sure some of the benefits of Sanger were accessible to others.”

Programme Lead, Discovery Science B.

Domain 4: Access to Patient Samples and Data (Power: Low | Autonomy: High)

Access to patient samples and clinical data depends on collaboration, trust, and ethical approval, forms of relational rather than formal power. Sanger’s autonomy lies in how it builds, sustains, and sometimes withdraws from these partnerships, even as external governance structures limit direct control over access. Researchers were acutely aware of their dependence on clinical partners, describing these relationships as simultaneously enabling and constraining.

“You have to go through the data access committee... the data doesn’t all come from one source—it might be twenty different sources. You’ve got to go through twenty different data access committees. It’s a faff, but not impossible... eventually, you will get the data.”

Programme Lead, Discovery Science A.

Some participants also reflected on the legacy of unethical data practices in global health research, particularly in African contexts, shaping current caution and procedural oversight.

“I was shocked by the strong feelings about the way science had been done in Africa—people came, did genetic profiling... what they called ‘helicopter science’.”

Senior Scientist, Translational Informatics.

Access, then, is not purely technical but deeply ethical, dependent on legitimacy and institutional accountability.

Domain 5: Working with collaborators (Power: Medium | Autonomy: High)

Collaboration is central to Sanger's work. In academic settings, the Institute often leads through expertise, infrastructure, and prestige.

"We built an international consortium—started with a London meeting of about a hundred scientists, convened a committee, and grew from there."

Programme Lead, Discovery Science

Yet in partnerships with government or industry, its influence is mediated by competing priorities. From my observations, collaboration at Sanger is both a form of collective creativity and a terrain of subtle hierarchy.

"It is a blocker, right, so geopolitical stigmatising of disease and geopolitics, closing borders and preventing access is the biggest problem we have."

Group Leader, Infectious Biology.

Sanger scientists have considerable discretion in initiating collaborations, particularly in discovery science where regulatory constraints are lighter. I found this autonomy often relies on personal initiative and trust networks, especially in global partnerships.

Domain 6: Regulating Ethical Behaviour (Power: Low | Autonomy: Medium)

Ethical practice at Sanger operates within national and international frameworks, but daily interpretation happens locally. I found that researchers often talk about ethics as culture rather than rule, something performed through example and mentorship.

"We teach them [PhDs and Postdocs] to make good figures and write good papers. We also create an environment where how you do science—and how we talk about science—is as important as the science itself."

Programme Lead, Discovery Science B.

Sanger's governance systems formalise this to an extent, but the real power lies in shared norms. Behaving ethically emerges as a set of everyday internal cultural practices, distinct

but related to ethical regulation and oversight mechanisms that formally govern those practices.

Sanger's compliance systems are well established, but the space for reinterpretation is limited. I found that while researchers appreciate internal discretion, they also acknowledge that external frameworks ultimately set the boundaries. Ethical freedom here means finding room for judgement within established rules.

"You don't tell your children they can do anything—like jump on the settee. You have to set some sort of ground rules."

Group Leader, Developmental Biology.

Ethical behaviour is sustained through trust, mentoring, and social norms rather than rigid codes. However, autonomy is limited because external regulation and funding priorities shape what "ethical" means in practice. I observed tension between individual integrity and institutional constraint, especially when funders' procedural requirements clashed with local ethical reasoning.

"Funding agencies are blind to local requirements. Worst case, they drive malpractice by prioritising permissions over proper processes... people do the wrong things due to grant constraints."

Group Leader, Infectious Disease.

Domain 7: Training the Workforce (Power: High | Autonomy: High)

Training appeared as one of Sanger's most profound forms of influence. It shapes both skills and values, embedding a distinctive approach to data, ethics, and collaboration.

"Training at Sanger looks good on a CV—you can get strong publications and move to the next stage of your career."

Programme Lead, Discovery Science B.

Training emerged as both a gift and a projection of institutional identity, a way of exporting Sanger's methods and worldview.

"It's a place to undertake training, grow a network, and bring those elements together—data, function, interactions with patients and a big clinical community."

Programme Lead, Discovery Science A.

“It’s very international at the PhD and postdoc level—people come for training from all over the world... there’s now more diversity among British researchers, which is a good thing.”

Programme Lead, Discovery Science B.

Domain 8: Ways of Working with Multidimensional Data (Power: High | Autonomy: Medium)

Sanger’s methodological choices are internally governed. Its power lies in technical mastery; autonomy is mediated by collective norms and funder expectations. I reflected that limits here are less about capacity than institutional will, and the willingness to challenge established paradigms.

“It’s on institutions, and then on society... institutions need to reflect what they extract from wider society, and how they use that to shape the research and training environment.”

Programme Lead, Discovery Science B.

These practices are not simply displays of power or assertions of autonomy. They are relational, causes and consequences of Sanger’s position within a dense ecosystem of partners, funders, and publics. Given the recurring themes of power and powerlessness in this thesis, these acts stand out as the most revealing: moments where values, constraints, and institutional capacity visibly meet.

As Figure 53 summarises how power and autonomy are distributed across 8 domains of institutional activity. What emerges is a differentiated landscape: while some domains are marked by both high power and high autonomy, such as training the workforce or data sharing, others reveal mismatches between influence and independence.

RELATIONAL UNDERSTANDING OF POWER AND AUTONOMY IN KEY GENOMIC RESEARCH BEHAVIOURS AT WELLCOME SANGER INSTITUTE

■ Power Score ■ Autonomy score

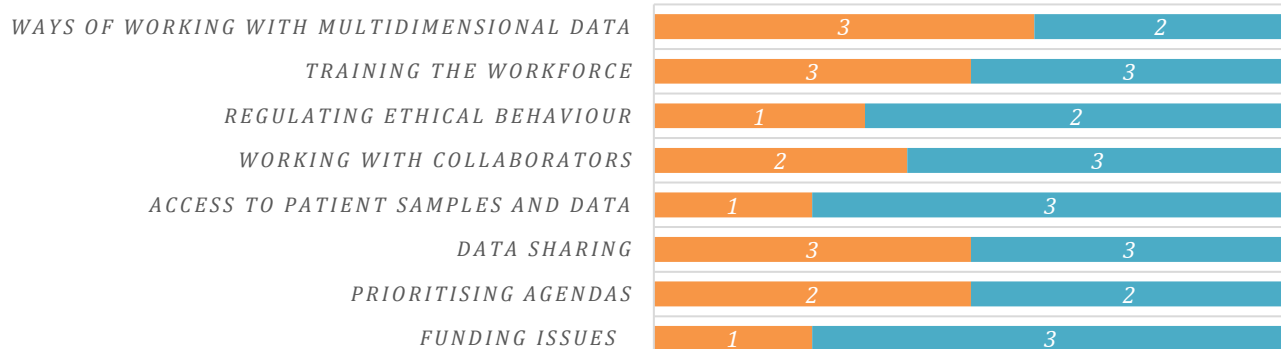


Figure 53: Relational understanding of power and autonomy in genomic research behaviours at the Wellcome Sanger Institute. Bar diagram showing score for low power or autonomy is 1, medium scores are 3 and higher scores are 5.

The domain of Funding Research is characterised by relatively low power but high autonomy. While Sanger cannot directly control the broader funding landscape or dictate its funders' strategic priorities, it retains flexibility in how those priorities are interpreted and implemented internally. This autonomy manifests in decisions about which projects to pursue, how to distribute internal resources, and how to align infrastructural commitments with broader scientific trends.

The domain of regulating ethical behaviour, meanwhile, holds low power but medium autonomy, indicating that while ethical practices are sustained through cultural norms, professional expectations, and collective identity, researchers' capacity to redefine or act independently of external ethical frameworks remains limited. In other words, Sanger can to some extent govern its own conduct but cannot unilaterally determine what counts as ethical across the broader research environment. Neither does it hold any regulating behaviour over other organisations passed its collaborative agreements.

Data Sharing emerges as a domain of *high power and autonomy*, illustrating Sanger's leadership in open data initiatives and infrastructural development. A balance of power and autonomy in prioritising agendas reflect the moderate freedom to innovate and communicate within and outside of Sanger, their ideas shape broader scientific agendas yet depends on

external dynamics such as editorial gatekeeping, institutional hierarchies, and the politics of academic prestige.

What becomes visible in this comparative view is the unevenness of Sanger's institutional positioning. High power is not always matched by high autonomy, and vice versa. Where the two align, the Institute appears confident and embedded in its operations. Where they diverge, more precarious forms of institutional agency emerge, spaces where discretion must be negotiated, and influence is exercised indirectly or through relational means.

These diagrams do not prescribe how power or autonomy ought to function. Instead, they invite reflection on the multiple ways authority is exercised, independence cultivated, and collaboration navigated across a scientific institution.

The final section now turns outward: how Sanger scientists understand power and autonomy in relation to other actors when three key ethical questions are examined. Building on the experiential, communicative dynamics traced here, the analysis considers using this deep institutional understanding around categorisation and data.

7.4.2 Power, Autonomy, and Ethical Orientations in the Genomic Ecosystem

By taking the emergence of power and autonomy as key indicators of institutional agency and the themes raised in Chapters 5 and 6, this section examines how these configurations shift when the ethical orientation of research changes. It focuses on the questions explored in this thesis by orienting around three approaches: improving ancestral data diversity, reforming categorisation practices and positional social justice as a core research area.

Drawing on a the definition of social justice presented earlier by analysing how Sanger navigates these domains, this section aims to reveal how institutional power and autonomy are mobilised in response to, or in tension with, broader commitments to social justice.

7.4.2.1 Improving Ancestral Data Diversity

When discussing improving ancestral data, the interviews made clear that researchers felt held to the power of others for the funding, time, and expertise needed to address data

diversity. Government remained a visible enabler, reflecting the political sensitivities of international collaboration and the need for public-sector anchoring (see section 6.1).

The power they held was relational, shifting with partners and generally increased when they became part of consortia or initiatives. By contrast, autonomy to join or convene groups was seen more positively: collaborations were usually described as matters of timing, trust, and opportunity. Finding the “right people” was organic and often opportunistic rather than centrally directed (see section 6.2 *Choosing a team*).

Participation at this scale requires expertise and institutional backing that are unevenly distributed across global genomics. Inclusion, therefore, depends on the redistribution of power and privilege as much as on welcoming language; funders and governance bodies often sit above consortia in making this real. Projects centred on ancestral diversity were, unsurprisingly, international in scope, often involving industry alongside academic partners. Interviewees highlighted a partial narrowing of the distance between sectors when the shared aim of diversifying data was explicit (See section 6.1 Open sources and equal benefits).

Crucially, improving data diversity is costly, financially and institutionally. Support is necessary for infrastructure, governance, and local leadership, particularly in African contexts. Even where initial sample banking is funded, analytical capacity often lags, illustrating the gap between aspiration and capacity.

In this landscape, power over diversity efforts is dispersed and negotiated, distributed across a web of funders, governments, and research institutions. Autonomy, meanwhile, resides in the capacity of scientists and organisations to convene, align, and persist within systems that are, by design, uneven, demanding collaboration, compromise, and strategic adaptation rather than unilateral action.

7.4.2.2 Reforming Categorisation Practices

Sanger scientists’ accounts describe an increase in institutional power, reflecting their technical authority and influence in shaping genomic standards, but a decrease in autonomy, given the broader social, ethical, and regulatory constraints surrounding population classification (see section 6.3).

Labels serve important scientific functions for example, stratifying data, detecting associations, and ensuring reproducibility but they also operate within a larger ecosystem of

expectations (see section 5.3). Funders, journals, policymakers, and public audiences all look to these categories to signify something beyond the science itself. As a result, the act of categorisation becomes both a technical task and a social negotiation, one that continually exposes the boundaries of scientific autonomy.

At Sanger, researchers expressed a strong sense of ownership over their methods, confident in their ability to justify the categories they used. Yet this autonomy did not fully extend beyond the scientific domain. Within the wider ecosystem, social framings of race, ethnicity, and ancestry remain deeply embedded in institutional practice. Scientists recognised the enduring *power of society*, through policy, regulation, and public discourse (see section 5.3.3), to shape how these categories are understood, which in turn constrains how science can be conducted.

Scientific power in categorisation is tempered by ethical constraint, and how autonomy in method collides with responsibility in meaning-making. This tension provides the bridge to the next section, which explores how these forces play out when researchers attempt to implement social justice as a core research aim, not merely as an ethical obligation, but as a defining orientation for genomic science itself.

7.4.2.3 Positioning Social Justice as a Core Research Aim

The final configuration, brings together the previous analyses of data diversity and categorisation, turning attention to how values such as equity and fairness are operationalised within the institutional life. Across interviews, many researchers expressed a genuine desire for social justice to shape their scientific legacy. Yet, this aspiration was tempered by a recurring concern: that such commitments risk becoming performative rather than transformative (see section 6.3).

Crucially, this gap was not framed as deliberate resistance but as a result of structural and cultural inertia. Scientists recognised that Sanger, like many research organisations, struggles to translate its aspirational goals into embedded practice.

Rather than cynicism, this was voiced as a call for institutional alignment: to ensure that values such as equity and inclusion are not simply symbolic but built into decision-making structures, funding mechanisms, and scientific priorities. The pattern observed in the power/autonomy maps reinforces this: Sanger's sense of power remains high, but its autonomy in pursuing social justice appears lower than in other domains. Despite this,

researchers still perceive themselves as culturally influential, able to shape thinking and practices beyond their own walls, even when internal change feels constrained.

Ethical formation and research culture are intertwined: social justice is not only a research goal but a pedagogical one. It is enacted through mentorship, training, and the subtle transmission of values within teams (see section 6.1 Social and ethical concerns). This observation situates social justice as structurally contingent, dependent on how collaboration, funding, and accountability are designed.

While Sanger retains significant autonomy compared to commercial entities (6.3), it operates within economic and strategic hierarchies that shape what can be prioritised. For example, Sanger's centralised funding model, and where significant financial authority is vested in a single director.

At the same time, several interviews illustrated how institutional power and autonomy can be harnessed to drive reform and collaboration. The building of international scientific consortia, initiated and coordinated from within Sanger, is key to its operations, and one they believe can be embedded with equity at its core (1.2.5, 6.2). Such collaboration can demonstrate how Sanger's prestige and infrastructure can amplify global capacity-building. This form of power, mobilised through cooperation rather than control, embodies a different expression of autonomy: one grounded in credibility, reputation, and the ability to convene others.

7.5 Final Thoughts

This chapter has shown that power within the genomic ecosystem is relational and diffuse, circulating through expertise, networks, and legitimacy rather than through simple hierarchy. Autonomy, in turn, is made and remade through practice, through the everyday acts of negotiation, discretion, and alignment that enable research to continue. These processes reveal not only how institutions act, but how they are themselves shaped by the moral, political, and scientific environments they inhabit.

The discussion that follows builds on these insights to consider what they reveal about responsibility, justice, and the redistribution of power in contemporary genomics. It turns from the description of institutional life to a critical interpretation of how structures of authority might be reconfigured, asking how an institution such as Sanger can align its considerable influence with its ethical ambitions, embedding social justice not as a peripheral ideal but as a central principle of scientific practice.

8 Discussion: Power as the Foundation of Justice in Genomics

8.1 Introduction

In this chapter, using the institutional practices and settings at Wellcome Sanger Institute, I bring together a final analysis of how race, ethnicity and ancestry (REA) are conceptualised and operationalised within contemporary genomic research. I argue that questions of justice in genomics are fundamentally questions about power. They concern who can define populations, who is rendered visible in data, whose knowledge is treated as credible, and whose interests are ultimately served by how genomic science is organised.

In my analysis of the literature and practice, the hope for more equitable research has increasingly been premised on the need for better and more diverse data. Responses have often focused on quantitative measures of diversity, carrying an assumption that more data will produce better outcomes. Yet, as I have shown throughout this thesis, the persistent discourse of the underrepresentation of the world's most genetically diverse populations is not merely a technical problem of missing data. Rather, it reflects deeper and uneven operations of power in how populations are categorised, measured, and understood. The widening gap between European and non-European samples is tied to classificatory practices that reproduce racialised hierarchies, to methodological decisions that favour simplicity over complexity, and to institutional structures that reward particular kinds of evidence and expertise.

My ethnographic engagement revealed that diversity initiatives at Sanger, although prominent in institutional rhetoric, often leave intact the underlying epistemic and structural conditions that produced inequity in the first place. Scientists expressed a desire to improve ancestral representation and to move away from race as a biological category, yet their practices remained tethered to inherited labels, Eurocentric reference datasets and funding models that prioritise statistically efficient designs over justice-oriented ones. Drawing on Critical Race Theory, the analysis traced how race operates as a social construct that is institutionally embedded and frequently made invisible through the language of ancestry and genetic diversity. Critical Realism provided a way to distinguish between the empirical

expressions of these dynamics, the regularities they produce, and the deeper mechanisms that sustain them.

The findings demonstrate that avoiding racial terminology in genomics does not eliminate racialised thinking. It tends instead to shift it into less visible forms, for example, through the use of continental ancestry categories, geopolitical labels, and notions of genetic similarity that are presented as neutral. Through CRT and CR, my analysis reveals that these notions are in fact structured by colonial histories and the normativity of whiteness. Genetic ancestry is often promoted as a scientifically robust alternative to race, yet its deployment through broad and homogenising labels often echoes past inequities. With these approaches, technical fixes are offered, but standards such as thresholds for variant frequency or definitions of adequate sample size are not only statistical choices. They are decisions shaped by who is included in reference datasets and who is absent from them.

This chapter is framed within an epistemology of social justice that understands a reliance on objectivity and neutrality as insufficient safeguards against the reproduction of racialised hierarchies.

In the sections that follow, I synthesise the findings of this thesis through three interrelated modes of power: conceptual, methodological, and structural. I begin by examining conceptual power, showing how race and ethnicity persist through classificatory practices that shape meaning, visibility, and legitimacy in genomic science. I then turn to methodological power, analysing how analytic standards, data infrastructures, and research designs concentrate authority and constrain ethical possibilities. Lastly, structural power situates these practices within institutional, funding, and regulatory arrangements that organise which kinds of genomics are possible. Throughout the chapter, I draw on the lived experiences of scientists and my own positionality to illuminate how these forms of power are encountered, negotiated, and sometimes resisted in practice.

However, as justice in genomics requires active engagement with the histories, power relations and social structures that shape knowledge production, I call for a fundamental re-examination of how populations are defined, analysed and represented. I argue that questions relating to social justice are not supplementary ethical questions but central scientific problems. I therefore conclude the chapter by considering how power might be reconfigured to pursue justice in genomics and introduce the JUST DATA framework.

8.2 Conceptual Power

Conceptual power of science operates through the categories, descriptors, and interpretive logics used to describe human variation. It is exercised in what becomes thinkable and sayable about difference, and in how race endures in genomic practice even when formally rejected, risking sustaining long-standing inequities for racialised and underserved groups under the guise of neutrality.

In the sections that follow, I trace how race is reintroduced through classificatory routines, managed through discomfort and avoidance, and made visible through lived experience and the ethical tensions that accompany categorisation.

8.2.1 Reintroducing Race in Genomic Practice

Ethnographic fieldwork conducted at Sanger illustrates how race and ethnicity are subtly yet persistently reintroduced across multiple stages of the genomic research process, revealing the contours of a deeply embedded system that requires both critical scrutiny and intentional disruption.

The classification and labelling of population groups within genomic studies is far from standardised. Methodologies for determining groupings vary, and the labels applied, whether based on geographic origin, ethnicity, or race, are inconsistently used. When researchers do not establish labels themselves, they often adopt those used by external datasets such as the UK Biobank, 23andMe, the 1000 Genomes Project, the Office for National Statistics, or NHS records (Aung et al., 2023; Malawsky et al., 2023; Thygesen et al., 2023). This inconsistency underscores the complexities and contradictions inherent in defining race and ethnicity within genomic research.

Against this background, categories persist not because they offer empirical insight into human genetic variation, but because they serve socio-political functions. In the British context, race is a product of intersecting histories of empire, migration, institutionalisation, and resistance (chapter 1.2). These interwoven histories are often not considered in genomic research in the UK which is essential if we are to dismantle the structures that perpetuate racial inequality today.

Science has at times positioned itself as capable of distilling and interpreting the meaning of racial categories that have been socially constructed (Lala & Feldman, 2024; Smedley & Smedley, 2005; Stuurman, 2000). Despite consequential questions of the place of racial constructs in genomics, they sometimes continue through genuine curiosity, and at other times through institutional arrogance. The International HapMap Project offers a clear

example of how race can become embedded in the architecture of genomic research, even when not explicitly invoked. As Ossorio and Duster, (2005) warned,

“the [hapmap] Project could have pernicious effects merely by studying genetic variation in human subject groups that evoke racial categories, regardless of the intentions of the researchers” (Ossorio & Duster, 2000 p. 119).

The contemporary elusiveness of race is sometimes taken as evidence of its conceptual instability, a rationale used to justify its removal from the scientific domain and its relegation to the social (Rutherford, 2020). As chapter 2.3.1 outlines, the attempted displacement of race as a biological concept has a long history within genetics (Timmer, 2023). From Lewontin, (1972) to Rutherford, (2020), a broad scientific consensus affirms that race has no biological basis. Studies of human genetic variation have consistently undermined the idea of discrete racial categories, rejecting such classifications as scientifically unsound (1000 Genomes Project Consortium et al., 2015; Collins, 2004; Venter J. Craig et al., 2001; Yudell et al., 2016)

Yet the very instability of race as a biological concept, and the long history of attempts to distance science from it, have not eradicated its presence. Instead, these attempts have produced a conceptual terrain marked by ambiguity and ambivalence . It is within this terrain that discomfort, avoidance, and selective erasure emerge, dynamics that shape how scientists talk about race, how they classify populations, and how they navigate the political and ethical stakes of their work. In fact ethnographic work by Fujimura & Rajagopalan, (2011) illustrates how genomic scientists have navigated these tensions, often engaging with racial categories while attempting to distance themselves from their social and historical weight.

8.2.2 Discomfort and Avoidance

Through this research, race was regularly cast as a social problem to be managed rather than a construct continually reproduced through research practices, categorisations, and data infrastructures. Conversations with scientists were often complex and contradictory. At times, I struggled to accept their inability, or perhaps their reluctance, to acknowledge the limitations of their frameworks. Yet I found myself extending to many of them a degree of empathy they did not always extend to themselves, not least because it was clear that they were negotiating ambiguities they had not been trained to confront.

Public engagement work was conducted alongside these interviews. Participation in public-facing events on genetics and ethics, facilitating deliberations with patient and public groups, and collaborating with community projects was a crucial counterpoint (Appendix C). It helped stabilise the critical distance necessary for analysis while revealing how closely the uncertainties voiced by scientists mirrored those circulating in wider society. Although scientists often positioned themselves as in service to the public, the questions they struggled with about race and identity exposed how deeply embedded these classifications are in everyday life (chapter 5.2.3).

Across the interviews and as captured in Chapter 5.2: *Power in communication*, discussions of race were frequently avoided or tightly circumscribed by institutional unease. There was a strong sense that even broaching the topic risked legitimising divisive or regressive ideologies. This discomfort reflected a broader academic tendency to sidestep or suppress racial discourse entirely.

This evasion was not confined to interpersonal dynamics. It was also evident in institutional practices. Some participants described receiving direct instructions to remove the word “race” from their work, even when they themselves were uncertain about its relevance or meaning.

Critical Race Theory identifies this reluctance as colour-blindness, a professed ignorance of race that, in practice, functions as a method of upholding systemic racism (Bonilla-Silva, 2022; Leonardo, 2009). The scientific erasure of race in language, while retaining its operational function through ancestry and genetic proxies, thus represents less a break from historical forms of racialisation than a strategic reconfiguration of them.

Despite these attempts to avoid the term, the power of race as a lived social construct continually challenged such evasions. As the next sections illustrate, neither scientists nor the public nor I, as a researcher embedded in these interactions, could fully escape the force of racialised experience. This confrontation with lived reality not only exposes the limits of avoidance but also underscores why engaging critically with race remains essential to any meaningful analysis of genomic practice.

8.2.3 Lived Experience as Legitimate Analytic Frames

Though I did not expect geneticists to offer the moral clarity on issues of race and identity that eludes wider society, it became clear that the power of racial concepts was both deep and inescapable within the scientific space. Lived experience provides an essential

counterpoint to scientific abstraction. The meanings and consequences of racial classification are not only conceptual matters within genomics but are lived, negotiated and felt by those who inhabit scientific spaces. My fieldwork, therefore, required attending not only to what scientists said about race, but to how race shaped the everyday conditions in which they and I operated.

My own positionality was never peripheral; it shaped the questions I was able to ask, the responses I received and the meanings I drew from the silences and deflections that arose. There was, it seemed, an emotional texture to the work because of this. I was not merely observing dynamics of racialisation; I was implicated in them.

Racialisation was experienced unevenly among scientists, shaped by nationality and migrant status. While some peers only encountered racialisation after coming to the UK, my upbringing as a second-generation immigrant meant that racialised identity was already an established feature of my personal and professional life.

These tension-laden perspectives reinforced the necessity of holding academic analysis alongside the lived realities of race in the UK. Racial categories therefore continue to permeate social life, and scientific discourse neither escapes nor resolves this. Instead, it reflects and refracts these wider forces, much like patterns observed in genetic data mirror the social and historical conditions that shape them.

My research underscored the importance of embedding minoritised perspectives within genomic science not as symbolic gestures of inclusion, but as intellectual and ethical contributions that challenge, unsettle and broaden the conceptual frameworks through which the field understands human difference.

Lived experience of myself and others reveals that race is not an abstract concept operating elsewhere, but a force that actively shapes scientific encounters, assumptions and interpretive habits. Recognising this is crucial for interpreting how the logics of race materialise in the categories and population descriptors used within genomic research.

8.2.4 Persistence of Racial Logics in Genomic Categories

The empirical material revealed that racial logics did not simply linger in genomics; they were continually regenerated through everyday classificatory routines, publication practices and inherited data structures. What follows traces how these mechanisms operated in practice, showing how race persisted.

As section 2.2.1 discusses religion, and Christianity in particular, played a central role in shaping racial ideologies. Despite considerable diversity across colonised populations, Europeans constructed a racial hierarchy in which they positioned themselves as the apex of civilisation. This worldview was normalised through political power, missionary activity, and assumed intellectual authority (J. K. Carter, 2008; Cox, 2002; Ekechi, 1971; Goetz, 2016; Harris, 2004; L. Young, 2009). It is within this ideological lineage that the ethical validity of data collected from such historically embedded power imbalances is questioned.

An analysis of the digital discourse in journal publications circulated on campus during the period of this study revealed a persistent entanglement of racial and ethnic labels within genomic publications. These categories were frequently used as non-genetic descriptors for population groups. As Chapter 4 highlighted, the use of familiar racial terminology, such as *Black* and *White*, was widespread. While some researchers employed these terms to indicate ethnicity, others acknowledged their racial connotations. *White* was frequently associated with individuals of *European descent*, often labelled as *White British* or similar designations tied to geographical origin.

Ethnicity was frequently seen as connected to, and therefore problematised with, race, not necessarily as a dimension of human variation but as a term that evokes discomfort and unintended associations. This suggests that while these categories are analytically convenient, they are also deeply fraught, laden with connotations that undermine their scientific neutrality.

The rejection of race as a scientifically valid category was juxtaposed with the largely uncontested use of concepts such as genetic ancestry and genetic similarity. While these are defined as dimensions distinct in scale, resolution, and analytical purpose from the ideas of race and often ethnicity, their deployment through broad continental groupings undermined this distinction.

The endurance of geo-political labels to describe genetic ancestry and similarity is not necessarily because they are epistemologically sound, but because they offer methodological expediency, an expediency that comes at the expense of both ethical integrity and epistemic robustness. As a result, race persists not always in language or as a coherent biological reality but as a constellation of practices, assumptions, and expectations embedded within the architecture of genomic research. As shown in the earlier ethnographic literature reviewed even when scientists avoid explicit racial terminology, the distinction between race and ancestry frequently collapses in practice (see section 2.7 and 5.5.4). These analyses of precision medicine projects show how these “slippages” are embedded in

data structures and analytic routines, revealing the depth at which racial logics persist. (Bentz et al., 2024; Gutin, 2019).

Scholars such as Leonardo, (2009) and Bonilla-Silva, (2022) argue that whiteness operates as an invisible yet normative force in knowledge production, dictating the standards for what is considered valid, credible, or even perceptible within academic and scientific contexts. In genomic science, this normativity is encoded in the tools, categories, and methodological priorities that dominate the field. They are presented as innocuous means of grouping populations, yet they resonate uncomfortably to me with historical legacies of empire, eugenics, and scientific racism.

At Sanger conversations, especially those regarding data quality, statistical power, and publication potential, were frequently anchored in the use of European reference data. This data was sometimes considered standard or “clean,” and at other times “dominant” and “well-powered”, operating as a silent benchmark, reinforcing the racialised norms embedded in genomic methodologies. To interrogate these practices is not to disparage science, but to strengthen it. Exposing how whiteness and power operate through scientific classification enables us to reimagine a genomics rooted in reflexivity, justice, and epistemic humility; one that actively questions the boundaries it draws, rather than reifying them

Researchers frequently negotiated this tension in both formal and informal discourses. On one hand, there was an uncritical comfort with the use of continental ancestry groupings as descriptors of human populations. At the same time, some participants expressed frustration with narrow conceptions of diversity. Notably, however, this viewpoint often emphasised that non-genetic determinants of health, such as environmental factors, were omitted from the current discourse on diversity. Criticisms of concepts of diversity did not address how even existing models of genetic variation themselves perpetuate reductive understandings of human difference.

Even where scholars such as Rebbeck et al. (2022) argue that “inferred ancestry must be distinguished from race” (p. 890), acknowledging that the latter is historically tied to social hierarchies of power and privilege, they frequently stop short of proposing truly alternative frameworks for representing genetic diversity. Similarly, there was little discussion of improved methodologies for studying human variation that move beyond existing categorisation, falling short of calls from Lewis et al. (2022) to “use concepts and tools that acknowledge that ancestry is continuous ” (p. 251).

These patterns, as revealed through my ethnographic and discursive analysis, show how genomic categories continued to draw upon and reproduce racial logics even when framed by participants as scientific, neutral, or purely methodological. What my findings expose are

not isolated slips but systemic tendencies. They raise difficult questions about responsibility, the limits of scientific neutrality, and the ethical vocabularies researchers mobilise to account for their decisions.

8.2.5 Ethics, responsibility and communication

A recurring pattern across this research was that the language of ‘ethics’ appeared frequently in institutional and interpersonal discourse, yet its practical application was uneven, contingent, and often overshadowed by methodological and structural demands.

Beauchamp and Childress’s principles of biomedical ethics (Beauchamp and Childress, 2019) are intended to function collectively as a safeguard for ethical practice. In the context of this study, concerns about maleficence centred on the potential harm caused by misclassification or reductionist groupings. In contrast, beneficence was often invoked to justify categorisation as a means of discovering group-specific insights or improving clinical outcomes. Autonomy appeared largely in symbolic terms; participants were sometimes led to believe they had influence over their classifications, even though, in practice, groupings were typically predefined by researchers. The principle of justice, concerned with equitable access, representation, and outcomes, was frequently cited but seldom operationalised in any substantive way. Emanuel et al., (2000) Outline seven requirements for evaluating the ethics of clinical research, two of which are particularly justice-oriented: fair subject selection and a favourable balance between risks, benefits, and the distribution of research gains. These justice-oriented principles require researchers to avoid exploiting vulnerable groups, to ensure that the burdens and benefits of participation are equitably distributed, and to justify inclusion on scientific rather than structural grounds. In genomic research, however, these ideals are often balanced by pragmatic priorities such as statistical power, data availability, and institutional expectations.

These dynamics resonate with Reardon’s analysis of how genomics and ethics intersect, where normative commitments to justice and accountability are continually shaped and constrained by the organisational and epistemic structures of the field (Reardon, 2009, 2017). My findings extend this critique by showing how ethical principles do not operate as stable or abstract ideals. Instead, they are negotiated, diluted, or sidelined in the routine classificatory and analytical work of genomic practice. In this sense, the postgenomic condition that Reardon describes is lived as much as theorised, manifest in the technical practices that subtly reproduce racial logics while claiming to move beyond them. It is at this point that my ethnographic material diverges from a description of the condition itself, illustrating how such ethical tensions materialise within everyday scientific practice.

In practice, bioethical and clinical research principles were routinely invoked in institutional rhetoric; they were refracted through the pragmatic imperatives of genomic research: statistical power, data standardisation, and publishing pressures. Rather than providing a coherent ethical architecture, they often operated unevenly and tactically, aligning with institutional priorities and output-driven goals. This pattern, evident across my fieldwork, demonstrates how ethical decision-making becomes treated as contingent yet quickly overridden by the methodological and structural demands of contemporary genomics.

How scientists describe race, ancestry, and ethnicity matters, not only to research participants but also to public interpretation. Interviews for this study revealed that researchers' discomfort with race is often framed around fear of public backlash rather than critical engagement with its scientific use (chapter 5.3.3). Such anxieties, while understandable, also point to a deeper institutional avoidance. Race remains both conceptually fraught and politically charged. Yet ignoring it altogether does not neutralise its impact; in fact, it may inadvertently entrench existing inequities by allowing them to go unchallenged (Bonilla-Silva, 2022; Gutin, 2019; Leonardo, 2009).

Some formal guidance exists for the use of race, ethnicity, and ancestry in research (Khan et al., 2022; National Academies of Sciences, 2023). Khan et al. (2022) In their guidance, underscore the far-reaching consequences of these terminological choices not only for the internal validity of research, but also for how findings are communicated to clinicians, policymakers, and the public. They caution that without clear, context-sensitive definitions and frameworks for interpreting population descriptors, genomic science can unintentionally perpetuate misunderstanding and inequity. It has also been suggested that more precise, critically informed terminology could mitigate the risk of misinterpretation and prevent the repurposing of genomic findings in ways that reinforce health inequalities (Panofsky & Bliss, 2017; Yudell et al., 2016). Further, as discussed in Section 5.3.3 and Chapter 6, engagement with existing guidance and resources was often limited by ignorance or unfamiliarity, as well as by the pressures of getting science done and a sense that such work fell outside participants' perceived roles (Samuel, 2024).

My engagement, however, highlights that addressing the justification of choices in communication requires more than careful word choice; it demands sustained scrutiny of how racialised categories are constructed, by whom, and with what authority. Such interrogation exposes not only the ethical obligations attached to naming but also the unequal power to define, delimit, and normalise population categories, a dynamic that becomes even more pronounced in the methodological architectures examined in the following section.

8.3 Methodological Power

Population categories make comparison and analysis possible, but they also simplify diversity, produce inclusion and exclusion, and carry ethical consequences for representation. While scientists often acknowledged these tensions, responsibility for their effects was frequently displaced. For marginalised populations in particular, these methodological practices show how power operates through research methods, shaping both scientific outcomes and the meanings attached to human difference.

As section 5.3.2 explores, scientific classification influences the equity of genomic research outcomes. This influence operates not only through direct outputs such as disease associations or pharmacogenomic insights, but also through less visible processes including the formation of inclusion and exclusion criteria, the ethical tensions surrounding data diversity, and the prioritisation of analytical clarity over social complexity.

Despite increasing recognition of the need for diversity, European populations remain disproportionately overrepresented in genomic research (Bentley et al., 2017; Corpas, et al., 2025). Problems of mislabelling and limited harmonisation across datasets also endure in the field (Corpas, et al., 2025), raising concerns about data stewardship and epistemic responsibility. These structural conditions formed the backdrop against which the significance of the methodological decisions examined in this study emerged.

The categorisation of human groups in genomic research is a site of methodological necessity and ethical tension. These processes employed are not neutral; they reflect a stratified hierarchy of acceptability that intersects with broader systems of power and privilege.

Scientific classification in this context becomes a form of boundary work. This concept, introduced by Gieryn (1983), refers to the discursive strategies used by scientists to delineate what counts as legitimate science and who qualifies as authoritative. Boundary work maintains intellectual demarcations that separate science from non-science, often in ways that protect the autonomy and prestige of the scientific field.

Star & Griesemer (1989) notion of boundary objects complements this understanding by describing categories or tools flexible enough to be interpreted across different social worlds yet stable enough to retain coherence within scientific institutions. Scientific categories are

therefore not neutral descriptors but sites where authority, meaning, and inclusion are negotiated.

In genomics, boundary work is particularly visible in decisions about how to classify human populations, whether through ethnicity, genetic ancestry, or genetic similarity (see section 4.2). Scientists often seek to separate scientific classifications from their social and political histories, even when those classifications draw upon earlier racialised schemes (Schramm & Rottenburg, 2012)

Framing “race” as socially constructed can be used to bracket off problematic social usages while leaving genetic population categories intact, a move mirroring the dynamics of boundary work (Gieryn, 1983; Schramm & Rottenburg, 2012). My own deep reading of scientific texts, institutional documents, and interview accounts demonstrates how such rhetorical manoeuvres permeate routine classificatory practice and governance.

Classification here, therefore, reflects boundary work in action (see section 5.3). It stabilises scientific authority by positioning population groupings as biologically meaningful even when they align with non-biological usage or politically loaded categories. Genetic ancestry, in particular, functions as a boundary object: mutable enough to operate across varied settings yet presented as an objective and apolitical measure (see section 5.2.2).

As Epstein (2007) and Fujimura & Rajagopalan (2011) note, the naming and framing of population categories in genomics both reflect and reproduce existing racial and social logics. These decisions are not merely technical; they are acts of epistemic authority. By choosing which forms of difference to foreground and which to occlude, researchers construct and legitimise certain groupings while rendering others invisible.

8.3.1 Techniques of Categorisation

A longstanding critique highlights that continental ancestry categories oversimplify genetic diversity and reproduce racial and colonial assumptions (Ali-Khan et al., 2011; Fujimura & Rajagopalan, 2011; Tishkoff & Kidd, 2004). These conventions persist within genomic workflows, reinforcing the historical legacies outlined earlier and shaping the methodological terrain encountered in my fieldwork.

A clearer picture emerges when examining how these systems operate in practice. Clustering techniques underpin the scientific conceptualisation of human variation, yet their outputs are routinely assigned geopolitical labels. As shown in section 4.3, publications overwhelmingly employ continental categories such as European, African, East Asian, South

Asian, West Asian, Oceanian, and American, often described as “major” or “super populations”. These labels appear statistical but rest on long-standing political and geographic assumptions.

Techniques such as PCA can inadvertently reproduce ethnic or racial groupings. As discussed in Chapter 4, PCA is ostensibly used to detect and correct for population structure, yet in practice, it is also used to infer or assign ancestry. Computational clustering, therefore, becomes a proxy for categorising samples into groups that map closely onto socially recognisable labels. Methods framed as neutral corrections thus contribute to the reification of distinctions that carry deep social meaning.

Although some participants described PCA as “traditional”, suggesting it may eventually be superseded, it remained central in explicit and implicit discussions of how groups are made and labelled. Alternative methods do exist, including those that model complex ancestral dynamics or rely on allelic distributions rather than predefined population labels. Techniques such as ancestral recombination graphs (ARGs), discussed in section 2.5.2, and newer approaches aimed at removing “crude” labels were cited as innovations, yet they appeared only marginally in the everyday discourse analysed in Chapter 4 and were mentioned only briefly and implicitly across discussions of scientific practice, power and methods analysed extensively in Chapters 5 and 6. This uneven uptake indicates that while alternatives are available, the techniques that become routine are shaped by constraints, expectations, and institutional norms (Dauda et al., 2023). As this research demonstrated, choices are always made in the context of data shaped by social, political, and historical forces.

For instance, thresholds such as the five per cent minor allele frequency (MAF), used to define “common” variants, exemplify how seemingly technical decisions can entrench unequal knowledge production. Largely derived from European samples, these thresholds have become de facto standards that marginalise variants relevant to underrepresented groups (Fatumo et al., 2022; Martin et al., 2019). This exclusion constitutes a form of epistemic injustice, privileging Eurocentric paradigms while rendering other forms of variation illegible (Popejoy & Fullerton, 2016; Sirugo et al., 2019).

Against the backdrop of persistent European overrepresentation in genomic datasets, what is at stake is not simply numerical imbalance but the power relations that shape how genomic knowledge is produced and legitimised. Dataset composition influences interpretability, accuracy, and the visibility of different populations in downstream analyses, reinforcing the methodological asymmetries outlined above.

Study recruitment practices further rely on categorical impositions. Predefined labels, which are not always statistical in origin, determine eligibility and representation (see section 4.2). Attempts to diversify genomic research hinge on representational inclusion, yet such inclusion is constrained by the categories through which it is operationalised (see section 5.3.2). Researchers frequently adopt labels provided by earlier teams, thereby inheriting prior classificatory decisions.

These labels shape what becomes scientifically visible. They influence research questions, the interpretation of findings, and the translation of genomic research into clinical and public health contexts (Popejoy & Fullerton, 2016; Sirugo et al., 2019). Structural constraints deepen these issues. As shown in section 6.1, well-resourced UK cohorts emerged from sustained investment, whereas comparable infrastructures are often absent in settings with higher proportions of non-European ancestry populations. Collaborations beyond established cohorts, therefore, require funding mechanisms not well supported by current grant schemes.

Scientists are thus working with data already shaped by exclusions. Low representation reflects not only statistical underpowering but also the exclusionary nature of recruitment strategies and labelling conventions. As a result, “diverse” datasets may still carry overlooked biases arising from the omission of individuals who do not fit established categories.

8.3.2 EDI: Exclusion, Data Practices, and Inequity

The continued use of inherited population and classificatory systems reflect long-standing structural and epistemic investments. While they may enable statistical analysis, they also obscure lived realities and reinforce reductive or harmful narratives (Morning, 2011; Panofsky & Bliss, 2017). Recruitment protocols, dataset composition, and visualisation tools manifest downstream in recognisable patterns. Variants once classified as pathogenic are reinterpreted when more diverse reference data become available. Polygenic risk scores underperform for individuals whose ancestry is poorly represented, and gene discovery remains concentrated within heavily sequenced populations. What counts as *sufficient*, *rare*, or *significant* is shaped by who is included in reference datasets and who is absent. Technical conventions concerning sample size and representativeness are therefore inseparable from the social and historical conditions structuring genomic evidence.

Datasets may appear inclusive while concealing patterned exclusions. Examining provenance, recruitment strategies, and labelling conventions, along with transparent reporting of limitations, is essential. In some contexts, degrees of missingness or imposed homogeneity should be evaluated as indicators of structural exclusion or empirical traces of marginalisation rather than neutral absences.

In section 6.2, researchers reflected on the assumption that open-source resources automatically translate into equitable access. Recognition that open data infrastructures can reproduce inequities unless supported by meaningful engagement, resourcing, and equitable governance again illustrates how routine and even assumed ethical practices shape patterns of inclusion and exclusion.

The pursuit of improved scientific accuracy aligns with the ethical imperative to include underrepresented populations and reveals what I term methodological interest convergence. Correcting biases becomes necessary not only for equity but also for scientific optimisation. Gains for marginalised groups thus emerge not in opposition to institutional priorities but because achieving those priorities requires greater inclusion.

Methodological routines underpinning genomic research are deeply entangled with classification, representation, and access. These practices stabilise inherited assumptions about human difference and reproduce longstanding inequities, even when framed as neutral or technically necessary. Efforts to refine categorisation and broaden representation reveal how scientific optimisation and equity concerns increasingly converge, yet these gains occur within infrastructures shaped by asymmetries of power, resources, and authority.

The production of genomic knowledge becomes inseparable from the structural conditions governing who is included, how categories are defined, and whose perspectives shape the interpretation of variation.

8.4 Structural Power

This interrogation did not stop at the level of categories and concepts. To understand why racialised practices persist in genomic science, it was necessary to move from individual decision-making to the institutional and structural contexts that enable or constrain more equitable research. Scientific knowledge is not produced in a vacuum; it emerges within an ecosystem shaped by funding structures, institutional hierarchies, peer review protocols, publication incentives, and national and global policy agendas. These forces do not simply

influence research priorities; they condition which questions are asked, who asks them, and whose interests are ultimately served.

Critical realism offered a particularly generative framework for understanding this ecosystem, urging attention not only to what is visible or measurable (the empirical), but also to the systemic enablers and constraints (the actual and the real) that underlie those empirical manifestations. Diversity and inclusion were widely acknowledged as important, yet researchers felt constrained by structural pressures to publish quickly, secure competitive grants, and adhere to conventional markers of rigour.

As discussed in Chapter 7, and as visualised in the system map (Figure 48), the centre of genomic research at Sanger comprised scientific practices focused on data generation, analysis, and interpretation. These were embedded within broader collaborative networks that included public and private-sector partners. Encircling this core were multiple layers of structural influence, including academic collaborators, clinical institutions, funding bodies, regulators, policy makers, and the public. Constraints were therefore actively maintained through institutional practices around funding, publishing, and ethical oversight that tended to favour technical efficiency over structural reflexivity.

8.4.1 Infrastructures of Legitimacy: Journals, Descriptors, and Institutional Hierarchies

One important dimension of structural power was the continued reliance on Eurocentric reference datasets, whose exclusionary effects are increasingly documented (Fatumo et al., 2022). Researchers who sought to expand the representativeness of their studies often encountered formidable barriers, including restricted access to diverse samples and complex legal and ethical protocols (see Chapter 6).

The choices scientists made about population descriptors are shaped not only by personal ideology but also by broader structural forces, including editorial expectations and disciplinary conventions (A. Lewis et al., 2022; Mauro et al., 2022; Sankar et al., 2015). Khan et al. (2022) similarly observe that guidance often stops short of reshaping the incentives that govern day-to-day research practice. Early in my PhD, I engaged directly with these debates by attending the first public National Academies of Sciences Engineering and Medicine (NASEM) meeting. In addition, when I later facilitated a workshop on equity in data with an international group of scientists at Sanger (see section 6:1). A conversation with one of the NASEM report editors highlighted a central issue: the guidelines being developed

were deeply influenced by US-specific classificatory histories, and there was genuine uncertainty about how well these frameworks would travel beyond a Euro-American context. This UK-based case highlights how ostensibly global guidance interacts with distinct national histories of race and governance.

This encounter echoed a point raised in the literature review: while debate about population descriptors in genomics is global (Popejoy et al., 2018) the literature remains overwhelmingly Euro-American in scope. Perspectives from outside these regions and from scholars working within alternative classificatory systems are strikingly underrepresented, a pattern itself a product of structural exclusion.

Publishing norms emerged as a key site where scientific value, career progression, and structural power intersected. Decisions made by journals about what to publish and where have profound implications for researchers' trajectories, funding prospects, and institutional standing. Editorial practices often prioritised methodological standardisation and statistical robustness, leaving limited space for reflexivity, equity, or critical engagement with race and power. Initiatives such as the Journal of the American Medical Association (JAMA) editorial on the reporting of race and ethnicity seek to impose clearer standards, yet also reveal how inconsistent practices remain across journals (Flanagin et al., 2021).

Journals, as gatekeepers of scientific legitimacy, can either passively reflect existing norms or work to reshape them, for example by establishing enforceable guidelines, requiring transparency about population descriptors, and promoting interdisciplinary scholarship that centres equity and justice (Khan et al., 2022; National Academies of Sciences, 2023). As my analysis of the genomic ecosystem showed, institutional power and autonomy are not always aligned, and where they diverge, important tensions emerge in how collaborative science is negotiated and enacted (see Chapter 7).

8.4.2 Funding and Policy

While funders often espouse values of diversity and inclusion, their stated commitments are not always matched by infrastructural support. Interviews repeatedly highlighted funding structures as a central barrier. Scientists described the difficulty of pursuing data diversity and equity-centred research within grant programmes not designed to support such goals (see section 6.1). Building large, well-characterised cohorts in the United Kingdom, such as UK Biobank, has required sustained and substantial investment. Comparable infrastructure was often absent for non-European ancestry populations or was located in settings with far

fewer resources. Efforts to establish collaborations or collect samples from underrepresented populations thus required funding models that current schemes were poorly equipped to provide.

Some funders, particularly newer philanthropic actors such as the Chan Zuckerberg Initiative, were described as rare sources of support for data diversity and inclusion-oriented agendas, for example through calls explicitly focused on ancestries underrepresented in genomic research (see section 6.1). Yet such initiatives remained the exception rather than the norm. Many researchers were motivated by a desire to produce inclusive and ethical science, but found themselves constrained by funding models and institutional logics that rewarded other priorities. The disjuncture between personal commitment and structural constraint reinforced the sense that researchers were navigating an ecosystem shaped more by inertia than by vision.

Regulatory frameworks added another layer of complexity (see sections 7.2.1 and 7.2.3). Oversight was exercised through a diffuse assemblage of government bodies, ethics committees, and professional associations. Researchers often experienced these systems as informal and distant, rooted in tacit practices rather than clearly articulated mechanisms. This did not necessarily signal a lack of regulation, but rather the layered and opaque nature of accountability. Critically, many ethical review processes prioritised issues such as consent and privacy while sidelining questions of justice (see section 6.1). Although these dimensions are vital, they do not address the structural and historical inequalities that shape research participation and vulnerability. Initiatives that centred community engagement, transparent data governance, and critical pedagogy were present, albeit unevenly. Given its stature and resources, the Sanger Institute has both the capacity and the responsibility to lead in this space.

While Sanger cannot directly reshape the global funding environment or dictate funder priorities, it does retain a degree of institutional flexibility in how those priorities are interpreted and operationalised. This autonomy is expressed in decisions about which projects to prioritise, how internal resources are allocated, and how institutional infrastructures align with broader scientific and societal trajectories (see section 7.2.2). As a globally respected hub for genomic research, Sanger occupies a uniquely influential position: it can help redefine what constitutes excellence in genomics, broadening this to include reflexivity, inclusivity, and accountability.

My research intersected with the Institute's obligations, driven by funder mandates, to develop policies on data diversity (see section 1.2.5 and 6.3). Yet I had limited engagement

with how institutional culture would need to change for such policies to be effective. Despite this, my PhD was at times framed as a lever for transformation (Appendix G1 and G2). As a consequence of my embeddedness in the field, I discovered, sometimes indirectly, that my work was referenced in internal meetings as part of the Institute's endeavours to improve data diversity. As my work unfolded, this offered less reassurance than expected, because it exposed both the weight of expectation placed on the project and the degree to which the scale and complexity of the issue were underestimated.

I also found myself invited into decision-making spaces as a presumed diversity and equity expert. Although I brought substantial knowledge from my work as a genetic counsellor and engaged in equity work in clinical genetics, I did not possess the same depth of technical expertise as many of the scientists around me. At times, the expectation that I would resolve these dilemmas felt less like collaboration and more like a subtle deflection of responsibility. While often couched in respect, these moments suggested a quiet displacement of discomfort onto my role.

My empathy for colleagues enabled me to recognise this vulnerability. Attending both to their human limitations and to the structural constraints they navigated opened space to imagine how social justice might function meaningfully within the genomic ecosystem, not simply as an external critique but as a practice embedded in its structures. The pursuit of benevolent outcomes in science is therefore not merely a matter of good intentions. It requires alignment between individual commitments, institutional infrastructures, and policy frameworks.

8.4.3 Access, Power, and Data Solidarity

As touched on in discussing methodological power practices of data sharing are often presented as inherently beneficial, yet they are deeply entangled with structural power. At institutions like Sanger, open-source data sharing remained a foundational principle (Ross-Hellauer et al., 2020), but inequities in access persisted. Researchers based in well-resourced institutions were far more likely to possess the computational capacity, training, and infrastructural support needed to navigate and exploit large genomic datasets. Those without such resources were effectively excluded from full participation in genomic research (Bezuidenhout et al., 2017).

As discussed in Chapter 6, scientists observed that only a relatively small group of highly trained individuals with significant computing power and coding expertise could use large-scale open data in practice. For many others, especially those working in under-resourced

settings, such data remained technically “open” yet materially out of reach. These comments point to a profound ethical challenge: data access is not only a technical matter but a question of justice, closely tied to legacies of colonial science, ongoing economic inequality, and the concentration of intellectual capital in

The ability to collect, store, govern, and openly share genomic data is itself an act of power, one that remains heavily concentrated in well-funded institutions in the high-income countries. These institutions not only control technical infrastructures but also shape the standards, categories, and narratives through which genomic data become legible and legitimate. In this sense, the global distribution of data does not reflect shared ownership or democratic participation, but rather the continuation of extractive and centralised models of knowledge production.

Barbara Prainsack’s concept of data solidarity is particularly relevant here. In the later stages of this research, a meeting with Prainsack helped to crystallise and extend reflections already emerging from my fieldwork about how data, benefit, and justice are linked (Prainsack et al., 2022). Data solidarity shifts attention away from questions solely of data ownership towards the uses to which data are put and who benefits from those uses. Applied to genomic diversity, this perspective challenges the assumption that increasing the diversity of datasets automatically leads to more equitable outcomes. If data from underrepresented communities are incorporated into global resources without meaningful benefit returning to those communities, then inclusion risks reproducing familiar extractive logics.

Ethical practice, viewed through data solidarity, requires more than broadening participation. It demands using insights gained from data to inform more just methodological and governance decisions: for instance, developing better classification systems, deliberately oversampling marginalised populations where appropriate, or choosing not to sample certain groups where risks or histories of harm make participation ethically fraught.

Thus, the pursuit of data diversity in genomics cannot be reduced to encouraging researchers to recruit “more diverse” samples or to deposit data in open repositories. It must confront deeper systemic questions about power, history, and structural inequality that shape which forms of knowledge are produced, by whom, and for whose benefit (Benjamin, 2019).

True collaboration and participation must be grounded in autonomy and reciprocity rather than institutional benevolence. If genomic research is to be ethically legitimate, it must

recognise and actively challenge the structures of power. Without these transformations, efforts towards inclusion risk entrenching inequality, reaffirming rather than disrupting dominant logics of scientific authority.

8.4.4 Power, Positionality, and the Case for a Justice Framework

The preceding sections demonstrated how classificatory practices, methodological choices, and institutional norms shape the production of genomic knowledge. Yet understanding these dynamics requires recognising the broader conditions under which scientists work. Questions of inclusion and justice cannot be examined solely in relation to datasets or methodological standards. They are also anchored in the relational and organisational dynamics that structure genomic research. Power circulates through institutions as much as through algorithms, and positionality influences not only who is present, but whose perspectives are legitimised, whose concerns are amplified, and whose absences are normalised. This section, therefore, brings these dynamics together to show why a more explicit justice framework is required.

Diversity and equity emerged not as discrete themes but as threads running through classification practices, methodological choices, and institutional structures. Sanger was frequently described by interviewees as a diverse space. Yet this narrative did not align straightforwardly with my own experience as a Black researcher, often one of very few *Black* people in genomic settings. Diversity appeared concentrated in junior, training, and administrative roles, with leadership remaining predominantly *White* and male.

These dynamics were not incidental. They reflected what this research has shown more broadly: that race and power are not confined to categories within datasets, but are embedded in institutional culture, collaboration patterns, and leadership structures. Whiteness operated as an unmarked norm, shaping which questions were prioritised, which populations were centred, and which ethical concerns were considered legitimate. At the same time, discourses of diversity and equity in genomics frequently blurred the distinction between visible diversity in teams and deeper structural change. Data diversity and workforce diversity were sometimes treated as interchangeable achievements, creating a false sense of progress even where systemic inequalities in access, authorship, and benefit sharing persisted.

Questions of power and autonomy were also visible in how scientists navigated classification and ethics. Researchers had considerable influence over how categories were applied in

practice, yet their capacity to disrupt inherited racial or ethnic labels was constrained by publishing norms, editorial expectations, and ambiguous guidance on population descriptors. Similarly, while ethical approval and informed consent were routinely invoked, these were often treated as procedural checkpoints rather than occasions for deeper reflection on structural inequality, reciprocity, or benefit sharing. In practice, scientists operated within systems that positioned ethics and diversity-related concerns as adjacent to, rather than constitutive of, rigorous science.

Power in genomics is relational and layered. It is exercised through institutional hierarchies, funding models, publishing infrastructures, and everyday interpersonal dynamics, as well as through classificatory schemes and technical standards. Autonomy is similarly constrained: researchers may wish to challenge racialised logics, but their efforts are shaped by incentives that prioritise speed, scale, and conventional markers of impact over reflexivity, equity, or redistribution.

It is within this context that the need for a more explicit Genomic Social Justice Framework becomes clear. I therefore introduce the JUST DATA framework as a justice-oriented model for rethinking the design, interpretation, and governance of genomic research. Building on Critical Race Theory, decolonial thought, and Critical Realism, the framework places ethics, power, and participation at the centre of genomic practice. It is intended to complement existing guidance by offering researchers and institutions concrete ways to embed equity and accountability throughout the research lifecycle, treating social justice not as a peripheral ideal but as a core condition of scientific integrity.

9 Operationalising Justice from the Start: The JUST DATA Framework

9.1 Introduction

Tracing the journey of human data, and the ways in which it has been systematically shaped and narrowed, makes clear that methodological and statistical practices cannot be separated from the conceptual and structural forces that organise genomic research. The preceding discussion has shown how racialised categories persist through classificatory conventions, how methodological norms reproduce Eurocentric defaults, and how institutional structures shape what counts as rigour, inclusion, and scientific value. These dynamics together underscore that efforts to diversify genomic datasets will remain limited unless grounded in an explicit social justice approach.

The JUST DATA framework responds to this need. It offers a practical, justice-oriented model for embedding equity principles into genomic research, designed to support scientists in working ethically and rigorously with human population datasets and their production. It positions justice not as an external consideration, but as integral to how data are assembled, labelled, analysed, and communicated. The need for such a framework is especially salient in the UK context, where racial discourse within scientific and public domains is often subtle, euphemised, or displaced, yet the underlying operations of categorisation, exclusion, and power within genomics remain comparable to those observed globally. A UK lens reveals a particular challenge: equity work is frequently interpreted as relevant only where overt racism is visible, obscuring the structural mechanisms through which racialised assumptions endure in scientific practice. The JUST DATA framework speaks directly to this gap.

Unlike frameworks such as FAIR (Rocca-Serra et al., 2023) and CARE (Carroll et al., 2020) or critical interventions that call for the expansion or reorientation of ethics in genomics, such as Unbounding ELSI (Adsit-Morris et al., 2023), this framework is intentionally situated and operational. While FAIR and CARE articulate important principles for data stewardship and governance, and Unbounding ELSI offers a powerful critique of how ethics has been historically bounded by the priorities of genomics itself, these approaches do not specify how justice is undermined or reproduced through everyday classificatory, methodological, and institutional decisions. JUST DATA is designed to intervene at the specific institutional, methodological, and narrative decision points through which genomic inequities are

produced and stabilised in practice, including population categorisation, agenda setting, data harmonisation, authorship, and communication. Grounded in ethnographic analysis of a UK genomic institution, the framework translates justice from an abstract aspiration into a set of actionable commitments that respond to the particular histories, organisational dynamics, and racialised silences that structure genomic research in context.

Although rooted in critical race theory, the principles of this framework extend beyond categories of race alone. They apply to any data derived from marginalised or underserved populations, and to any scientific context in which historical inequality shapes who is represented, who is excluded, and whose interests are served.

The framework is built around eight interlinked pillars that correspond to the points of structural vulnerability identified in this thesis. None of these pillars operate in isolation; each is dependent on the others, and each illustrates how equity in genomics requires transformation rather than mere representation. The JUST DATA framework therefore functions as a 360-degree evaluative tool, intended to support both the development of new actions and the critical assessment of existing research practices. As shown below, each pillar is underpinned by four foundational actions: reflective, informed, sincere, and equitable. These actions form the iterative process through which justice-oriented decision-making must move. They are not linear or prescriptive steps, but cycles of ongoing dialogue and evaluation, designed to generate action, assess its impact, and refine future efforts. Crucially, they require that institutions and researchers remain open to critique, responsive to new insights, and prepared to confront inherited norms.

This is not a checklist. It is a transformative approach to how human data are conceived of, accessed, analysed, and shared. It recognises that scientific practices are never neutral, and that achieving justice in genomics demands rethinking how power, knowledge, and ethics intersect across every stage of the research process. If enacted with intention, the JUST DATA framework has the potential not only to improve data diversity but to reshape the values that define excellence within genomic science.

9.2 Architecture of the JUST DATA Framework

The JUST DATA framework builds from the empirical findings presented in this thesis, which show that inequities in genomics arise not only from methodological decisions or data limitations, but from the interaction of conceptual norms, institutional logics, and historical

power structures. The framework therefore integrates ethical inquiry, technical rigour, and structural awareness into a single model capable of guiding both analysis and action.

At the centre of the framework sit eight interlinked pillars, each addressing a distinct point of vulnerability within the genomic research ecosystem: the assumptions that shape scientific judgement; the structures and incentives that organise research; the narratives through which genomics communicates risk and difference; the mechanisms through which trust and accountability are enacted; and the infrastructures that determine who can meaningfully participate in genomic science. While each pillar responds to a discrete set of challenges, none operate independently. They function relationally, reflecting the interdependence of social, ethical, and technical decisions in genomic practice.

These pillars rest upon four foundational actions - Reflective, Informed, Sincere, and Equitable - which form the iterative cycle through which justice-oriented practice must pass. These actions constitute the conditions necessary for the pillars to be applied with integrity. They are not sequential steps, but ongoing commitments to critical awareness, contextual knowledge, transparent communication, and redistribution of power.

A central premise of the JUST DATA framework is that justice cannot be assessed only at the level of outcomes. Equity is not something that can be retroactively added to research outputs; it must be built into the conceptual, methodological, and institutional processes through which genomic knowledge is formed. For this reason, the framework is positioned at the beginning of scientific decision-making, not at the end. It functions as a guide for the foundational choices that shape research design, data generation, categorisation, analysis, and communication. Power and social justice are therefore not themes that accompany the framework, but the conditions that underpin it. The alignment of justice and methodology is what allows genomic research to produce equitable outputs, rather than simply testing equity after the fact.

The framework is intentionally cyclical rather than linear. Genomic research rarely follows rigid stages, and neither do the ethical concerns that accompany it. Instead, the JUST DATA framework offers a structure for revisiting decisions, interrogating assumptions, and refining practices as projects evolve. It is not a compliance tool or checklist. It is a conceptual anchor designed to guide researchers and institutions in evaluating the justice implications of their choices, from project inception to publication, data sharing, and long-term impact.

Figure 58 illustrates this architecture: the eight pillars form the body of the framework, while the four foundational actions operate as continuous forces shaping how each pillar is

enacted. The framework is designed to sit within ongoing institutional processes such as research design, funding applications, ethics review, analytic decision-making, and communication strategies. It can be used to initiate new actions or to critically assess existing ones.

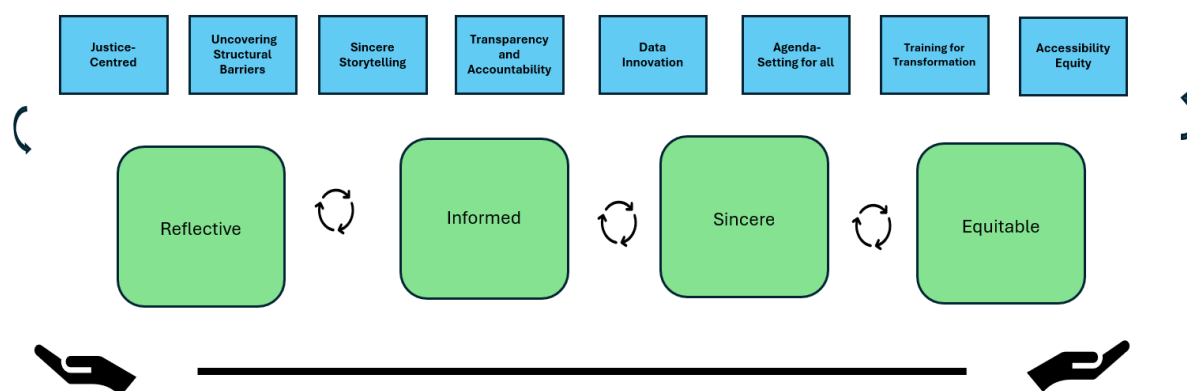


Figure 58: JUST DATA principles that support the four foundational actions of the framework.

In the sections that follow, each foundational action and each pillar is described in detail. For every pillar, the discussion is structured around four elements:

- **Aim** - what the pillar aims to address.
- **Why it matters** - the structural, methodological, or conceptual risks if the pillar is ignored.
- **Empirical grounding** - how issues identified in this thesis demonstrate the need for the pillar.
- **What it involves** - the concrete practices, behaviours, and institutional commitments required to enact the pillar in everyday research contexts.

Because power is unequally distributed within genomic institutions, justice requires not only identifying these asymmetries but actively redistributing decision-making authority, epistemic legitimacy, and material resources. These components form a framework that seeks not only to improve representation in genomic datasets, but to transform the cultural, institutional, and conditions under which genomic knowledge is produced.

9.2.1 Foundational Actions of the JUST DATA Framework

The JUST DATA Framework is built on four foundational actions that operate as the conditions through which justice must be enacted in genomic science: reflective, informed, sincere, and equitable.

Each foundational action consolidates an empirical strand developed earlier: conceptual slippage and classificatory persistence (Chapters 4 and 8.2), communicative discomfort and displaced responsibility (Chapter 5), methodological standards and inherited Eurocentric defaults (Chapter 6 and 8.3), and institutional incentives and infrastructures (Chapter 7 and 8.4).

These actions precede and shape methodological and analytical decisions. They recognise that justice cannot be secured by adjusting outputs at the end of a project. Instead, it must be embedded at the point where scientific judgements are formed, classifications are chosen, collaborations are negotiated, and data practices are designed. Each action responds directly to the structural dynamics illuminated in this research, particularly the operations of power, the persistence of inherited categories, and the institutional tendencies that narrow the possibilities for ethical and inclusive genomic science.

Crucially, these actions are not linear steps. They form an iterative cycle, guiding researchers and institutions through ongoing processes of questioning, assessment, and refinement. They require scientists to remain receptive to critique, open to discomfort, and committed to addressing the deeper historical and political conditions that shape genomic research. Together, they create the foundation upon which the eight pillars of the JUST DATA Framework rest. Without them, the pillars risk becoming technical aspirations rather than drivers of structural change.

Reflective

Reflective practice involves a sustained engagement with the assumptions that underlie genomic research. It requires researchers to interrogate the conceptual frameworks, inherited categories, and normative logics embedded within scientific training and methodological choices. As demonstrated in this thesis, classificatory practices in genomics are shaped by historical racial science, geopolitical defaults, and institutional expectations that elevate clarity and efficiency over complexity and nuance.

This action builds directly on my analysis of how classificatory habits and the normativity of whiteness shape what becomes thinkable in genomic practice, even when race is formally disavowed. (Chapters 4, sections 5.2 and 8.2.4.).

These pressures can lead scientists to reproduce racialised patterns of categorisation even when they recognise their limitations.

Reflection is required here as a one-off ethical gesture, but also as a continuous process of asking how race, power, and inequality shape the scientific imagination. Reflective action positions justice as a scientific responsibility and demands that researchers continually re-examine the conceptual scaffolding of their work.

Informed

Informed action requires that research decisions be grounded in both technical expertise and context-sensitive knowledge. This includes historical awareness, engagement with lived experiences, and familiarity with the social and political contexts from which data emerge.

To be informed is therefore to recognise that scientific competence alone is insufficient. It responds to the gaps I observed in researchers' historical and ethical grounding, where race was treated as external to science and guidance was often unknown, unused, or experienced as outside participants' roles (as discussed in section 5.3.3; 8.2.2; 8.2.5 and Chapter 6).

It requires integrating insights from social sciences, humanities, and communities affected by research. This action challenges the epistemic hierarchies that have historically marginalised non-biological forms of knowledge within genomics and positions interdisciplinary learning as essential to scientific rigour. It also acknowledges that informed decision-making is necessary for developing methodological approaches that do not reproduce historical inequities.

Sincere

Sincere action calls for transparency, accountability, and candour in how scientific decisions are justified and communicated. Sincere here is an intentional choice chosen as a call to both genuine and serious attention.

This action is grounded in my findings that ethics and governance were often treated as procedural checkpoints, while responsibility for the downstream consequences of classification and communication was frequently displaced (sections 8.2.5, 8.3, 5.2, and 5.3).

Such proceduralism can obscure the deeper questions of justice, reciprocity, and structural inequality that accompany work with human population data.

Sincerity requires scientists and institutions to aim beyond minimal compliance. It includes acknowledging the limitations of datasets, the exclusions built into recruitment and sampling strategies, and the historical forces shaping the availability of particular forms of data.

Sincere action is central to building trustworthiness in genomic research, particularly for communities who have historically been harmed or marginalised by scientific institutions. It embeds ethical responsibility not as a bureaucratic obligation but as a public commitment to openness, humility, and accountability.

Equitable

Equitable action is grounded in the principle that justice in genomics demands the redistribution of power, opportunity, and material resources. It recognises that representation alone cannot resolve the structural inequalities that shape who participates in genomic science and who benefits from it. This study revealed inequities particularly for researchers in countries with lower financial capacity and resources than the UK. It draws on the structural constraints documented across funding, publishing, and data infrastructures, where inclusion was pursued within Eurocentric defaults and unequal capacity, limiting who can lead, interpret, and benefit from genomic research (as analysed in sections 6.1, 6.2; sections 8.4.1 to 8.4.3 and Chapter 7).

Equitable action requires confronting these asymmetries directly. It includes supporting collaborations that build capacity rather than extract data, ensuring that communities can meaningfully influence research agendas, and addressing the institutional hierarchies that limit participation and leadership for marginalised groups. It demands not only fair distribution of benefits, but fair distribution of decision-making authority. Equitable action is essential because it ensures that justice is not treated as an aspirational value but as a practical commitment embedded in the structures of genomic research itself.

These four foundational actions constitute the ethical grounding of the JUST DATA Framework. They establish the conditions under which justice can shape scientific practice from the outset, ensuring that the pillars of the framework are not adopted superficially but enacted in ways that meaningfully address power, inequity, and the historical legacies that structure genomic research. They provide the lens through which methodological, institutional, and communicative decisions must be evaluated, and create the iterative cycle

through which research practices can evolve towards greater justice, responsibility, and inclusivity.

9.2.2 Pillars of the JUST DATA Framework

1. Justice-Centred Habits (J)

Aim

This pillar places justice at the centre of scientific reasoning. It calls for a shift from episodic ethical reflection to a sustained practice of examining the assumptions that underpin genomic work, including the racialised logics, normative defaults, and inherited categories embedded in training and methodology.

Why it matters

Without justice-centred habits, scientific practices risk reproducing the biases, omissions, and classificatory legacies documented throughout this thesis. Embedding justice into everyday decision-making ensures that methodological choices are guided by structural awareness, not convenience or tradition. This pillar establishes the ethical orientation necessary for all other elements of the framework to function meaningfully

Empirical grounding

- Scientists acknowledged tensions around racialised categories but felt constrained by institutional norms.
- Reflexive engagement appeared informally but was not widely institutionalised. ref
- Leadership homogeneity limited opportunities for justice-oriented cultural transformation.

What it involves

Justice-centred habits require researchers and institutions to confront the epistemic conditions in which genomic science operates, to recognise the histories that shape present-day data practices, and to cultivate cultures where questioning entrenched norms is a collective responsibility rather than a personal burden. It positions reflexivity not as an

optional add-on, but as a fundamental scientific habit that guides how problems are framed, how populations are conceptualised, and how findings are interpreted.

2. *Uncovering Institutional Barriers (U)*

Aim

This pillar focuses on identifying the structural and organisational conditions that shape scientific possibility.

Why it matters

Institutional structures profoundly shape research outcomes. Unless these barriers are made visible and addressed, justice-oriented intentions will be continually overridden by systemic constraints. This pillar shifts responsibility from individuals to institutions and underscores that equity requires structural change, not isolated acts of inclusion.

Empirical grounding

- Funding models and publication incentives prioritised speed and novelty over reflexivity and equity.
- Researchers were structurally tied to inherited datasets and restrictive editorial norms.
- Power remained centralised in homogenous leadership structures that influenced agenda-setting and collaboration.

What it involves

It asks institutions to scrutinise how their funding structures, recruitment patterns, leadership demographics, authorship norms, and ethical governance processes enable or constrain justice-oriented research. Uncovering institutional barriers means recognising that inequity persists not because individuals fail to care, but because systems reward certain forms of productivity, knowledge, and authority while marginalising others. It requires institutions to examine how default practices become naturalised, how decision-making power is concentrated, and how scientific agendas are shaped by forces that are not neutral.

3. *Storytelling and Power Analysis (S)*

Aim

This pillar foregrounds scientific communication and recognises that genomic narratives can reinforce racialised interpretations or challenge them, depending on how they are crafted.

Why it matters

Narratives shape how genomic research is understood by clinicians, policymakers, communities, and the public. Unexamined narratives risk reinforcing racialised thinking and obscuring the social and historical processes that shape genetic variation. This pillar ensures that communication becomes an ethical practice, grounded in transparency, accuracy, and accountability.

Empirical grounding

- Publications often used racial or ethnic labels without contextual explanation.
- Ethical and historical considerations were often omitted from published work.
- Simplified or deterministic interpretations were sometimes amplified through institutional communication channels.

What it involves

Genomic research does not simply generate findings; it tells stories about populations, risk, identity, and human difference. Storytelling and power analysis require scientists to interrogate the narratives they construct, to be transparent about how categories were defined, which assumptions informed analytical choices, and what has been omitted or simplified. It calls for careful attention to language, context, and interpretation, ensuring that research communication does not naturalise social hierarchies or reproduce inequities

4. *Trust, Transparency and Accountability (T)*

Aim

This pillar emphasises that trust in genomic research must be actively cultivated through transparency and accountability rather than assumed as a by-product of neutrality or technical competence.

Why it matters

Without accountability, trust remains fragile and easily undermined, particularly for communities historically excluded or harmed by genomics. This pillar ensures that institutions move beyond procedural ethics toward relational responsibility, strengthening the legitimacy, transparency, and social value of genomic research.

Empirical grounding

- Ethical processes are more often treated as procedural obligations rather than substantive engagements.
- Researchers expressed uncertainty about who held responsibility for justice-oriented decisions.
- Communities and collaborators had limited visibility within institutional accountability structures.

What it involves

It calls for governance structures that extend beyond formal ethics review to include community oversight, transparent institutional decision-making, and internal mechanisms that hold researchers responsible for the social implications of their work. Shared accountability recognises that trust is relational; it is built when institutions acknowledge past harms, demonstrate openness in their practices, and distribute responsibility across leadership, researchers, and broader governance systems. This pillar promotes a research culture where accountability is visible, collective, and grounded in ethical responsibility.

5. Data Innovation (D)

Aim

This pillar focuses on reimagining the technical and statistical foundations of genomic science to promote equity, accuracy, and inclusivity.

Why it matters

Genomic findings are only as reliable as the categories, samples, and algorithms that shape them. Without equity-driven innovation, genomic science risks reinforcing the very biases it

seeks to correct. This pillar ensures that methodological reform becomes a central pathway to justice by aligning technical accuracy with ethical responsibility.

Empirical grounding

- Existing population descriptors often failed to capture lived identities or demographic realities.
- Scientists described methodological constraints tied to inherited datasets and dominant statistical tools.
- Institutions rarely recognised the development of equity-focused methods as legitimate scientific work

What it involves

It calls for innovation that does not simply increase representation but interrogates the conceptual and structural biases embedded in population categorisation, sampling strategies, and analytical models. Data innovation requires scientists to develop tools that quantify diversity, detect bias, and challenge inherited assumptions about human variation. Data integrity involves transparency in how descriptors are selected, how limitations are understood, and how methodological choices reflect ethical and contextual considerations. This pillar situates methodological reform as a core scientific responsibility, not an optional ethical enhancement.

6. *Agenda-Setting for All (A)*

Aim

This pillar asserts that justice in genomics cannot be achieved if research priorities are defined solely by funders, institutional leaders, or traditional scientific agendas.

Why it matters

Without inclusive agenda-setting, genomic research risks reinforcing global inequities and failing to address the needs of those most affected by its outcomes. This pillar supports a shift toward shared authority, enabling research that is both ethically grounded and scientifically relevant to diverse populations.

Empirical grounding

- Community perspectives not visible in agenda-setting processes.
- Collaborations across less resourced financial contexts contributes to inequity of outcomes.
- Researchers described the dominance of funder-driven priorities over equity-oriented goals.

What it involves

It calls for structurally embedding the perspectives of communities historically marginalised or underrepresented in genomic data. Agenda-setting for all requires meaningful participation from marginalised voices at the earliest stages of project development, extending through funding applications, ethics review, analysis, and dissemination. It draws from participatory research traditions and emphasises co-creation of scientific priorities, ensuring that genomic research addresses real-world needs rather than simply aligning with existing power structures

7. Training for Transformation (T)

Aim

This pillar recognises that long-term structural change in genomics depends on education that integrates ethical, historical, and social perspectives alongside technical expertise.

Why it matters

Without comprehensive education, justice remains aspirational rather than actionable. This pillar ensures that researchers have the tools, knowledge, and institutional support needed to apply the framework, enabling cultural change that can sustain justice-centred scientific practice across generations.

Empirical grounding

- Training on race, ethics, and classification was inconsistent and or felt absent.
- Ethical responsibility was disproportionately placed or taken up by junior or marginalised staff.

- Scientists expressed uncertainty about how to navigate the ethical and social dimensions of their methodological choices.

What it involves

Training for transformation calls for embedding critical genomics education across all levels of scientific work, including leadership, early-career researchers, administrative staff, ethics boards, and communication teams. It challenges institutions to move beyond optional workshops or isolated modules and instead cultivate a sustained culture of learning. This pillar emphasises that scientists must understand the historical legacies of race science, the ethics of data practices, and the epistemic dimensions of categorisation to engage meaningfully with justice-oriented research.

8. Accessibility and Equity (A)

Aim

This pillar highlights that equity in genomics requires more than the availability of data; it requires equitable capacity to use, interpret, and benefit from genomic resources.

Why it matters

Technical openness without equitable capacity reinforces global hierarchies in knowledge production. This pillar ensures that genomic data systems are designed to support fair participation and benefit-sharing, preventing the reproduction of extractive or exclusionary models of scientific practice.

Empirical grounding

- Researchers in lower-resourced countries faced structural barriers to meaningful data use.
- Open data initiatives did not account for disparities in computational capacity or training.
- Data governance frameworks often reproduce existing inequities by contributing to methodological choices because of differences in access.

What it involves

Accessibility and equity in data use call for infrastructure, training, computational resources, and governance structures that support meaningful participation across diverse research environments. This pillar challenges simplistic assumptions about open access and emphasises that access must be understood relationally: who can work with the data, who can act on it, and who remains excluded.

9.2.3 Bringing the Pillars Together: How JUST DATA Operates as a Cycle

The JUST DATA Framework is designed as an integrated, iterative model in which the eight pillars and four foundational actions work together to shape ethical, methodological, and institutional decisions in genomic research. Rather than functioning as isolated components, the pillars form a coherent structure that addresses the conceptual, methodological, and structural challenges identified throughout this thesis. When viewed alongside the framework graphic (Figure 59), their interdependence becomes clear. Justice-centred science emerges not from adopting a single pillar, but from understanding how each reinforces the others within a continuous cycle of practice.

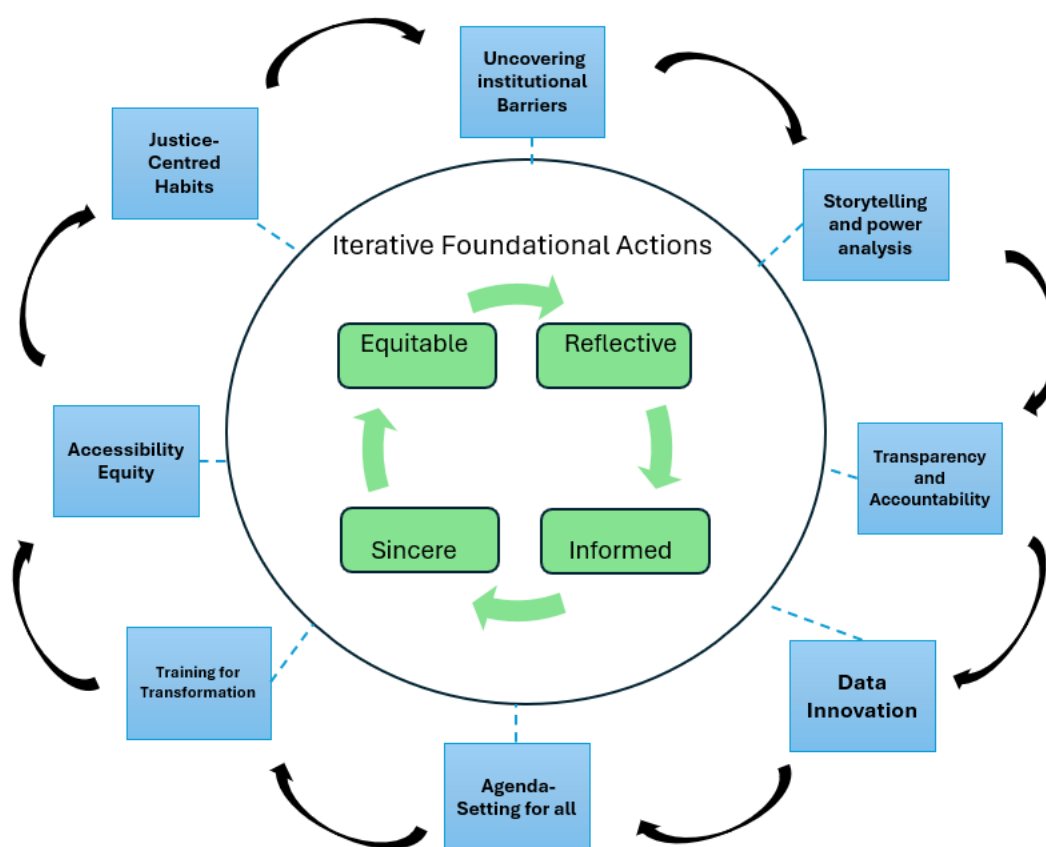


Figure 59 shows the interdependence of pillars and action as a continuous cycle.

At the centre of this model are the four foundational actions: reflective, informed, sincere, and equitable. These actions operate across every pillar, guiding the choices researchers and institutions make when framing research questions, selecting categories, designing methods, communicating findings, and building collaborations. They ensure that the framework intervenes at the start of the scientific process, shaping the assumptions, habits, and structural conditions that precede methodological decisions. By repeatedly engaging in these actions, researchers build the capacity to recognise how power operates in data production and to embed justice directly into scientific reasoning.

The eight pillars provide the practical architecture for this process. Each addresses a distinct point of structural vulnerability in the genomic ecosystem, ranging from classification practices and institutional constraints to communication, collaboration, education, and data access. Taken together, they address the full lifecycle of genomic research: how data are conceptualised, generated, categorised, analysed, and circulated. The cycle therefore

reflects the iterative nature of scientific work itself. As projects evolve, so too must the ethical and methodological choices that shape them. The framework provides a means to revisit decisions, refine assumptions, and identify where justice has been sidelined or constrained.

The graphic illustrates how this cycle loops back on itself. Researchers may enter at any pillar, but work becomes transformative only when the full cycle is engaged. For example, innovations in statistical methods (Data Innovation and Integrity) cannot realise their ethical potential without justice-centred habits, transparency in communication, equitable agenda-setting, and meaningful access to data. Similarly, efforts to involve marginalised communities require institutional reform, training for transformation, and an honest analysis of where power sits within existing systems. The cycle makes it clear that justice in genomics is not a discrete outcome but an ongoing process requiring coordinated attention across multiple dimensions of scientific practice.

In sum, the JUST DATA Framework provides a way of thinking and working that embeds justice into the everyday fabric of genomic research. Bringing the pillars and foundational actions together into a cyclical model offers a practical route for addressing the inequities and power dynamics that shape genomic science and for guiding the development of research practices that are both methodologically robust and socially responsible.

9.3 Limitations and Future Research

This thesis has developed a justice-oriented analysis of genomic research through a UK-based institutional case study. Its strength lies in the close ethnographic and critical examination of how race, ethnicity and ancestry are conceptualised, operationalised and negotiated within a major genomic research institution. However, this situated focus also creates important limitations. The findings should not be read as universally generalisable to all genomic institutions, national contexts or research cultures. Rather, the thesis offers an empirically grounded account of how power, classification and equity operate within a specific institutional and geopolitical setting.

A central limitation is that the research is UK-focused. The analysis is shaped by the histories, policy environment, institutional arrangements and racial formations of the United Kingdom. Race and ethnicity are not stable or universal categories; they are produced through particular histories of empire, migration, governance and social classification. The ways in which race is avoided, displaced or reintroduced through ancestry in the UK may therefore differ from how similar processes unfold in the United States, continental Europe, Africa, Asia, Latin America or other genomic research settings. The UK focus is justified

because it enables detailed examination of a context where race is often made less explicit, while racialised logics remain embedded in scientific and institutional practice. However, future research should explore how the JUST DATA framework travels across different national, regulatory and historical contexts, and whether its principles require adaptation in settings with different classificatory systems, community relationships and histories of scientific extraction.

A second limitation is that this thesis is context-specific. The empirical work is grounded in one institutional environment and in the practices, discourses and structures encountered there. While the Wellcome Sanger Institute is globally influential and therefore a valuable site for examining the organisation of genomic science, it cannot represent the full diversity of genomic research institutions. Other settings may have different funding arrangements, governance structures, community engagement models, leadership cultures or relationships with clinical translation. This specificity is not a weakness; it reflects the methodological commitment of the thesis to understanding how power operates in practice, within concrete institutional arrangements. Nonetheless, future research could test, extend and refine the framework through comparative studies across multiple institutions, including clinical genomics services, biobanks, commercial genomics companies, community-led research initiatives and research settings outside high-income countries.

A third limitation concerns the thesis's primary focus on race, ethnicity and ancestry. This focus was necessary because the research question centred on the persistence of racialised logics in genomic classification and because race remains a particularly powerful and historically loaded organising construct within genomics. The thesis therefore deliberately foregrounds race, racialisation, whiteness and the politics of ancestry. However, this means that intersectionality and the wider range of human diversity were not examined in equal depth. Dimensions such as disability, sex, gender, sexuality, class, religion, migration status, language, geography, age and socioeconomic position also shape participation in research, access to genomic medicine, experiences of classification and the distribution of benefit and harm. These dimensions do not operate separately from race but intersect with it in complex ways. Future research should therefore examine how the JUST DATA framework can be applied intersectionally, and how its pillars might need to be expanded or adapted to account for overlapping forms of marginalisation.

Relatedly, the framework has not yet been empirically tested against other measures or dimensions of diversity. JUST DATA was developed from the findings of this thesis and is therefore grounded in the empirical realities of race, ethnicity, ancestry and institutional power in genomics. Its broader relevance lies in its claim that justice must be embedded at

the level of research design, categorisation, governance, analysis and communication. However, future work should examine whether the framework is equally useful when applied to other forms of exclusion or underrepresentation, such as disability inclusion in genomic research, gendered patterns of diagnosis, socioeconomic barriers to participation, geographic inequities in data access, or the exclusion of linguistically minoritised communities. Such work would help determine whether JUST DATA functions as a general justice framework for genomics or whether it requires distinct adaptations for different domains of inequality.

Future research could also evaluate the framework in practice. This might involve piloting JUST DATA within funding applications, ethics review processes, cohort design, data access committees, publication guidelines or institutional strategy. Researchers could assess whether the framework changes the questions asked at the start of projects, improves transparency around population descriptors, strengthens community accountability, or supports more equitable collaborations. Longitudinal studies would be particularly valuable for examining whether the framework influences institutional culture over time, rather than being adopted superficially as another statement of values. Its usefulness should also be assessed by those most affected by genomic inequities, including communities whose data are collected, researchers working in lower-resourced settings, and professionals involved in translating genomic findings into care.

Although this thesis does not claim to offer a universal solution to all inequities in genomics. It offers a situated, theoretically informed and empirically grounded framework for recognising how justice is shaped by power, classification, institutional structures and methodological choices. Its future value will depend on whether it can be critically adapted, tested and refined across different contexts and forms of diversity. In this sense, JUST DATA should be understood not as a finished model, but as a starting point for building genomic science that is more reflective, accountable and responsive to the full complexity of human difference.

9.4 Conclusions

This thesis has shown that genomic science is shaped by practices of classification, methodological norms, and institutional structures that reproduce longstanding patterns of inequality. The JUST DATA Framework developed here offers a practical way to intervene in these processes, positioning justice not as an outcome to be checked at the end of research,

but as a principle that must guide scientific decisions from their earliest stages. By bringing together eight pillars and four foundational actions, the framework reframes excellence, rigour, and responsibility in genomic research as inseparable from social justice.

The implications for genomic practice and policy are significant. Embedding justice into research design challenges institutions to reconsider how they define merit, allocate resources, structure collaborations, and engage with the communities whose data they depend on. It aligns with global movements that call for more responsible and inclusive science yet attends to the specific histories and power relations that shape genomic research in the United Kingdom. Institutions such as Sanger are well positioned to adopt models like JUST DATA, not only to improve the quality and relevance of their work, but to help lead the field towards a more equitable future.

Justice in genomics cannot be achieved through representation alone. It requires confronting the structural and epistemic conditions that shape how human difference is defined, measured, and valued. This thesis argues that genomic science must engage with these challenges directly, recognising that every methodological choice is also an ethical one. The promise of genomics lies not only in its ability to map variation, but in its potential to contribute to a more just scientific and social world. The responsibility now lies with researchers, institutions, and policy-makers to act with intention, to question inherited assumptions, and to build systems of knowledge that reflect and respect the full diversity of human experience.

The contribution of this thesis is therefore twofold: it demonstrates how structural and methodological forces shape inequities within genomic science in a UK institution, and it offers a framework through which justice can be operationalised from the outset of scientific decision-making

Representation can be celebrated while substantive inclusion remains limited, and scientific authority continues to be shaped by the norms and comfort of whiteness. This reality underscores why justice in genomics must move beyond representation and toward a redistribution of power, knowledge, and opportunity. The JUST DATA framework is an invitation to this work: to reimagine genomic science not only as a technical endeavour, but as a collective ethical project grounded in responsibility, solidarity, and care.

10 Appendices

10.1 Appendix A Ethics documents

Ethics Approval was granted by the Faculty of Education Research Ethics Committee (REC) and the Higher Degrees Office of the University of Cambridge.

10.2 Appendix A1

Faculty of Education

Research Ethics Review Process - Principles

The Faculty of Education Research Ethics Review Process follows stages and principles already well embedded within Faculty practices.

The Faculty of Education Research Ethics Review Process recognizes the diversity of research carried out in the Faculty. This includes research designs involving, but not limited to, qualitative research, quantitative research, mixed-methods research, theoretical research, research conducted on or with existing artefacts (e.g., creative works, literary sources, historical document analysis, literature reviews, or meta-analyses), secondary data analysis, and others.

The Faculty of Education Research Ethics Review Process aims to ensure that important yet ethically risky research can still be conducted, with due consideration for a range of issues.

Underpinning the Faculty of Education Research Ethics Review Process are five key research ethics principles: Autonomy; Beneficence; Nonmaleficence; Justice; and Independent Review.

The Faculty of Education Research Ethics Review Process follows guidelines and best practices currently outlined in the *Ethical Guidelines for Educational Research* of the British Educational Research Association (BERA), and any future updates.

Research Ethics Review Process

The Faculty's Research Ethics Review process

The Faculty of Education Research Ethics Review Process follows the principles outlined above and consists of **three stages**.

All three stages are underpinned by the initial completion of this Research Ethics Review Form and assembly of all accompanying documentation, as detailed in the Submission Checklist on the last page of the Form.

Stage 1 consists of the **submission** of the Form and all accompanying documentation to an independent knowledgeable person of standing, **and a discussion** with this person. This should be an individual in the Faculty of Education with relevant research expertise who would be deemed to be independent of the research being reviewed, that is who would not be deemed to have a conflict of interest with either the researcher(s) making the submission, or the research being submitted for review.

This Stage 1 discussion may result in one of two outcomes:

1. Ethical endorsement for the research granted (potentially after the submission of further supporting documentation) if no further ethical concerns are raised; or
2. Recommendation for Stage 2 review (potentially after the submission of further supporting documentation) if minor ethical concerns or clarifications are raised (at the discretion of the person undertaking the review).

Stage 2 consists of a **discussion** of any ethical dimensions of your research as evident from Stage 1 with the initial independent knowledgeable person **and another individual**. This individual should be an expert in either the substantive topic of the research, and/or its methodological design as relevant to the range of ethical dimensions to be discussed.

This Stage 2 discussion may result in one of two outcomes:

1. Ethical endorsement for the research granted (potentially after the submission of further supporting documentation) if no further ethical concerns are raised; or
2. Recommendation for Stage 3 review (potentially after the submission of further supporting documentation) if unresolved or complex ethical issues remain.

Stage 3 consists of the lead researcher obtaining formal ethical endorsement through the Faculty of Education's Standing Panel for Ethics, convened as per Policy R001/2019.

In line with Policy R001/2019, Stage 3 may result in one of three outcomes:

1. Ethical endorsement for the research granted (potentially after the submission of further supporting documentation) if no further ethical concerns are raised; or
2. Recommendation for the submission to be forwarded to a School or University level ethics committee; or
3. Ethical endorsement for the research not granted. This means that the research cannot be undertaken in its submitted form. If a subsequent submission is to be made, full reconsideration of the research design and its ethical implications must be carried out in advance, and a new form completed and put through all three stages above.

After ethical endorsement has been obtained

Once ethical endorsement has been obtained, the form and all accompanying documentation should be lodged with the appropriate office, by email, as follows:

For undergraduate students, this is the ethics coordinator of the relevant paper.

For graduate students, this is the Graduate Office: graduate@educ.cam.ac.uk. Please also observe any route-specific arrangements communicated to route students and staff.

For staff, this is the Research Office: Research-Office@educ.cam.ac.uk. Each endorsed ethics submission lodged with the Research Office will be assigned a number, which will be communicated to the lead researcher/principal investigator.

In the event of changes to a study that has already received ethical endorsement

On occasion, studies may undergo changes after ethical endorsement has been received.

In this case, the primary investigator of the study (as named in the original ethics submission) must revise the originally submitted ethics form with updated information, retaining any information that has remained the same and updating all answers to reflect the new study parameters. If the change relates to the primary investigator changing, then the new primary investigator must note this change and revise the originally submitted form.

The revised form should then enter the research ethics review process. In the interest of robust review, the process should start from the beginning, with Stage 1. There is no guarantee that ethical endorsement will be given at the same Stage as for the initial submission.

Constituencies for whom The Faculty of Education Research Ethics Review Process does not apply

The Faculty of Education Research Ethics Review Process applies to all constituencies in the Faculty of Education except PGCE students and PPD students, whose courses have specific arrangements for ethical review and securing ethical endorsement. These processes are aligned with the above principles but recognize the needs and nature of research carried out in these programmes.

10.3 Appendix A2

Research Ethics Review Form

1. Title of the Project
Race, Ethnicity and Genetic Ancestry in Genomic Research

2. Please Confirm	
<i>Projects involving pharmaceuticals, drugs, nutritional supplements or collecting human tissue CAN NOT be reviewed by the Faculty of Education. If you cannot answer YES to both questions below, please liaise with the Research Office for suitable alternatives.</i>	
This research does not include the administration of pharmaceuticals, drugs, nutritional supplements or related compounds.	☒ Yes
This research does not include the collection of human tissue samples (e.g., saliva, hair, urine, blood, breast milk).	☒ Yes

3. Primary Investigator			
<i>The primary investigator is the person who has overall responsibility for the study. If you are a PGCE or PPD student, please consult your respective course manager or course leader for the appropriate ethics review form to complete.</i>			
Name	Sasha Henriques		
Email	SH2161@cam.ac.uk	Phone	07946311060
Faculty of Education Role (select any that apply)	☐ University Teaching Officer ☐ Senior Teaching Associate ☒ Student	☐ Senior Research Associate ☐ Teaching Associate ☐ Visiting Scholar	☐ Research Associate ☐ Other staff (describe below) ☐ Other (describe below)
If Other staff/Other, please describe:			
If Primary Investigator is a student, then please indicate level and relevant assignment:			
☐ Undergraduate non-Dissertation coursework	☐ Undergraduate Dissertation	☐ Masters Dissertation	☐ Doctoral Pilot study ☐ Doctoral study ☒ Other (describe below)
If Other, please describe:			

4. Supervisor / Host			
<i>If the primary investigator is a student or visitor, then their supervisor (for students) or host (for visitors) must be listed here. The supervisor/host will be asked at the end of the form to confirm that they have reviewed the ethics submission with the student/visitor.</i>			
Name	Anna Middleton		
Email	am2624@cam.ac.uk	Phone	01223492391

Faculty of Education Role	☐	University Teaching Officer	☐	Senior Research Associate	☐	Research Associate
	☐	Senior Teaching Associate	☐	Teaching Associate	☒	Other staff (please describe)
	☐	Doctoral Student	☐	Other (please describe)		
If Other staff/Other, please describe:						

5. Co-Investigators

List the names of all other researchers involved in the study. Individuals collecting the data must be appropriately trained to handle foreseeable adverse events, such as a participant being injured or becoming emotionally distressed.

Please copy and paste rows to match the number of co-investigators involved in the study. Please provide all requested details for all co-investigators. For large projects or projects with many Co-Investigators, you may attach a separate file with all relevant details.

Name	N/A
Email	
University / Organisation	
Department/Centre	
Role	
Relevant Training / Experience	

6. Start and End Dates

The start and end dates of the research.

Start dates must precede direct engagement with participants.

For students, start dates will normally refer to the start of fieldwork but may be earlier than that; and end dates will normally refer to the deadline for the submission of the assignment (e.g. dissertation deadline for undergraduate or Masters students, end of registration period for doctoral students).

If exact dates are unavailable, please give approximate dates and explain why.

Studies with a duration of longer than 5 years must submit for ethical review every 5 years. If the study duration is longer than 5 years, please indicate the formal end date and ensure that a further ethical review submission is made.

Start Date	18 Jan 2022	End Date	18 Jan 2025
Provide explanation if exact dates not known			

7. Purpose and Rationale

Briefly describe the purpose and rationale of the research. Please limit the response for this question to 200 words.

If you have a formal study proposal which outlines these clearly, you may submit that instead of filling in this section, but it must be clear what the purpose is.

See Study Protocol

8. Methods and Procedures

Describe what will happen in the research. Any studies involving new data collection should make clear exactly what will happen to participants. Include information about any interventions, interview schedules, duration, order and frequency of assessments. Additional and relevant material can be submitted with this form (e.g., questionnaires, supporting information, grant proposals, etc.) Please limit the response for this question to 1000 words.

If you have a formal study proposal which outlines these clearly, you may submit that instead of filling in this section, but it must be clear what the methods and relevant procedures are.

See Study Protocol

9. Project Funding

Indicate any sources of funding for the research. Graduate students awarded scholarship(s) for their course should list that source as a funder here.

(select all that apply)

- | | | |
|--|--|---|
| ☐ Non-funded | ☐ University funds | ☐ College funds |
| ☐ UK Research Council | ☐ Charitable Trust | ☒ Corporate Funding |
| ☐ UK Government | ☐ Non-UK Government | |

☐ Other funding (please specify):

If the research is funded, then name the specific funder(s) and the grant award number(s), where available.

10. Research Location

Where will the research be conducted?

Will this research occur within the United Kingdom? ☒ Yes ☐ No

Will this research occur in any countries outside of the United Kingdom? ☐ Yes ☒ No

If Yes, then list all countries:

Where will the research be conducted? Please select all that apply

☐ University of Cambridge

If University of Cambridge is selected, then specify:

- | |
|--|
| ☐ Faculty of Education |
| ☐ other University Department / Centre (please specify) |
| ☐ a Cambridge College (please specify) |
| ☐ other University location (please specify) |

☐ other university ☐ primary / secondary school ☐ nursery school or childcare centre ☒ a community location (e.g., library, community centres, etc.) Please specify here: ☒ a professional setting (e.g., a company, institution, charity, etc.) Please specify here: ☐ participant's home (or student room at University)	
If Participant's home is selected, then will research team enter participant's home?	
☐ Yes ☐ No	
☐ other location. Please specify here:	
<i>If the research will occur in an international location, then are there any ethical frameworks that need to be adhered to? Note – when a research project falls under multiple ethical frameworks, then all should be adhered to; in cases where the frameworks differ, then the stricter guidance should be followed.</i>	
<i>If the research will occur in the primary place of work for any researchers not employed by the University, then identify who the gatekeeper(s) of the research are. Please list their position(s) briefly below and, where this is not self-evident, describe the nature of their relationship with those on whose behalf they are giving 'informed consent'. You as the researcher cannot act as the gatekeeper. You may attach a separate file with all relevant details if there are more than one gatekeepers.</i>	
Gatekeeper Name	1) Jack Monaghan 2) Shamita Sharmacharja 3) Lisa Anderson
Gatekeeper position(s) as relevant for the research location	1) Science Engagement and Enrichment Manager, Wellcome connecting science 2) Curator, Wellcome Collection 3) Managing Director, Black Cultural Archive
Gatekeeper Email	1) jack.monaghan@wellcomeconnectingscience.org 2) Lisa.Anderson@bcaheritage.org.uk 3) s.sharmacharja@wellcome.org
Gatekeeper Phone	1) +44 (0) 01223 495332 2) 02076118822 3) +44 (0) 20 3757 8513
Will the research team inform people in the research context of the identity of the gatekeeper?	
☒ Yes ☐ No	

11. Artefacts and Secondary Data	
<i>Will the research involve existing artefacts, already published findings or secondary data analysis?</i>	
Does the research involve artefacts (e.g., creative works, literary sources, historical document analysis, literature reviews, meta-analyses)?	☐ Yes ☒ No
Does the research involve secondary data analysis of an existing data set?	☐ Yes ☒ No
If the research involves artefacts, then:	
Identify the source of the artefact(s):	

Describe any agreements needed to source and use the artefact(s) for the purposes of the research:	
What assurances do you have that the artefact(s) were generated ethically?	
If the research involves secondary data analysis, then:	
Identify the source of the secondary data:	
Describe any user agreements needed to access the data for the purposes of the research:	
What assurances do you have that the original data were collected ethically?	

12. Human Participants	
<i>Will the research involve the collection of new data from human participants?</i>	
Does the research involve new data collected from human participants?	☒ Yes ☐ No
<i>If the research involves new data collection from human participants, then answer the following questions about the participants:</i>	
How many participants will be involved in your research?	6 interviewed. (ethnography observation over 50 scientists and public groups)
What are the ages of the participants involved?	15 – 25+
How will participants be selected?	Scientists are selected purposefully based on their scientific research and stage of career. Public participants are those that are actively engaging with art projects around genetic identity and ethics.
What strategies are in place in terms of selecting participants equitably in terms of gender, race, ethnicity, other protected characteristics, or any other characteristics of relevance to the research?	With the small number of scientists, equitable recruitment is initially employed through institutional policy and further attempted by the researcher's selection approach of openness, and fair accessibility to be part of the study.
Will participants be excluded from the research for any reason? If yes, then justify why.	
Recruitment Procedures	
<i>If the research involves new data collection from human participants, then describe how potential participants will be identified and how access to contact information will be obtained. Describe how participants will be recruited. Submit all advertising materials (posters, emails, letters etc.) with this form. Please limit the response for this question to 200 words (unless there are substantial ethical issues to discuss).</i>	
Scientists will specifically include genomic researchers who encounter race, ethnicity and ancestry through the datasets in their work. On the genome campus this includes those who use GWAS, are involved in human variation or genetic interpretation of human data. The pilot study was carried out with scientists from one particular scientific group I have used this to identify further scientists to fit the profile and these will be approached through email introduction and open seminars about the study on the research campus.. Career stage of scientists: The profile of participants will consist of those more senior researchers and heads of research groups. I have identified these as the group who have the most influence over large research questions, funding applications and overview of publications. Within the immersion of scientists however I will be observing and communicating with scientists across career levels of research groups. Although in depth interviews are not planned with all these individuals I will have substantial contact with them.	

Crucial data to address research question three in particular around the ecosystem of genomic research will be accessible through this method.

The public in this case consists of those engaged with genetics research around race, ethnicity, and ancestry who are attending events or related to the Genetic Automata exhibition which includes data categorisation around race, ethnicity and genetic ancestry

This will include invited participants from the Sickle Cell and Thalassemia community and from genetic youth forums where race, ethnicity and ancestry have already been encountered.. A selected sample (two-three) participants at each public event may be asked to take part in semi-structured interviews. The ages of the participants will vary across this sample to provide access to the data from a broader range of public views. However, this data will be used to better inform my reflexivity and interpretations rather than a representative sample of the societal views.

Informed Consent

If the project involves new data collection from human participants, then describe how informed consent will be obtained.

Informed consent is the process where the participants are provided detailed information about the purpose of the research, the risks and benefits to them as participants, what will be expected of them, and then given the opportunity to agree to participate or not. Consent documents and scripts must be written in a language and at a level the subjects will understand. Participants should be given the right to withdraw their consent at any time and for any reason without consequences.

How long will you allow potential participants to decide whether to take part? What arrangements have been made for people who might not adequately understand verbal explanations or written information given in English, or who have special communication needs?

The researcher is responsible for minimizing coercion and undue influence. **Coercion** occurs when there is an overt or implicit threat of harm presented in order to obtain participation, such as when a subject will lose access to certain services if they decline participation, when a student will experience reprisal or disapproval from an instructor, or when an employee will experience reprisal or disapproval from a supervisor. **Undue influence** can occur when there is an offer of an excessive or inappropriate reward to secure participation, such as a large cash payment or other gift.

Describe the procedures used for informed consent for this study. If consent is from **adult participants**, then give details of who will take consent and how it will be done. If informed consent will be sought from **vulnerable groups** (e.g., those with learning difficulties, incarcerated populations) or if **the researcher holds power over participants** (e.g., classroom teacher, line manager, historical colonial structures, etc.), then say how you will ensure that consent is voluntary and fully informed.

If you are recruiting **children or young adults** (aged under 18 years) specify the age-range of participants and describe the arrangements for seeking informed consent from a person with parental responsibility. If you intend to provide children under 16 with information about the study and seek agreement, outline how this process will vary according to their age and level of understanding.

If the research design does not involve informed consent (e.g., covert observation of people in public spaces, posts from social media, online data harvesting, etc.), please provide a clear justification for this decision to deny participant autonomy; you should outline below how participant beneficence, nonmaleficence and justice will be withheld.

Voluntary informed consent will be obtained from all participants in writing. The consent forms and information sheets for both parts of the study are provided in Appendix B of the study protocol.

Consent for interviews will be collected via a consent form that will be distributed with an information sheet prior to the interview, and discussed at the start of the interview. All research participants will be told the purpose of the project and asked to consent to audio recording.

Participants will be provided with contact details for Sasha Henriques, Prof Middleton and Prof Arday. The information and consent process will ensure, as far as possible, that participants understand the scope of the study. as described below. They will be told that they can withdraw from the project at any time if they wish to do so.

Consent will be treated as dynamic and continual, as willingness to enter into a reciprocal relationship and not as equivalent to wanting to participate to any part of the study. Individual consent to observation will be sought at the beginning of each session, or if circumstances change (e.g. someone enters the room during the observation).

Possible public exposure of individuals and teams working on sensitive subjects will be paid attention to and mitigated as much as possible. Data may not be included in published analysis for example or avoidance of using direct quotes from participants. Where possible, such decisions will be negotiated in dialogue with participants themselves and colleagues with experience of conducting research on sensitive topics.

Anonymity and confidentiality

Interviews and transcripts de-identified will be used only in pseudonymised form – both to ensure they are protected from any harm associated with what they say, and to facilitate frank and open discussion. Although anonymity is aimed for it cannot entirely be guaranteed. Genetics is a highlight specific and unique field, failure to maintain anonymity may also be a source of professional caution for the researcher and participant.

Participants will be informed of the implications at the beginning of the study. To protect their confidentiality, extra caution will be used at the stage of reporting. Identifying details of individual researchers will be completely removed from interviews, notes as well as documentary and photographic evidence, and any information that could possibly be used to identify the teams will only be included when strictly necessary to convey an import point. Analysis of data and interviews will be thematic and any mention of the scientific studies or roles that participants have that could be identifying will be described in terms of the themes that emerge from the data which will more general and less identifiable. While the researcher will adhere to these measures, participants themselves they might decide to relinquish their anonymity (e.g. to be co-authors on research outputs or to respond to the research)

Written Consent

If the study involves new data collection from human participants, please indicate whether written consent will be used.

If the study does not involve the collection of new data from human participants, then you may skip this section.

Will consent be written? ☒ Yes ☐ No

If yes, then include a consent form when submitting this form.

If no, then describe and justify an alternative procedure (verbal, electronic etc.):

Will assent procedures be used? ☐ Yes ☒ No

If the research involves assent (e.g., studies involving children where parents have given written consent), then provide an assent script when submitting this form; and also describe assent procedures:

Participant Compensation

If the study involves new data collected from human participants, then describe any compensation (e.g., payment, gifts, course credit) given to participants. Compensation for participation is allowable if it is not excessive or inappropriate.

Will participants be compensated in any way for their involvement in this research? ☐ Yes ☒ No

If yes, then justify the type and amount. Describe how compensation will be disbursed, including for

participants who withdraw from the study. Identify the funding source for the compensation (e.g., personal, Faculty funding, College funding).

Participant Debriefing

Notes: If the study involves new data collection from human participants, then what will participants be told about the research at the end of the study? Submit any participant debriefing information with this form.

13. Vulnerable Populations

Indicate if this research will specifically include participants who are deemed vulnerable in research. Vulnerable participants are usually in circumstances which prevent them from providing fully informed consent. Any studies specifically involving vulnerable populations need to carefully consider the ethics of the consent process and how participant autonomy and the right to withdraw consent will be protected.

Does the research specifically involve recruiting participants with protected characteristics?

- ☐ None. This study does not specifically involve vulnerable populations.
- ☒ Individuals under 18 years of age ☐ Individuals with developmental disability or delays
- ☐ Incarcerated individuals ☐ Medical patients (physical or mental health)
- ☐ Individuals deemed unable to provide their own consent and not in any category above (please specify in the space below)
- ☐ Another group which may be deemed vulnerable in the context of your research (please specify in the space below)

If the research involves participants from vulnerable populations, then describe how participant autonomy, beneficence, nonmaleficence and justice will be upheld. Please limit the response for this question to 200 words (unless there are substantial ethical issues to discuss).

14. Sensitive Topics or Approaches in Research

Indicate if this research will specifically involve topics deemed sensitive for research.

Does the research involve any topics deemed sensitive for research?

- ☐ None. The research does not specifically involve topics deemed sensitive for research.
- ☐ Abuse / neglect / harassment / bullying ☐ Research involving manipulation of the participants' attitudes, opinions, or self-esteem
- ☐ Traumatic events ☐ Risks to participants (e.g., exposure to infectious disease, increased anxiety/stress, speaking against a totalitarian government, etc.)
- ☐ Politics / political attitudes / religious beliefs ☐ Participants do not give consent to be in the research (e.g., covert observation of people in public spaces, posts from social media, harvesting online data, etc.)
- ☒ Discrimination / racism ☐ Deception or withholding any information about the study from participants

☐ Sexuality, sexual images, sex education ☐ Repeated testing or research sessions, over and above what participants might normally experience	☐ Specifically involves populations who have been historically marginalised ☐ Other. Please specify:
If Other, please specify:	
<i>If the study involves any sensitive topics for research, then then provide a rationale for why those issues are included and how risks to participants will be alleviated. Describe how participant autonomy, beneficence, nonmaleficence and justice will be upheld</i> <i>Describe any discomfort or inconvenience that participants may experience. Include information about procedures that for some people could be physically stressful or might impinge on the safety of participants (e.g., noise levels, visual stimuli, distractions, physical activity including performances or activities of any kind that may be challenging) or that for some people could be psychologically stressful (e.g., mood induction procedures, tasks with high failure rate, repetitive questioning or testing,). Describe how a participant debriefing form provided at the end of the study will clarify any deception or provide resources for participants who have found the research topic disturbing or stressful.</i> <i>Please limit the response for this question to 200 words (unless there are substantial ethical issues to discuss).</i>	
Because of the recruitment of participants across different ethnic groups, some may identify as from communities who have been impacted by the harms and colonisation and racism in the past and present. This is distinct in some ways from the experience of the researcher who is also embedded in the subject from an academic perspective. Although there may be benefits for participants to engage in research that opens the dialogue around the science that dispels typological thinking and racist ideology they may be discomfort and trauma from these discussions that the researcher is unable to address. Therefore psychological support has been secured to provide additional support to all participants if these issues arise.	

15. Conflicts of Interest	
<i>Declare whether there are any potential conflicts of interest in this research.</i>	
One or more members of the research team is conducting research on their own place of work (e.g., a teacher conducting research in their own school, Cambridge researchers conducting research about Cambridge, an employee conducting research on their own organisation).	☐ Yes ☒ No
One or more members of the research team or the funding body could financially benefit from the results of this research.	☐ Yes ☒ No
<i>If yes for any of the above questions, then provide details of the conflict of interest and how it will be mitigated. Please limit the response for this question to 200 words (unless there are substantial conflicts of interest to discuss).</i> <i>Please note that the section below engages with power dynamics, including those emerging from coloniality, researcher and participant positionality, and other relevant issues.</i>	

16. Power Dynamics	
<i>Indicate whether there are any differential power structures existing between the researcher and the potential participants.</i>	
There are potential power structures that could influence participants or the research.	☒ Yes ☐ No

One or more members of the research team has power over the participants (e.g., is their classroom teacher, is their line manager)	☐ Yes	☒ No
<i>If the researcher has power over the participants (e.g., is their classroom teacher, line manager, is positioned to benefit from colonial legacies, etc.), then give a clear rationale for how participant autonomy and beneficence will be upheld. What practices of mutual respect and reciprocity are included in the research? How will the research team attend to local norms? If relevant, how are relationships configured to minimise the dynamics of coloniality? Please limit the response for this question to 200 words (unless there are substantial ethical issues to discuss).</i>		
The researcher is a qualified genetic counsellor and represents a large academic institution. There may be a power imbalance in discussions with non-scientific audiences. To address this public groups are selected for their understanding of genomic issues and their experience of race, ethnicity and ancestry in relation to this. This will be achieved by recruitment from public participants who are involved with genetic engagement projects or charity groups for genetic conditions strongly associated with ethnic or ancestry groups eg. Sickle Cell Disease. The researcher will also take part in workshops and prior to any observation to help provide information non-scientific groups may need.		

17. Data Protection and Data Security Please note that if any human participants are involved in this research, this information must be provided and relevant documents (that is, participant information sheet including a data protection notice) must be submitted with this application. This is a legal requirement under UK GDPR. <i>What arrangements are there for data security during and after the study? Describe how the data will be used, both during and after the research. Indicate whether it will be disseminated through publication, presentation or other means, and in what form. Describe where and how the data will be stored.</i> <i>If relevant, indicate the types of personal data that will be recorded and confirm its collection is necessary and proportionate to the aims of the study.</i> <i>Describe any risks associated with any personal data that is collected, shared, or analysed for the purposes of this research – including audio or video recordings.</i> <i>Datasets involving high-risk data should seek advice from the University's Data Protection Officer. High-risk data includes personal data that, if there were a breach, could be extremely harmful to an individual's health, safety, professional or personal life; the profiling of children or vulnerable individuals; direct collection of data without consent; combining or processing data in a way that could have significant effects for individuals.</i> <i>Please confirm that you have, or are in the process of, setting up appropriate data sharing agreements where relevant. Please also confirm that no sharing of identifiable and/or non-anonymised participant will occur before such an agreement is in place.</i> <i>Digital data stored on a computer requires compliance with the General Data Protection Regulation. Please indicate if you have discussed this with a relevant Data Protection Officer and describe any special circumstances that have been identified from that discussion.</i> <i>Please confirm that all applicants have received GDPR training. Say who will have access to participants' personal data during the study and for how long personal data will be stored or accessed after the study has ended, and whether data will be transferred or stored outside the UK.</i> <i>Please outline also any perceived risks in relation to data security, including any issues around potential unauthorized access, corruption, or theft of the digital information generated through the research, especially if the research is to be undertaken in a digital medium.</i> <i>Full information regarding University policies on data protection and data security are available here: https://www.information-compliance.admin.cam.ac.uk/data-protection</i> <i>Please limit the response for this question to 200 words (unless there are substantial ethical issues to discuss).</i>
Please see study protocol Coded/de-identified transcripts, fieldnotes, documents and pictures will be stored for analysis in the Atlas.ti software and within the Atlas.ti web analysis platform. Data communicated to the Atlas.ti web platform is encrypted using TLS authentication and the platform is GDPR compliant (https://atlastihelp.helpscoutdocs.com/article/168-security-at-atlas-ti-web). Data will be transferred to transcribers using their secure data transfer systems. All recordings and transcripts will be identified only by a code, and all identifying information within interviews will be redacted after transcription. Personal information - in the form of the names and contact details of interviewees - will be stored in a separate, password protected Word .docx file. This file will be deleted 12 months after the end of the project. Interview data will be stored and backed up through the university OneDrive, and backed up monthly to an external hard disk drive stored in the office of the lead investigator in the faculty of Education, University of Cambridge. Access to personal information

will only be available to the direct research team (Prof Arday, Prof Middleton and Sasha Henriques) and such personal information (which will consist of names and contact details) will be stored in a password-protected Word .docx file.

Other collaborators at the Kavli Centre will have access to pseudonymised/coded interview transcripts shared using OneDrive and/or the Atlas.ti Web platform.

18. Research Transparency and Reproducibility

Will this study adhere to any open science or open research practices?

Has there been or will there be a pre-registration linked to this research? ☐ Yes ☒ No ☐ N/A

If yes, then provide the link to the pre-registration:

Do you plan to share the data openly (e.g., via Open Science Foundation, Apollo, UK Data Service, etc.) ☒ Yes ☐ No ☐ N/A

If yes, then what plans are in place to protect the identity of the participants?

Long-term preservation of data will be achieved through the UK Data service data repository. This will consist of de-identified transcripts or identifiable transcripts where this has been requested by interviewees. Fieldnotes and other ethnographic data will not be shared due to the difficulty of de-identifying this dataset.

19. High-Risk Research

The University normally insures and indemnifies researchers and departments against risks associated with specific research studies. The University. Insurance cover for studies not deemed high-risk, that is not involving any procedures, interventions, treatments, scans, samples, drugs, etc. (e.g. questionnaires, interviews, data analysis only) is provided under the University's Public Liability and Professional Indemnity Insurance policies. The Faculty of Education will make regular submissions to the University of lists of research studies that are not deemed high risk. These do not require any further action on behalf of the Primary Investigator.

If you deem the research to be high risk, please note that a separate submission for insurance purposes must be made once ethical endorsement has been secured. The 'what happens next' section below outlines the procedure in this case.

Please consult the University policy at the following link and make a determination around the high-risk nature of the research.

<https://www.insurance.admin.cam.ac.uk/insurance-guidance/human-volunteer-studies-and-clinical-trials>

According to your reading of the above policy, do you deem this research to be high risk? ☐ Yes ☒ No

20. Submission Checklist

Confirm what other materials are being submitted with this form.

Please note that if any human participants are involved in this research, participant information sheets to include a data protection notice must be provided alongside the application. This is a legal requirement.

Grant Proposal ☐ Yes ☐ No ☒ Not Applicable

Recruitment Materials ☒ Yes ☐ No ☐ Not Applicable

Participant Information Sheet(s) ☒ Yes ☐ No ☐ Not Applicable

Consent Form(s)	☐ Yes	☐ No	☒ Not Applicable
Assent Script(s) or Form(s)	☐ Yes	☐ No	☒ Not Applicable
Research Materials: Questionnaires	☒ Yes	☐ No	☐ Not Applicable
Research Materials: Interview protocols or schedules	☒ Yes	☐ No	☐ Not Applicable
Research Materials: Other. Please specify here:	☐ Yes	☐ No	☒ Not Applicable
List of co-investigators if space above insufficient	☐ Yes	☐ No	☒ Not Applicable
Other documents. Please specify here:	☐ Yes	☐ No	☒ Not Applicable

21. Primary Investigator Signature and Assurances

By signing below as the Primary Investigator, I certify that to the best of my knowledge:

1. The information provided for this project is correct.
2. I understand that new data collection cannot commence until I have submitted confirmation that this research is covered by University Liability Insurance (or that it is exempt).
3. I agree to conduct this research as described in the attached supporting documents and no other procedures will be used.
4. I will not implement any changes to the protocol (procedures, personnel, etc.), including modifications requested by the funding agency, prior to receiving written approval from the Faculty.
5. I will comply with national and University policies for conducting ethical research.
6. I will be responsible for ensuring that the work of my co-investigator(s)/student researcher(s) complies with this protocol.
7. Any unexpected or otherwise significant events in the course of this study will be promptly reported to the Faculty.
8. I understand that any noncompliance associated with this protocol can result in disciplinary action under the University Research Ethics as well as the Academic Integrity policy of the University.

Signature: 

22. Supervisor / Host Signature and Assurances

By signing below as research student supervisor or visiting scholar host, I certify that to the best of my knowledge:

1. I have had a discussion with the student / visiting scholar about ethical issues related to this study.
2. I have read this full application; it reflects my understanding of what will occur in the research.
3. I have confirmed with the student that they understand that they cannot begin the research until they have submitted confirmation that this research is covered by University Liability Insurance (or that it is exempt).
4. If the student / visiting scholar reports to me any unexpected or otherwise significant ethical events in the course of this study, then I will support them (as needed) in reporting those events to the Faculty.
5. If I am concerned about the ethics at any stage of this project, then I will raise them with the student / visiting scholar as well as the course leadership or research office (as appropriate).

Signature: 

23. Co-Investigator(s) Assurances

By signing below as the Primary Investigator, I confirm that all co-Investigator(s) agree to the following co-investigator assurance:

As co-investigator, I certify that to the best of my knowledge:

1. The information provided for this project is correct

2. *I agree to conduct this research as described in the attached supporting documents and no other procedures will be used.*
3. *I will not implement any changes to the protocol (procedures, personnel, etc.), including modifications requested by the funding agency, prior to receiving expressed permission from the Primary Investigator.*
4. *I will comply with national and University policies for conducting ethical research.*
5. *Any unexpected or otherwise significant events in the course of this study will be promptly reported to the Primary Investigator.*
6. *I understand that any noncompliance associated with this protocol can result in disciplinary action under the University Research Ethics as well as the Academic Integrity policy of the University.*

Signature:

What happens next

Once this form has been completed and all relevant documentation (as per the Submission Checklist above) assembled, please refer to the three stages of the Research Ethics Review Process outlined at the start of this document.

Once the Research Ethics Review Process has completed and ethical endorsement has been received, consult the Risk Assessment information to ascertain whether you must undertake a risk assessment relating to your research.

If you have deemed your research to be high risk according to University rules in response to Question 19, you will need to make a separate submission for insurance purposes, only once ethical endorsement has been secured. Information about this process may be secured from the Research Office (for staff, Research-Office@educ.cam.ac.uk), the Graduate Office (for graduate students, graduate@educ.cam.ac.uk), or the ethics coordinator for the relevant paper (for undergraduate students) and at the link included in Question 19. Once this step has been completed (if relevant), the study may commence.

10.4 Appendix A3



Dear Sasha Henriques

Thank you for submitting your Faculty of Education Personal Risk Assessment.

This letter is to confirm that your risk assessment has been processed and approved by the Faculty of Education. If the circumstances of your research change, you will need to submit a new risk assessment.

We ask that you keep in regular contact with your supervisor and inform the Faculty Risk Team of any situations or changes in your local area that may affect your research or your personal safety.

Yours sincerely,



Patrick Boydell

Departmental
Safety Officer
Faculty of
Education
184 Hills Road
Cambridge
CB28PQ

2022-06-15

10.5 Appendix B Information and Consent

Public participants who volunteered at events were also provided with information sheets and consent/assent forms

10.6 Appendix B1

Participant information Sheet



Sanger Scientist Observation and Interview Information sheet

Title: Race, Ethnicity and Genetic Ancestry in Genomic Research.

This information sheet is about a project that you can choose to take part in. It explains how personal information (data) about you will be processed if you participate.

Before you decide to take part, please read this information sheet carefully. You can have someone help you to read this information if you need to. You can also discuss the project with others if you wish.

If you have any questions, you can contact us using the details at the end of this document.

What is the purpose of the project?

There are challenges around the use of human population categories, race, ethnicity, and ancestry in genomic research. These include how these terms are defined and implemented and how this impacts our understanding of genes and their function. Importantly, the historical and current misuse that surrounds this complexity has resulted in unequal outcomes of genomic research for individuals of different ethnic and geographic populations.

This research aims to investigate what makes genomic research practice around data categorisations 'socially just' when issues around race, ethnicity and ancestry are encountered. The project will look at how genomic researchers and the public who engage with them use these concepts. The understanding of the systems that underpin ethical choices around population descriptors will be used to inform science policy.

What does the project involve?

This study will use an ethnographic approach to understand the experiences and perspectives of the issues surrounding race and ethnicity in genomic research practice. This will include:

1. Multiple interviews of selected scientists at the Wellcome Sanger Institute over the course of six months
2. Six-month observation of scientists in their professional environment at the Wellcome Sanger Institute (e.g., general team meetings and seminars, workshops/meetings around data categorisation)
3. Observation of three public events to observe how the public connects with ideas around race, ethnicity, and ancestry in genomics.
4. Semi-structured interviews with a sample of the public at each event to discuss themes emerging from the analysis of the scientists' data.

Interviews will generally be conducted online over Zoom or in person. Recordings will be made and transcribed and stored on a password-protected laptop. Recordings will be deleted once the analysis of this project is complete (January 2025). Group settings such as meetings and public events may also be recorded when feasible.

The aim is to understand the experience of scientists encountering and dealing with race, ethnicity and genetic ancestry in their research.

Why have I been invited?

You have been asked to take part because you have been identified as having relevant expertise, possibly because of your involvement in research.

What will happen if I decide to take part?

If you agree to participate, you will be interviewed over the course of the next six months three times. Interviews will generally be conducted online over Zoom or in person and will last up to 60 minutes. The time of interviews may vary depending on your availability or your answers.

With your consent, observations of your teams and meetings will also take place. Field notes will be qualitative notes recorded during or soon after observations. They will be factual information (times, dates, behaviour observed, comments by participants) and reflections and interpretations.

Interviews and attendance at meetings and events may be recorded via camera or Zoom. These recordings will be used to support observations from field notes, and the researcher will ask permission before any recordings are taken.

What happens if I do not want to take part or if I change my mind?

You can withdraw from the project and thereby withdraw your consent without giving a reason. If you are unsure about continuing, you can discuss your concerns with the researcher or simply inform them of your decision to withdraw.

You are free to withdraw at any time and do not have to give a reason. If at any point you decide to withdraw, all data concerning you (e.g. recordings, notes or transcripts) will be destroyed to prevent further use. However, data already included in publications or reports can not be withdrawn.

If you decide to withdraw your consent after your opinions have been used, we will inform you where necessary that you have withdrawn your consent. But we won't be able to withdraw your data if it has already been made public (e.g. in a written publication) prior to that point.

Will my involvement be kept confidential?

All information collected will be treated as strictly confidential. No personal data about individual participants will be included in the notes and transcripts or in any outputs (e.g. article, presentation) resulting from the study.

Who will carry out the research?

The research will be conducted by Sasha Henriques, a PhD Student based at the Faculty of Education at the University of Cambridge. Supervised by Prof. Anna Middleton, Cambridge University.

What will happen to the data?

Your personal details will be removed from the data and will not be shared outside of the core research team.

The data will be analysed and used in the preparation of a PhD thesis, academic papers, policy recommendations and public-facing outputs. Summary interview content, verbatim quotes from interviews and excerpts from the notes may be used in publications or other academic outlets such as conferences or meetings. These will not contain your personal details and will be attributed to you using broad descriptors (e.g. Young person at the event).

Hand-written notes will be kept on secure premises. Digital files will be kept encrypted on the Cambridge University OneDrive and downloaded to the researcher's external drive. You will be identified by a code rather than your name. The anonymised data will be shared with the research team at Wellcome Connecting Science and the Kavli Centre for Ethics at the University of Cambridge. Recordings will be deleted once the analysis of this project is complete (January 2025).

Who is funding the research?

This doctoral project is funded by the Sanger Institute Director's Office (Associate Director: Julia Wilson) and is part of the Black PhD Fellowship programme delivered by the Head of Equality, Diversity and Inclusion at the Sanger Institute (Dr Saher Ahmed).

Who is responsible for this project?

The institutional sponsor for this research is the University of Cambridge.

This research has been developed according to the ethics guidelines of the British Educational Research Association and reviewed by the research ethics review process of the Faculty of Education, University of Cambridge. It is the responsibility of Prof. Anna Middleton and Sasha Henriques at Cambridge University.

If you decide to take part in the project, you will be asked to sign a consent form to confirm your agreement to participate and for the Project Team to use your personal information as set out below.

What if something goes wrong?

It is unlikely that there are any health or safety risks with your involvement in this project, given the historical and current inequalities and atrocities associated with the use of race and ethnicity in science. Therefore, the questions asked and topics covered may be uncomfortable for you. Although we will attempt to protect your anonymity, the unique nature of your research or your place of work may impair the extent to which this can be ensured.

If you are unhappy about the way you have been treated or wish to make a complaint, you can contact the GRL Research Governance team on researchgovernance@sanger.ac.uk or the PhD supervisor Anna Middleton am25624@cam.ac.uk.

How do I take part?

If you decide to take part you will be asked to sign a consent form.

What if I want to know more about the research before deciding whether to take part?

Contact the research team for more information:

Chief Investigator: Sasha Henriques

sh2161@cam.ac.uk

Sasha Henriques C/O

University of Cambridge, Sidney Sussex College, Sidney Street, Cambridge CB2 3HU

Or the two PhD supervisors: Professor Anna Middleton am2624@cam.ac.uk and/or Professor Jason Arday jaa80@cam.ac.uk, University of Cambridge

For further general information about the University of Cambridge's use of your personal data as a participant in a research study, please see: <https://www.information-compliance.admin.cam.ac.uk/data-protection/research-participant-data>

If you have taken part and been affected by any of the issues discussed and would like to discuss things further, a qualified counsellor is also available to you.

Please contact _____

Thank you for reading this information sheet and for considering taking part in this research.

10.7 Appendix B2

Scientists consent form



Race, Ethnicity and Genetic Ancestry in Genomic Research: Interview consent

I confirm that I have read and understand the information sheet for the above study and have had the opportunity to ask questions.

I understand that my participation is voluntary and that I am free to withdraw at any time, without giving any reason.

I consent to the researcher observing and recording me/my team at the lab and during conferences, meetings and other activity that is relevant to my/my team's work. I understand that the researcher will take notes of what I/my team say and do, and might gather documentary and photographic evidence of the work being done. I understand that all the data (notes, documents and pictures) will be anonymised (unless stated otherwise below) and that I can ask the research to leave or stop the observations at any point.

I understand that I am being asked to participate in an interview and I consent to that interview being audio or video recorded and transcribed.

I give permission to be identified as the source of any data from/about me (including quotes from my interview, notes and other documentary or photographic evidence) that are used and published in materials for research and public audiences.

OR

I give permission for de-identified, anonymised data from/about me (including quotes from my interview, notes and other documentary or photographic evidence) to be used and published in materials for research and public audiences.

I give permission for the transcripts of my interview (and no other type of data) to be deposited in the University of Cambridge data repository for use by other researchers in the future, according to my preferences on anonymisation above.

I agree to take part in this study.

.....
NAME OF PARTICIPANT

.....
DATE

.....
SIGNATURE

.....
NAME OF PERSON

.....
DATE

.....
SIGNATURE

10.8 Appendix B3

Public Information (adult) sheet



Public information sheet and consent form

Title: Race, Ethnicity Ancestry in Genomic (genetic) Research.

This information sheet is about a project that you can choose to take part in. It explains how personal information (data) about you will be processed if you participate.

Before you decide to take part, please read this information sheet carefully. You can have someone help you to read this information if you need to. You can also discuss the project with others if you wish.

If you have any questions, you can contact us using the details at the end of this document.

Background

In genetic studies, humans are sometimes put in groups (populations) to work out their differences and similarities. Ways to group humans include race, ethnicity, or ancestry.

These groups mean different things and are not always accurate measures of how we are the same or different. The way these groups have been used in research can cause unequal benefits and some people being mistreated.

This research aims to study if and how social justice (doing the best for society) is part of grouping human data in genetic research. The project will look at how genetic researchers and the public understand this. This understanding will be used to inform policy and guidance recommendations at the Wellcome Sanger Institute.

What does the study involve?

The researcher aims to understand the issues surrounding race and ethnicity in genomic research practice and to do this they will:

- 1) Carry out multiple interviews of selected scientists at the Wellcome Sanger Institute over the course of six months
- 2) Undertake a six-month observation of scientists in their professional environment at the Wellcome Sanger Institute (e.g., general team meetings and seminars, workshops/meetings around data categorisation)
- 3) Observe public events to observe how the public connects with ideas around race, ethnicity, and ancestry in genomics.
- 4) Carry out semi-structured interviews with a sample of the public at these events to discuss themes emerging from the analysis of the scientists' data.

Interviews will generally be conducted online over Zoom or in person. Recordings will be made of the interviews and stored on a password-protected laptop. Group settings such as meetings and public events may also be recorded when feasible.

The aim is to understand the experience of scientists encountering and dealing with race, ethnicity and genetic ancestry in their research.

Why have I been invited?

You have been asked to take part because you have been identified as having relevant experience or knowledge of how race, ethnicity and genetic ancestry are used in genetics.

What does participation involve?

You will be asked to sign a consent form. The researcher will take field notes of observations and conversations with you while you attend an art exhibition around genetics and identity.

Conversations may be recorded via Dictaphone, camera or Zoom. These recordings will be used to support observations from field notes.

Field notes will be factual information (times, dates, behaviour observed, comments by participants) and the researcher's reflections and interpretations.

At any point, you can ask the researcher to leave or stop taking notes. You are welcome to review the notes and you can decide to take things off the record at any moment up to the point of analysis.

You may be invited to take part in formal and informal interviews about your experiences and perspectives.

Formal interviews will be audio-recorded, or if conducted online, video-recorded. The interview will take place at a time that is convenient to you. Ideally, it would be face-to-face but can be conducted by telephone, Zoom or Teams. The duration of interviews will vary depending on the time available and the length of your answers, but we expect most interviews to take around 45 minutes to 1 hour.

At any moment up to the point of analysis, you can request to review the transcripts/notes of your interview and decide to take things off the record.

Can I withdraw from the study?

Yes, you are free to withdraw at any time and do not have to give a reason. If at any point you decide to withdraw, all data concerning you (e.g. notes or transcripts) will be destroyed.

Due to the nature of the observations, it might not be possible to destroy all data concerning the activities that you took part in (e.g. data about your contribution to an event will be removed but notes about the rest of the event will be retained).

If you have any concern, you can speak with the researcher who will do their utmost to meet your request.

Will my involvement be kept confidential?

All information collected will be treated as strictly confidential. No personal data about individual participants will be included in the notes and transcripts or in any outputs (e.g. article, presentation) resulting from the study.

Who will carry out the research?

The research will be conducted by Sasha Henriques, a PhD Student based at the Faculty of Education at the University of Cambridge. Supervised by Prof. Anna Middleton, Cambridge University.

What will happen to the data?

Your personal details will be removed from the data and will not be shared outside of the core research team.

The data will be analysed and used in the preparation of a PhD Thesis, academic papers, policy recommendations and public-facing outputs. Summary interview content, Verbatim quotes from interviews and excerpts from the notes may be used in publications or other academic outlets such as conferences or meetings. These will not contain your personal details and will be attributed to you using broad descriptors (e.g. Young person at the event).

Hand-written notes will be kept in secure premises. Digital files will be kept encrypted on the Cambridge University OneDrive and downloaded to the researchers external drive. You will be identified by a code rather than your name. The anonymised data will be shared with the research team at Wellcome Connecting Science and the Kavli Centre for Ethics at the University of Cambridge. Recordings will be deleted once the analysis of this project is complete (January 2026).

Who is funding the research?

The research is funded by Genome Research Limited, Hinxton, UK

Who is responsible for this project?

This research has been developed according to the ethics guidelines of the British Educational Research Association and reviewed by the research ethics review process of the Faculty of Education, University of Cambridge. It is the responsibility of Prof. Anna Middleton and Sasha Henriques at Cambridge University.

If you decide to take part in the project, you will be asked to sign a consent form to confirm your agreement to participate and for the Project Team to use your personal information as set out below.

What if something goes wrong?

We do not anticipate major risk from taking part. However, If you are worried about this research, or if you are concerned about how it is being conducted, you can contact the Graduate Office of Cambridge University. graduate@educ.cam.ac.uk

How do I take part?

If you decide to take part you will be asked to sign a consent form.

What if I want to know more about the research before deciding whether to take part?

Contact the research team for more information:

Chief Investigator: Sasha Henriques

sh2161@cam.ac.uk

Sasha Henriques C/O

University of Cambridge, Sidney Sussex College, Sidney Street, Cambridge CB2 3HU

Or the two PhD supervisors: Professor Anna Middleton am2624@cam.ac.uk and/or Professor Jason Arday jaa80@cam.ac.uk , University of Cambridge

If you have taken part and been affected by any of the issues discussed and would like to discuss things further, a qualified counsellor is also available to you.

Please contact the person organising this event for further details

Thank you for reading this information sheet and for considering taking part in this research.

10.9 Appendix B4

Public Consent (Adult) form



Race, Ethnicity and Genetic Ancestry in Genomic Research: Public Observation and interview consent

Initial

I confirm that I have read and understand the information sheet for the above study and have had the opportunity to ask questions.

☐

I understand that my participation is voluntary and that I am free to withdraw at any time, without giving any reason.

☐

I consent to the researcher observing and recording this event. I understand that the researcher will take notes of what may say and do. I understand that all the data (notes, documents and pictures) will be anonymised (unless stated otherwise below) and that I can ask the research to leave or stop the observations at any point.

☐

I understand that I am being asked to participate in an interview and I consent to that interview being audio or video recorded and transcribed.

☐

I give permission to be identified as the source of any data from/about me (including quotes from my interview, notes and other documentary or photographic evidence) that are used and published in materials for research and public audiences.

☐

OR

I give permission for de-identified, pseudonymised data from/about me (including quotes from my interview, notes and other documentary or photographic evidence) to be used and published in materials for research and public audiences.

☐

I give permission for the transcripts of my interview (and no other type of data) to be deposited in UK Data services data repository for use by other researchers in the future, according to my preferences on pseudonymisation above.

☐



I agree to take part in this study.

NAME OF PARTICIPANT
SIGNATURE

DATE

NAME OF PERSON TAKING CONSENT

DATE

10.10 Appendix B5

Public events notice



Notice that will be shown on the Event details for all public events

Dear attendees, please note that recording and observation will occur at this event as part of academic research. The researcher will carry out the recording using a secure mobile device. Recording will only be of the background of the event and people moving around the space. Fieldnotes will be taken by the researcher as part of the observation. Please notify the event organisers if you do not wish to be recorded.

Research project title: **Race and Ethnicity in Genomic Research**

Research investigator: Sasha Henriques

Summary:

I am a PhD student at the University of Cambridge. This research aims to study if and how social justice (doing the best for society) is part of grouping human data in genomic research. The project will look at how genomic researchers and the public understand this. The understanding of this will be used to inform science policy.

Further information

- Recordings from this event will be stored on The University of Cambridge password-protected laptops and external drive.
- Recordings will be deleted once the analysis of this project is complete (January 2025).
- You have the right to ask for any footage/recording to be withdrawn from the research at any time.

- Anonymised field notes of the event will be taken by **Sasha Henriques**.
- Access to filmed footage and field notes will be limited to Sasha Henriques and academic colleagues and researchers with whom she might collaborate as part of the research process.
- Any summary content or direct quotations that are made available through academic publications or other academic outlets will be anonymised, and care will be taken that other information in the interview that could identify you is not revealed.

Contact Information

If you have any further questions or concerns about this study, please contact Sasha Henriques sh2161@cam.ac.uk

Or the two PhD supervisors, Professor Anna Middleton, am2624@cam.ac.uk or Professor Jason Arday, jaa80@cam.ac.uk

Sasha Henriques C/O, University of Cambridge, Sidney Sussex College, Sidney Street, Cambridge CB2 3HU

If you are worried about this research or if you are concerned about how it is being conducted, you can contact Professor Anna Middleton at am2624@cam.ac.uk.

For information on Cambridge University's use of your personal data as a participant in a research study, please see:

<https://www.information-compliance.admin.cam.ac.uk/data-protection/research-participant-data>

10.11 Appendix B6

Public Information (Under 18) sheet



Public information sheet and consent form

About this information sheet

This information sheet is about a research project that you can choose to take part in. It explains what the project is about and how information about you will be used if you decide to take part.

Please read this information carefully. You can ask an adult you trust, such as a parent, carer, or teacher, to help you read and understand it. You can also talk about the project with other people if you want to.

If you have any questions at any time, you can contact the research team using the details at the end of this document.

Taking part is your choice.

Background

In genetic research, people are sometimes put into groups to help scientists understand similarities and differences between humans. These groups can be called race, ethnicity, or ancestry.

These words can mean different things to different people and they are not always accurate ways of describing who we are. In the past, using these kinds of groups in research has sometimes led to unfair treatment or unequal benefits for some people.

This research looks at whether ideas about social justice, meaning doing what is fair and responsible for society, are part of how human genetic data is grouped and used. The project explores how scientists and members of the public think about race, ethnicity, and ancestry in genetics.

What is learned from this research will be used to help improve guidance and policies at the Wellcome Sanger Institute.

What does the study involve?

The researcher wants to understand how race, ethnicity, and ancestry are talked about and used in genomic research. To do this, they will:

Carry out interviews with scientists working at the Wellcome Sanger Institute over a period of six months

Observe scientists in their normal working environments, such as meetings, seminars, or workshops about data and research practices

Observe public events where genetics, race, ethnicity, or identity are discussed

Carry out interviews with some members of the public at these events to talk about ideas that come up during the research

Interviews will usually take place online using Zoom, or in person. Interviews will be recorded and stored securely on a password protected device. Some group activities, such as meetings or public events, may also be recorded when possible.

The aim is to understand people's experiences and views, not to test or judge them.

Why have I been invited?

You have been invited because you are attending, or have taken part in, an event or activity linked to genetics, identity, race, ethnicity, or ancestry, and your experiences or views may help the researcher understand how these topics are understood by young people.

What does participation involve?

If you decide to take part, you will be asked to give your agreement, called assent. A parent or carer with parental responsibility will also be asked to give consent.

The researcher may take notes while observing activities at an event, such as an art exhibition related to genetics and identity. These notes may include factual details, such as what happens at the event, and the researcher's reflections.

Some conversations may be recorded using audio, video, or online tools such as Zoom. These recordings are used only to help the researcher remember what was said.

You can ask the researcher to stop taking notes or recording at any time. You can also ask to see what has been written about you and ask for parts to be taken off the record at any point before the data is analysed.

You may also be invited to take part in an interview to talk about your experiences or views. Interviews usually last between 45 minutes and one hour and will be arranged at a time that suits you. Interviews can take place in person, online, or by phone.

Can I withdraw from the study?

Yes. Taking part is completely voluntary.

You can change your mind and stop taking part at any time without giving a reason. If you withdraw, any information that identifies you will be destroyed.

Because some parts of the research involve group activities or events, it may not always be possible to remove all information about an event as a whole. However, anything that could identify you personally will be removed wherever possible.

If you are unsure or concerned, you can talk to the researcher, who will do their best to respect your wishes.

Will my involvement be kept confidential?

Yes. Your information will be treated as confidential.

Your name and other identifying details will not be included in notes, transcripts, or publications. When quotes or summaries are used, they will be labelled using general descriptions, such as “young person at the event”.

Who will carry out the research?

The research is being carried out by Sasha Henriques, a PhD student in the Faculty of Education at the University of Cambridge.

The project is supervised by Professor Anna Middleton, University of Cambridge.

What will happen to the data?

Your personal details will be removed from the data and replaced with a code. Only the core research team will have access to the data.

The data will be used for a PhD thesis, academic publications, policy recommendations, and public facing outputs. Short quotes or summaries may be used, but these will not identify you.

Handwritten notes will be stored securely. Digital files will be encrypted and stored on secure University of Cambridge systems.

Anonymised data may be shared with research teams at Wellcome Connecting Science and the Kavli Centre for Ethics at the University of Cambridge.

Recordings will be deleted once the analysis is complete, expected in January 2026.

Who is funding the research?

The research is funded by Genome Research Limited, Hinxton, UK.

Who is responsible for this project?

This project follows the ethics guidelines of the British Educational Research Association and has been reviewed through the Faculty of Education research ethics process at the University of Cambridge.

Responsibility for the project lies with Professor Anna Middleton and Sasha Henriques at the University of Cambridge.

What if something goes wrong?

The research is not expected to cause harm or distress. However, if you are worried about the study or how it is being carried out, you or your parent or carer can contact:

Graduate Office, University of Cambridge

Email: graduate@educ.cam.ac.uk

If any of the topics discussed raise difficult feelings, a qualified counsellor is available. Please ask the person organising the event for details.

How do I take part?

If you decide to take part, you will be asked to sign an assent form. Your parent or carer will also be asked to sign a consent form.

What if I want to know more before deciding?

You or your parent or carer can contact the research team:

Sasha Henriques sh2161@cam.ac.uk

Or the two PhD supervisors, Professor Anna Middleton, am2624@cam.ac.uk or Professor Jason Arday, jaa80@cam.ac.uk

Sasha Henriques C/O, University of Cambridge, Sidney Sussex College, Sidney Street, Cambridge CB2 3HU

If you are worried about this research or if you are concerned about how it is being conducted, you can contact

Professor Anna Middleton at am2624@cam.ac.uk.

For information on Cambridge University's use of your personal data as a participant in a research study, please see:

<https://www.information-compliance.admin.cam.ac.uk/data-protection/research-participant-data>

10.12 Appendix B7

Assent form



Race, Ethnicity and Genetic Ancestry in Genomic Research: Public Observation and interview assent form

Initial

I confirm that I have read and understand the information sheet for the above study and have had the opportunity to ask questions.

☐

I understand that my participation is voluntary and that I am free to withdraw at any time, without giving any reason.

☐

I consent to the researcher observing and recording this event. I understand that the researcher will take notes of what may say and do. I understand that all the data (notes, documents and pictures) will be anonymised (unless stated otherwise below) and that I can ask the research to leave or stop the observations at any point.

☐

I understand that I am being asked to participate in an interview and I consent to that interview being audio or video recorded and transcribed.

☐

I give permission to be identified as the source of any data from/about me (including quotes from my interview, notes and other documentary or photographic evidence) that are used and published in materials for research and public audiences.

☐

OR

I give permission for de-identified, pseudonymised data from/about me (including quotes from my interview, notes and other documentary or photographic evidence) to be used and published in materials for research and public audiences.

☐

I give permission for the transcripts of my interview (and no other type of data) to be deposited in UK Data services data repository for use by other researchers in the future, according to my preferences on pseudonymisation above.

☐

I agree to take part in this study.



NAME OF YOUNG PERSON

DATE

SIGNATURE

NAME OF PERSON TAKING ASSENT

DATE

SIGNATURE

10.13 Appendix B9

Parent/Carer Consent form



Race, Ethnicity and Genetic Ancestry in Genomic Research: Public Observation and interview Parent /Carer consent form

Initial

I confirm that I have read and understand the information sheet for the above study and have had the opportunity to ask questions.

☐

I understand that my child's participation is voluntary and that they may withdraw at any time without giving a reason, and without any negative consequences.

☐

I understand that the researcher may observe and record the event and may take notes about what my child says and does. I understand that my child may ask the researcher to stop observing or recording them at any time.

☐

I understand that data collected about my child, including notes, recordings, photographs, or documents, will be anonymised where possible, unless stated otherwise below.

☐

I understand that my child may be invited to take part in an interview and I consent to that interview being audio or video recorded and transcribed.

☐

I give permission for information from/about my child, including quotes from interviews, notes, or images, to be used in research and public outputs where they are described in general terms and not identifiable (for example, "young person at the event")

☐

OR

I give permission for only de-identified or pseudonymised information from/about my child to be used in research and public outputs, so that they cannot be recognised.

☐

I give permission for the written transcript of my child’s interview only (and no other data) to be deposited in a UK Data Services repository for use by other researchers in the future, in line with the anonymisation choice above.

☐

I confirm that I have parental responsibility for the young person named below.

☐

I consent to my child taking part in this study

☐

NAME OF YOUNG PERSON	DATE	SIGNATURE
NAME OF PARENT / CARER	DATE	SIGNATURE
NAME OF PERSON TAKING ASSENT	DATE	SIGNATURE

10.14 Appendix C Table of Public Facing Events

I continued to participate in public-facing events on genetics and ethics, facilitating deliberations with patient and public groups, and collaborating with community projects

Date	Activity	Description
28.7.22	Utopia Now	A public panel event to celebrate the launch of a zine created by young people. The zine was the culmination of workshops embedded in the community, which explored themes, hopes, and fears for the future. I was included in a group of scientists invited to engage with the young people about the important social and ethical questions involved in our research/ Views and questions were also shared by the invited guests in the audience, who were family, friends and members of the public who attended the theatre.
13.9.22	UK Citizens' Jury on Genome Editing	People with lived experience of genetic disease provided insight into how those most directly affected understand the benefits, risks, and potential wider harms associated with emerging genomic technologies. Through a four day citizens' jury, participants deliberated on whether there are circumstances under which UK law should permit intentional genome editing of human embryos for serious genetic conditions. These discussions foregrounded how experiential knowledge can be integrated into policy making, and how ethical considerations are shaped by the perspectives of those living with inherited genetic conditions.
07.06.23	Film: <i>Genetic Automata: God Mode</i>	Worked the artists to support research into the scientific and social themes explored in the film. This involved exploring the campus, taking images for the film and facilitating talks with scientists, activity. As part of the process of translating questions of race, technology, and power into a public-facing creative form, the experience informed my curiosity around how genomic narratives are constructed and received beyond academic settings.

Date	Activity	Description
16.08.23	Wellcome Collection Youth Panel: Exploring Genetic Automata	Hosted at Sanger, I spent a day with the young panel while they explored the campus. Informal and formal talks were held with young people, their responsible adults and parents about genetics, ethics and representation in science. This engagement allowed exploration of themes of how genomic authority accepted, questioned and reinterpreted by publics who are often interested but may be excluded from formal scientific debate.
15.09.23	Wellcome Collection: Passing Down	This evening event was hosted at the Wellcome Collection. I participated in a public panel discussion accompanying the Genetic Automata programme. Situating genomic research within artistic, historical, and community narratives that challenge dominant scientific framings of race and inheritance was a significant aspect throughout my study.
13–21.09.23	Ada Lovelace Institute and Nuffield Council Public Dialogue	As part of a professional advisory panel, I volunteered to participate in a deliberative public dialogue on AI-powered genomic healthcare. By observing how the public reasons about governance, risk, and justice when engaging with emerging genomic technologies, it is challenging to interpret how and why scientists could engage with these same subjects
08.10.23–28.01.24	Black Cultural Archives Exhibition: Challenging the Narrative	As part of this public exhibition on Black resistance to scientific bias, I contributed through consultation and visited the gallery on two occasions after the exhibition opened to observe and discuss visitors' responses to its themes. Conversations with the public provided insight into how experiences of racialisation, power, and the UK context of scientific authority were understood and articulated, directly informing and reinforcing key analytic themes developed in this thesis.
24.01.24	Festival of Genomics and Biodata Panel	This public genomics event contributed to broader thinking about the wider genomics ecosystem. As part of a panel on health inequities in genomics, I engaged with an audience of scientists, clinicians, and patient advocates. The discussion highlighted how research practices connect to wider structural inequalities in genomics and healthcare,

Date	Activity	Description
		reinforcing the relevance of my thesis to ongoing debates across research, clinical, and advocacy contexts.
25.01.24	Black Cultural Archives: In Conversation	I participated in a public conversation on genetics, eugenics, and institutional responsibility. This engagement reinforced my thesis by connecting contemporary genomics to longer histories of racial harm and resistance.
09.02.24	Queen Mary University of London Teaching	I spoke with undergraduate History of Science students visiting the <i>Genetic Automata</i> exhibition. Their questions and experiences of the exhibition shaped my analysis of how historical injustice, previously discussed with scientists in interviews, was understood, learned about, and valued in relation to research directions and the distribution of benefits.
21.03.24	Mixed Museum: Brown Babies	I spoke with descendants of Black US servicemen about DNA testing and race. This engagement informed my thesis by foregrounding lived experiences that complicate scientific narratives of ancestry.
13.05.24	Pint of Science Festival	I delivered a public talk on race, categorisation, and ethics in genetics. This engagement tested how my thesis arguments resonated with non-specialist audiences.

10.15 Appendix D Interview Topics Guide

Overview

Interviews were semi-structured and exploratory. The guide below outlines the core topic areas that informed interviews across the study. Questions were adapted over time in response to emerging themes, prior interviews, and ongoing ethnographic observation.

1. Introduction and record consent at each interaction

2. Background and Role

- Can you describe your current role and responsibilities?
- How would you describe your field of research?
- What are the primary aims or goals of your work?
- In an ideal scenario, what do you think your field might achieve?
- How do you describe yourself - use the descriptors you find comfortable do not present as a tick box but part of the conversation

3. Research Practice and Challenges

- What are some of the biggest challenges in your work?
- Do these challenges tend to be technical, organisational, or collaborative
- What challenges do you see for your institution?

24. What challenges do you see for science more broadly?

1. Diversity, Populations, and Categorisation

- Does lack of diversity in data or samples arise in your work?
- What does a “diverse population” mean to you in a research context?
- How do you understand and use terms such as race, ethnicity, ancestry, and population?
- How are population categories constructed in your field – question about how they work
- When are population labels used, avoided, or removed, and why?
- What are the scientific implications of excluding populations without appropriate controls?

- What are the implications for the populations themselves?

4. Ethics in Research Practice

- How do you define ethics in the context of your work?
- Where did you learn about ethics as a scientist?
- What ethical guidance do you rely on in day-to-day research practice?
- What ethical principles guide your use of human population data?
- How do ethical issues arise unexpectedly in your work?
- What does “overly cautious ethics” mean to you, if anything?
- How do you assess whether collaborators work to similar ethical standards?
- What practices enable ethical behaviour, or allow unethical practices to persist?

5. Responsibility, Harm, and Governance

- Do you think genetics or genomic research has responsibilities beyond technical accuracy?
- In what ways might genomic research cause harm, intended or unintended?
- Does your work generate knowledge that could be misused?
- What mechanisms exist to prevent or manage such risks?
- How are responsibilities distributed across individuals, teams, and institutions?

6. History, Society, and Social Change

- Why do you think race remains an evocative category in science?
- Do you think historical knowledge is important for ethical scientific practice?
- How do changes in society shape what is considered ethical in research?
- How do you understand the relationship between science and social justice?
- Do you see social justice as relevant to your own work? If so, how?

7. Translation, Benefit, and Impact

- Who do you think ultimately benefits from your research?

- How do research outcomes help populations that are under-represented in datasets?
- What methods exist to support smaller or marginalised populations within large-scale studies?
- What barriers limit the use of these methods in practice?
- How close do you feel your work is to clinical or public benefit?

8. Follow-up and Reflection (where applicable)

- What has been happening since we last spoke?
- Have issues around data, diversity, or ethics arisen recently?
- Are there topics we discussed previously that you have since reflected on differently?
- Are there ethical questions in your field that you think are routinely avoided or deferred?
- How do you think patients or publics view the kind of research you do, and does that matter to you? the research environment.

Closing

- Is there anything we have not discussed that you think is important?

10.16 Appendix E Library process for Sanger early publication review

The library process involves systematic searches of PubMed, Europe PMC, Web of Science, Wellcome Open Research, and bioRxiv/medRxiv. Searches are conducted primarily in the author address or affiliation fields, in order to capture outputs from the following: the Wellcome Sanger Institute, Wellcome Connecting Science, the Genomic Surveillance Unit, Health Data Research UK (Cambridge only), Open Targets, the European Bioinformatics Institute (EMBL-EBI), and Elixir.

Coverage is deliberately broad. The library aims to capture the entire publication output of these units, with only minor supplementation from non-staff papers occasionally suggested by colleagues. Staff suggestions, however, were rare. Published articles were comprehensively covered, but coverage of preprints was acknowledged to be patchy, and book chapters or conference abstracts were excluded. Overall, coverage of peer-reviewed research was considered strong and systematic.

Saved search strategies were maintained across databases to ensure consistent capture. The table 2. below summarises the main search formulations used by the library in 2023.

Database	Search Strategy (Affiliation/Author fields)	Coverage Focus
PubMed	"Sanger Institute" OR "Sanger Institute COVID-19 Surveillance Team" OR "Genomic Surveillance Unit" OR "Genomic Pathogen Surveillance" OR "Health Data Research UK" OR "Open Targets" OR "Connecting Science" OR CB10 1SA/CB10 1HH (postal codes)	Sanger, HDR-UK (Cambridge), Connecting Science, Open Targets
EMBL-EBI	"CB10 1SD" OR "EMBL-EBI" OR "European Bioinformatics" OR "Elixir" OR "Open Targets" (Cambridge)	EMBL-EBI, Elixir, Open Targets
Web of Science	Combinations of OG/OO/AD/SG fields for "Wellcome Trust Sanger Institute", "Connecting Science", "Health Data Research UK", "Genomic Pathogen Surveillance Unit", "Open Targets", "Sanger COVID-19 Surveillance"	Sanger, HDR-UK, Open Targets, Surveillance Units
Europe PMC	"Sanger Institute" OR "Genomic Surveillance Unit/Pathogen Surveillance" OR "Health Data Research UK" OR "Open Targets" OR "Genomic Surveillance Laboratory Platforms" OR "Wellcome Sanger Institute COVID-19 Surveillance Team"	Sanger, HDR-UK, Open Targets, Surveillance Labs
Wellcome Open Research	Tree of Life data notes separately tracked	Wellcome-funded outputs
bioRxiv / medRxiv	Manual searches for Sanger-related terms	Preprints

Table2. Main search formulations used by the Wellcome Sanger Institute library in 2023 to find weekly publications

10.17 Appendix F Wellcome Sanger Institute papers

Link to papers for situational context and not included in the main literature review.

<https://docs.google.com/spreadsheets/d/1mVHMa4rRPfEjAy5s-qZam-UMfhKuf5padvIjZjqxWp4/edit?usp=sharing>

10.18 Appendix G Sanger fellowship proposal and diverse data commitment documents

10.19 Appendix G1

Proposal for fellowship * '###' redacted



Board paper number: ####

Board Meeting: #####

Author: Saher Ahmed/Anna Middleton

To be called for paper: None

Proposal for a new Post-Doctoral Fellowship and Doctoral project exploring race, equity and research good practice

Category of paper (<i>Science / Strategic / Oversight / Operations</i>)	Operations
--	-------------------

Board Action (<i>Approve / Note / Discuss and comment</i>)	Approve
---	----------------

	GRL Meeting Paper Journey	Date	Status
1	Proposal for Post-doctoral Fellowship (circulated & discussed with XXXX) Doctoral project exploring race, equity and research good practice (XXX)	20 April 2021	Complete
2	To OB for discussion/approval	May 2021	Complete
3	BoM update and agreement		Pending

Executive Summary

There are two strands to this proposal:

Proposal 1 – Post-doctoral Fellowship aimed at people from a Black British background

It is proposed that we initiate a new Post-Doctoral Fellowship aimed at people from a Black British background. This will run in a similar way to the Janet Thornton Fellowship, with at least one Fellowship awarded every year (see below). This scheme is not intended to replace work that we need to do to improve our processes to ensure that excellence, regardless of background, can be supported, brought in and nurtured at Sanger. This is positive action to acknowledge and recognise the disadvantage that our Black colleagues face in all walks of life, including academia, due to discrimination and racism. To initiate our own scheme would demonstrate that we are actively trying to be agents for positive change and setting a standard for others. We will set-up a steering group with both internal and external representation to provide advice and help shape the scheme and the Fellows will have full access to mentorship, support and training to enable them to flourish. We will also explore complimentary partnerships with external organisations.

Proposal 2 – PhD project exploring race, equity and research good practice

A new joint collaborative project across GRL has been set-up between the Sanger and Connecting Science (XXXX's team) to support a doctoral researcher (PhD candidate), to explore whether it is possible to create race and ethnicity categories that are helpful for genomic science, and culturally acceptable for different global populations. This project will also investigate what type of studies and analyses should take ethnicity into consideration and how, and if there are projects that should not.

Key to the approach is providing an opportunity for a doctoral researcher from a Black or other under-represented Black, Asian and Minority, Ethnic (BAME) background to both develop their career in this field, and to utilise their lived experience as part of a minority group, to help connect with specific communities so that voices of under-represented research participants can be heard.

Wellcome Connecting Science Society and Ethics Research Group supervises PhD students in the social sciences at the University of Cambridge, however, as these candidates are not doing molecular projects they are currently ineligible to receive support through the current Sanger PhD programme and so are all personally funded by individuals or externally funded (e.g. via NIHR). Within the next QQ, the Society and Ethics Research group have already costed funding for 1 PhD student and within this proposal, are looking to finance 2 BAME PhD students, to start in October 2022.

Background

Racial inequalities persist in society and we see this mirrored in Higher Education, where there is a lack of diversity amongst senior leaders and an under-representation of BAME people progressing within the sector. When we disaggregate the data, we see that Black British people are particularly disadvantaged and this is propagated along the academic pipeline - from degree awarding gaps at undergraduate level, through to fewer numbers at postgraduate and professoriate levels and lower success rates with funding councils and granting agencies (see Appendix A for a data snapshot).

The significant bias in genetic research datasets towards 'White European' samples has been recognised, with many global calls to address this. However if we acknowledge that race is predominantly a social construct, then the term 'White' has little scientific value as a descriptor, and European could include individuals from a wide range of backgrounds, irrespective of race. For example - "I'm White Northern European" or "I have Ashkenazi Jewish ancestry" or "I'm mixed race", could all refer to the same person. Within the BAME doctoral research projects, we are looking to explore whether it is possible to create race and ethnicity categories that are helpful for genomic science, and culturally acceptable for different global populations. This project will also investigate what type of studies and analyses should take ethnicity into consideration and how, and if there are projects that should not. This will be supported by the Institute's Research Culture Lead, Hayley Clissold and Head of Policy, Sarion Bowers.

Processes

It is envisaged that the Post-Doctoral Fellowship scheme will run in a similar way to the Janet Thornton Fellowship where we build on the candidates' skills and experiences and co-create a project. E.g. candidates would apply to a broad project outline set by Sanger Faculty and if shortlisted, work with them to develop a project proposal. Shortlisted candidates will be interviewed by a panel, including members of Faculty, the steering group and the Head of EDI. Unsuccessful candidates will be offered the opportunity to be mentored and supported to apply again.

The BAME Doctoral scheme is a partnership between the Sanger Institute and Wellcome Connecting Science Society and Ethics Research Group and will enable candidates to complete a social sciences in the context of genomics PhD project awarded through the Faculty of Education, University of Cambridge. The projects will be co-created between the EDI team, Wellcome Connecting Science and the Policy team at Sanger Institute.

Finances with CFO Note on Funding

Post-doctoral Fellowship:

Recruiting 1 Fellow each year, by Year 3, we will have 3 Fellows in post each year going forwards:

Cost for 1 FTE	Year 1	Year 2	Year 3	Year 4	Year 5	
Salary	£37,000	£38,110	£39,253	£40,431	£41,644	
On costs	£10,360	£10,670	£10,990	£11,321	£11,660	
Consumables	£18,000	£18,000	£18,000	£18,000	£18,000	
Laptop (year 1 only)	£2,000	£2,000	£2,000	£2,000	£2,000	
Travel	£1,500	£1,500	£1,500	£1,500	£1,500	
	£68,860	£70,280	£71,743	£73,251	£74,804	
Projected costs for a rolling programme	Year 1 - 1 Fellow in post	Year 2 - 2 Fellows in post	Year 3 - 3 Fellows in post	Year 4 - 3 Fellows in post	Year 4 - 3 Fellows in post	Total
Costs	£68,860	£138,560	£211,229	£215,753	£220,411	£854,814
Mentoring/careers support/steering group honorarium costs	£6,000	£6,000	£6,000	£6,000	£6,000	£30,000
Total						£884,814

Doctoral Fellows

To ensure that financial barriers do not deter promising BAME candidates, with family or other responsibilities, from applying, we intend to offer a more generous stipend equivalent to the salary of an early career researcher, in addition to a research budget, and university fees over three years. The rationale for this stipend has been discussed with the Wellcome EDI team and they have offered their full endorsement. We have also discussed this with the Faculty of Education, University of Cambridge, where the PhD candidates will be registered, and in line with their own positive action programmes, this enhanced stipend is also supported. We also already have one BAME PhD student who is registered to start their PhD studies this year, who will be receiving the same enhanced stipend (the funding for this has already been committed by Sanger DO via Julia Wilson, Sarion Bowers and Connecting Science DO via Julian Rayner). Thus precedent has been set and we are asking, as a one-off, one-off for 2 additional PhD students only under this specific scheme.

Recruiting 2 PhD students FTE in Year 2 only (i.e. to start in the 2022 October academic year)

Cost for 2 FTE	Year 1	Year 2	Year 3	Year 4	Year 5	
Stipend	0	60,000	60,000	60,000	0	
University fees	0	20,000	20,000	20,000	0	

Research costs	0	20,000	20,000	20,000		
Laptop (year 1 only)	0	4,000	0	0	0	
Travel	0	2,000	2,000	2,000	0	
Mentoring/careers support/steering group honourarium costs	0	6,000	6,000	6,000	0	
Total	0	112,000	108,000	108,000	0	£328,000

Total costs over 5 years for both programmes = £1,218,814

CFO Note on Funding

Although there was no provision for this project in the QQ Submission to Wellcome in Dec19, the full £1.2m was included as a funding “pressure” in the June BOM review of funding for the next QQ. At that meeting it was decided to accept the additional cost of this project along with others “at risk” if necessary without clear sight of funding source.

Next Steps

With Operations Board approval, we will engage with BoM to sign off the proposals and work with the Finance team to secure funding and will set-up and mobilise the steering group to help develop the approaches and processes. We have strong support from Sanger Faculty and enthusiasm from external senior Black academics who we have connected with to explore both of these concepts.

10.20 Appendix G2

Board of Management paper on call for data diversity (XXX = redacted)



ARP paper number: x Board Meeting: xxxx

Author: xxxx To be called for paper: xxxx

Title here

Category of paper <i>(Science / Strategic / Oversight / Operations)</i>	xxx
Board Action <i>(Approve / Note / Discuss and comment)</i>	xxx

Purpose

The conditions of our core grant award from Wellcome requires that we develop a strategy around data diversity by May 2024. This paper will provide a high-level outline of the key components of such a strategy and an outline plan for development. The board is asked to note this paper and provide feedback.

Background

The Wellcome Sanger Institute is one of the largest producers and holders of genomic data in the world. Since the Human Genome Project, the question of whose genome(s) should be sequenced, to represent humanity, has been a recurring theme. As genomics has scaled, projects have moved to exponentially larger numbers of genomes. Internationally, countries have been increasing their genomics capacity with high income countries having significantly greater genomics capacity than counterparts in low and middle income countries. As genomics capacity has developed, research in genomics has encountered the

same issues with recruiting and engaging participants from under-represented groups that research in general faces. This concentration of genomic capacity and bias in participant recruitment has resulted in the majority of global genomic data being derived from participants of European-descent. The precise situation within Sanger's own data is unknown at this point, but the assumption is that our data reflects this general bias; noting that there have been projects specifically focussing on different populations and communities, such as the Born in Bradford study and studies of tribal groups in Papua New Guinea.

As well as a bias towards European participants, research has historically been biased towards men, with women often being entirely excluded from participating in research for a variety of reasons including fluctuations in hormone cycles and possible effects on future and existing pregnancies.

While it is likely participation is less of an issue within genomics, although barriers to women participating in research remain, if analyses are not sex-disaggregated important biological differences are not captured, to the detriment of women's health.

The lack of "diversity" in research, and genomics in particular, has become an increasingly pressing issue within the community. As genomic research moves into real world applications, often partnered with other big data innovations such as AI, it has become an important issue of social justice that the biases in data are addressed in order to prevent or avoid exacerbating harms in communities already under-served by research. Wellcome is seeking to address this issue in the research it funds and as a result one condition of our award states,

*"The Sanger Institute generates large-scale genomic and molecular data for the global scientific and research community. The Institute is asked to develop and implement an organizational strategy around data diversity, which encompasses (1) considering the representativeness, generalizability and diversity of the data it generates based on the populations, cohorts and questions that are being addressed, and (2) identifying gaps and implementing solutions for improving data representativeness, generalizability and diversity where necessary. A draft strategy should be submitted by the Sanger Institute to Wellcome by **24 May 2024**, and an approved plan submitted by **22 November 2024**."*

Proposed plan

The question being asked by Wellcome is not straightforward and is open to interpretation and debate. While the initial framing of the requirement is around "diversity in data", the specific focus on the representativeness and generalisability of the data takes us beyond experimental design and into scientific strategy.

As such, there will be two broad components to the strategy. The first component can loosely be considered "experimental design", ie the tools and guidelines for how we design and conduct our research to create a "diverse" research output with findings that are applicable to the populations that need to benefit from the research. The second component

is our scientific strategy, ie what research questions are we asking. The first component can help shape the second component, through the provision of tools and frameworks to support decision making. Our research can never be considered representative, diverse and generalizable if we only address diversity in participant recruitment but do not use that information in our analyses. Similarly diverse research participant recruitment is of limited use if we are not asking research questions that represent a range of populations and cohorts.

The Policy Team, within the Director's Office, will take responsibility for the first component, and the rest of the paper will largely focus on this aspect. The science strategy is set every five years and for now is outside the immediate scope of this project.

Developing a Strategy - Experimental Design

In order to develop tools, frameworks and guidelines to direct experimental design and produce "diverse data" we will need to work through four high level steps:

Definitions – we will need to agree a set of definitions for what is meant by "representative", "generalizable" and "diverse", and how they apply to experimental design. We must ensure that there is agreement on terminology. Some of these terms may vary in their meaning based on the context they are being used in, making it important to have agreement on their use and meaning.

Baseline – in order to identify gap and propose solutions we will need to develop a baseline measurement of what our current position is. This presents a lot of challenges, but we propose to develop a pragmatic approach which considers broad categories to provide a basic outline. This should allow us to identify gaps and allow us to loosely measure improvements without being overly burdensome.

Tools and frameworks – there is already significant activity in this space. XXXXX the GA4GH Policy Analyst who sits in the Sanger Policy Team co-leads a working group working on diversity in data. Sasha Henriques is a PhD student in Anna Middleton's group who is being funded by the DO to work on representation in data. Genomics England have a partnership strategy to support organisations working on these questions to join up and work together. We will explore what tools and resources are already available to the community that we can utilise, before developing any of our own.

Policies and Guidelines – the end product of this work will be to develop policies and guidelines for Sanger Institute research and researchers to support experimental design for diverse data and ensures our research is representative and generalizable.

Considerations and Recommendations

Sex and Gender – the majority of the work being undertaken already in the field looking at diversity in data is considering the issues of race, ethnicity and ancestry. However, sex and gender is an extremely important issue and one that is largely neglected. The scientific rationale for considering biological sex is strong and significantly less complex than for ethnicity and ancestry.

Recommendation: we use biological sex as a pilot for developing a strategy on experimental design.

Scientific Racism – one of the most challenging aspects of considering race, ethnicity and ancestry in genomic research is considering how and when categories that may be linked to race are used in our own research and also how that research could be misused by others. There is a tension between the social construct of race and the use of categories for race and ethnicity, which are often self-identified, for genetic analyses.

Recommendation: we work with our Race Equity Network, and other relevant stakeholders including faculty and BAME staff and students to identify actions we can take to prevent misuse of our data and counter scientific racism. We should also work with these groups to develop a strong justification for gathering diverse data.

Representation – in developing a strategy on diversity in data we must ensure that relevant stakeholders are consulted and engaged. This applies to both questions of biological sex or gender and ancestry and ethnicity and is critical where we seek to prevent harm through discrimination or counter historic social injustice.

Recommendation: identify and evaluate different possibilities for stakeholder engagement

e.g. advisory board versus focus groups.

Wellcome – although this work is a condition of our funding, Wellcome have not given much indication of their expectations in this space. Ongoing discussions will be essential to ensure our work meets their expectations and we should seek to influence their views.

Discussions at other meetings and boards – within GRL reporting structure

Although this has not been discussed at other boards and meeting at this point, engaging with Connecting Science, the EDI team and different internal and external groups, including the EDI Forum will be an integral part of this work.

Risks

Failure to develop a strategy on diversity in data would be a breach of our funding conditions, resulting in sanctions or reduced funding in future.

Potential Conflicts of Interest

N/A

Conclusions and Next steps Conclusions

The Sanger Institute must develop a strategy on diversity in data by May 2024. We propose a series of steps to help us develop the tools and guidelines that underpin a strategy, and we will start by piloting these steps with sex and gender.

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